26016 Concourse viewing platform

Appendix - IDEA StatiCa Connection reports, all r 8.5 m joints (current design)

Compiled 13 Nov 2026. One IDEA StatiCa 26.0 detailed report per joint, generated with the IDEA Connection API from the calculated .ideaCon projects (stored CBFEM results, no re-analysis). Plates = plastic strain against the 5 % limit; bolts and welds = utilisation. Status from each report's own Summary page. ULS combinations 101-130, 135, 141-148, 151-154 (critical cases per joint). Level 1 A and B joints are on the v19m forces (no flange plates, ring-end Mz released); strut joints and all Level 2 joints are still on the v15 forces. Not included: L2-R12-E1_000 (v14 only), superseded variants (see inventory.csv).

Joint	Location	Model / force version	Plates %	Bolts %	Welds %	Status	Page
L1-R8-A_000	Level 1, r 8.5 m, 0 deg	v19m, no flange plates	0.0	13.6	73.3	PASS	2
L1-R8-A_080	Level 1, r 8.5 m, 80 deg	v19m, no flange plates	0.0	13.4	62.5	PASS	13
L1-R8-A_120	Level 1, r 8.5 m, 120 deg	v19m, no flange plates	0.0	13.0	74.7	PASS	24
L1-R8-A_160	Level 1, r 8.5 m, 160 deg	v19m, no flange plates	0.0	13.7	76.5	PASS	35
L1-R8-A_200	Level 1, r 8.5 m, 200 deg	v19m, no flange plates	0.0	13.3	64.2	PASS	46
L1-R8-A_240	Level 1, r 8.5 m, 240 deg	v19m, no flange plates	0.0	13.1	72.0	PASS	57
L1-R8-A_320	Level 1, r 8.5 m, 320 deg	v19m, no flange plates	0.0	11.7	63.8	PASS	68
L1-R8-B_040	Level 1, r 8.5 m, 40 deg	v19m, no flange plates	0.2	18.1	73.9	PASS	79
L1-R8-B_280	Level 1, r 8.5 m, 280 deg	v19m, no flange plates	0.6	17.8	93.8	PASS	91
L1-R8-S_020	Level 1, r 8.5 m, 20 deg	v15 as-is, strut 10 deg ring	0.1	26.7	98.2	PASS	103
L1-R8-S_100	Level 1, r 8.5 m, 100 deg	v15 as-is, strut 10 deg ring	0.1	27.0	98.1	PASS	116
L1-R8-S_140	Level 1, r 8.5 m, 140 deg	v15 as-is, strut 10 deg ring	0.1	22.0	98.1	PASS	129
L1-R8-S_180	Level 1, r 8.5 m, 180 deg	v15 as-is, strut 10 deg ring	0.1	28.6	98.1	PASS	142
L1-R8-S_220	Level 1, r 8.5 m, 220 deg	v15 as-is, strut 10 deg ring	0.1	21.8	98.1	PASS	155
L1-R8-S_300	Level 1, r 8.5 m, 300 deg	v15 as-is, strut 10 deg ring	0.7	29.6	98.3	PASS	168
L1-R8-S_340	Level 1, r 8.5 m, 340 deg	v15 as-is, strut 10 deg ring	0.1	25.4	98.2	PASS	181
L1-R8-T_060	Level 1, r 8.5 m, 60 deg	v15 as-is, strut 10 deg ring	0.4	27.2	98.4	PASS	194
L1-R8-T_260	Level 1, r 8.5 m, 260 deg	v15 as-is, strut 10 deg ring	1.7	25.9	99.3	PASS	207
L2-R8-A_000	Level 2, r 8.5 m, 0 deg	v15 as-is, haunch flush M20	0.0	0.9	19.1	PASS	220
L2-R8-A_020	Level 2, r 8.5 m, 20 deg	v15 as-is, haunch flush M20	0.0	0.9	18.9	PASS	232
L2-R8-A_040	Level 2, r 8.5 m, 40 deg	v15 as-is, haunch flush M20	0.0	0.7	18.9	PASS	244
L2-R8-A_100	Level 2, r 8.5 m, 100 deg	v15 as-is, haunch flush M20	0.0	2.1	29.4	PASS	256
L2-R8-A_120	Level 2, r 8.5 m, 120 deg	v15 as-is, haunch flush M20	0.0	1.2	29.8	PASS	268
L2-R8-A_140	Level 2, r 8.5 m, 140 deg	v15 as-is, haunch flush M20	0.0	1.2	29.2	PASS	280
L2-R8-A_160	Level 2, r 8.5 m, 160 deg	v15 as-is, haunch flush M20	0.0	1.1	28.3	PASS	292
L2-R8-A_180	Level 2, r 8.5 m, 180 deg	v15 as-is, haunch flush M20	0.0	1.1	28.2	PASS	304
L2-R8-A_200	Level 2, r 8.5 m, 200 deg	v15 as-is, haunch flush M20	0.0	1.2	23.7	PASS	316
L2-R8-A_220	Level 2, r 8.5 m, 220 deg	v15 as-is, haunch flush M20	0.0	1.2	23.9	PASS	328
L2-R8-A_240	Level 2, r 8.5 m, 240 deg	v15 as-is, haunch flush M20	0.0	2.3	28.5	PASS	340
L2-R8-A_300	Level 2, r 8.5 m, 300 deg	v15 as-is, haunch flush M20	0.0	0.6	19.2	PASS	352
L2-R8-A_320	Level 2, r 8.5 m, 320 deg	v15 as-is, haunch flush M20	0.0	1.0	18.8	PASS	364
L2-R8-A_340	Level 2, r 8.5 m, 340 deg	v15 as-is, haunch flush M20	0.0	0.9	18.7	PASS	376
L2-R8-B_060	Level 2, r 8.5 m, 60 deg	v15 as-is, haunch flush M20	0.0	6.0	31.2	PASS	388
L2-R8-B_080	Level 2, r 8.5 m, 80 deg	v15 as-is, haunch flush M20	0.0	8.6	86.2	PASS	401
L2-R8-B_260	Level 2, r 8.5 m, 260 deg	v15 as-is, haunch flush M20	0.0	7.7	65.4	PASS	414
L2-R8-B_280	Level 2, r 8.5 m, 280 deg	v15 as-is, haunch flush M20	0.0	1.8	27.4	PASS	427

Project data

Project name	L1-R8-A-000
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring

Name	SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

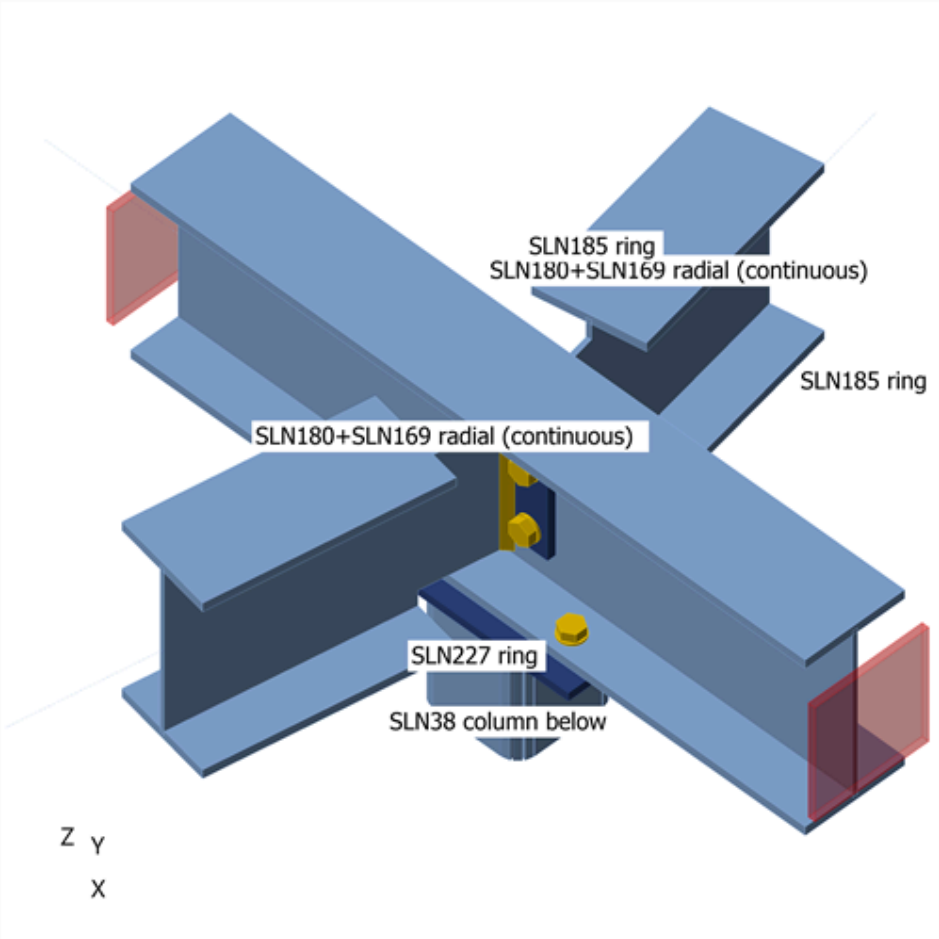
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN180+SLN169 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN38 column below	2 - SHS150x150x10	0.0	90.0	180.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN180+SLN169 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN180+SLN169 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN38 column below / begin		Position	0
SLN185 ring / end		Position	0
SLN227 ring / end		Position	0



SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.3	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	18.9	-0.2	-25.8	0.0	0.0	0.0
C LC_103	SLN38 column below / Begin	133.1	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	7.8	0.5	-24.0	0.0	0.0	0.0
	SLN227 ring / End	7.5	-0.1	-24.2	0.0	0.0	0.0
C LC_104	SLN38 column below / Begin	132.3	0.0	-5.0	0.0	-5.9	0.0
	SLN185 ring / End	21.3	0.6	-24.0	0.0	0.0	0.0
	SLN227 ring / End	21.0	-0.1	-24.2	0.0	0.0	0.0
C LC_105	SLN38 column below / Begin	132.9	3.5	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	11.8	0.4	-24.0	0.0	0.0	0.0
	SLN227 ring / End	7.8	-0.2	-24.2	0.0	0.0	0.0
C LC_106	SLN38 column below / Begin	133.1	-3.5	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	8.5	0.5	-24.0	0.0	0.0	0.0
	SLN227 ring / End	11.7	-0.1	-24.2	0.0	0.0	0.0
C LC_107	SLN38 column below / Begin	4.0	0.0	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-10.3	0.2	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-10.3	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN38 column below / Begin	2.3	0.0	-8.4	0.0	-5.8	0.0
	SLN185 ring / End	12.5	0.1	-0.5	0.0	0.0	0.0
	SLN227 ring / End	12.5	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN38 column below / Begin	3.6	7.0	0.0	0.0	-0.2	0.0
	SLN185 ring / End	-4.7	-0.1	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-11.6	-0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN38 column below / Begin	3.9	-7.0	0.0	0.0	-0.1	0.0
	SLN185 ring / End	-11.0	0.0	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-4.2	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN38 column below / Begin	90.6	0.0	-0.6	0.0	-2.3	0.0
	SLN185 ring / End	-2.3	0.5	-16.2	0.0	0.0	0.0
	SLN227 ring / End	-2.5	-0.1	-16.3	0.0	0.0	0.0
C LC_112	SLN38 column below / Begin	89.0	0.0	-8.9	0.0	-7.8	0.0
	SLN185 ring / End	21.8	0.4	-16.2	0.0	0.0	0.0
	SLN227 ring / End	21.7	0.0	-16.3	0.0	0.0	0.0
C LC_113	SLN38 column below / Begin	90.2	7.0	-0.5	0.0	-2.2	0.0
	SLN185 ring / End	4.0	0.2	-16.2	0.0	0.0	0.0
	SLN227 ring / End	-3.2	-0.2	-16.3	0.0	0.0	0.0
C LC_114	SLN38 column below / Begin	90.5	-7.0	-0.5	0.0	-2.1	0.0
	SLN185 ring / End	-2.4	0.3	-16.2	0.0	0.0	0.0
	SLN227 ring / End	4.3	0.0	-16.3	0.0	0.0	0.0
C LC_115	SLN38 column below / Begin	11.3	0.0	-0.1	0.0	-0.4	0.0
	SLN185 ring / End	-8.7	0.2	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-8.7	0.0	-1.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN38 column below / Begin	9.6	0.0	-8.5	0.0	-5.9	0.0
	SLN185 ring / End	14.2	0.1	-1.7	0.0	0.0	0.0
	SLN227 ring / End	14.2	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN38 column below / Begin	10.9	7.0	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-3.1	0.0	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-9.9	-0.1	-1.8	0.0	0.0	0.0
C LC_118	SLN38 column below / Begin	11.2	-7.0	0.0	0.0	-0.2	0.0
	SLN185 ring / End	-9.3	0.0	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-2.6	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN38 column below / Begin	141.6	0.0	-0.8	0.0	-3.4	0.0
	SLN185 ring / End	10.9	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	10.5	-0.1	-25.8	0.0	0.0	0.0
C LC_120	SLN38 column below / Begin	141.2	0.0	-5.0	0.0	-6.1	0.0
	SLN185 ring / End	24.5	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	24.2	-0.2	-25.8	0.0	0.0	0.0
C LC_121	SLN38 column below / Begin	141.7	3.5	-0.8	0.0	-3.3	0.0
	SLN185 ring / End	14.9	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	10.9	-0.2	-25.8	0.0	0.0	0.0
C LC_122	SLN38 column below / Begin	141.8	-3.5	-0.8	0.0	-3.3	0.0
	SLN185 ring / End	11.5	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	14.8	-0.1	-25.8	0.0	0.0	0.0
C LC_123	SLN38 column below / Begin	107.7	0.0	-0.7	0.0	-2.7	0.0
	SLN185 ring / End	3.7	0.5	-19.4	0.0	0.0	0.0
	SLN227 ring / End	3.5	-0.1	-19.5	0.0	0.0	0.0
C LC_124	SLN38 column below / Begin	106.8	0.0	-9.0	0.0	-8.2	0.0
	SLN185 ring / End	28.2	0.6	-19.4	0.0	0.0	0.0
	SLN227 ring / End	28.0	-0.1	-19.5	0.0	0.0	0.0
C LC_125	SLN38 column below / Begin	107.7	7.0	-0.6	0.0	-2.6	0.0
	SLN185 ring / End	10.2	0.3	-19.4	0.0	0.0	0.0
	SLN227 ring / End	2.9	-0.2	-19.5	0.0	0.0	0.0
C LC_126	SLN38 column below / Begin	108.0	-7.0	-0.6	0.0	-2.5	0.0
	SLN185 ring / End	3.8	0.3	-19.4	0.0	0.0	0.0
	SLN227 ring / End	10.4	0.0	-19.5	0.0	0.0	0.0
C LC_127	SLN38 column below / Begin	28.3	0.0	-0.2	0.0	-0.8	0.0
	SLN185 ring / End	-2.7	0.3	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-2.7	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN38 column below / Begin	27.5	0.0	-8.6	0.0	-6.3	0.0
	SLN185 ring / End	20.5	0.3	-4.9	0.0	0.0	0.0
	SLN227 ring / End	20.6	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN38 column below / Begin	28.4	7.0	-0.2	0.0	-0.7	0.0
	SLN185 ring / End	3.1	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-3.9	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN38 column below / Begin	28.7	-7.0	-0.1	0.0	-0.6	0.0
	SLN185 ring / End	-3.2	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	3.6	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN38 column below / Begin	141.7	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	16.6	0.6	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN227 ring / End	16.2	-0.2	-25.8	0.0	0.0	0.0
C LC_141	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.1	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.1	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.6	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.5	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	18.2	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	17.8	-0.1	-25.8	0.0	0.0	0.0
C LC_146	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	20.4	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	20.1	-0.2	-25.8	0.0	0.0	0.0
C LC_147	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.4	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	18.8	-0.2	-25.8	0.0	0.0	0.0
C LC_148	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.2	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	19.1	-0.2	-25.8	0.0	0.0	0.0
C LC_151	SLN38 column below / Begin	91.2	0.0	-1.7	0.0	-7.0	0.0
	SLN185 ring / End	6.6	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.4	-0.1	-11.5	0.0	0.0	0.0
C LC_152	SLN38 column below / Begin	79.2	0.0	0.8	0.0	3.3	0.0
	SLN185 ring / End	21.0	0.5	-14.2	0.0	0.0	0.0
	SLN227 ring / End	20.7	-0.1	-14.2	0.0	0.0	0.0
C LC_153	SLN38 column below / Begin	141.4	0.0	-0.8	0.0	-3.4	0.0
	SLN185 ring / End	19.3	0.6	-21.0	0.0	0.0	0.0
	SLN227 ring / End	19.0	-0.2	-20.7	0.0	0.0	0.0
C LC_154	SLN38 column below / Begin	90.5	0.0	-1.7	0.0	-6.9	0.0
	SLN185 ring / End	6.7	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.5	-0.1	-11.5	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.6 < 100%	OK
Welds	73.3 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN180+SLN169 radial (continuous)-bfl 1	12.7	C LC_120	149.8	0.0	45.1	OK
SLN180+SLN169 radial (continuous)-tfl 1	12.7	C LC_153	72.2	0.0	0.0	OK
SLN180+SLN169 radial (continuous)-w 1	7.2	C LC_125	243.2	0.0	29.8	OK
SLN38 column below	10.0	C LC_120	168.8	0.0	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_120	28.8	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_135	23.5	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_120	166.7	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_120	27.4	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_121	21.7	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_120	123.6	0.0	0.0	OK
End plate column below SLN38 column below	12.0	C LC_108	135.7	0.0	17.5	OK
End plate SLN227 ring	12.0	C LC_109	35.7	0.0	5.9	OK
End plate SLN185 ring	12.0	C LC_125	35.5	0.0	5.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

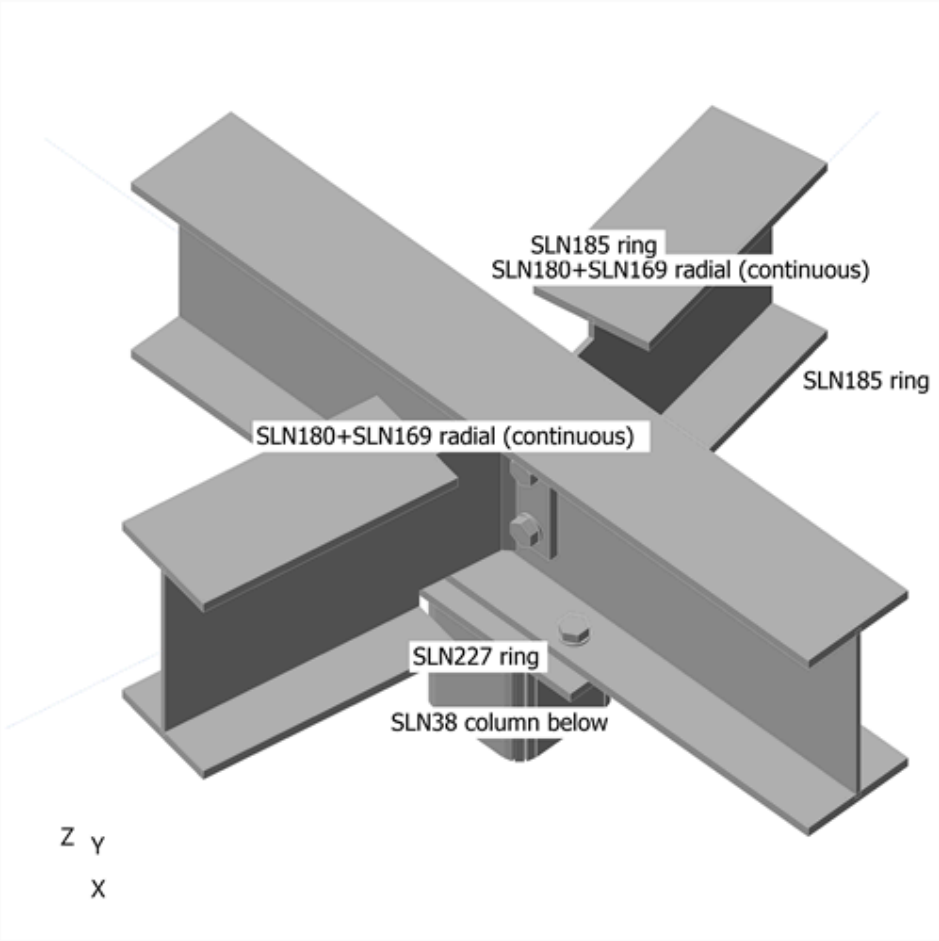
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

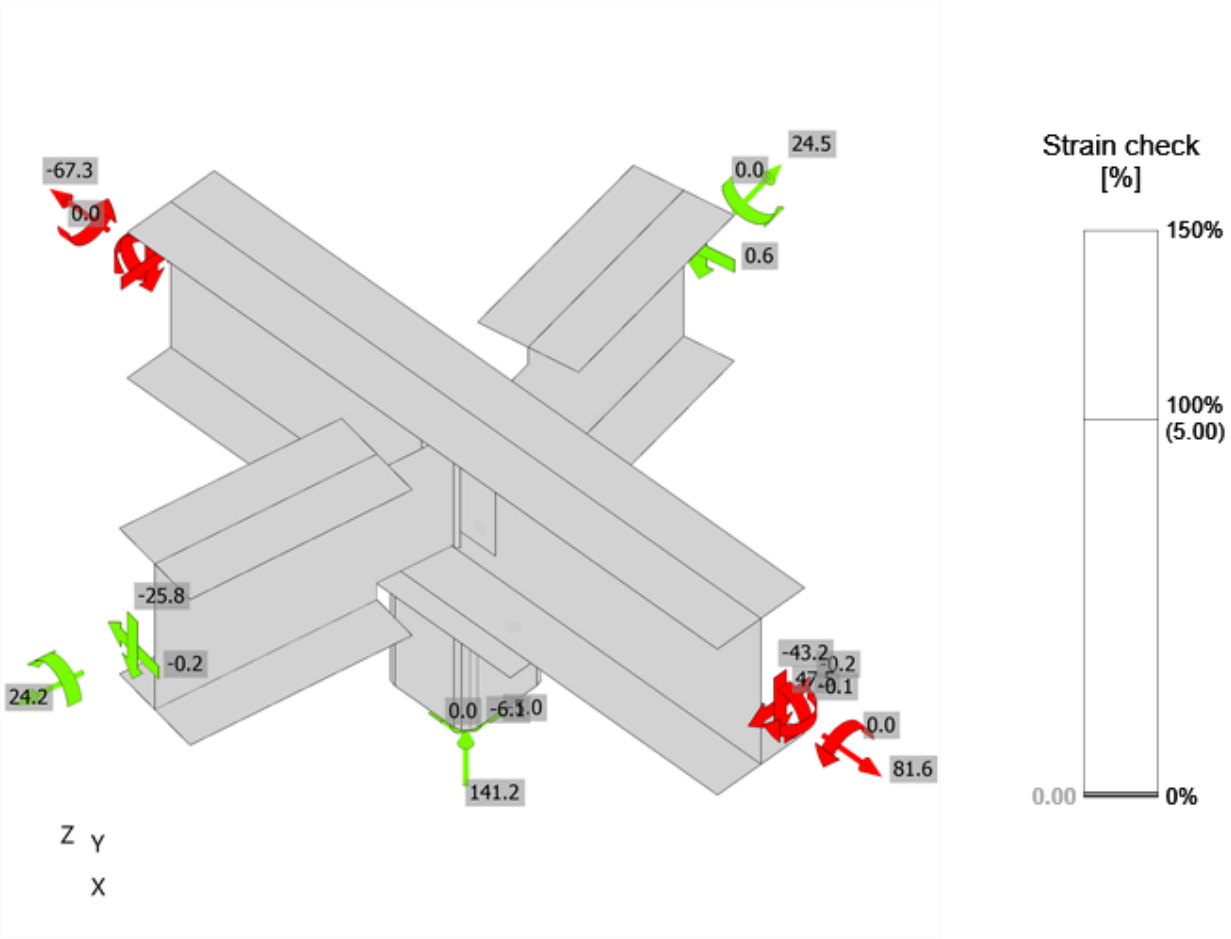
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN38 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

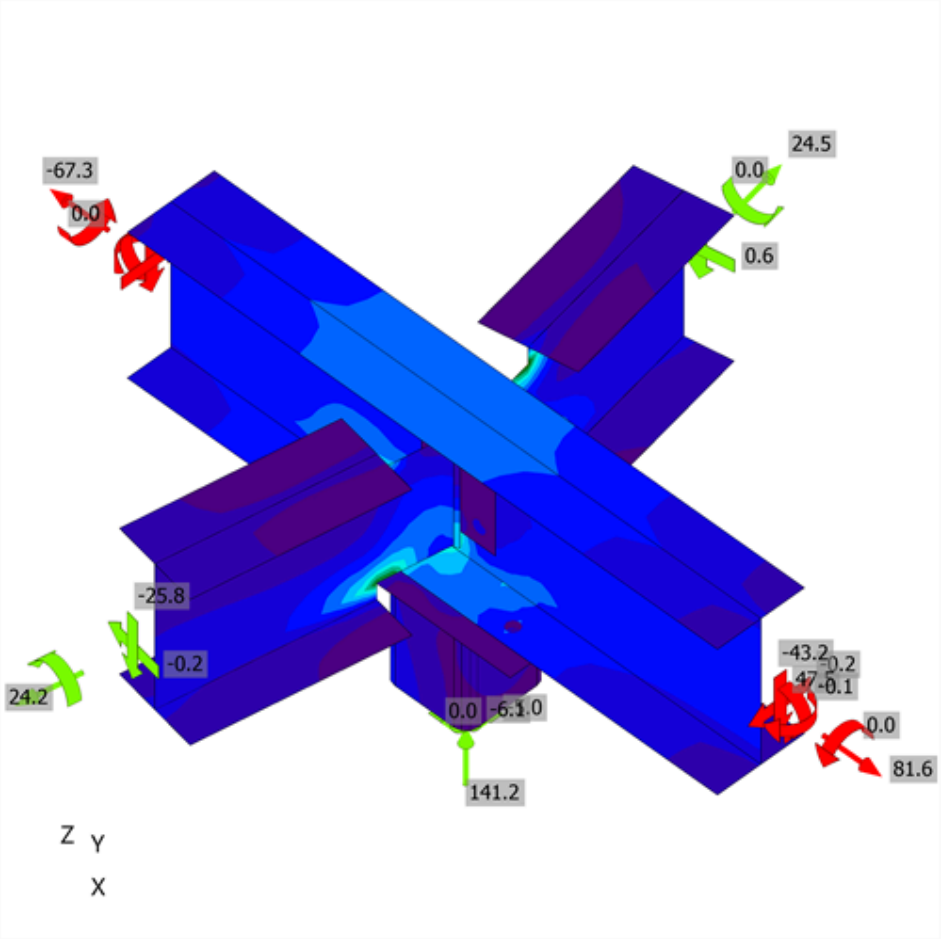
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

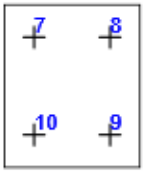


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_124	0.4	4.9	173.9	0.3	5.2	5.4	OK	OK
	B3	M20 8.8 - 1	C LC_108	19.2	2.5	98.1	13.6	2.7	12.4	OK	OK
	B4	M20 8.8 - 1	C LC_108	19.2	2.5	98.0	13.6	2.7	12.4	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.4	4.8	173.9	0.3	5.1	5.4	OK	OK
	B7	M20 8.8 - 1	C LC_124	0.2	0.3	123.8	0.1	0.5	0.4	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.0	0.7	0.5	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.9	0.5	123.8	1.3	0.5	1.5	OK	OK
	B10	M20 8.8 - 1	C LC_125	1.8	0.8	123.8	1.3	0.9	1.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN38 column below	SLN38 column below	◢ 4.2 ◣	521	C LC_120	272.8	0.0	-134.9	-136.9	-0.4	67.4	37.2	OK	OK
		◢ 4.2 ◣	521	C LC_120	296.7	0.0	-149.8	147.9	0.4	73.3	41.1	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_125	97.7	0.0	5.8	7.9	-55.7	24.1	14.7	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_122	94.9	0.0	4.6	3.2	-54.6	23.5	13.9	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_122	82.6	0.0	-22.0	-17.9	42.3	20.4	12.5	OK	OK
		◢ 4.2 ◣	149	C LC_122	98.2	0.0	-18.3	24.5	-50.0	24.3	15.2	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_121	100.0	0.0	-18.6	-22.9	51.9	24.7	15.8	OK	OK
		◢ 4.2 ◣	149	C LC_121	81.9	0.0	-22.3	19.8	-41.0	20.2	11.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-080
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring

Name	SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

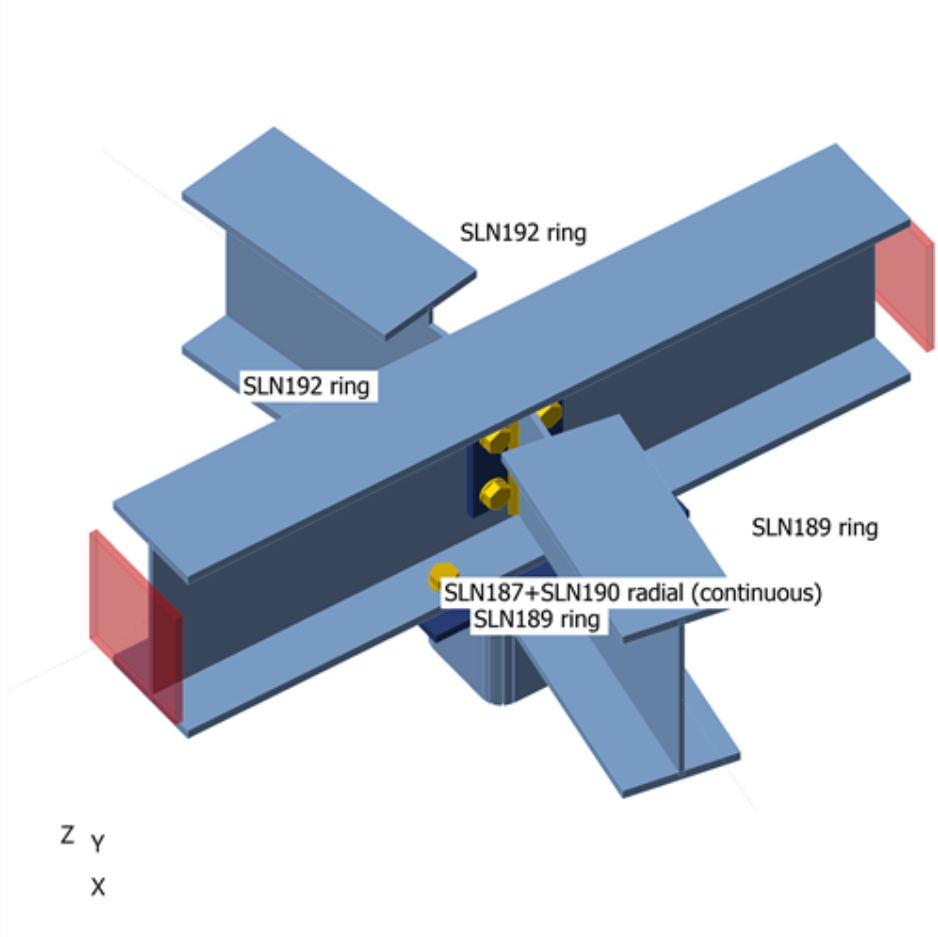
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN187+SLN190 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN42 column below	2 - SHS150x150x10	0.0	90.0	-100.0	0	0	0
SLN189 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN187+SLN190 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN187+SLN190 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN42 column below / begin		Position	0
SLN189 ring / end		Position	0
SLN192 ring / end		Position	0



SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN42 column below / Begin	133.1	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	28.0	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN42 column below / Begin	124.6	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	17.5	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	17.3	0.1	-24.0	0.0	0.0	0.0
C LC_104	SLN42 column below / Begin	124.4	-3.4	-1.1	0.0	-1.9	0.0
	SLN189 ring / End	21.2	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	17.1	0.2	-24.0	0.0	0.0	0.0
C LC_105	SLN42 column below / Begin	123.6	0.6	-4.5	0.0	-4.1	0.0
	SLN189 ring / End	30.1	-0.2	-24.2	0.0	0.0	0.0
	SLN192 ring / End	30.3	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN42 column below / Begin	124.9	0.0	-0.4	0.0	-1.6	0.0
	SLN189 ring / End	14.8	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	14.5	0.2	-24.0	0.0	0.0	0.0
C LC_107	SLN42 column below / Begin	3.2	0.0	0.0	0.0	0.0	0.0
	SLN189 ring / End	-7.7	0.0	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-7.9	-0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN42 column below / Begin	2.8	-6.9	-1.5	0.0	-1.0	0.0
	SLN189 ring / End	-1.6	0.1	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-8.6	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN42 column below / Begin	1.3	1.2	-8.2	0.0	-5.4	0.0
	SLN189 ring / End	14.8	0.0	-0.5	0.0	0.0	0.0
	SLN192 ring / End	15.8	-0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN42 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	-11.6	-0.1	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-11.4	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN42 column below / Begin	84.5	0.0	-0.2	0.0	-1.0	0.0
	SLN189 ring / End	5.6	0.0	-16.4	0.0	0.0	0.0
	SLN192 ring / End	5.4	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN42 column below / Begin	84.1	-6.9	-1.7	0.0	-1.9	0.0
	SLN189 ring / End	12.2	0.0	-16.4	0.0	0.0	0.0
	SLN192 ring / End	4.8	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN42 column below / Begin	82.5	1.2	-8.5	0.0	-6.4	0.0
	SLN189 ring / End	29.0	-0.1	-16.4	0.0	0.0	0.0
	SLN192 ring / End	29.8	0.0	-16.2	0.0	0.0	0.0
C LC_114	SLN42 column below / Begin	85.1	0.0	-0.3	0.0	-1.2	0.0
	SLN189 ring / End	1.3	-0.1	-16.4	0.0	0.0	0.0
	SLN192 ring / End	1.2	0.2	-16.2	0.0	0.0	0.0
C LC_115	SLN42 column below / Begin	10.1	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	-5.5	0.0	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-5.6	0.0	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN42 column below / Begin	9.7	-6.9	-1.5	0.0	-1.0	0.0
	SLN189 ring / End	0.7	0.1	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-6.3	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN42 column below / Begin	8.1	1.2	-8.2	0.0	-5.5	0.0
	SLN189 ring / End	17.1	0.0	-1.8	0.0	0.0	0.0
	SLN192 ring / End	18.1	-0.1	-1.7	0.0	0.0	0.0
C LC_118	SLN42 column below / Begin	10.7	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	-9.3	-0.1	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-9.2	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN42 column below / Begin	132.9	0.0	-0.4	0.0	-1.5	0.0
	SLN189 ring / End	20.9	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	20.7	0.2	-25.6	0.0	0.0	0.0
C LC_120	SLN42 column below / Begin	132.7	-3.4	-1.1	0.0	-2.0	0.0
	SLN189 ring / End	24.7	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	20.5	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN42 column below / Begin	132.1	0.6	-4.5	0.0	-4.2	0.0
	SLN189 ring / End	33.8	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	34.0	0.2	-25.6	0.0	0.0	0.0
C LC_122	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.6	0.0
	SLN189 ring / End	18.1	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	17.8	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN42 column below / Begin	101.1	0.0	-0.3	0.0	-1.1	0.0
	SLN189 ring / End	12.4	0.0	-19.6	0.0	0.0	0.0
	SLN192 ring / End	12.2	0.1	-19.4	0.0	0.0	0.0
C LC_124	SLN42 column below / Begin	100.7	-6.9	-1.7	0.0	-2.1	0.0
	SLN189 ring / End	19.2	0.0	-19.6	0.0	0.0	0.0
	SLN192 ring / End	11.7	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN42 column below / Begin	99.5	1.2	-8.5	0.0	-6.6	0.0
	SLN189 ring / End	36.4	-0.2	-19.6	0.0	0.0	0.0
	SLN192 ring / End	37.2	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN42 column below / Begin	101.2	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	7.9	-0.1	-19.6	0.0	0.0	0.0
	SLN192 ring / End	7.7	0.2	-19.4	0.0	0.0	0.0
C LC_127	SLN42 column below / Begin	26.7	0.0	-0.1	0.0	-0.2	0.0
	SLN189 ring / End	1.3	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	1.2	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN42 column below / Begin	26.3	-6.9	-1.5	0.0	-1.2	0.0
	SLN189 ring / End	7.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	0.6	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN42 column below / Begin	25.1	1.2	-8.3	0.0	-5.7	0.0
	SLN189 ring / End	24.5	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	25.5	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN42 column below / Begin	26.8	0.0	-0.1	0.0	-0.4	0.0
	SLN189 ring / End	-2.8	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	-2.6	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	28.4	-0.1	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.9	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	9.1	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.6	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	27.7	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_146	SLN42 column below / Begin	133.0	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	28.3	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN42 column below / Begin	133.0	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	29.1	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	28.8	0.3	-25.6	0.0	0.0	0.0
C LC_148	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	26.8	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	26.4	0.1	-25.6	0.0	0.0	0.0
C LC_151	SLN42 column below / Begin	81.1	0.0	-1.3	0.0	-5.3	0.0
	SLN189 ring / End	12.6	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.3	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN42 column below / Begin	78.3	0.0	0.9	0.0	3.6	0.0
	SLN189 ring / End	28.5	-0.1	-14.1	0.0	0.0	0.0
	SLN192 ring / End	28.3	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN42 column below / Begin	133.4	0.0	-0.4	0.0	-1.7	0.0
	SLN189 ring / End	28.1	-0.1	-20.6	0.0	0.0	0.0
	SLN192 ring / End	27.7	0.2	-21.1	0.0	0.0	0.0
C LC_154	SLN42 column below / Begin	80.3	0.0	-1.3	0.0	-5.1	0.0
	SLN189 ring / End	12.7	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.4	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.4 < 100%	OK
Welds	62.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN187+SLN190 radial (continuous)-bfl 1	12.7	C LC_121	128.1	0.0	38.2	OK
SLN187+SLN190 radial (continuous)-tfl 1	12.7	C LC_153	65.5	0.0	0.0	OK
SLN187+SLN190 radial (continuous)-w 1	7.2	C LC_124	230.9	0.0	26.8	OK
SLN42 column below	10.0	C LC_125	141.9	0.0	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_121	29.2	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_121	135.2	0.0	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_121	29.0	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_147	134.1	0.0	0.0	OK
End plate column below SLN42 column below	12.0	C LC_109	126.9	0.0	17.5	OK
End plate SLN189 ring	12.0	C LC_124	34.4	0.0	3.3	OK
End plate SLN192 ring	12.0	C LC_124	34.9	0.0	5.7	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

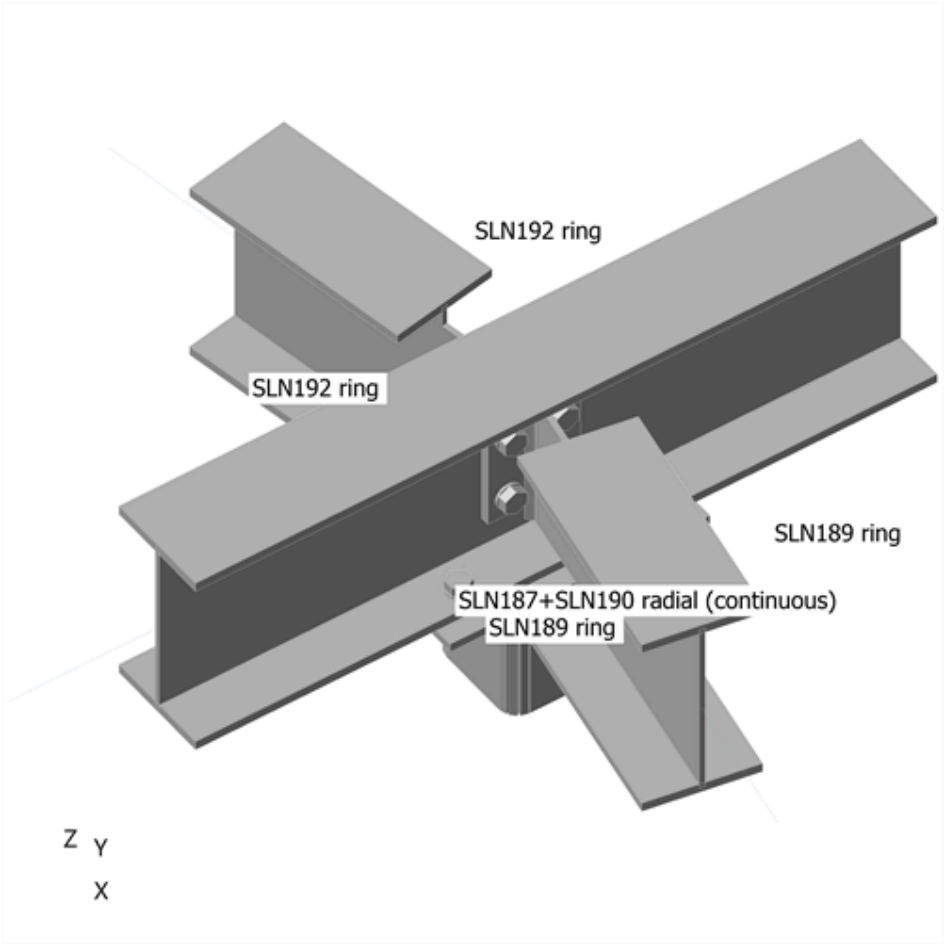
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

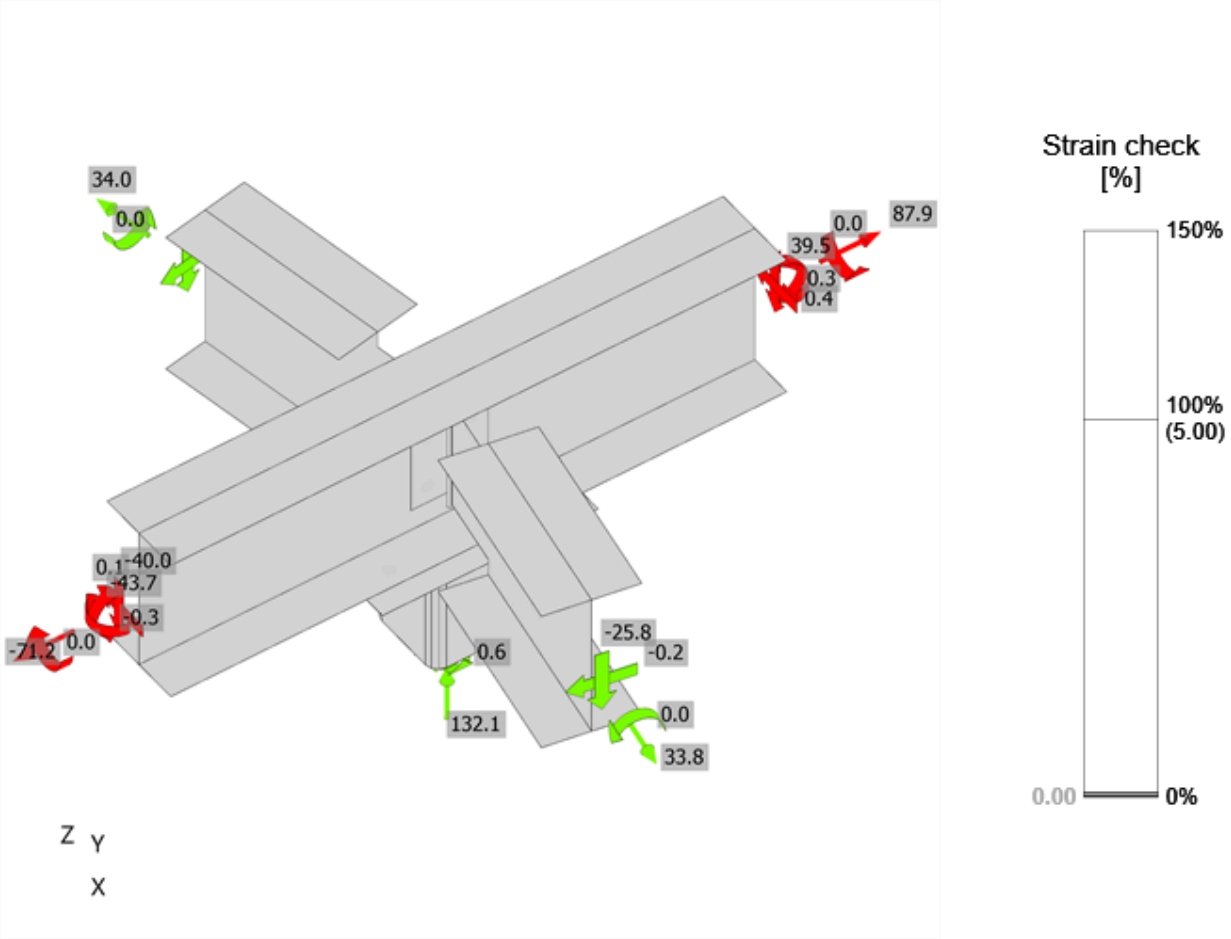
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN42 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

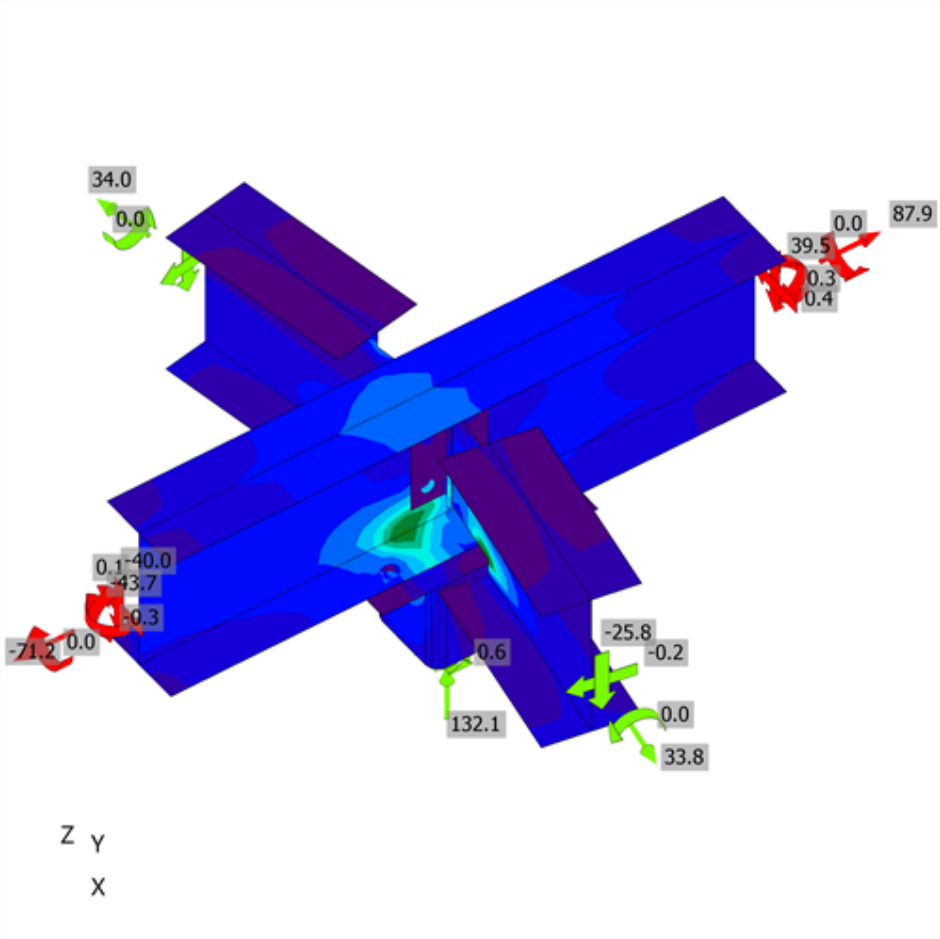
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

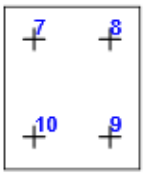


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_125	0.4	4.2	171.0	0.3	4.5	4.7	OK	OK
	B3	M20 8.8 - 1	C LC_109	18.9	2.7	91.6	13.4	2.9	12.4	OK	OK
	B4	M20 8.8 - 1	C LC_109	17.7	2.3	110.5	12.5	2.5	11.4	OK	OK
	B5	M20 8.8 - 1	C LC_125	0.4	4.6	176.2	0.3	4.9	5.1	OK	OK
	B7	M20 8.8 - 1	C LC_124	0.5	0.2	72.9	0.4	0.3	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_124	0.7	0.3	62.0	0.5	0.4	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_124	1.8	0.5	65.3	1.3	0.5	1.5	OK	OK
	B10	M20 8.8 - 1	C LC_124	1.7	0.7	123.8	1.2	0.8	1.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN42 column below	SLN42 column below	◢ 4.2 ◣	521	C LC_121	237.0	0.0	-115.7	-113.8	-35.9	58.6	32.1	OK	OK
		◢ 4.2 ◣	521	C LC_121	252.9	0.0	-121.3	123.5	34.3	62.5	34.7	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_120	94.2	0.0	10.8	-13.2	52.4	23.3	14.9	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_121	89.3	0.0	27.6	35.3	-34.0	22.1	14.8	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_120	79.4	0.0	-20.8	-17.4	40.7	19.6	12.1	OK	OK
		◢ 4.2 ◣	149	C LC_120	94.5	0.0	-18.6	23.6	-48.0	23.3	14.6	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_121	90.8	0.0	-17.7	-23.7	45.6	22.4	13.9	OK	OK
		◢ 4.2 ◣	149	C LC_121	76.8	0.0	-21.0	16.9	-39.1	19.0	11.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-120
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring

Name	SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

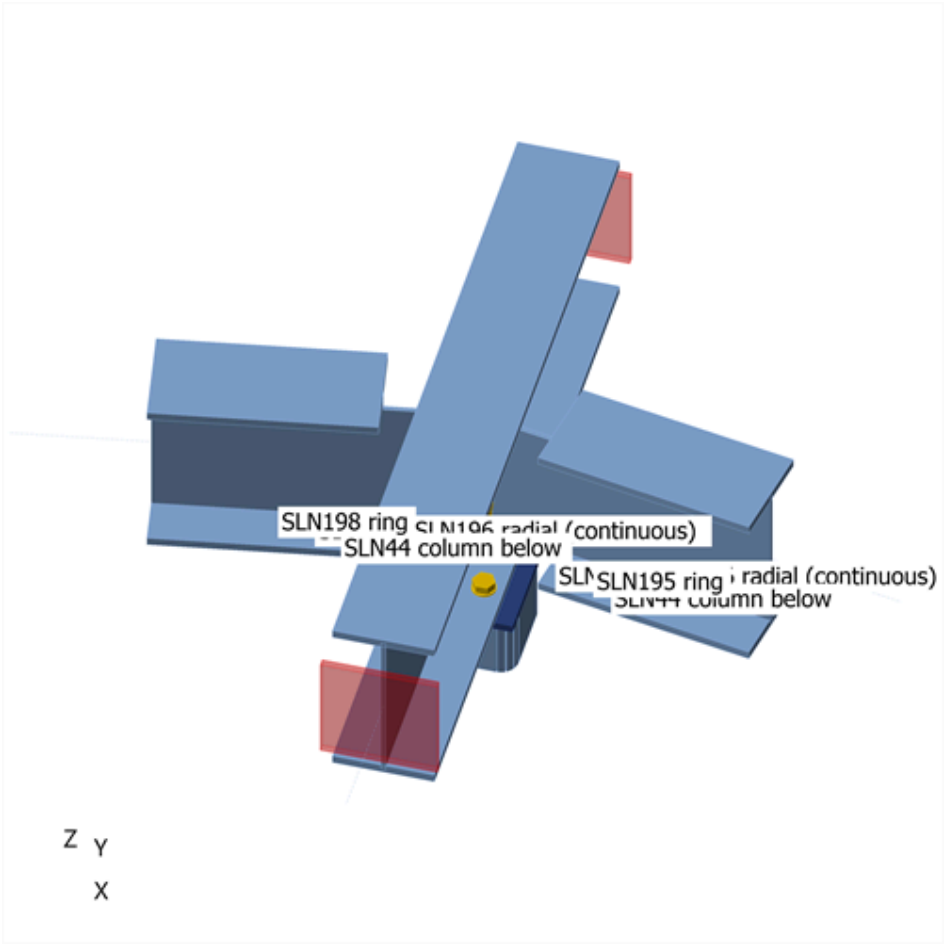
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN194+SLN196 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN44 column below	2 - SHS150x150x10	0.0	90.0	-60.0	0	0	0
SLN195 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN194+SLN196 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN194+SLN196 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN44 column below / begin		Position	0
SLN195 ring / end		Position	0
SLN198 ring / end		Position	0



SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.8	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	27.7	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.8	0.2	-26.0	0.0	0.0	0.0
C LC_103	SLN44 column below / Begin	132.3	3.0	-2.9	0.0	-4.9	0.0
	SLN195 ring / End	21.7	-0.2	-24.2	0.0	0.0	0.0
	SLN198 ring / End	25.0	0.2	-24.4	0.0	0.0	0.0
C LC_104	SLN44 column below / Begin	132.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	16.3	-0.1	-24.2	0.0	0.0	0.0
	SLN198 ring / End	16.2	0.2	-24.4	0.0	0.0	0.0
C LC_105	SLN44 column below / Begin	132.0	-1.8	-4.5	0.0	-5.8	0.0
	SLN195 ring / End	29.5	-0.2	-24.2	0.0	0.0	0.0
	SLN198 ring / End	27.7	0.2	-24.4	0.0	0.0	0.0
C LC_106	SLN44 column below / Begin	132.7	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	14.9	-0.1	-24.2	0.0	0.0	0.0
	SLN198 ring / End	15.2	0.2	-24.4	0.0	0.0	0.0
C LC_107	SLN44 column below / Begin	3.1	6.1	-4.2	0.0	-3.0	0.0
	SLN195 ring / End	-0.3	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	5.7	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN44 column below / Begin	3.7	0.0	-0.1	0.0	-0.2	0.0
	SLN195 ring / End	-9.3	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	-9.3	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN44 column below / Begin	2.3	-3.5	-7.3	0.0	-5.0	0.0
	SLN195 ring / End	14.2	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	10.6	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN44 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN195 ring / End	-11.0	-0.1	-0.5	0.0	0.0	0.0
	SLN198 ring / End	-10.9	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN44 column below / Begin	89.5	6.1	-4.8	0.0	-5.3	0.0
	SLN195 ring / End	13.4	-0.2	-16.3	0.0	0.0	0.0
	SLN198 ring / End	19.6	0.1	-16.5	0.0	0.0	0.0
C LC_112	SLN44 column below / Begin	90.1	0.0	-0.6	0.0	-2.5	0.0
	SLN195 ring / End	3.8	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	3.7	0.2	-16.5	0.0	0.0	0.0
C LC_113	SLN44 column below / Begin	88.7	-3.5	-7.8	0.0	-7.2	0.0
	SLN195 ring / End	28.2	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	24.6	0.1	-16.5	0.0	0.0	0.0
C LC_114	SLN44 column below / Begin	90.2	0.0	-0.6	0.0	-2.5	0.0
	SLN195 ring / End	1.7	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	2.0	0.2	-16.5	0.0	0.0	0.0
C LC_115	SLN44 column below / Begin	10.4	6.1	-4.3	0.0	-3.2	0.0
	SLN195 ring / End	2.0	-0.1	-1.8	0.0	0.0	0.0
	SLN198 ring / End	8.0	0.0	-1.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN44 column below / Begin	11.0	0.0	-0.1	0.0	-0.4	0.0
	SLN195 ring / End	-7.0	0.0	-1.8	0.0	0.0	0.0
	SLN198 ring / End	-7.1	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN44 column below / Begin	9.6	-3.5	-7.3	0.0	-5.2	0.0
	SLN195 ring / End	16.5	0.0	-1.8	0.0	0.0	0.0
	SLN198 ring / End	12.9	0.0	-1.8	0.0	0.0	0.0
C LC_118	SLN44 column below / Begin	11.1	0.0	-0.1	0.0	-0.4	0.0
	SLN195 ring / End	-8.8	-0.1	-1.8	0.0	0.0	0.0
	SLN198 ring / End	-8.6	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN44 column below / Begin	141.2	3.0	-3.0	0.0	-5.1	0.0
	SLN195 ring / End	25.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.5	0.2	-26.0	0.0	0.0	0.0
C LC_120	SLN44 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	19.8	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	19.6	0.2	-26.0	0.0	0.0	0.0
C LC_121	SLN44 column below / Begin	140.9	-1.8	-4.5	0.0	-6.1	0.0
	SLN195 ring / End	33.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	31.3	0.3	-26.0	0.0	0.0	0.0
C LC_122	SLN44 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	18.2	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	18.5	0.2	-26.0	0.0	0.0	0.0
C LC_123	SLN44 column below / Begin	107.2	6.1	-4.9	0.0	-5.7	0.0
	SLN195 ring / End	20.4	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	26.7	0.2	-19.7	0.0	0.0	0.0
C LC_124	SLN44 column below / Begin	107.5	0.0	-0.7	0.0	-2.9	0.0
	SLN195 ring / End	10.7	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	10.5	0.2	-19.7	0.0	0.0	0.0
C LC_125	SLN44 column below / Begin	106.7	-3.5	-7.9	0.0	-7.7	0.0
	SLN195 ring / End	35.5	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	31.8	0.2	-19.7	0.0	0.0	0.0
C LC_126	SLN44 column below / Begin	107.4	0.0	-0.7	0.0	-2.9	0.0
	SLN195 ring / End	8.3	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	8.7	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN44 column below / Begin	28.1	6.1	-4.4	0.0	-3.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	15.1	0.1	-5.0	0.0	0.0	0.0
C LC_128	SLN44 column below / Begin	28.4	0.0	-0.2	0.0	-0.8	0.0
	SLN195 ring / End	-0.2	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	-0.3	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN44 column below / Begin	27.6	-3.5	-7.4	0.0	-5.6	0.0
	SLN195 ring / End	23.8	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	20.1	0.0	-5.0	0.0	0.0	0.0
C LC_130	SLN44 column below / Begin	28.3	0.0	-0.2	0.0	-0.8	0.0
	SLN195 ring / End	-2.2	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	-2.0	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	28.1	-0.2	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / End	28.2	0.2	-26.0	0.0	0.0	0.0
C LC_141	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.9	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.5	0.3	-26.0	0.0	0.0	0.0
C LC_146	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	27.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.1	0.2	-26.0	0.0	0.0	0.0
C LC_147	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.8	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.7	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	26.6	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	26.8	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN44 column below / Begin	90.2	0.0	-1.8	0.0	-7.2	0.0
	SLN195 ring / End	12.4	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.4	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN44 column below / Begin	78.6	0.0	0.7	0.0	3.0	0.0
	SLN195 ring / End	28.4	-0.2	-14.1	0.0	0.0	0.0
	SLN198 ring / End	28.4	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN44 column below / Begin	140.9	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	27.8	-0.2	-21.1	0.0	0.0	0.0
	SLN198 ring / End	27.9	0.2	-20.9	0.0	0.0	0.0
C LC_154	SLN44 column below / Begin	89.6	0.0	-1.8	0.0	-7.1	0.0
	SLN195 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.5	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.0 < 100%	OK
Welds	74.7 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN194+SLN196 radial (continuous)-bfl 1	12.7	C LC_121	149.0	0.0	44.7	OK
SLN194+SLN196 radial (continuous)-tfl 1	12.7	C LC_153	71.9	0.0	0.0	OK
SLN194+SLN196 radial (continuous)-w 1	7.2	C LC_123	251.8	0.0	36.5	OK
SLN44 column below	10.0	C LC_121	168.2	0.0	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_121	29.0	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_122	20.2	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_121	138.2	0.0	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_121	28.9	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_122	20.7	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_147	138.0	0.0	0.0	OK
End plate column below SLN44 column below	12.0	C LC_109	130.5	0.0	17.0	OK
End plate SLN195 ring	12.0	C LC_123	31.4	0.0	6.1	OK
End plate SLN198 ring	12.0	C LC_123	34.2	0.0	3.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

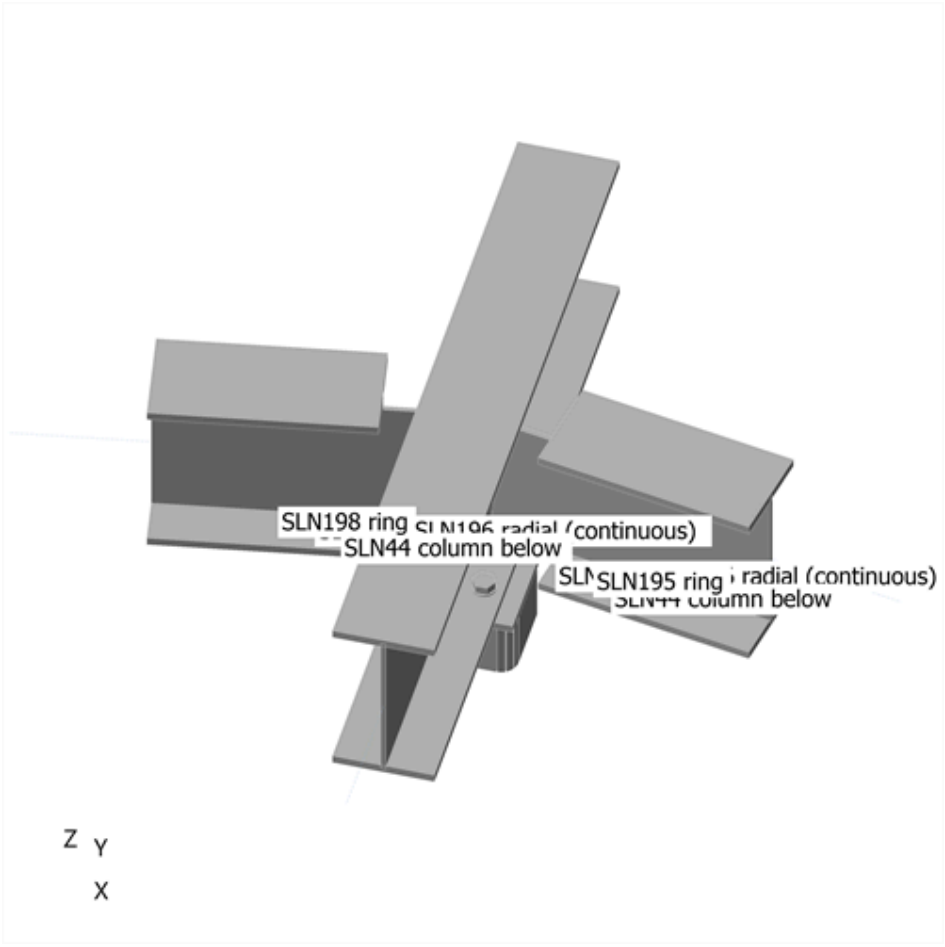
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

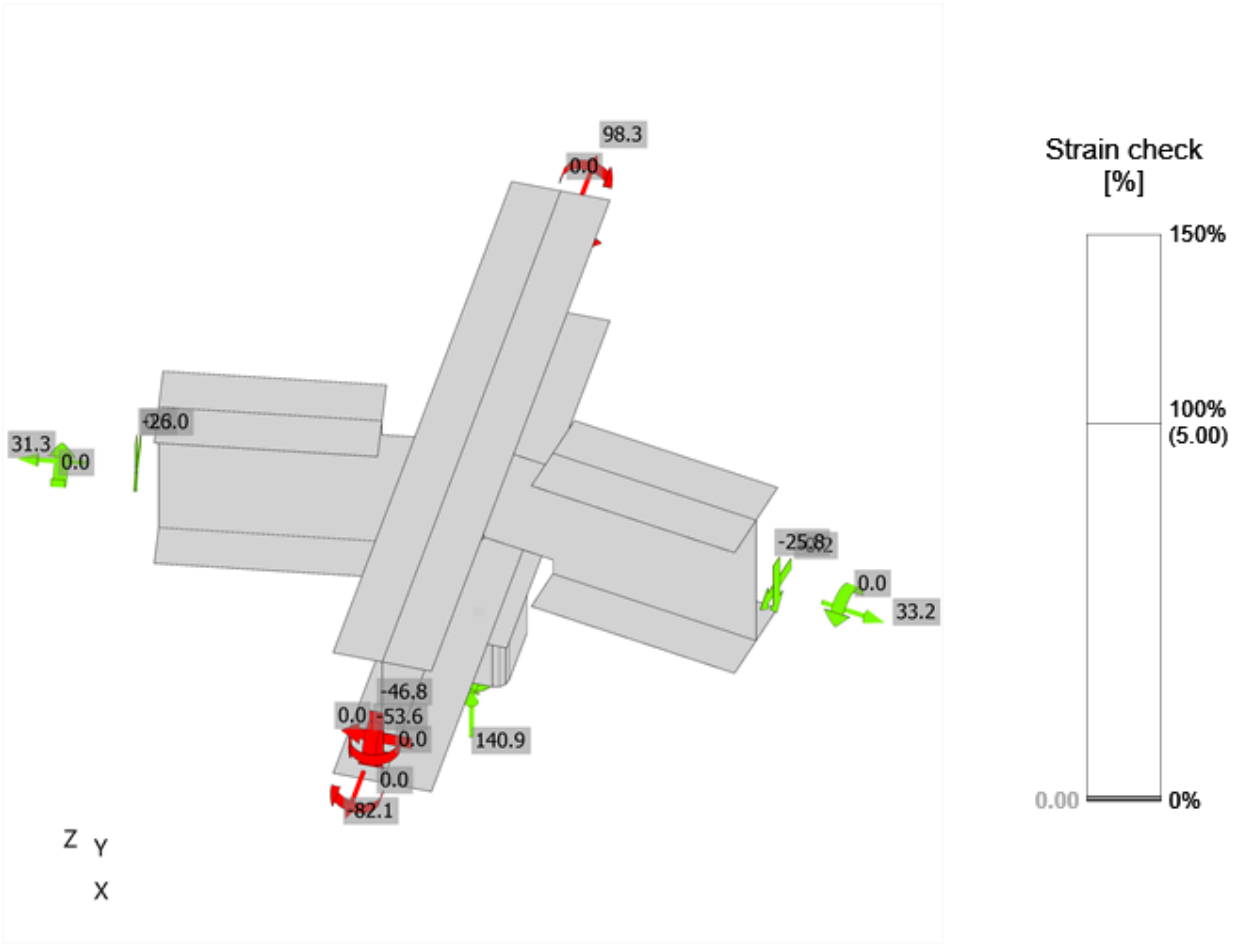
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN44 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

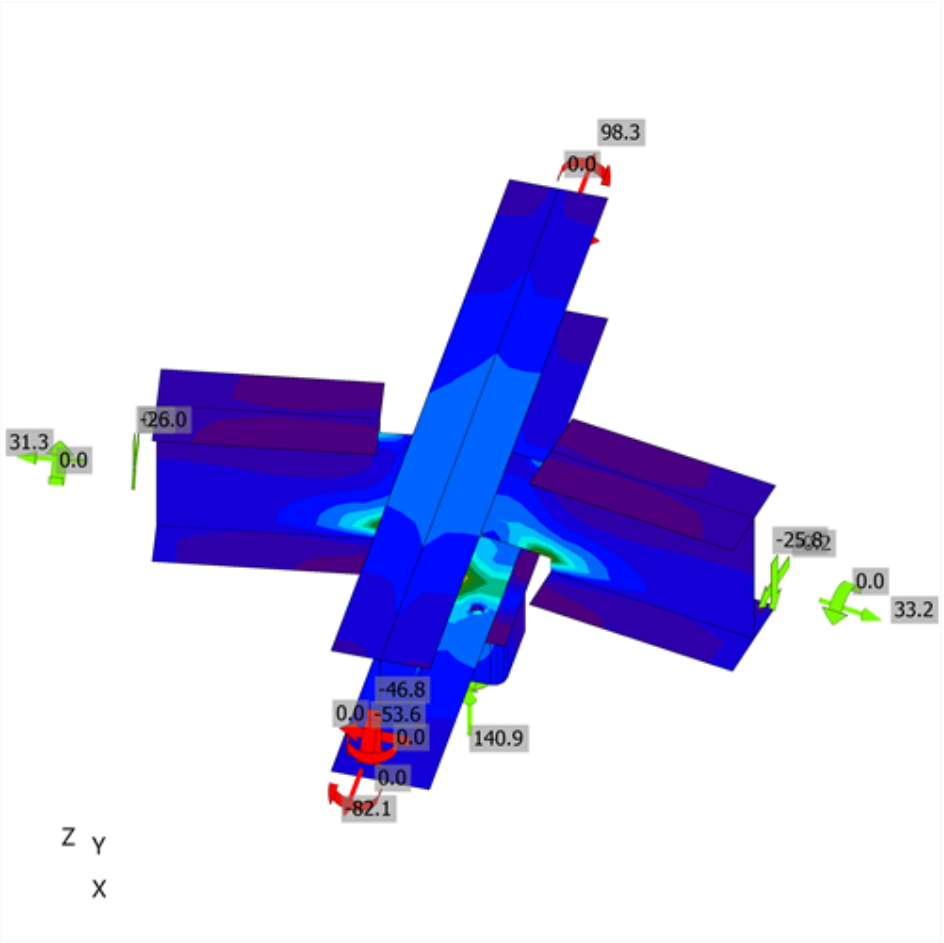
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

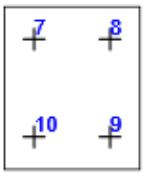


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_125	0.4	5.1	186.9	0.3	5.4	5.6	OK	OK
	B3	M20 8.8 - 1	C LC_109	14.9	1.8	156.4	10.5	1.9	9.5	OK	OK
	B4	M20 8.8 - 1	C LC_109	18.3	2.8	82.0	13.0	3.4	12.3	OK	OK
	B5	M20 8.8 - 1	C LC_123	0.3	4.8	196.0	0.2	5.1	5.3	OK	OK
	B7	M20 8.8 - 1	C LC_123	0.5	0.3	62.7	0.3	0.4	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_123	0.6	0.4	69.9	0.5	0.6	0.8	OK	OK
	B9	M20 8.8 - 1	C LC_123	1.7	0.4	67.8	1.2	0.4	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.7	0.9	123.8	1.2	1.3	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN44 column below	SLN44 column below	◢ 4.2 ◣	521	C LC_121	278.1	0.0	-137.0	-138.6	18.0	68.7	36.8	OK	OK
		◢ 4.2 ◣	521	C LC_121	302.5	0.0	-151.4	149.7	-21.2	74.7	40.7	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_119	93.1	0.0	27.3	-34.6	38.0	23.0	13.7	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_119	99.5	0.0	12.2	15.0	-55.0	24.6	15.8	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_121	79.1	0.0	-22.5	-18.2	39.8	19.5	11.7	OK	OK
		◢ 4.2 ◣	149	C LC_121	95.1	0.0	-17.8	24.4	-48.1	23.5	14.7	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_119	102.1	0.0	-18.4	-24.7	52.5	25.2	16.0	OK	OK
		◢ 4.2 ◣	149	C LC_119	86.0	0.0	-24.0	20.1	-43.3	21.3	12.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-160
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring

Name	SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

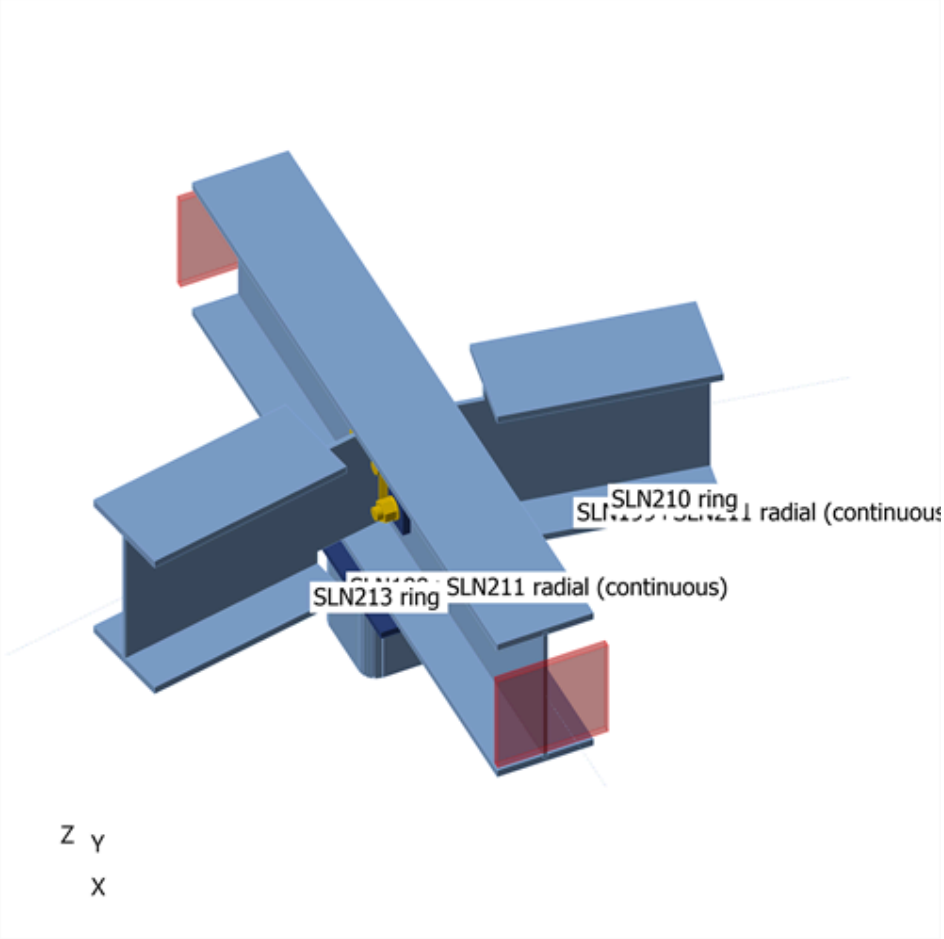
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN199+SLN211 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN46 column below	2 - SHS150x150x10	0.0	90.0	-20.0	0	0	0
SLN210 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN199+SLN211 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN199+SLN211 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN46 column below / begin		Position	0
SLN210 ring / end		Position	0
SLN213 ring / end		Position	0



SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN46 column below / Begin	131.9	1.2	-4.8	0.0	-6.0	0.0
	SLN210 ring / End	27.9	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	28.9	0.2	-24.1	0.0	0.0	0.0
C LC_104	SLN46 column below / Begin	132.7	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	15.6	-0.1	-24.0	0.0	0.0	0.0
	SLN213 ring / End	15.3	0.2	-24.1	0.0	0.0	0.0
C LC_105	SLN46 column below / Begin	132.5	-3.3	-2.3	0.0	-4.4	0.1
	SLN210 ring / End	24.2	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	20.4	0.2	-24.1	0.0	0.0	0.0
C LC_106	SLN46 column below / Begin	132.7	0.0	-0.8	0.0	-3.4	0.0
	SLN210 ring / End	16.0	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	16.0	0.1	-24.1	0.0	0.0	0.0
C LC_107	SLN46 column below / Begin	2.2	2.4	-7.9	0.0	-5.4	0.0
	SLN210 ring / End	10.7	0.1	-0.5	0.0	0.0	0.0
	SLN213 ring / End	13.1	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN46 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-9.8	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-10.0	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN46 column below / Begin	3.3	-6.6	-2.9	0.0	-2.1	0.0
	SLN210 ring / End	4.4	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-2.3	0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN46 column below / Begin	3.7	0.0	0.0	0.0	-0.2	0.0
	SLN210 ring / End	-10.1	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-10.1	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN46 column below / Begin	88.7	2.4	-8.5	0.0	-7.6	0.0
	SLN210 ring / End	24.8	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	27.1	0.1	-16.2	0.0	0.0	0.0
C LC_112	SLN46 column below / Begin	90.3	0.0	-0.6	0.0	-2.5	0.0
	SLN210 ring / End	3.0	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	2.7	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN46 column below / Begin	89.7	-6.6	-3.5	0.0	-4.3	0.1
	SLN210 ring / End	18.2	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	11.2	0.2	-16.2	0.0	0.0	0.0
C LC_114	SLN46 column below / Begin	90.1	0.0	-0.6	0.0	-2.4	0.0
	SLN210 ring / End	3.0	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	3.1	0.0	-16.2	0.0	0.0	0.0
C LC_115	SLN46 column below / Begin	9.6	2.4	-8.0	0.0	-5.6	0.0
	SLN210 ring / End	13.1	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	15.4	0.0	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN46 column below / Begin	11.2	0.0	-0.1	0.0	-0.4	0.0
	SLN210 ring / End	-7.6	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	-7.8	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN46 column below / Begin	10.6	-6.6	-3.0	0.0	-2.3	0.0
	SLN210 ring / End	6.7	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	0.0	0.1	-1.7	0.0	0.0	0.0
C LC_118	SLN46 column below / Begin	11.0	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-7.9	-0.1	-1.7	0.0	0.0	0.0
	SLN213 ring / End	-7.8	-0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN46 column below / Begin	140.9	1.2	-4.8	0.0	-6.2	0.0
	SLN210 ring / End	31.4	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	32.5	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN46 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN210 ring / End	19.0	-0.1	-25.6	0.0	0.0	0.0
	SLN213 ring / End	18.6	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN46 column below / Begin	141.3	-3.3	-2.3	0.0	-4.6	0.1
	SLN210 ring / End	27.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	23.9	0.3	-25.6	0.0	0.0	0.0
C LC_122	SLN46 column below / Begin	141.4	0.0	-0.9	0.0	-3.6	0.0
	SLN210 ring / End	19.4	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	19.4	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN46 column below / Begin	106.6	2.4	-8.6	0.0	-8.1	0.0
	SLN210 ring / End	32.0	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	34.3	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN46 column below / Begin	107.4	0.0	-0.7	0.0	-2.9	0.0
	SLN210 ring / End	9.8	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	9.4	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN46 column below / Begin	107.4	-6.6	-3.6	0.0	-4.8	0.1
	SLN210 ring / End	25.3	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	18.2	0.2	-19.4	0.0	0.0	0.0
C LC_126	SLN46 column below / Begin	107.6	0.0	-0.7	0.0	-2.8	0.0
	SLN210 ring / End	9.8	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	9.9	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN46 column below / Begin	27.5	2.4	-8.1	0.0	-6.0	0.0
	SLN210 ring / End	20.2	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	22.6	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN46 column below / Begin	28.4	0.0	-0.2	0.0	-0.8	0.0
	SLN210 ring / End	-0.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	-1.1	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN46 column below / Begin	28.3	-6.6	-3.1	0.0	-2.7	0.0
	SLN210 ring / End	13.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	7.0	0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN46 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN210 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	-1.0	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.3	-0.2	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.7	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.9	-0.3	-25.6	0.0	0.0	0.0
	SLN213 ring / End	28.7	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	26.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	26.4	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.3	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.8	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.3	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.3	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN46 column below / Begin	90.1	0.0	-1.8	0.0	-7.3	0.0
	SLN210 ring / End	12.4	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.3	0.1	-11.6	0.0	0.0	0.0
C LC_152	SLN46 column below / Begin	79.0	0.0	0.8	0.0	3.0	0.0
	SLN210 ring / End	28.4	-0.2	-14.1	0.0	0.0	0.0
	SLN213 ring / End	28.2	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN46 column below / Begin	141.2	0.0	-0.9	0.0	-3.8	0.0
	SLN210 ring / End	28.0	-0.2	-20.8	0.0	0.0	0.0
	SLN213 ring / End	27.7	0.2	-20.8	0.0	0.0	0.0
C LC_154	SLN46 column below / Begin	89.6	0.0	-1.8	0.0	-7.2	0.0
	SLN210 ring / End	12.6	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.4	0.1	-11.6	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.7 < 100%	OK
Welds	76.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN199+SLN211 radial (continuous)-bfl 1	12.7	C LC_119	151.8	0.0	45.9	OK
SLN199+SLN211 radial (continuous)-tfl 1	12.7	C LC_153	72.6	0.0	0.0	OK
SLN199+SLN211 radial (continuous)-w 1	7.2	C LC_125	245.2	0.0	35.1	OK
SLN46 column below	10.0	C LC_119	174.4	0.0	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_119	28.6	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_145	135.1	0.0	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_119	29.0	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_120	20.3	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_119	141.6	0.0	0.0	OK
End plate column below SLN46 column below	12.0	C LC_107	133.6	0.0	18.0	OK
End plate SLN210 ring	12.0	C LC_125	33.6	0.0	2.0	OK
End plate SLN213 ring	12.0	C LC_125	32.9	0.0	5.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

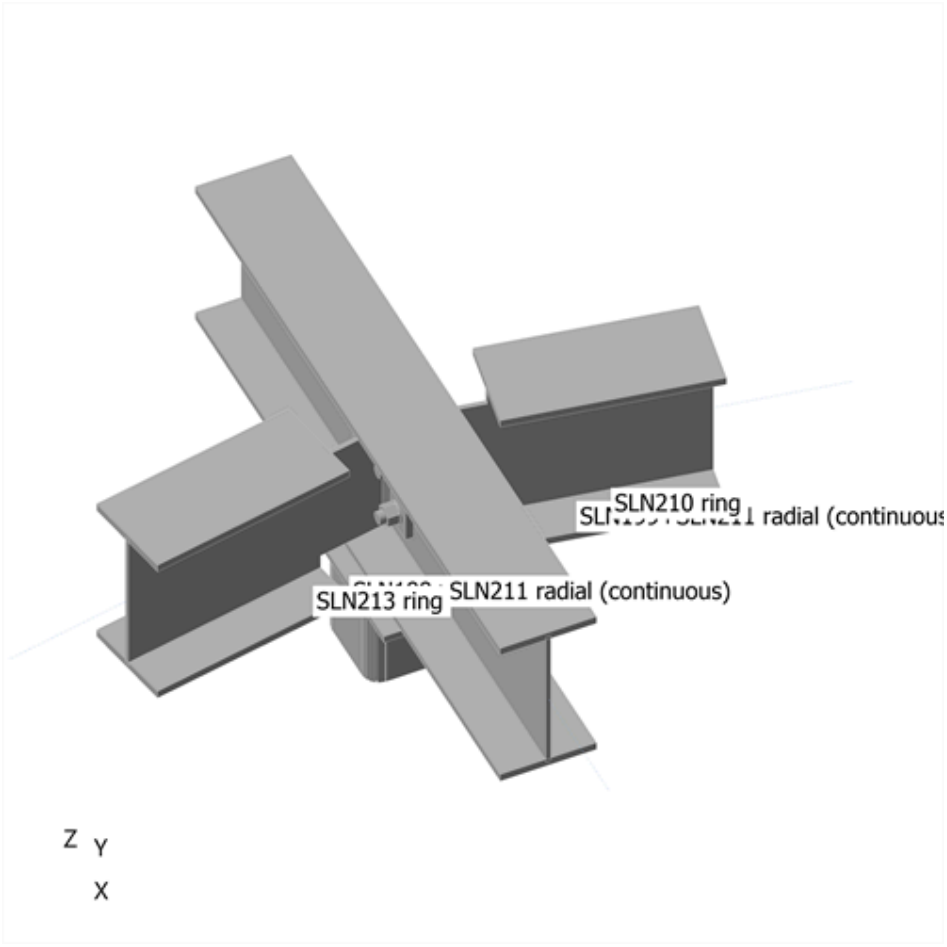
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

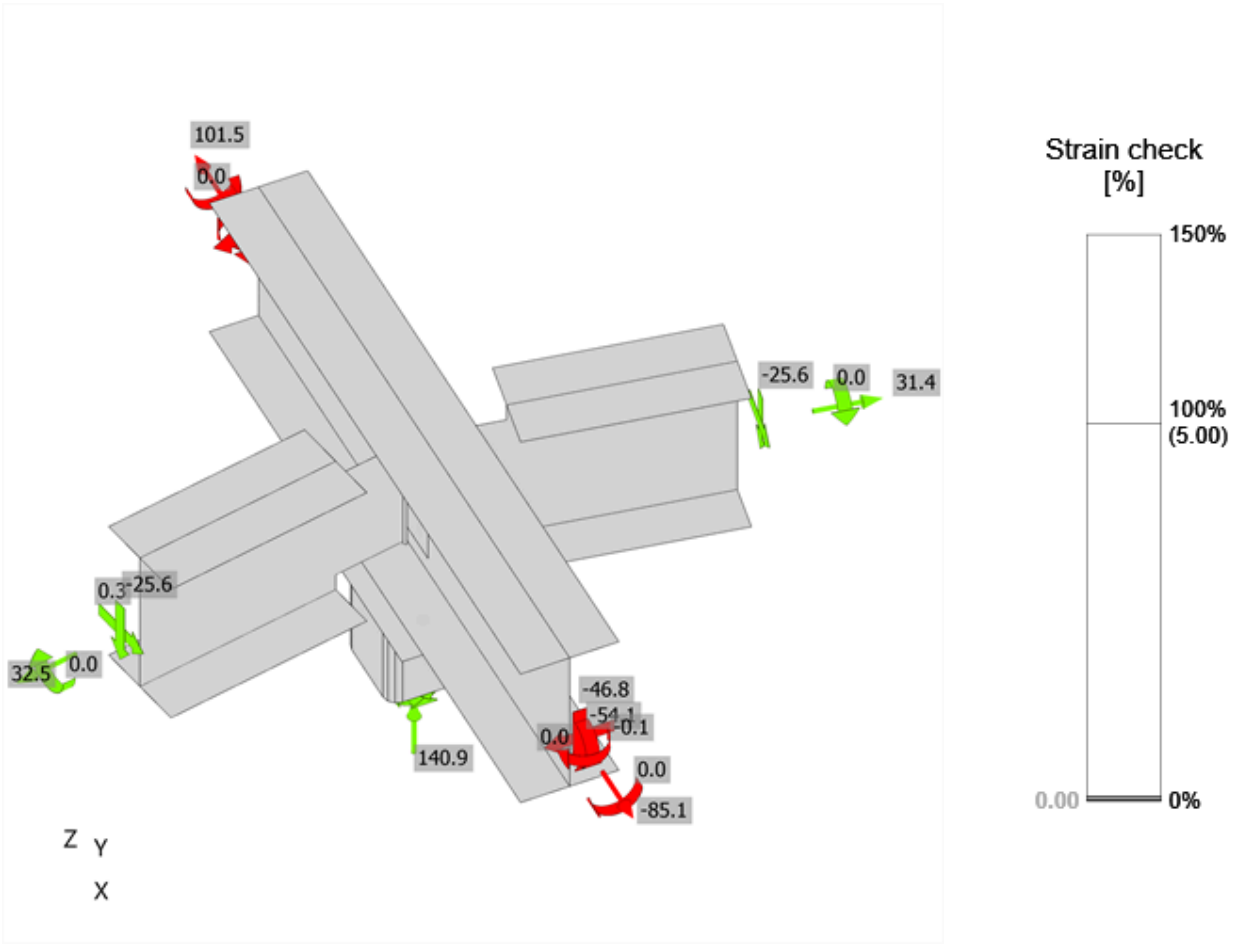
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN46 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

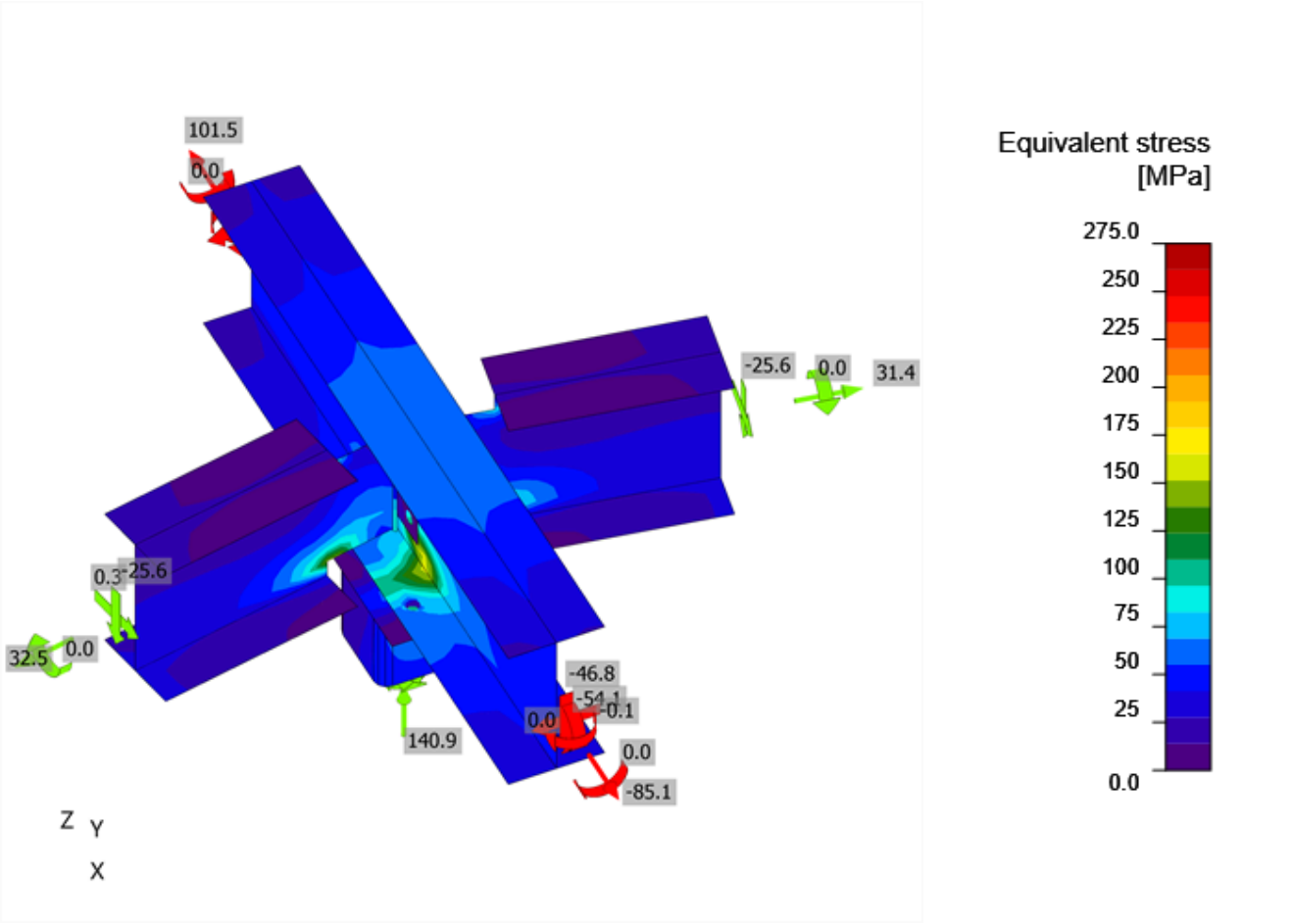
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

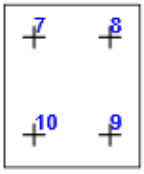


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.6	4.4	172.8	0.4	4.7	5.0	OK	OK
	B3	M20 8.8 - 1	C LC_107	19.3	2.8	85.9	13.7	3.3	12.7	OK	OK
	B4	M20 8.8 - 1	C LC_107	16.9	2.1	129.6	12.0	2.2	10.8	OK	OK
	B5	M20 8.8 - 1	C LC_123	0.4	5.1	181.3	0.3	5.4	5.6	OK	OK
	B7	M20 8.8 - 1	C LC_123	0.2	0.3	123.8	0.2	0.5	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.1	0.8	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.7	0.4	63.0	1.2	0.5	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_125	1.7	0.9	123.8	1.2	1.2	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN46 column below	SLN46 column below	◢ 4.2 ◣	521	C LC_119	285.3	0.0	-139.0	-137.0	-43.6	70.5	38.9	OK	OK
		◢ 4.2 ◣	521	C LC_119	309.6	0.0	-148.9	150.8	42.6	76.5	42.6	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_119	94.6	0.0	27.3	-35.6	38.3	23.4	14.6	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_119	93.3	0.0	11.3	15.1	-51.3	23.0	15.2	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_121	80.7	0.0	-22.5	-19.0	40.5	19.9	12.1	OK	OK
		◢ 4.2 ◣	149	C LC_121	95.7	0.0	-17.2	22.9	-49.3	23.6	15.0	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_119	97.7	0.0	-18.2	-24.6	49.7	24.1	15.2	OK	OK
		◢ 4.2 ◣	149	C LC_119	80.4	0.0	-22.6	18.4	-40.6	19.9	11.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-200
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring

Name	SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

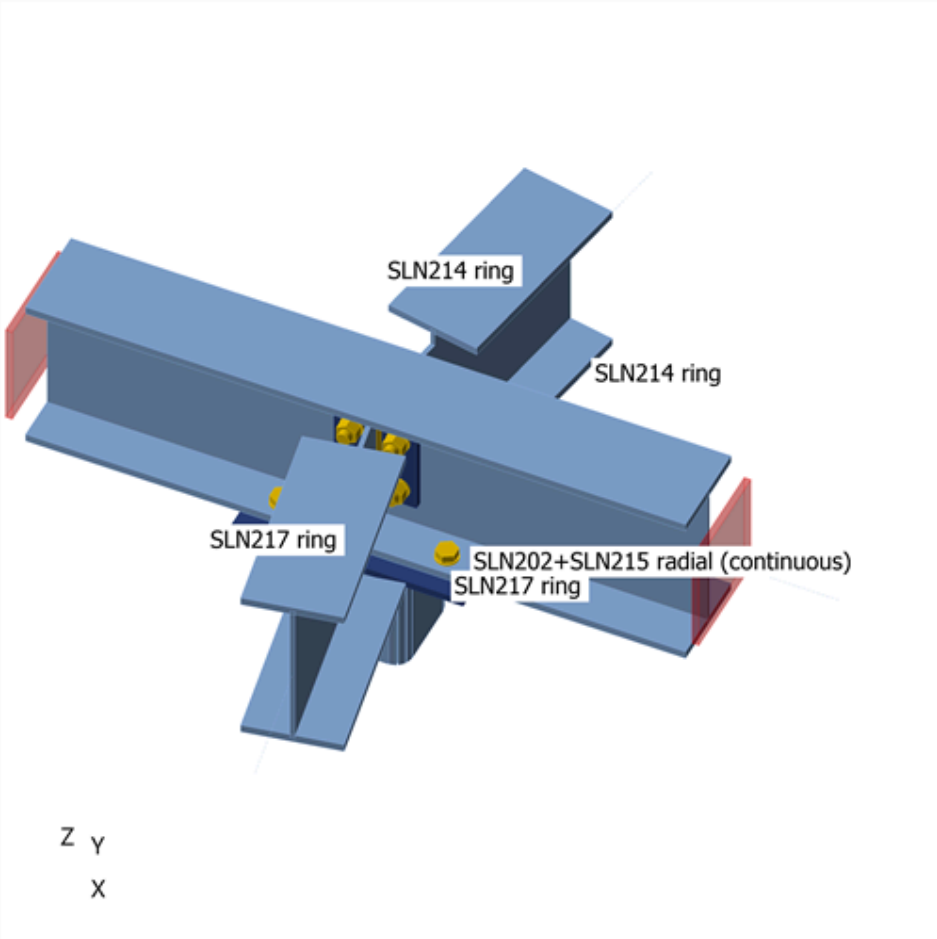
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN202+SLN215 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN48 column below	2 - SHS150x150x10	0.0	90.0	20.0	0	0	0
SLN214 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN202+SLN215 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN202+SLN215 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN48 column below / begin		Position	0
SLN214 ring / end		Position	0
SLN217 ring / end		Position	0



SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	27.7	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN48 column below / Begin	122.4	-1.2	-4.4	0.0	-4.4	0.0
	SLN214 ring / End	29.0	-0.2	-24.0	0.0	0.0	0.0
	SLN217 ring / End	27.6	0.3	-24.0	0.0	0.0	0.0
C LC_104	SLN48 column below / Begin	123.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	15.3	-0.1	-24.0	0.0	0.0	0.0
	SLN217 ring / End	15.2	0.2	-24.0	0.0	0.0	0.0
C LC_105	SLN48 column below / Begin	123.4	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	16.0	-0.1	-24.0	0.0	0.0	0.0
	SLN217 ring / End	15.7	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN48 column below / Begin	122.9	3.3	-1.9	0.0	-2.8	0.0
	SLN214 ring / End	20.5	-0.3	-24.0	0.0	0.0	0.0
	SLN217 ring / End	23.9	0.1	-24.0	0.0	0.0	0.0
C LC_107	SLN48 column below / Begin	1.3	-2.4	-7.9	0.0	-5.2	0.0
	SLN214 ring / End	13.1	0.2	-0.5	0.0	0.0	0.0
	SLN217 ring / End	10.7	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN48 column below / Begin	3.7	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	-10.0	0.0	-0.5	0.0	0.0	0.0
	SLN217 ring / End	-10.1	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN48 column below / Begin	3.3	0.0	0.0	0.0	-0.1	0.0
	SLN214 ring / End	-10.1	0.0	-0.5	0.0	0.0	0.0
	SLN217 ring / End	-10.2	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN48 column below / Begin	2.4	6.6	-2.9	0.0	-1.9	0.0
	SLN214 ring / End	-2.2	-0.1	-0.5	0.0	0.0	0.0
	SLN217 ring / End	4.5	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN48 column below / Begin	81.7	-2.4	-8.2	0.0	-6.4	0.0
	SLN214 ring / End	27.2	0.0	-16.2	0.0	0.0	0.0
	SLN217 ring / End	24.7	0.2	-16.2	0.0	0.0	0.0
C LC_112	SLN48 column below / Begin	84.2	0.0	-0.4	0.0	-1.5	0.0
	SLN214 ring / End	2.7	-0.1	-16.2	0.0	0.0	0.0
	SLN217 ring / End	2.5	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN48 column below / Begin	83.8	0.0	-0.3	0.0	-1.3	0.0
	SLN214 ring / End	3.1	-0.1	-16.2	0.0	0.0	0.0
	SLN217 ring / End	2.8	0.1	-16.2	0.0	0.0	0.0
C LC_114	SLN48 column below / Begin	82.8	6.6	-3.2	0.0	-3.1	0.0
	SLN214 ring / End	11.3	-0.2	-16.2	0.0	0.0	0.0
	SLN217 ring / End	18.0	0.0	-16.2	0.0	0.0	0.0
C LC_115	SLN48 column below / Begin	8.1	-2.4	-7.9	0.0	-5.3	0.0
	SLN214 ring / End	15.4	0.1	-1.7	0.0	0.0	0.0
	SLN217 ring / End	13.1	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN48 column below / Begin	10.5	0.0	-0.1	0.0	-0.3	0.0
	SLN214 ring / End	-7.8	0.0	-1.7	0.0	0.0	0.0
	SLN217 ring / End	-7.9	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN48 column below / Begin	10.1	0.0	0.0	0.0	-0.2	0.0
	SLN214 ring / End	-7.9	0.0	-1.7	0.0	0.0	0.0
	SLN217 ring / End	-8.0	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN48 column below / Begin	9.2	6.6	-2.9	0.0	-2.0	0.0
	SLN214 ring / End	0.1	-0.1	-1.7	0.0	0.0	0.0
	SLN217 ring / End	6.7	-0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN48 column below / Begin	130.8	-1.2	-4.4	0.0	-4.5	0.0
	SLN214 ring / End	32.6	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	31.2	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN48 column below / Begin	131.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	18.7	-0.1	-25.5	0.0	0.0	0.0
	SLN217 ring / End	18.5	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN48 column below / Begin	131.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	19.5	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	19.0	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN48 column below / Begin	131.2	3.3	-1.9	0.0	-2.9	0.0
	SLN214 ring / End	24.0	-0.3	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN48 column below / Begin	98.6	-2.4	-8.2	0.0	-6.7	0.0
	SLN214 ring / End	34.4	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	31.8	0.3	-19.4	0.0	0.0	0.0
C LC_124	SLN48 column below / Begin	100.2	0.0	-0.4	0.0	-1.7	0.0
	SLN214 ring / End	9.4	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	9.2	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN48 column below / Begin	100.1	0.0	-0.4	0.0	-1.5	0.0
	SLN214 ring / End	10.0	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	9.5	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN48 column below / Begin	99.4	6.6	-3.2	0.0	-3.4	0.0
	SLN214 ring / End	18.3	-0.3	-19.3	0.0	0.0	0.0
	SLN217 ring / End	25.1	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN48 column below / Begin	25.0	-2.4	-7.9	0.0	-5.5	0.0
	SLN214 ring / End	22.6	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	20.2	0.2	-4.9	0.0	0.0	0.0
C LC_128	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.5	0.0
	SLN214 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	-1.3	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.4	0.0
	SLN214 ring / End	-1.0	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	-1.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN48 column below / Begin	25.8	6.6	-2.9	0.0	-2.2	0.0
	SLN214 ring / End	7.0	-0.2	-4.9	0.0	0.0	0.0
	SLN217 ring / End	13.8	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	28.1	-0.2	-25.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / End	27.9	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.7	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	28.8	-0.3	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.6	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	26.5	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	26.4	0.1	-25.6	0.0	0.0	0.0
C LC_147	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	27.4	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.0	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	28.0	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN48 column below / Begin	80.1	0.0	-1.3	0.0	-5.4	0.0
	SLN214 ring / End	12.3	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.3	0.1	-11.8	0.0	0.0	0.0
C LC_152	SLN48 column below / Begin	77.3	0.0	0.8	0.0	3.3	0.0
	SLN214 ring / End	28.3	-0.2	-14.5	0.0	0.0	0.0
	SLN217 ring / End	28.1	0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-2.1	0.0
	SLN214 ring / End	27.8	-0.2	-21.5	0.0	0.0	0.0
	SLN217 ring / End	27.6	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN48 column below / Begin	79.3	0.0	-1.3	0.0	-5.3	0.0
	SLN214 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.4	0.1	-11.8	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.3 < 100%	OK
Welds	64.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN202+SLN215 radial (continuous)-bfl 1	12.7	C LC_119	128.4	0.0	39.0	OK
SLN202+SLN215 radial (continuous)-tfl 1	12.7	C LC_153	64.0	0.0	0.0	OK
SLN202+SLN215 radial (continuous)-w 1	7.2	C LC_126	235.7	0.0	29.4	OK
SLN48 column below	10.0	C LC_119	143.3	0.0	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_119	28.7	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_120	19.9	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_145	135.8	0.0	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_119	28.8	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_120	20.1	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_119	140.7	0.0	0.0	OK
End plate column below SLN48 column below	12.0	C LC_107	132.9	0.0	17.4	OK
End plate SLN214 ring	12.0	C LC_126	34.4	0.0	7.7	OK
End plate SLN217 ring	12.0	C LC_126	34.8	0.0	1.1	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

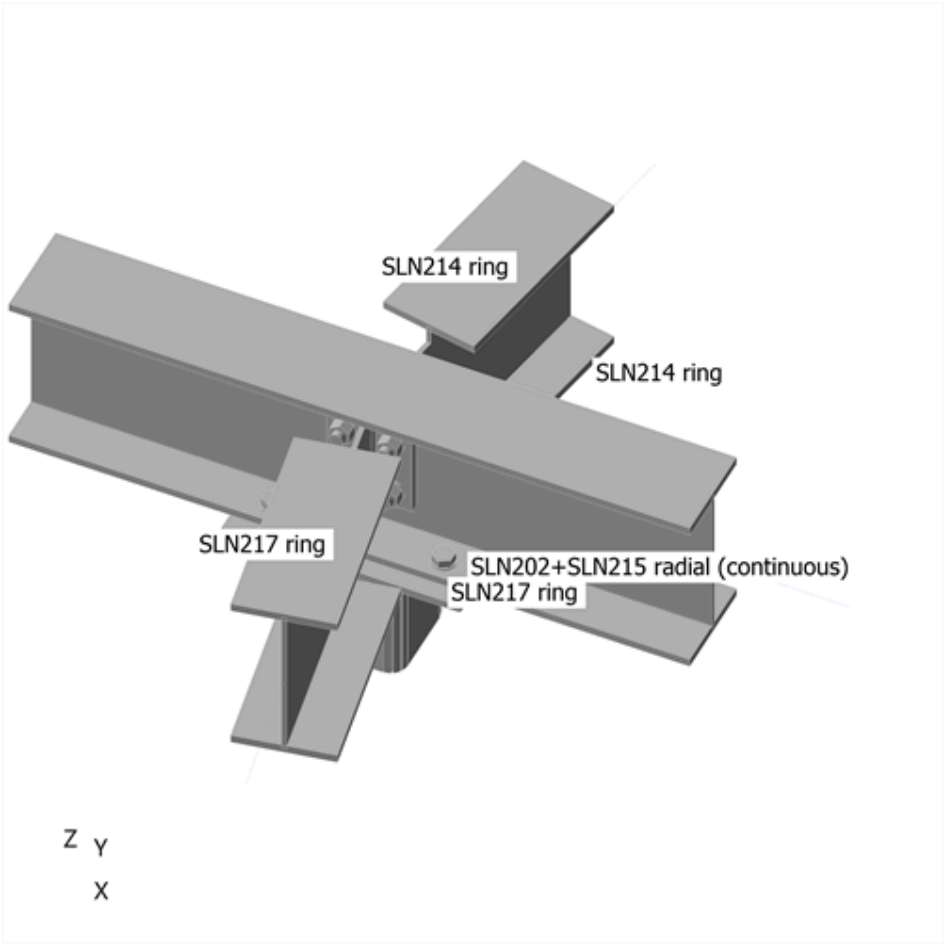
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

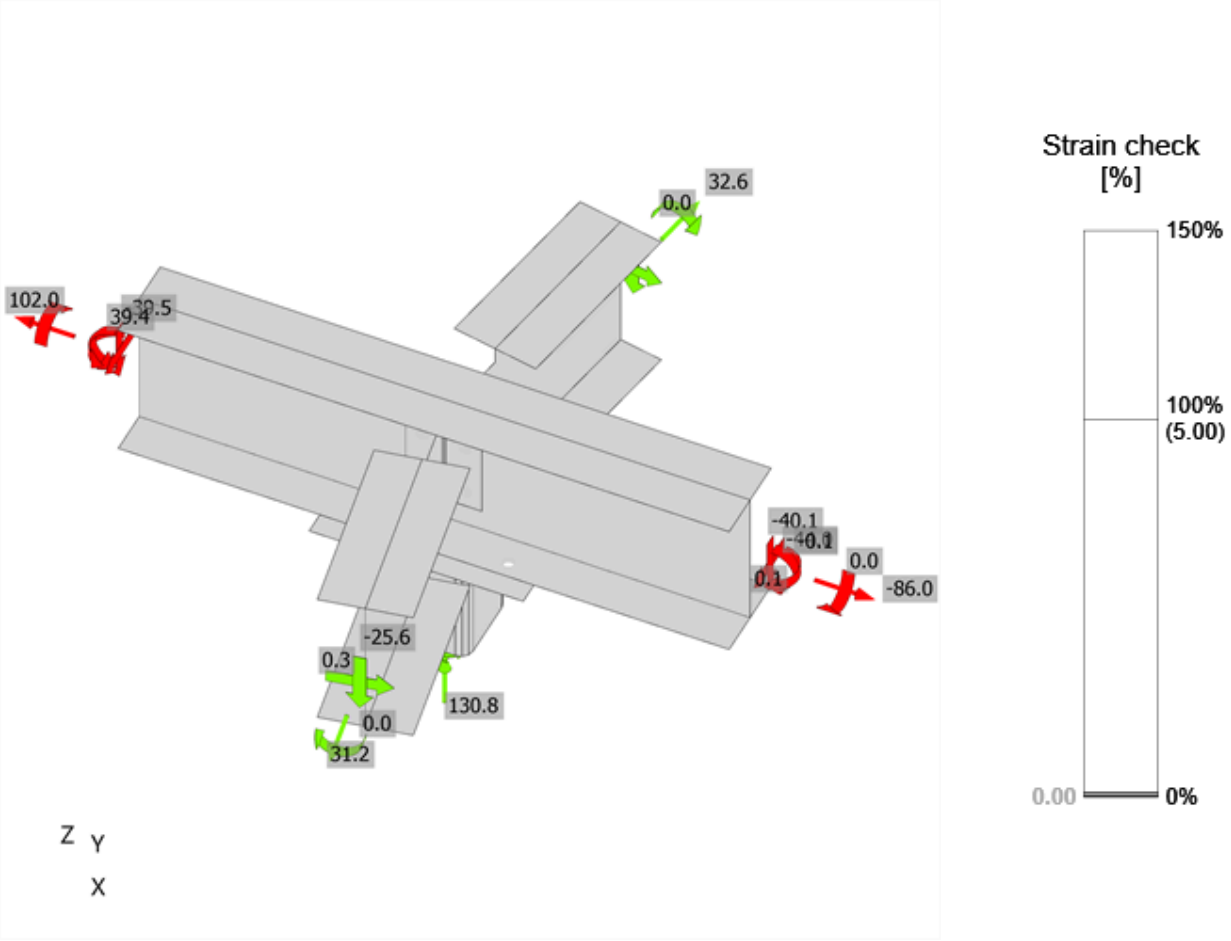
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN48 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

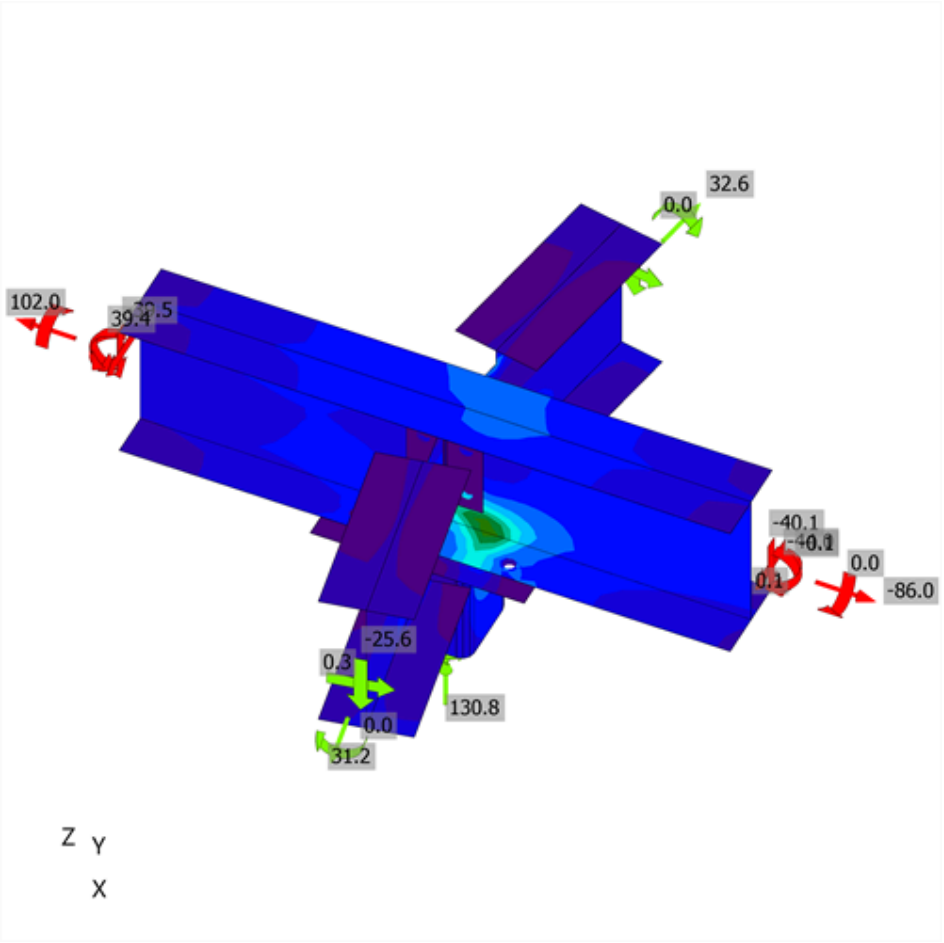
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

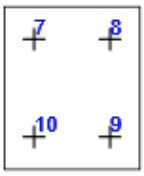


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_123	0.3	4.7	181.5	0.2	5.0	5.1	OK	OK
	B3	M20 8.8 - 1	C LC_107	16.4	2.1	130.9	11.6	2.2	10.5	OK	OK
	B4	M20 8.8 - 1	C LC_107	18.8	2.8	86.2	13.3	3.2	12.5	OK	OK
	B5	M20 8.8 - 1	C LC_126	0.2	4.3	189.8	0.2	4.5	4.6	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.5	0.2	63.8	0.4	0.3	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.7	0.3	62.8	0.5	0.5	0.7	OK	OK
	B9	M20 8.8 - 1	C LC_126	1.8	0.4	72.9	1.3	0.5	1.4	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.7	0.8	123.8	1.2	1.0	1.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN48 column below	SLN48 column below	◢ 4.2 ◣	521	C LC_119	243.3	0.0	-117.5	-121.4	19.8	60.1	32.0	OK	OK
		◢ 4.2 ◣	521	C LC_119	260.0	0.0	-131.7	127.7	-20.8	64.2	34.7	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_122	86.7	0.0	31.5	26.0	-38.7	21.4	12.3	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	90.1	0.0	11.9	-9.1	50.8	22.3	14.3	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_119	71.6	0.0	-19.4	-15.1	36.8	17.7	10.8	OK	OK
		◢ 4.2 ◣	149	C LC_119	84.9	0.0	-16.8	22.9	-42.2	21.0	13.3	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	94.9	0.0	-18.9	-24.7	47.7	23.4	14.9	OK	OK
		◢ 4.2 ◣	149	C LC_122	78.9	0.0	-20.3	16.1	-41.0	19.5	12.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-240
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring

Name	SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

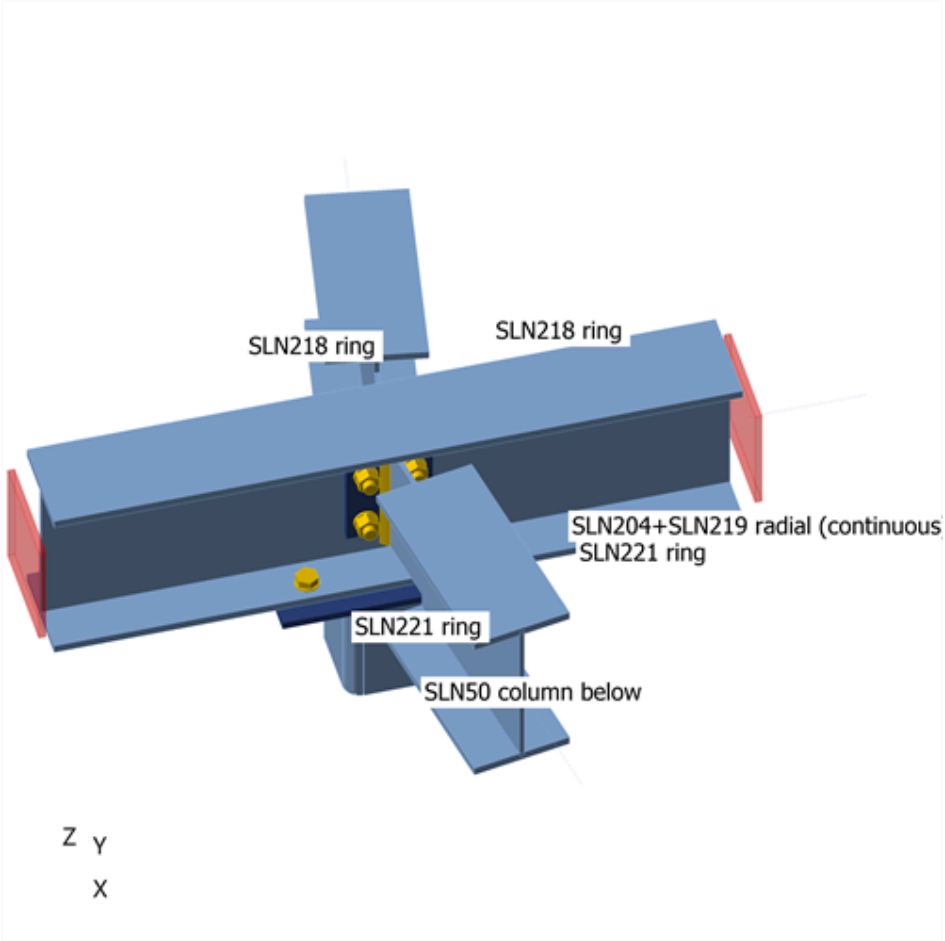
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN204+SLN219 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN50 column below	2 - SHS150x150x10	0.0	90.0	60.0	0	0	0
SLN218 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN204+SLN219 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN204+SLN219 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN50 column below / begin		Position	0
SLN218 ring / end		Position	0
SLN221 ring / end		Position	0



SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.7	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.8	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	27.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	27.8	0.2	-25.7	0.0	0.0	0.0
C LC_103	SLN50 column below / Begin	133.0	-3.0	-2.8	0.0	-4.5	0.0
	SLN218 ring / End	24.6	-0.1	-24.5	0.0	0.0	0.0
	SLN221 ring / End	21.8	0.2	-24.1	0.0	0.0	0.0
C LC_104	SLN50 column below / Begin	133.4	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	15.6	-0.1	-24.5	0.0	0.0	0.0
	SLN221 ring / End	16.1	0.1	-24.1	0.0	0.0	0.0
C LC_105	SLN50 column below / Begin	133.5	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	14.8	-0.2	-24.5	0.0	0.0	0.0
	SLN221 ring / End	15.0	0.1	-24.1	0.0	0.0	0.0
C LC_106	SLN50 column below / Begin	132.9	1.8	-4.4	0.0	-5.5	0.0
	SLN218 ring / End	27.3	-0.2	-24.5	0.0	0.0	0.0
	SLN221 ring / End	29.6	0.1	-24.1	0.0	0.0	0.0
C LC_107	SLN50 column below / Begin	2.7	-6.1	-4.2	0.0	-2.9	0.0
	SLN218 ring / End	5.7	0.2	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-0.3	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN50 column below / Begin	3.6	0.0	0.0	0.0	-0.2	0.0
	SLN218 ring / End	-9.8	0.0	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-9.9	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN50 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN218 ring / End	-10.9	-0.1	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-11.0	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN50 column below / Begin	2.5	3.5	-7.3	0.0	-5.0	0.0
	SLN218 ring / End	10.5	0.0	-0.5	0.0	0.0	0.0
	SLN221 ring / End	14.2	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN50 column below / Begin	89.7	-6.1	-4.7	0.0	-4.9	0.0
	SLN218 ring / End	19.3	0.1	-16.5	0.0	0.0	0.0
	SLN221 ring / End	13.5	0.2	-16.3	0.0	0.0	0.0
C LC_112	SLN50 column below / Begin	90.6	0.0	-0.5	0.0	-2.2	0.0
	SLN218 ring / End	3.0	-0.1	-16.5	0.0	0.0	0.0
	SLN221 ring / End	3.2	0.1	-16.3	0.0	0.0	0.0
C LC_113	SLN50 column below / Begin	90.8	0.0	-0.6	0.0	-2.3	0.0
	SLN218 ring / End	1.7	-0.2	-16.5	0.0	0.0	0.0
	SLN221 ring / End	1.8	0.0	-16.3	0.0	0.0	0.0
C LC_114	SLN50 column below / Begin	89.5	3.5	-7.8	0.0	-7.0	0.0
	SLN218 ring / End	24.3	-0.2	-16.5	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.0	-16.3	0.0	0.0	0.0
C LC_115	SLN50 column below / Begin	10.1	-6.1	-4.2	0.0	-3.0	0.0
	SLN218 ring / End	8.0	0.2	-1.8	0.0	0.0	0.0
	SLN221 ring / End	2.0	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN50 column below / Begin	11.0	0.0	-0.1	0.0	-0.3	0.0
	SLN218 ring / End	-7.5	0.0	-1.8	0.0	0.0	0.0
	SLN221 ring / End	-7.7	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN50 column below / Begin	11.2	0.0	-0.1	0.0	-0.4	0.0
	SLN218 ring / End	-8.7	-0.1	-1.8	0.0	0.0	0.0
	SLN221 ring / End	-8.8	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN50 column below / Begin	9.9	3.5	-7.3	0.0	-5.2	0.0
	SLN218 ring / End	12.8	0.0	-1.8	0.0	0.0	0.0
	SLN221 ring / End	16.5	0.0	-1.7	0.0	0.0	0.0
C LC_119	SLN50 column below / Begin	141.9	-3.0	-2.9	0.0	-4.6	0.0
	SLN218 ring / End	28.2	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	25.4	0.2	-25.7	0.0	0.0	0.0
C LC_120	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	18.9	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	19.5	0.1	-25.7	0.0	0.0	0.0
C LC_121	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	18.1	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	18.3	0.1	-25.7	0.0	0.0	0.0
C LC_122	SLN50 column below / Begin	141.9	1.8	-4.4	0.0	-5.7	0.0
	SLN218 ring / End	30.8	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	33.2	0.2	-25.7	0.0	0.0	0.0
C LC_123	SLN50 column below / Begin	107.6	-6.1	-4.8	0.0	-5.3	0.0
	SLN218 ring / End	26.5	0.0	-19.7	0.0	0.0	0.0
	SLN221 ring / End	20.6	0.2	-19.5	0.0	0.0	0.0
C LC_124	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.5	0.0
	SLN218 ring / End	9.7	-0.1	-19.7	0.0	0.0	0.0
	SLN221 ring / End	10.0	0.2	-19.5	0.0	0.0	0.0
C LC_125	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.6	0.0
	SLN218 ring / End	8.4	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	8.4	0.0	-19.5	0.0	0.0	0.0
C LC_126	SLN50 column below / Begin	107.5	3.5	-7.9	0.0	-7.4	0.0
	SLN218 ring / End	31.4	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	35.6	0.1	-19.5	0.0	0.0	0.0
C LC_127	SLN50 column below / Begin	28.0	-6.1	-4.3	0.0	-3.4	0.0
	SLN218 ring / End	15.2	0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.1	0.2	-4.9	0.0	0.0	0.0
C LC_128	SLN50 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN218 ring / End	-0.9	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	-0.9	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN50 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN218 ring / End	-2.0	-0.2	-5.0	0.0	0.0	0.0
	SLN221 ring / End	-2.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN50 column below / Begin	27.9	3.5	-7.4	0.0	-5.5	0.0
	SLN218 ring / End	20.0	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	23.8	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	27.8	-0.2	-26.1	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN221 ring / End	28.3	0.2	-25.7	0.0	0.0	0.0
C LC_141	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.9	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.9	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.5	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.5	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.9	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.0	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.2	-25.7	0.0	0.0	0.0
C LC_146	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.6	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	27.3	0.1	-25.7	0.0	0.0	0.0
C LC_147	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.4	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	26.7	0.1	-25.7	0.0	0.0	0.0
C LC_148	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.9	0.2	-25.7	0.0	0.0	0.0
C LC_151	SLN50 column below / Begin	90.1	0.0	-1.7	0.0	-7.0	0.0
	SLN218 ring / End	12.2	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.5	0.1	-11.7	0.0	0.0	0.0
C LC_152	SLN50 column below / Begin	79.3	0.0	0.8	0.0	3.3	0.0
	SLN218 ring / End	28.0	-0.2	-14.0	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.1	-13.9	0.0	0.0	0.0
C LC_153	SLN50 column below / Begin	141.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	27.5	-0.2	-20.7	0.0	0.0	0.0
	SLN221 ring / End	27.9	0.2	-20.6	0.0	0.0	0.0
C LC_154	SLN50 column below / Begin	89.5	0.0	-1.7	0.0	-6.9	0.0
	SLN218 ring / End	12.3	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.6	0.1	-11.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.1 < 100%	OK
Welds	72.0 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN204+SLN219 radial (continuous)-bfl 1	12.7	C LC_122	150.1	0.0	44.7	OK
SLN204+SLN219 radial (continuous)-tfl 1	12.7	C LC_153	72.5	0.0	0.0	OK
SLN204+SLN219 radial (continuous)-w 1	7.2	C LC_123	244.0	0.0	36.4	OK
SLN50 column below	10.0	C LC_122	165.1	0.0	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_122	28.9	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_121	20.7	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_122	136.9	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_122	29.0	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_120	20.0	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_122	132.8	0.0	0.0	OK
End plate column below SLN50 column below	12.0	C LC_110	130.9	0.0	17.2	OK
End plate SLN218 ring	12.0	C LC_123	33.9	0.0	1.5	OK
End plate SLN221 ring	12.0	C LC_123	30.6	0.0	5.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

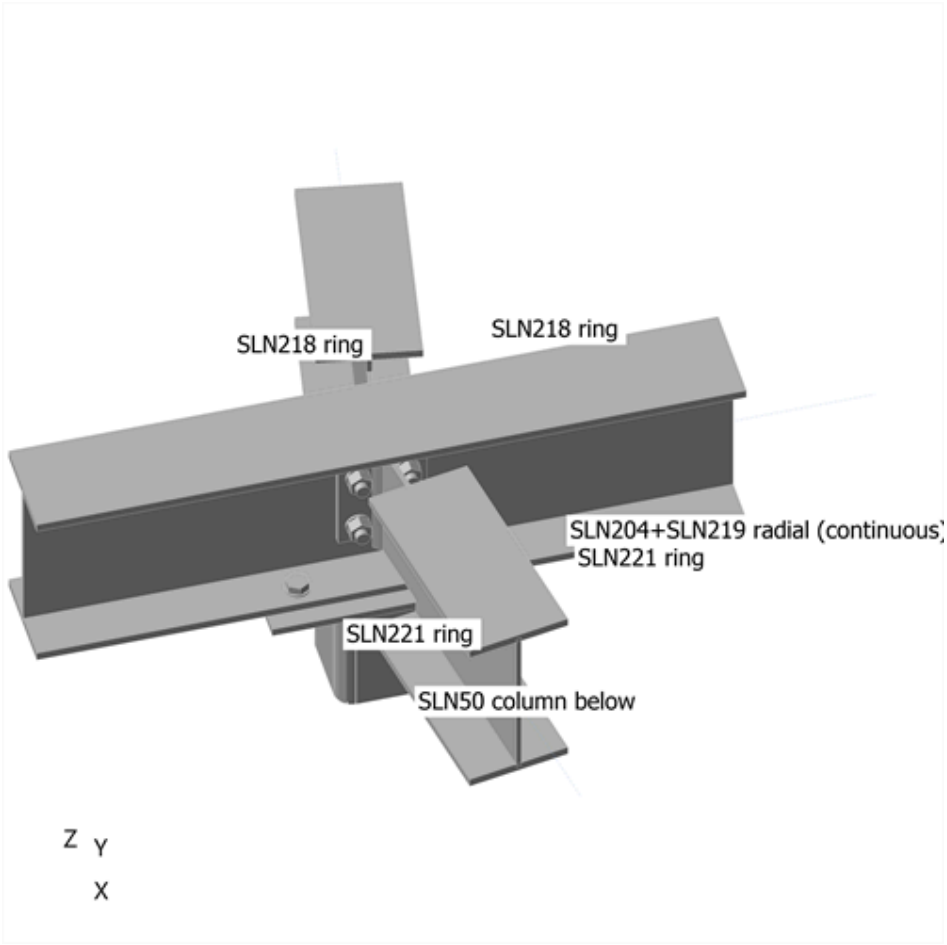
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

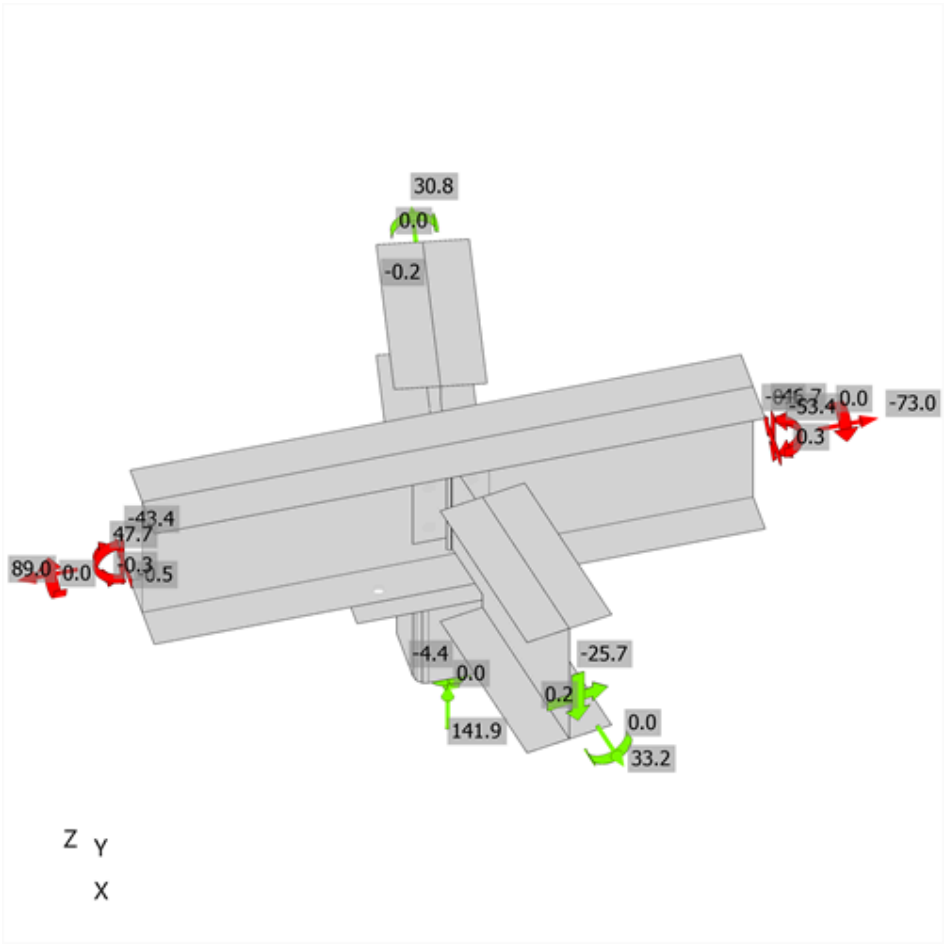
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN50 column below	150	C LC_154	0	5	0.0	OK

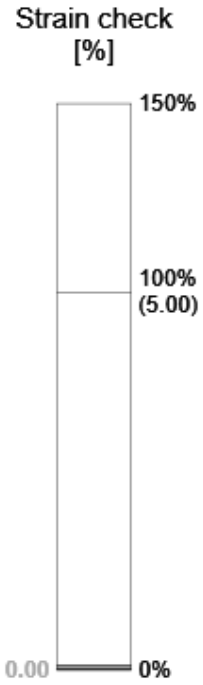
Symbol explanation

d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation

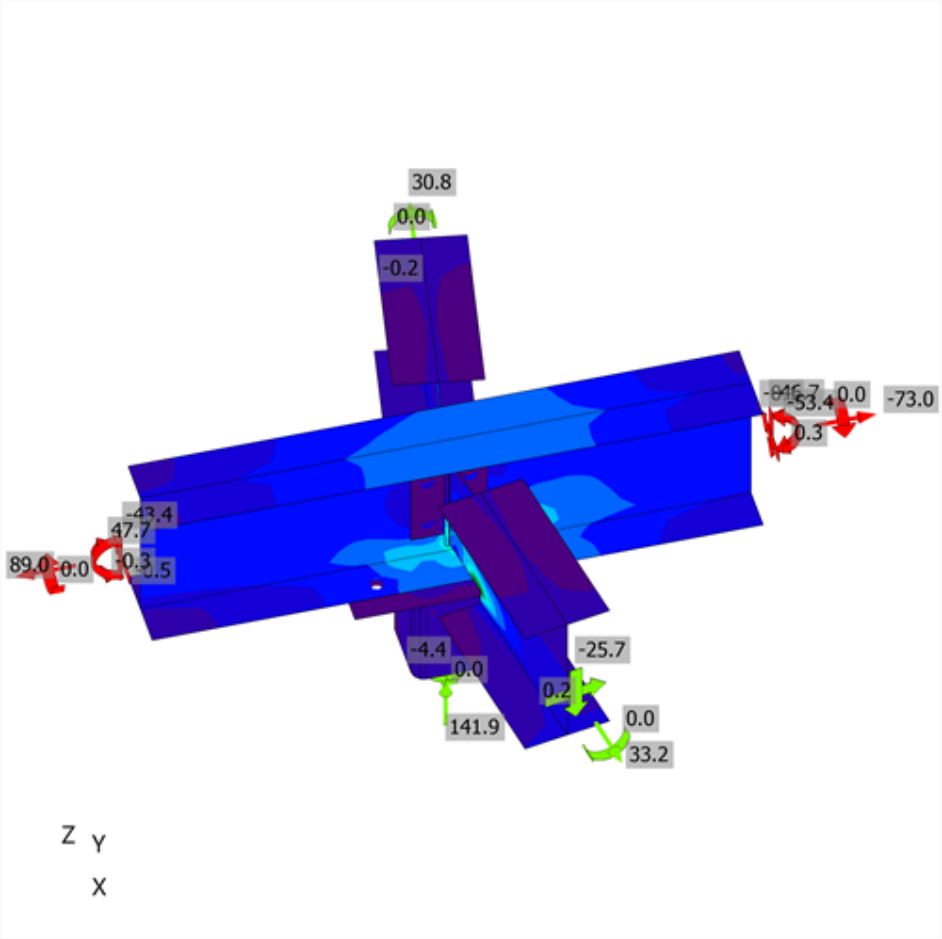


Overall check



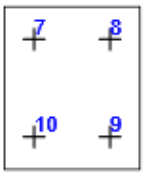


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_123	0.2	4.8	196.7	0.2	5.1	5.2	OK	OK
	B3	M20 8.8 - 1	C LC_110	18.4	2.8	81.9	13.1	3.4	12.3	OK	OK
	B4	M20 8.8 - 1	C LC_110	14.9	1.8	156.4	10.6	2.0	9.5	OK	OK
	B5	M20 8.8 - 1	C LC_126	0.4	5.0	186.8	0.3	5.4	5.6	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.4	0.3	123.8	0.3	0.5	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.6	0.4	123.8	0.4	0.7	0.8	OK	OK
	B9	M20 8.8 - 1	C LC_123	1.7	0.4	63.8	1.2	0.4	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.6	0.9	123.8	1.2	1.3	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN50 column below	SLN50 column below	◢ 4.2 ◣	521	C LC_122	268.1	0.0	-133.1	-134.3	0.0	66.2	36.8	OK	OK
		◢ 4.2 ◣	521	C LC_122	291.3	0.0	-146.5	145.3	2.1	72.0	40.4	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_119	92.9	0.0	9.7	7.2	-52.8	22.9	14.6	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_119	90.3	0.0	30.9	-24.2	42.6	22.3	12.9	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_119	81.7	0.0	-21.7	-17.5	42.0	20.2	12.4	OK	OK
		◢ 4.2 ◣	149	C LC_119	96.8	0.0	-18.0	24.2	-49.3	23.9	15.1	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_122	89.3	0.0	-16.4	-22.8	45.3	22.1	14.0	OK	OK
		◢ 4.2 ◣	149	C LC_122	77.2	0.0	-21.7	17.6	-39.0	19.1	11.6	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-A-320
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring

Name	SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

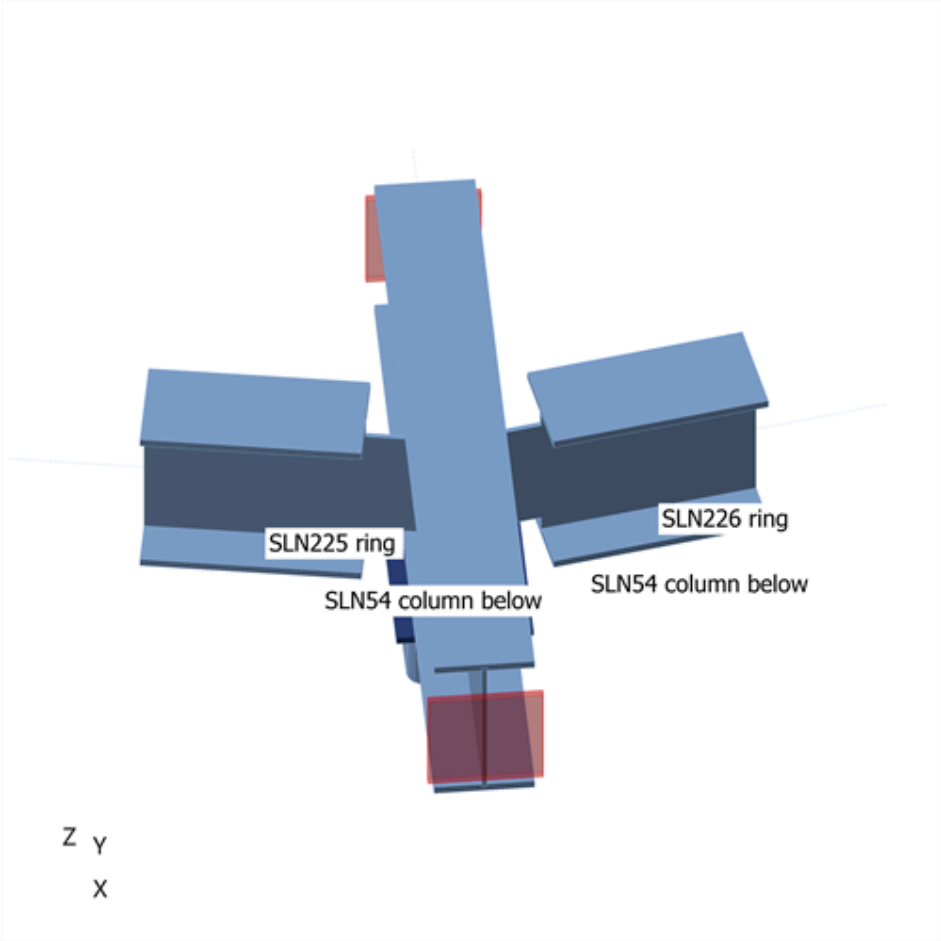
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN208+SLN224 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN54 column below	2 - SHS150x150x10	0.0	90.0	140.0	0	0	0
SLN225 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN208+SLN224 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN208+SLN224 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN54 column below / begin		Position	0
SLN225 ring / end		Position	0
SLN226 ring / end		Position	0



SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.3	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	19.2	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.9	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN54 column below / Begin	124.1	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	8.2	-0.4	-23.9	0.0	0.0	0.0
	SLN226 ring / End	7.8	0.2	-24.0	0.0	0.0	0.0
C LC_104	SLN54 column below / Begin	123.2	2.2	-3.6	0.0	-3.6	0.0
	SLN225 ring / End	17.5	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	19.6	0.1	-24.0	0.0	0.0	0.0
C LC_105	SLN54 column below / Begin	124.2	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	7.4	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	7.1	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN54 column below / Begin	123.1	-2.7	-3.1	0.0	-3.3	0.0
	SLN225 ring / End	18.7	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	15.7	0.2	-24.0	0.0	0.0	0.0
C LC_107	SLN54 column below / Begin	3.5	0.0	-0.1	0.0	-0.2	0.0
	SLN225 ring / End	-10.1	-0.1	-0.5	0.0	0.0	0.0
	SLN226 ring / End	-10.1	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN54 column below / Begin	1.8	4.5	-6.4	0.0	-4.3	0.0
	SLN225 ring / End	5.7	0.0	-0.5	0.0	0.0	0.0
	SLN226 ring / End	10.0	-0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN54 column below / Begin	3.8	0.0	0.0	0.0	-0.1	0.0
	SLN225 ring / End	-11.7	-0.1	-0.5	0.0	0.0	0.0
	SLN226 ring / End	-11.9	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN54 column below / Begin	1.6	-5.4	-5.4	0.0	-3.6	0.0
	SLN225 ring / End	8.0	0.0	-0.5	0.0	0.0	0.0
	SLN226 ring / End	2.8	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN54 column below / Begin	84.4	0.0	-0.3	0.0	-1.3	0.0
	SLN225 ring / End	-2.0	-0.3	-16.1	0.0	0.0	0.0
	SLN226 ring / End	-2.3	0.2	-16.2	0.0	0.0	0.0
C LC_112	SLN54 column below / Begin	82.7	4.5	-6.7	0.0	-5.3	0.0
	SLN225 ring / End	14.8	-0.3	-16.1	0.0	0.0	0.0
	SLN226 ring / End	19.0	0.0	-16.2	0.0	0.0	0.0
C LC_113	SLN54 column below / Begin	84.6	0.0	-0.3	0.0	-1.2	0.0
	SLN225 ring / End	-3.6	-0.4	-16.1	0.0	0.0	0.0
	SLN226 ring / End	-3.9	0.0	-16.2	0.0	0.0	0.0
C LC_114	SLN54 column below / Begin	82.5	-5.4	-5.6	0.0	-4.7	0.0
	SLN225 ring / End	17.1	-0.4	-16.1	0.0	0.0	0.0
	SLN226 ring / End	11.7	0.2	-16.2	0.0	0.0	0.0
C LC_115	SLN54 column below / Begin	10.3	0.0	-0.1	0.0	-0.3	0.0
	SLN225 ring / End	-8.5	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	-8.6	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN54 column below / Begin	8.7	4.5	-6.4	0.0	-4.3	0.0
	SLN225 ring / End	7.4	0.0	-1.7	0.0	0.0	0.0
	SLN226 ring / End	11.7	-0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN54 column below / Begin	10.6	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	-10.1	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	-10.3	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN54 column below / Begin	8.5	-5.4	-5.4	0.0	-3.7	0.0
	SLN225 ring / End	9.6	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	4.5	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN54 column below / Begin	132.1	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	11.2	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	10.8	0.2	-25.6	0.0	0.0	0.0
C LC_120	SLN54 column below / Begin	131.6	2.2	-3.6	0.0	-3.7	0.0
	SLN225 ring / End	20.7	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	22.7	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	10.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	10.2	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN54 column below / Begin	131.4	-2.7	-3.1	0.0	-3.4	0.0
	SLN225 ring / End	21.8	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.8	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN54 column below / Begin	100.4	0.0	-0.4	0.0	-1.4	0.0
	SLN225 ring / End	4.1	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	3.7	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN54 column below / Begin	99.3	4.5	-6.7	0.0	-5.5	0.0
	SLN225 ring / End	21.0	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	25.3	0.1	-19.4	0.0	0.0	0.0
C LC_125	SLN54 column below / Begin	100.7	0.0	-0.3	0.0	-1.4	0.0
	SLN225 ring / End	2.5	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	2.2	0.0	-19.4	0.0	0.0	0.0
C LC_126	SLN54 column below / Begin	99.1	-5.4	-5.7	0.0	-4.9	0.0
	SLN225 ring / End	23.3	-0.5	-19.3	0.0	0.0	0.0
	SLN226 ring / End	17.8	0.3	-19.4	0.0	0.0	0.0
C LC_127	SLN54 column below / Begin	26.3	0.0	-0.1	0.0	-0.4	0.0
	SLN225 ring / End	-2.5	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	-2.6	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN54 column below / Begin	25.3	4.5	-6.5	0.0	-4.5	0.0
	SLN225 ring / End	13.6	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	18.0	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN54 column below / Begin	26.7	0.0	-0.1	0.0	-0.4	0.0
	SLN225 ring / End	-4.0	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	-4.3	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN54 column below / Begin	25.0	-5.4	-5.5	0.0	-3.9	0.0
	SLN225 ring / End	15.8	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	10.6	0.2	-4.9	0.0	0.0	0.0
C LC_135	SLN54 column below / Begin	132.2	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	16.5	-0.6	-25.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / End	16.2	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.2	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.1	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN54 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN225 ring / End	6.5	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.5	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.2	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.5	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.4	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	18.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.0	0.1	-25.6	0.0	0.0	0.0
C LC_146	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	20.0	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	19.9	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	18.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.3	0.1	-25.6	0.0	0.0	0.0
C LC_148	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	20.0	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	19.6	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN54 column below / Begin	81.5	0.0	-1.3	0.0	-5.2	0.0
	SLN225 ring / End	6.5	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.3	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN54 column below / Begin	77.6	0.0	0.9	0.0	3.5	0.0
	SLN225 ring / End	21.0	-0.4	-13.9	0.0	0.0	0.0
	SLN226 ring / End	20.7	0.1	-14.1	0.0	0.0	0.0
C LC_153	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.8	0.0
	SLN225 ring / End	19.2	-0.5	-20.6	0.0	0.0	0.0
	SLN226 ring / End	19.0	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN54 column below / Begin	80.7	0.0	-1.2	0.0	-5.0	0.0
	SLN225 ring / End	6.6	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.5	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	11.7 < 100%	OK
Welds	63.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN208+SLN224 radial (continuous)-bfl 1	12.7	C LC_120	127.2	0.0	37.1	OK
SLN208+SLN224 radial (continuous)-tfl 1	12.7	C LC_153	64.7	0.0	0.0	OK
SLN208+SLN224 radial (continuous)-w 1	7.2	C LC_126	215.8	0.0	32.7	OK
SLN54 column below	10.0	C LC_120	138.5	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_122	28.2	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_121	22.8	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_122	158.2	0.0	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_120	27.1	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_119	21.3	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_122	123.1	0.0	0.0	OK
End plate column below SLN54 column below	12.0	C LC_108	117.2	0.0	15.4	OK
End plate SLN225 ring	12.0	C LC_126	30.1	0.0	1.3	OK
End plate SLN226 ring	12.0	C LC_124	28.3	0.0	0.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

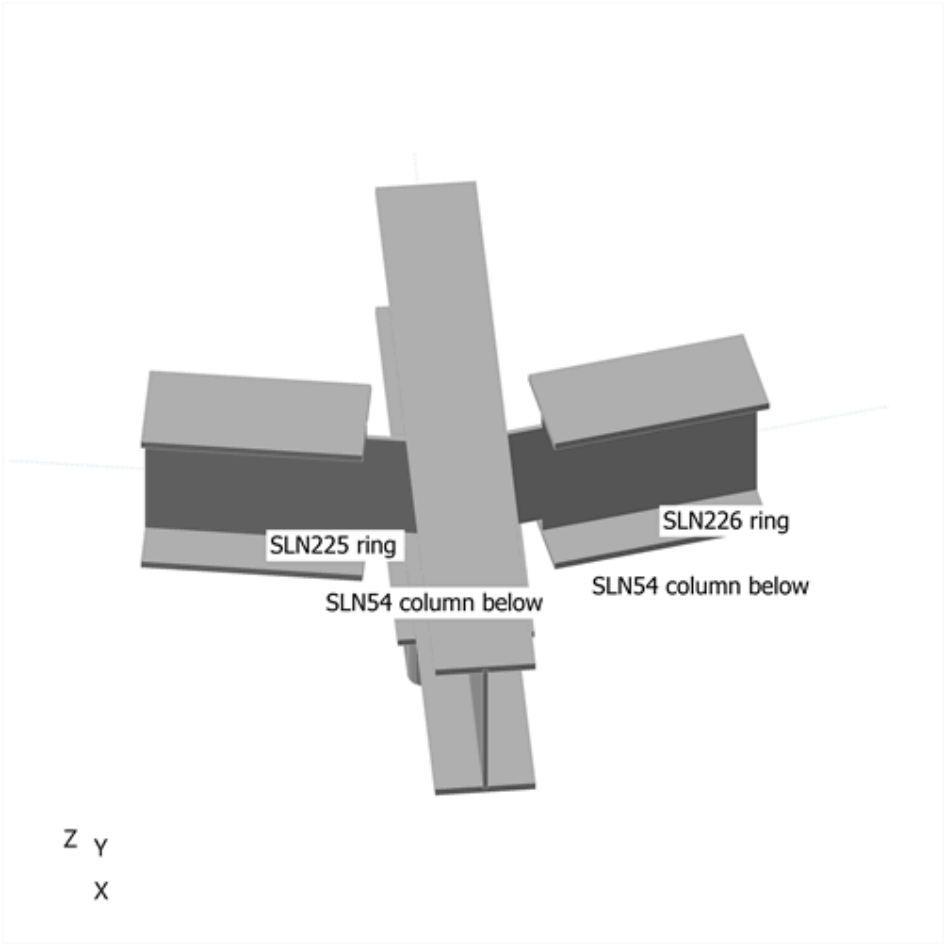
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

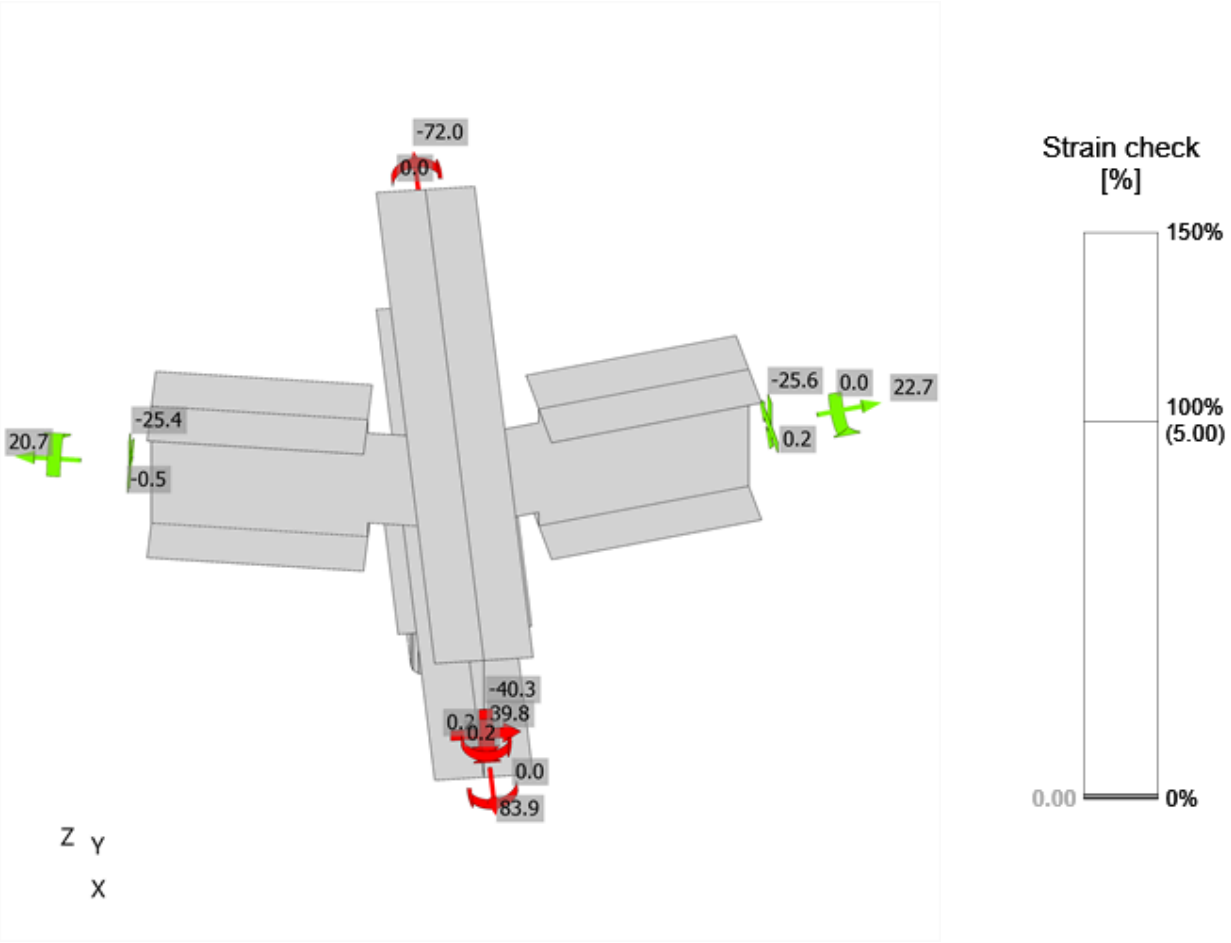
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN54 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

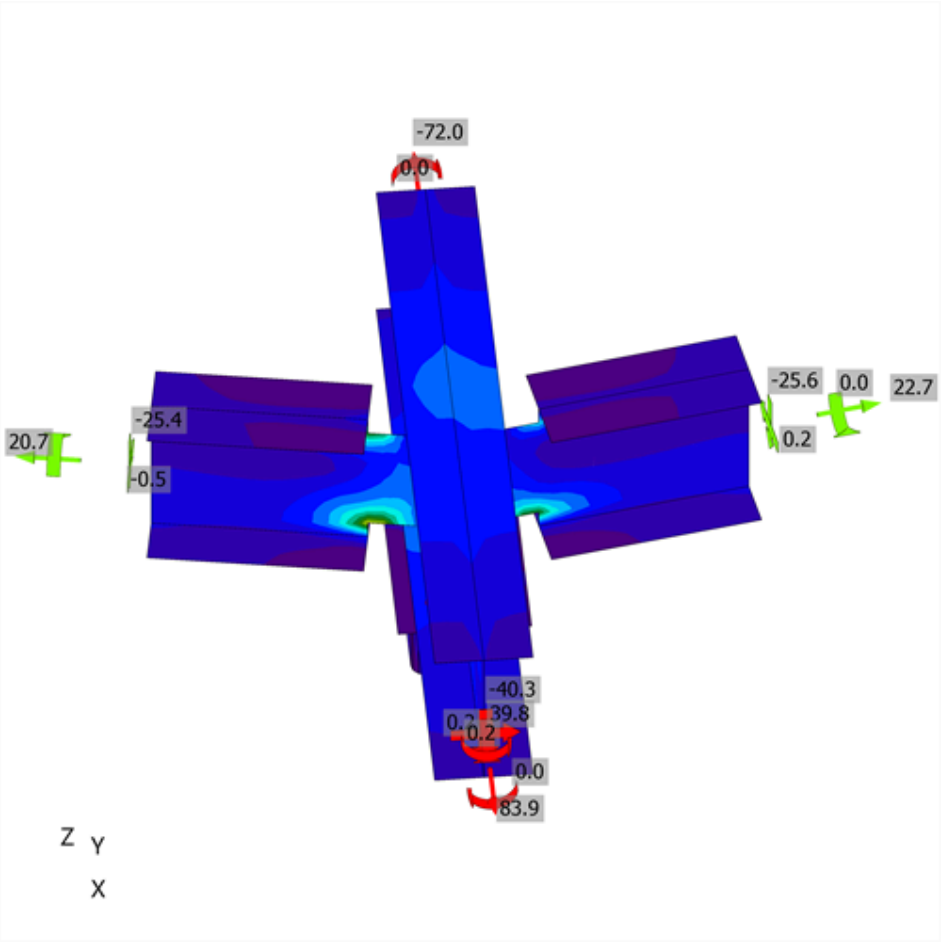
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

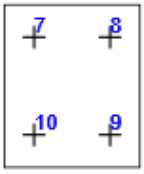


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_126	0.2	4.6	200.1	0.1	4.9	5.0	OK	OK
	B3	M20 8.8 - 1	C LC_108	16.5	2.8	79.4	11.7	3.5	11.3	OK	OK
	B4	M20 8.8 - 1	C LC_110	15.0	2.7	77.0	10.6	3.5	10.4	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.3	4.7	192.0	0.2	5.0	5.2	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.4	0.2	63.3	0.3	0.4	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.6	0.3	66.7	0.4	0.5	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_126	1.4	0.4	62.3	1.0	0.4	1.1	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.5	0.8	123.8	1.0	1.2	1.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN54 column below	SLN54 column below	◢ 4.2 ◣	521	C LC_120	243.0	0.0	-119.0	-115.8	-39.6	60.1	32.5	OK	OK
		◢ 4.2 ◣	521	C LC_120	258.3	0.0	-121.9	125.1	40.3	63.8	35.0	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_122	83.8	0.0	8.0	4.3	-48.0	20.7	13.1	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_120	85.2	0.0	7.9	-4.8	48.8	21.1	13.2	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_122	71.0	0.0	-17.9	-14.5	36.9	17.5	10.5	OK	OK
		◢ 4.2 ◣	149	C LC_122	90.6	0.0	-18.6	23.2	-45.7	22.4	14.5	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_120	90.4	0.0	-17.7	-23.7	45.4	22.3	14.1	OK	OK
		◢ 4.2 ◣	149	C LC_120	74.4	0.0	-19.3	15.0	-38.6	18.4	11.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-B-040
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring

Name	SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

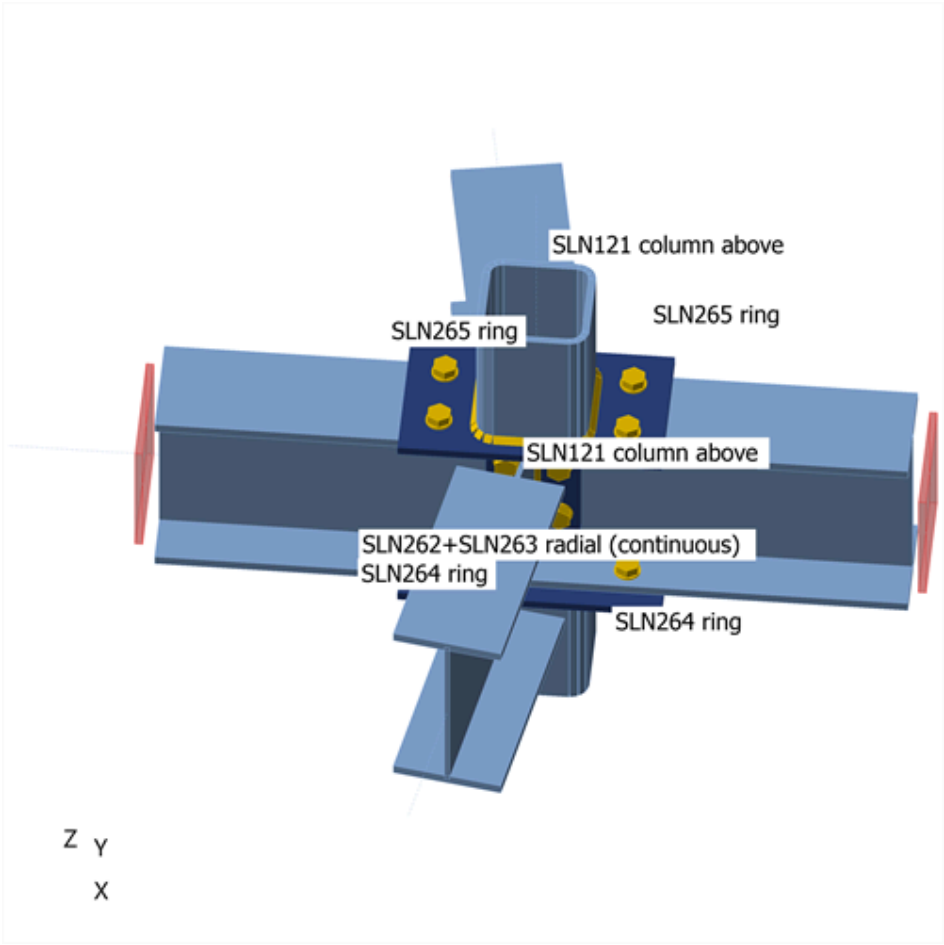
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN262+SLN263 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN40 column below	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN264 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN262+SLN263 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN262+SLN263 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN40 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN264 ring / end		Position	0
SLN265 ring / end		Position	0



SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.8	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.6	0.1
	SLN264 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.8	0.3	-26.0	0.0	0.0	0.0
C LC_103	SLN40 column below / Begin	220.8	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-61.8	0.1	1.7	-0.1	-1.8	0.2
	SLN264 ring / End	16.0	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	16.5	0.3	-24.4	0.0	0.0	0.0
C LC_104	SLN40 column below / Begin	227.1	-2.5	-3.2	0.0	-1.9	0.9
	SLN121 column above / End	-62.1	-0.2	1.4	-0.2	-0.9	-0.9
	SLN264 ring / End	28.1	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	25.4	0.3	-24.4	0.0	0.0	0.0
C LC_105	SLN40 column below / Begin	221.1	2.9	-2.8	0.0	-2.1	-1.0
	SLN121 column above / End	-62.1	0.3	1.6	-0.1	-1.5	1.0
	SLN264 ring / End	24.3	0.1	-24.0	0.0	0.0	0.0
	SLN265 ring / End	27.7	0.2	-24.4	0.0	0.0	0.0
C LC_106	SLN40 column below / Begin	226.8	0.1	0.1	0.0	0.2	-0.4
	SLN121 column above / End	-61.8	0.1	1.4	-0.1	-1.1	0.4
	SLN264 ring / End	14.5	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	14.9	0.2	-24.4	0.0	0.0	0.0
C LC_107	SLN40 column below / Begin	7.2	0.1	-0.1	0.0	-0.4	-0.3
	SLN121 column above / End	-5.1	0.1	0.1	-0.1	-0.1	0.3
	SLN264 ring / End	-8.7	0.1	-0.5	0.0	0.0	0.0
	SLN265 ring / End	-8.6	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN40 column below / Begin	19.6	-5.0	-6.2	0.0	-3.5	2.0
	SLN121 column above / End	-5.6	-0.5	-0.5	-0.1	1.5	-2.0
	SLN264 ring / End	11.9	0.2	-0.5	0.0	0.0	0.0
	SLN265 ring / End	6.3	0.0	-0.5	0.0	0.0	0.0
C LC_109	SLN40 column below / Begin	7.6	5.8	-5.4	0.0	-3.8	-1.9
	SLN121 column above / End	-5.7	0.5	0.0	0.1	0.3	1.8
	SLN264 ring / End	4.7	0.0	-0.5	0.0	0.0	0.0
	SLN265 ring / End	10.8	-0.2	-0.5	0.0	0.0	0.0
C LC_110	SLN40 column below / Begin	19.0	0.1	0.2	0.0	0.7	-0.5
	SLN121 column above / End	-5.0	0.1	-0.4	0.0	1.3	0.5
	SLN264 ring / End	-12.1	-0.1	-0.5	0.0	0.0	0.0
	SLN265 ring / End	-11.6	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN40 column below / Begin	149.7	0.1	-0.1	0.0	-0.5	-0.3
	SLN121 column above / End	-43.5	0.1	1.2	-0.1	-1.3	0.3
	SLN264 ring / End	4.0	0.2	-16.2	0.0	0.0	0.0
	SLN265 ring / End	4.4	0.3	-16.5	0.0	0.0	0.0
C LC_112	SLN40 column below / Begin	162.1	-5.0	-6.3	0.0	-3.7	2.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN121 column above / End	-43.9	-0.5	0.6	-0.2	0.4	-2.0
	SLN264 ring / End	25.8	0.3	-16.2	0.0	0.0	0.0
	SLN265 ring / End	20.3	0.2	-16.5	0.0	0.0	0.0
C LC_113	SLN40 column below / Begin	150.1	5.8	-5.5	0.0	-4.0	-1.9
	SLN121 column above / End	-44.0	0.5	1.0	0.0	-0.8	1.9
	SLN264 ring / End	18.5	0.0	-16.2	0.0	0.0	0.0
	SLN265 ring / End	24.8	0.0	-16.5	0.0	0.0	0.0
C LC_114	SLN40 column below / Begin	161.5	0.1	0.1	0.0	0.6	-0.6
	SLN121 column above / End	-43.4	0.1	0.7	-0.1	0.1	0.6
	SLN264 ring / End	0.7	0.0	-16.2	0.0	0.0	0.0
	SLN265 ring / End	1.3	0.2	-16.5	0.0	0.0	0.0
C LC_115	SLN40 column below / Begin	20.9	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-9.6	0.1	0.2	-0.1	-0.1	0.3
	SLN264 ring / End	-6.5	0.1	-1.7	0.0	0.0	0.0
	SLN265 ring / End	-6.3	0.2	-1.8	0.0	0.0	0.0
C LC_116	SLN40 column below / Begin	33.3	-5.0	-6.2	0.0	-3.5	2.0
	SLN121 column above / End	-10.1	-0.5	-0.4	-0.1	1.5	-2.0
	SLN264 ring / End	14.2	0.2	-1.7	0.0	0.0	0.0
	SLN265 ring / End	8.6	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN40 column below / Begin	21.3	5.8	-5.4	0.0	-3.8	-1.9
	SLN121 column above / End	-10.2	0.5	0.1	0.1	0.3	1.9
	SLN264 ring / End	7.0	0.0	-1.7	0.0	0.0	0.0
	SLN265 ring / End	13.1	-0.1	-1.8	0.0	0.0	0.0
C LC_118	SLN40 column below / Begin	32.7	0.1	0.2	0.0	0.8	-0.5
	SLN121 column above / End	-9.5	0.1	-0.3	0.0	1.3	0.5
	SLN264 ring / End	-9.9	-0.1	-1.7	0.0	0.0	0.0
	SLN265 ring / End	-9.4	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN40 column below / Begin	233.2	0.1	-0.1	0.0	-0.5	-0.3
	SLN121 column above / End	-65.9	0.1	2.1	-0.2	-2.4	0.3
	SLN264 ring / End	19.3	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	19.8	0.3	-26.0	0.0	0.0	0.0
C LC_120	SLN40 column below / Begin	239.2	-2.5	-3.2	0.0	-2.1	1.0
	SLN121 column above / End	-65.6	-0.2	1.4	-0.2	-0.9	-1.0
	SLN264 ring / End	31.7	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	29.0	0.3	-26.0	0.0	0.0	0.0
C LC_121	SLN40 column below / Begin	233.2	2.9	-2.8	0.0	-2.3	-1.0
	SLN121 column above / End	-65.6	0.3	1.6	-0.1	-1.5	1.0
	SLN264 ring / End	28.0	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	31.4	0.2	-26.0	0.0	0.0	0.0
C LC_122	SLN40 column below / Begin	239.2	0.1	0.0	0.0	0.1	-0.4
	SLN121 column above / End	-65.8	0.1	1.8	-0.1	-1.7	0.4
	SLN264 ring / End	17.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	18.1	0.3	-26.0	0.0	0.0	0.0
C LC_123	SLN40 column below / Begin	174.3	0.1	-0.2	0.0	-0.8	-0.5
	SLN121 column above / End	-51.6	0.1	1.9	-0.2	-2.6	0.5
	SLN264 ring / End	10.5	0.3	-19.4	0.0	0.0	0.0
	SLN265 ring / End	11.1	0.3	-19.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN40 column below / Begin	186.3	-5.0	-6.4	0.0	-4.1	2.1
	SLN121 column above / End	-50.9	-0.5	0.7	-0.2	0.3	-2.1
	SLN264 ring / End	33.0	0.2	-19.4	0.0	0.0	0.0
	SLN265 ring / End	27.4	0.3	-19.7	0.0	0.0	0.0
C LC_125	SLN40 column below / Begin	174.3	5.8	-5.6	0.0	-4.5	-2.0
	SLN121 column above / End	-51.1	0.5	1.1	0.0	-0.9	1.9
	SLN264 ring / End	25.8	0.0	-19.4	0.0	0.0	0.0
	SLN265 ring / End	32.2	0.1	-19.7	0.0	0.0	0.0
C LC_126	SLN40 column below / Begin	186.2	0.1	0.1	0.0	0.3	-0.6
	SLN121 column above / End	-51.4	0.1	1.3	-0.1	-1.0	0.6
	SLN264 ring / End	7.2	0.0	-19.4	0.0	0.0	0.0
	SLN265 ring / End	7.8	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN40 column below / Begin	45.5	0.1	-0.1	0.0	-0.6	-0.4
	SLN121 column above / End	-17.7	0.1	0.9	-0.1	-1.5	0.4
	SLN264 ring / End	0.0	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	0.3	0.2	-5.0	0.0	0.0	0.0
C LC_128	SLN40 column below / Begin	57.5	-5.0	-6.3	0.0	-3.9	2.1
	SLN121 column above / End	-17.1	-0.5	-0.3	-0.1	1.4	-2.1
	SLN264 ring / End	21.5	0.2	-4.9	0.0	0.0	0.0
	SLN265 ring / End	15.8	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN40 column below / Begin	45.5	5.8	-5.6	0.0	-4.3	-1.9
	SLN121 column above / End	-17.3	0.5	0.1	0.1	0.2	1.9
	SLN264 ring / End	14.3	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	20.5	-0.1	-5.0	0.0	0.0	0.0
C LC_130	SLN40 column below / Begin	57.4	0.1	0.1	0.0	0.5	-0.5
	SLN121 column above / End	-17.5	0.1	0.3	0.0	0.1	0.5
	SLN264 ring / End	-3.4	-0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	-2.9	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN40 column below / Begin	241.8	0.0	0.1	0.0	0.3	-0.1
	SLN121 column above / End	-65.7	0.0	1.5	-0.1	-1.1	0.1
	SLN264 ring / End	27.9	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.2	0.3	-26.0	0.0	0.0	0.0
C LC_141	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.5	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN40 column below / Begin	52.9	0.0	0.0	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN40 column below / Begin	52.9	0.0	0.0	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.6	0.1	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN265 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN40 column below / Begin	237.0	0.0	0.0	0.0	-0.1	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.7	0.1
	SLN264 ring / End	26.5	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.0	0.2	-26.0	0.0	0.0	0.0
C LC_146	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.7	0.0	1.7	-0.1	-1.5	0.1
	SLN264 ring / End	28.5	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.6	0.3	-26.0	0.0	0.0	0.0
C LC_147	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.5	0.1
	SLN264 ring / End	28.3	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.7	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.7	0.1
	SLN264 ring / End	26.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	26.9	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN40 column below / Begin	190.0	0.0	-0.7	0.0	-2.9	-0.1
	SLN121 column above / End	-66.3	0.0	3.4	-0.1	-6.0	0.1
	SLN264 ring / End	12.3	0.2	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.5	0.2	-11.3	0.0	0.0	0.0
C LC_152	SLN40 column below / Begin	125.3	0.0	1.3	0.0	5.3	-0.1
	SLN121 column above / End	-16.6	0.0	-2.6	-0.1	7.6	0.1
	SLN264 ring / End	28.2	0.1	-14.0	0.0	0.0	0.0
	SLN265 ring / End	28.3	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN40 column below / Begin	235.4	0.0	-0.1	0.0	-0.3	-0.1
	SLN121 column above / End	-65.0	0.0	1.7	-0.1	-1.8	0.1
	SLN264 ring / End	27.7	0.2	-20.4	0.0	0.0	0.0
	SLN265 ring / End	27.9	0.3	-20.3	0.0	0.0	0.0
C LC_154	SLN40 column below / Begin	190.1	0.0	-0.7	0.0	-2.8	-0.1
	SLN121 column above / End	-67.0	0.0	3.4	-0.1	-6.0	0.1
	SLN264 ring / End	12.4	0.1	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.6	0.2	-11.3	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	18.1 < 100%	OK
Welds	73.9 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN262+SLN263 radial (continuous)-bfl 1	12.7	C LC_120	190.8	0.0	54.5	OK
SLN262+SLN263 radial (continuous)-tfl 1	12.7	C LC_152	231.7	0.0	22.9	OK
SLN262+SLN263 radial (continuous)-w 1	7.2	C LC_124	275.4	0.2	46.7	OK
SLN40 column below	10.0	C LC_120	186.3	0.0	0.0	OK
SLN121 column above	10.0	C LC_152	193.3	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_120	28.9	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_122	19.9	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_135	141.1	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_120	29.2	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_122	20.7	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_120	146.7	0.0	0.0	OK
End plate column below SLN40 column below	12.0	C LC_120	146.1	0.0	48.1	OK
End plate column above SLN121 column above	12.0	C LC_152	184.9	0.0	23.3	OK
End plate SLN264 ring	12.0	C LC_125	87.8	0.0	38.3	OK
End plate SLN265 ring	12.0	C LC_124	93.1	0.0	40.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

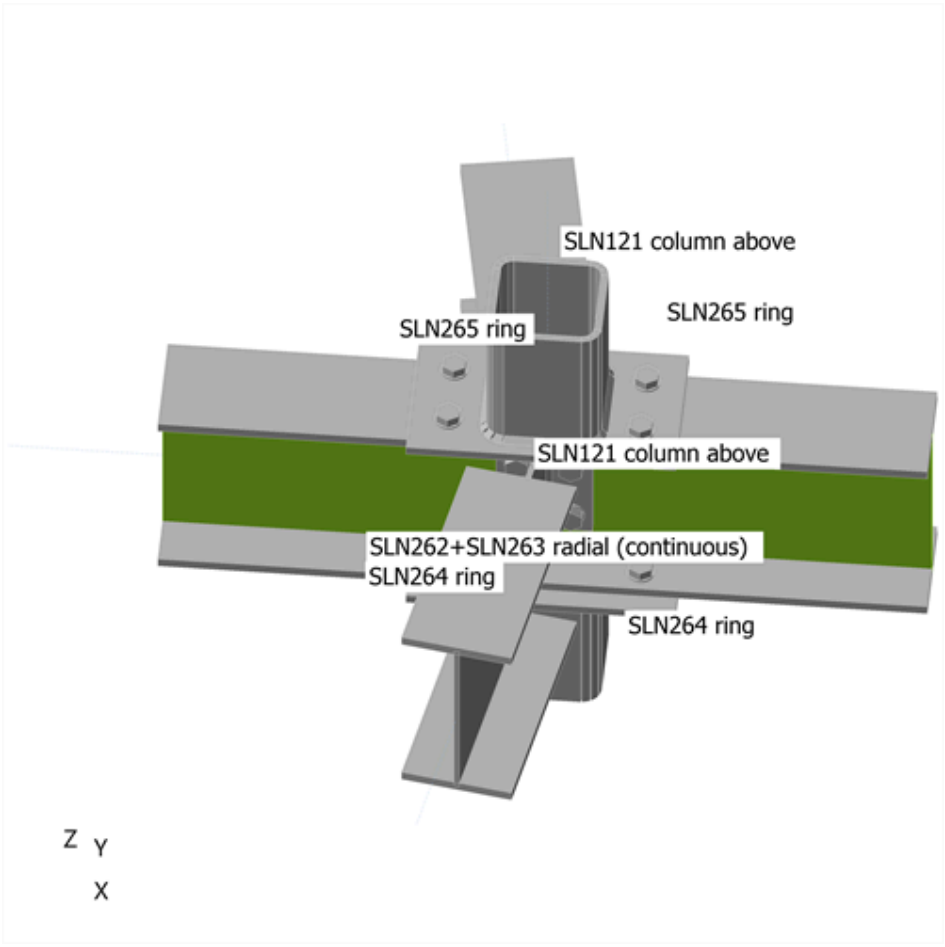
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

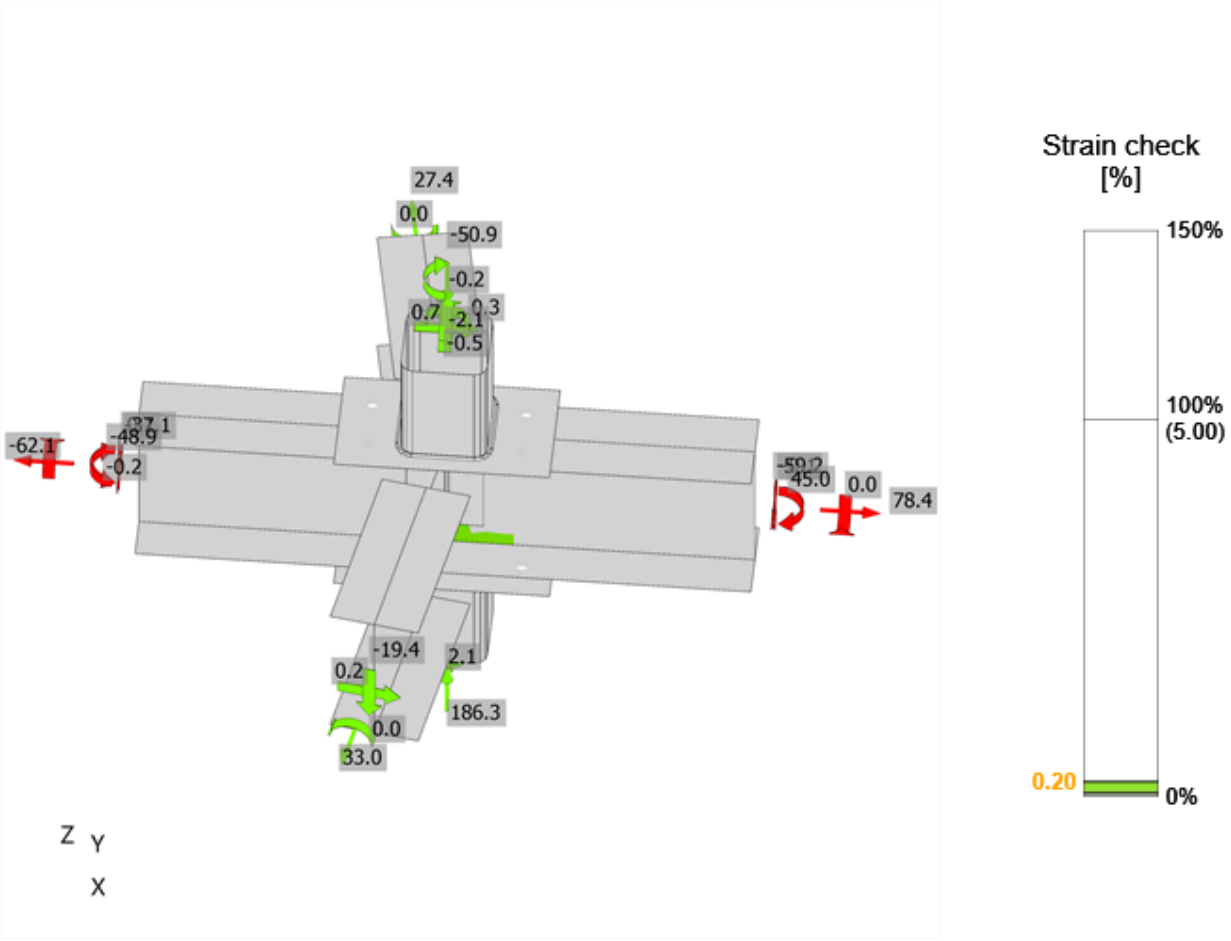
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN40 column below	150	C LC_154	0	5	0.0	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK

Symbol explanation

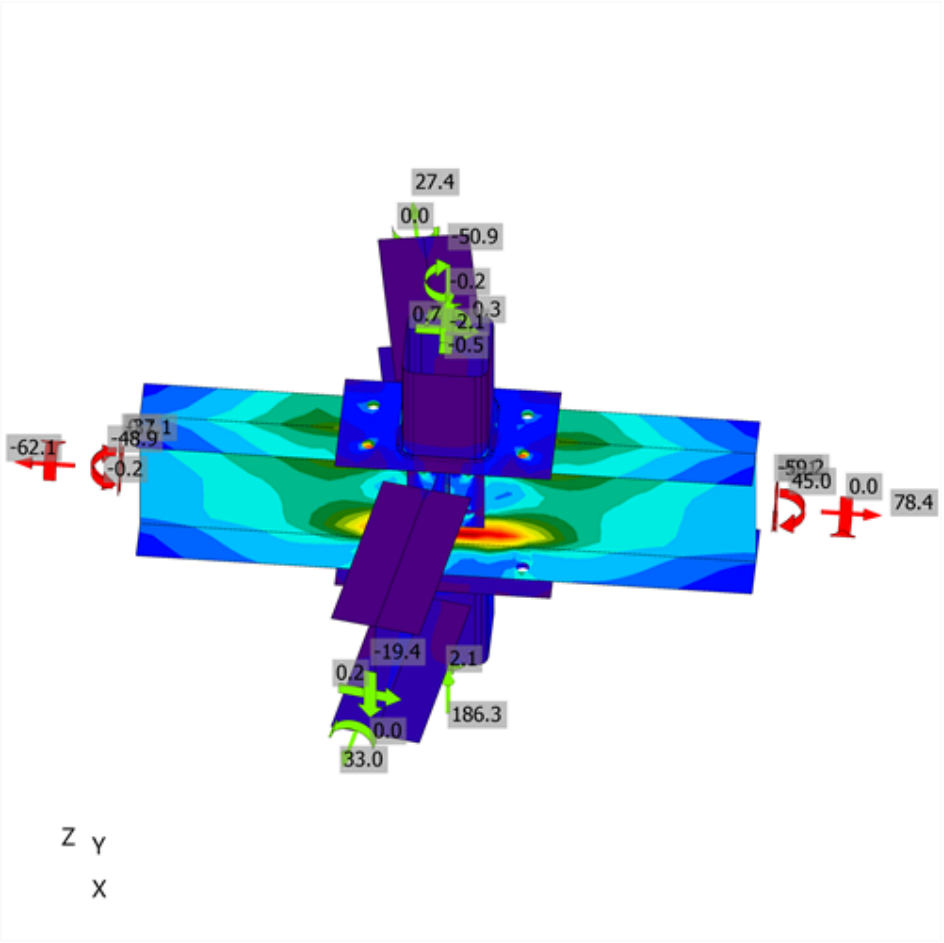
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

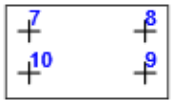
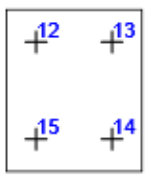


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_124	4.3	5.1	199.1	3.1	5.4	7.6	OK	OK
	B3	M20 8.8 - 1	C LC_108	10.9	2.3	156.4	7.7	2.4	7.9	OK	OK
	B4	M20 8.8 - 1	C LC_109	15.3	2.4	156.4	10.9	2.5	10.3	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.4	6.3	170.1	0.3	6.7	6.9	OK	OK
	B7	M20 8.8 - 1	C LC_125	7.5	3.2	156.4	5.3	3.4	7.2	OK	OK
	B8	M20 8.8 - 1	C LC_152	25.5	3.3	94.1	18.1	3.5	16.4	OK	OK
	B9	M20 8.8 - 1	C LC_152	25.0	3.4	88.3	17.7	3.9	16.3	OK	OK
	B10	M20 8.8 - 1	C LC_124	7.9	3.4	156.4	5.6	3.6	7.6	OK	OK
	B12	M20 8.8 - 1	C LC_128	5.0	0.9	72.9	3.5	1.2	3.5	OK	OK
	B13	M20 8.8 - 1	C LC_124	4.8	1.1	63.5	3.4	1.7	3.6	OK	OK
	B14	M20 8.8 - 1	C LC_128	4.2	0.9	72.3	3.0	1.0	3.1	OK	OK
	B15	M20 8.8 - 1	C LC_124	3.9	1.2	69.7	2.8	1.3	3.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN40 column below	SLN40 column below	◢ 5.6 ◣	521	C LC_120	258.3	0.0	-130.0	-128.7	6.1	63.8	35.2	OK	OK
		◢ 5.6 ◣	521	C LC_120	266.1	0.0	-132.1	133.4	3.0	65.8	36.7	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	521	C LC_154	134.0	0.0	-66.4	-66.3	-10.8	33.1	17.9	OK	OK
		◢ 5.6 ◣	521	C LC_152	146.8	0.0	104.1	-51.9	29.7	36.3	22.4	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_125	283.9	0.0	-9.2	9.3	163.5	70.1	47.4	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_124	299.2	0.0	-10.2	-9.4	-172.4	73.9	50.0	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_125	206.0	0.0	-56.5	-48.4	103.7	50.9	30.5	OK	OK
		◢ 4.2 ◣	149	C LC_125	240.8	0.0	-45.8	59.1	-123.0	59.5	35.7	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_124	257.7	0.0	-53.5	-64.8	130.3	63.7	37.7	OK	OK
		◢ 4.2 ◣	149	C LC_124	210.8	0.0	-55.4	47.9	-107.2	52.1	31.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-B-280
Project number	
Author	Claude for Wing
Description	
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring

Name	SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

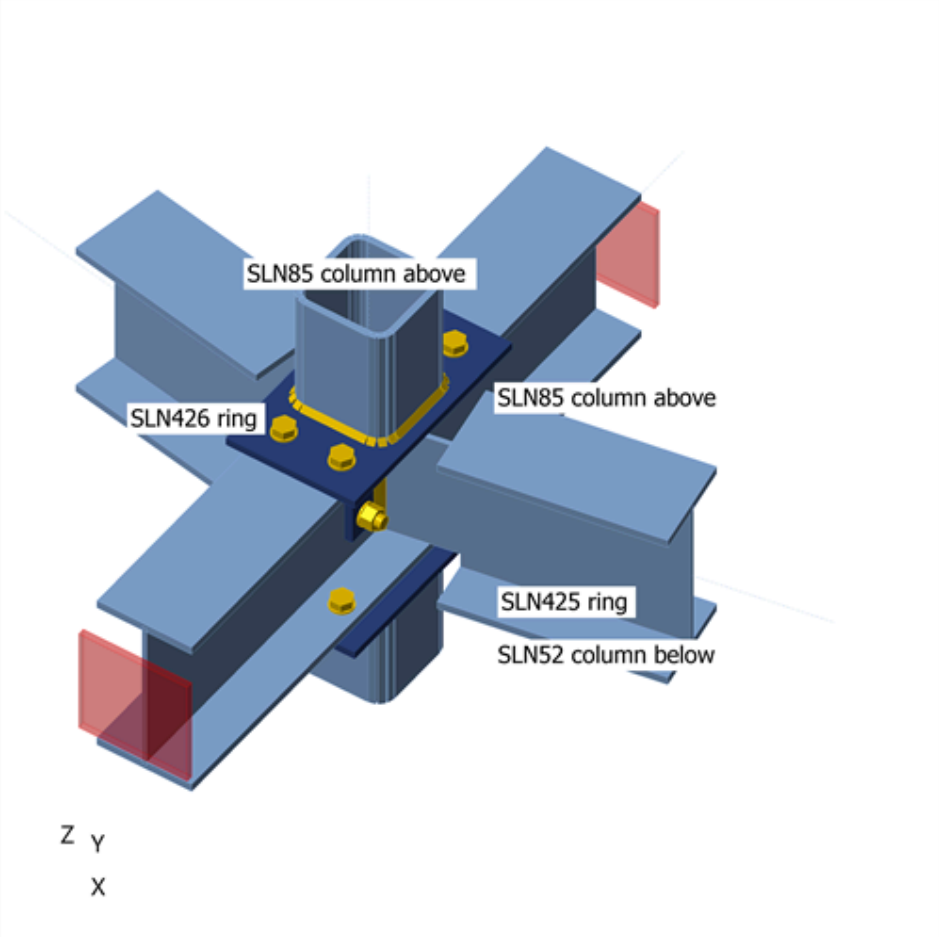
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN206+SLN222 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN52 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN85 column above	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	-180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN206+SLN222 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN206+SLN222 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN52 column below / begin		Position	0
SLN85 column above / end		Position	0
SLN425 ring / end		Position	0
SLN426 ring / end		Position	0



SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.3	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	0.0
C LC_103	SLN52 column below / Begin	222.6	0.0	-0.1	0.0	-0.3	-0.2
	SLN85 column above / End	-64.4	0.1	1.9	0.0	-2.0	0.3
	SLN425 ring / End	16.5	-0.1	-23.9	0.0	0.0	0.0
	SLN426 ring / End	16.8	-0.2	-23.9	0.0	0.0	0.0
C LC_104	SLN52 column below / Begin	230.0	3.8	-0.6	0.0	-0.1	-1.6
	SLN85 column above / End	-64.4	0.4	1.6	0.1	-1.0	1.6
	SLN425 ring / End	20.8	-0.1	-23.9	0.0	0.0	0.0
	SLN426 ring / End	16.6	-0.2	-23.9	0.0	0.0	0.0
C LC_105	SLN52 column below / Begin	226.4	-0.1	0.0	0.0	0.0	0.4
	SLN85 column above / End	-64.2	-0.1	1.8	0.1	-1.6	-0.4
	SLN425 ring / End	13.8	-0.2	-23.9	0.0	0.0	0.0
	SLN426 ring / End	14.3	-0.3	-23.9	0.0	0.0	0.0
C LC_106	SLN52 column below / Begin	226.0	-0.6	-4.1	0.0	-2.7	0.2
	SLN85 column above / End	-64.6	0.0	1.7	0.1	-1.4	-0.1
	SLN425 ring / End	29.7	-0.2	-23.9	0.0	0.0	0.0
	SLN426 ring / End	30.5	-0.2	-23.9	0.0	0.0	0.0
C LC_107	SLN52 column below / Begin	6.6	0.1	-0.1	0.0	-0.4	-0.6
	SLN85 column above / End	-5.6	0.1	0.2	-0.1	-0.3	0.6
	SLN425 ring / End	-8.5	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-8.9	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN52 column below / Begin	21.2	7.7	-1.2	0.0	0.0	-3.3
	SLN85 column above / End	-5.5	0.8	-0.5	0.0	1.6	3.3
	SLN425 ring / End	-1.1	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-9.4	0.0	-0.5	0.0	0.0	0.0
C LC_109	SLN52 column below / Begin	14.1	-0.2	0.0	0.0	0.2	0.7
	SLN85 column above / End	-5.1	-0.2	-0.2	0.1	0.7	-0.7
	SLN425 ring / End	-12.3	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-11.7	-0.2	-0.5	0.0	0.0	0.0
C LC_110	SLN52 column below / Begin	13.3	-1.3	-8.2	0.0	-5.3	0.2
	SLN85 column above / End	-6.0	-0.1	-0.1	0.0	0.8	-0.2
	SLN425 ring / End	15.0	-0.1	-0.5	0.0	0.0	0.0
	SLN426 ring / End	16.2	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN52 column below / Begin	150.4	0.1	-0.1	0.0	-0.5	-0.5
	SLN85 column above / End	-45.4	0.1	1.4	-0.1	-1.6	0.6
	SLN425 ring / End	4.4	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	4.3	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN52 column below / Begin	165.1	7.7	-1.2	0.0	-0.1	-3.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN85 column above / End	-45.3	0.8	0.7	0.0	0.3	3.3
	SLN425 ring / End	12.3	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	3.9	-0.1	-16.2	0.0	0.0	0.0
C LC_113	SLN52 column below / Begin	157.9	-0.2	0.0	0.0	0.1	0.7
	SLN85 column above / End	-45.0	-0.2	1.1	0.2	-0.6	-0.7
	SLN425 ring / End	0.2	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	0.9	-0.3	-16.2	0.0	0.0	0.0
C LC_114	SLN52 column below / Begin	157.1	-1.3	-8.2	0.0	-5.4	0.3
	SLN85 column above / End	-45.8	-0.1	1.1	0.1	-0.4	-0.2
	SLN425 ring / End	28.9	-0.2	-16.2	0.0	0.0	0.0
	SLN426 ring / End	30.2	-0.1	-16.2	0.0	0.0	0.0
C LC_115	SLN52 column below / Begin	20.5	0.1	-0.1	0.0	-0.3	-0.6
	SLN85 column above / End	-10.2	0.1	0.3	-0.1	-0.3	0.6
	SLN425 ring / End	-6.3	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-6.7	0.1	-1.7	0.0	0.0	0.0
C LC_116	SLN52 column below / Begin	35.1	7.7	-1.2	0.0	0.0	-3.3
	SLN85 column above / End	-10.2	0.8	-0.4	0.0	1.6	3.3
	SLN425 ring / End	1.2	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-7.2	0.0	-1.7	0.0	0.0	0.0
C LC_117	SLN52 column below / Begin	28.0	-0.2	0.1	0.0	0.2	0.7
	SLN85 column above / End	-9.8	-0.2	-0.1	0.1	0.7	-0.7
	SLN425 ring / End	-10.1	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-9.5	-0.2	-1.7	0.0	0.0	0.0
C LC_118	SLN52 column below / Begin	27.2	-1.3	-8.2	0.0	-5.2	0.2
	SLN85 column above / End	-10.6	-0.1	-0.1	0.0	0.7	-0.2
	SLN425 ring / End	17.3	-0.2	-1.7	0.0	0.0	0.0
	SLN426 ring / End	18.5	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN52 column below / Begin	234.9	0.0	-0.1	0.0	-0.5	-0.2
	SLN85 column above / End	-68.4	0.1	2.1	0.0	-2.3	0.2
	SLN425 ring / End	19.9	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	20.3	-0.2	-25.5	0.0	0.0	0.0
C LC_120	SLN52 column below / Begin	242.3	3.8	-0.7	0.0	-0.3	-1.6
	SLN85 column above / End	-68.2	0.4	1.7	0.1	-1.3	1.7
	SLN425 ring / End	24.2	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	20.1	-0.3	-25.5	0.0	0.0	0.0
C LC_121	SLN52 column below / Begin	238.8	-0.1	0.0	0.0	-0.1	0.5
	SLN85 column above / End	-68.5	-0.1	2.2	0.2	-2.3	-0.4
	SLN425 ring / End	17.0	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	17.6	-0.3	-25.5	0.0	0.0	0.0
C LC_122	SLN52 column below / Begin	238.1	-0.6	-4.2	0.0	-2.9	0.2
	SLN85 column above / End	-68.1	0.0	1.7	0.1	-1.3	-0.1
	SLN425 ring / End	33.3	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	34.2	-0.3	-25.5	0.0	0.0	0.0
C LC_123	SLN52 column below / Begin	175.1	0.1	-0.2	0.0	-0.9	-0.4
	SLN85 column above / End	-53.4	0.1	1.8	0.0	-2.2	0.5
	SLN425 ring / End	11.3	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	11.4	-0.1	-19.3	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN52 column below / Begin	189.8	7.7	-1.3	0.0	-0.5	-3.3
	SLN85 column above / End	-53.0	0.8	1.0	0.0	-0.1	3.4
	SLN425 ring / End	19.2	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	10.8	-0.2	-19.3	0.0	0.0	0.0
C LC_125	SLN52 column below / Begin	182.9	-0.2	0.0	0.0	-0.1	0.8
	SLN85 column above / End	-53.6	-0.2	1.9	0.2	-2.1	-0.7
	SLN425 ring / End	6.6	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	7.3	-0.3	-19.3	0.0	0.0	0.0
C LC_126	SLN52 column below / Begin	181.5	-1.3	-8.3	0.0	-5.9	0.2
	SLN85 column above / End	-52.9	0.0	1.1	0.1	-0.3	-0.2
	SLN425 ring / End	36.3	-0.2	-19.3	0.0	0.0	0.0
	SLN426 ring / End	37.6	-0.2	-19.3	0.0	0.0	0.0
C LC_127	SLN52 column below / Begin	45.2	0.1	-0.2	0.0	-0.7	-0.5
	SLN85 column above / End	-18.2	0.1	0.7	-0.1	-1.0	0.5
	SLN425 ring / End	0.5	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	0.3	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN52 column below / Begin	59.8	7.7	-1.3	0.0	-0.4	-3.4
	SLN85 column above / End	-17.9	0.9	-0.1	0.0	1.1	3.4
	SLN425 ring / End	8.1	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	-0.3	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN52 column below / Begin	52.9	-0.2	0.0	0.0	0.0	0.7
	SLN85 column above / End	-18.4	-0.2	0.8	0.1	-0.9	-0.7
	SLN425 ring / End	-3.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	-3.0	-0.2	-4.9	0.0	0.0	0.0
C LC_130	SLN52 column below / Begin	51.5	-1.3	-8.3	0.0	-5.7	0.2
	SLN85 column above / End	-17.8	0.0	0.0	0.0	0.8	-0.2
	SLN425 ring / End	24.7	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	25.9	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN52 column below / Begin	243.9	0.0	0.1	0.0	0.4	0.1
	SLN85 column above / End	-68.3	0.0	1.8	0.1	-1.3	0.0
	SLN425 ring / End	27.7	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	28.1	-0.3	-25.5	0.0	0.0	0.0
C LC_141	SLN52 column below / Begin	53.4	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN52 column below / Begin	53.4	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.2	0.0
	SLN425 ring / End	8.4	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.5	-0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN426 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN52 column below / Begin	239.2	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.0	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.5	-0.3	-25.5	0.0	0.0	0.0
C LC_146	SLN52 column below / Begin	239.4	0.0	0.0	0.0	0.1	0.1
	SLN85 column above / End	-68.3	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.5	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	0.0
C LC_147	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.9	0.0
	SLN425 ring / End	26.1	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	26.4	-0.2	-25.5	0.0	0.0	0.0
C LC_148	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.1	0.1
	SLN85 column above / End	-68.3	0.0	1.9	0.1	-1.7	0.0
	SLN425 ring / End	28.5	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	28.8	-0.3	-25.5	0.0	0.0	0.0
C LC_151	SLN52 column below / Begin	191.9	0.0	-0.7	0.0	-2.8	0.0
	SLN85 column above / End	-68.2	0.0	3.5	0.1	-6.1	0.0
	SLN425 ring / End	12.2	-0.1	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.4	-0.2	-11.9	0.0	0.0	0.0
C LC_152	SLN52 column below / Begin	127.3	0.0	1.3	0.0	5.4	0.2
	SLN85 column above / End	-18.0	0.0	-2.5	0.1	7.6	-0.1
	SLN425 ring / End	27.8	-0.1	-14.0	0.0	0.0	0.0
	SLN426 ring / End	28.0	-0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN52 column below / Begin	238.7	0.0	0.0	0.0	-0.2	0.1
	SLN85 column above / End	-67.6	0.0	1.9	0.1	-1.9	0.0
	SLN425 ring / End	27.4	-0.1	-20.6	0.0	0.0	0.0
	SLN426 ring / End	27.7	-0.3	-21.1	0.0	0.0	0.0
C LC_154	SLN52 column below / Begin	192.1	0.0	-0.7	0.0	-2.7	0.0
	SLN85 column above / End	-68.9	0.0	3.6	0.1	-6.1	0.0
	SLN425 ring / End	12.3	-0.1	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.5	-0.2	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.6 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	17.8 < 100%	OK
Welds	93.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN206+SLN222 radial (continuous)-bfl 1	12.7	C LC_120	185.4	0.0	51.6	OK
SLN206+SLN222 radial (continuous)-tfl 1	12.7	C LC_124	256.8	0.0	17.0	OK
SLN206+SLN222 radial (continuous)-w 1	7.2	C LC_124	276.3	0.6	40.4	OK
SLN52 column below	10.0	C LC_122	197.4	0.0	0.0	OK
SLN85 column above	10.0	C LC_108	192.7	0.0	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_122	28.9	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_121	20.0	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_135	133.7	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_122	29.3	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_121	20.5	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_122	144.3	0.0	0.0	OK
End plate column below SLN52 column below	12.0	C LC_122	149.7	0.0	52.2	OK
End plate column above SLN85 column above	12.0	C LC_152	183.4	0.0	22.8	OK
End plate SLN426 ring	12.0	C LC_124	120.2	0.0	36.2	OK
End plate SLN425 ring	12.0	C LC_128	132.9	0.0	31.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

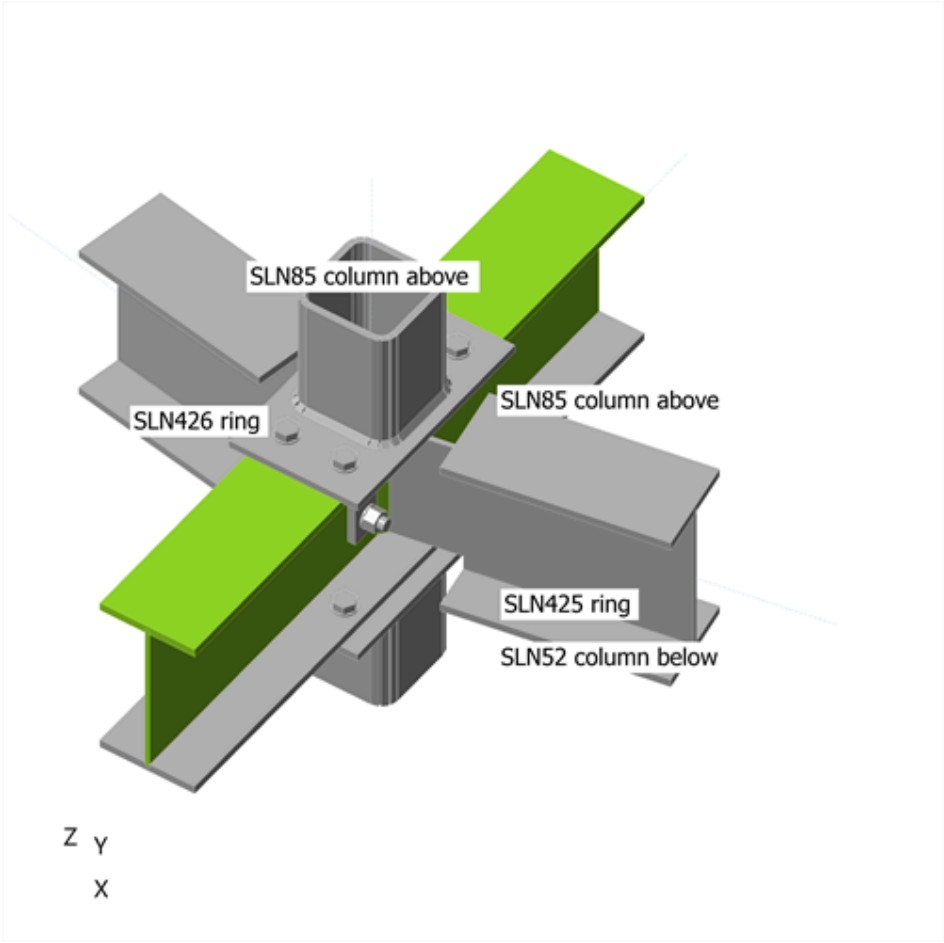
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

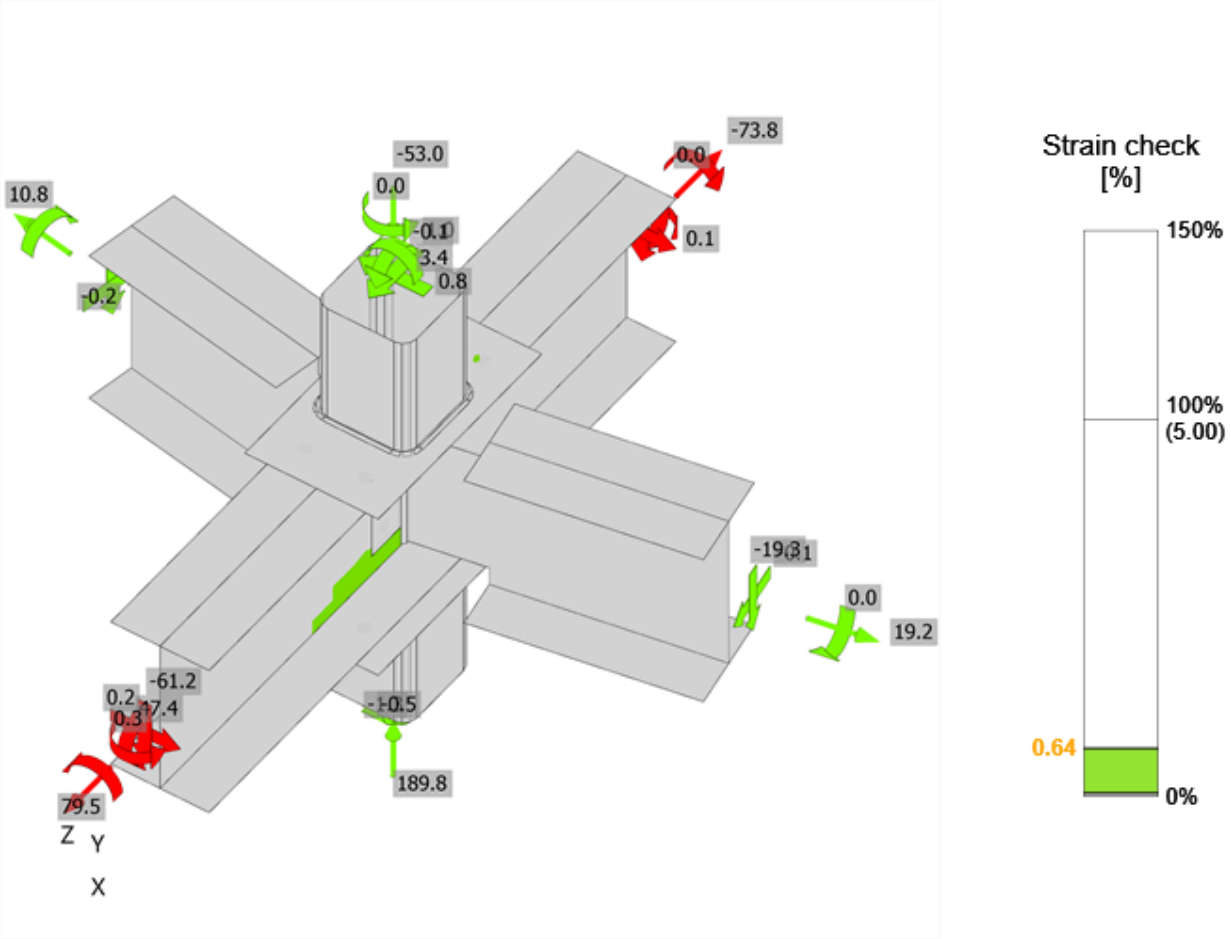
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN52 column below	150	C LC_154	0	5	0.0	OK
SLN85 column above	150	C LC_154	0	5	0.0	OK

Symbol explanation

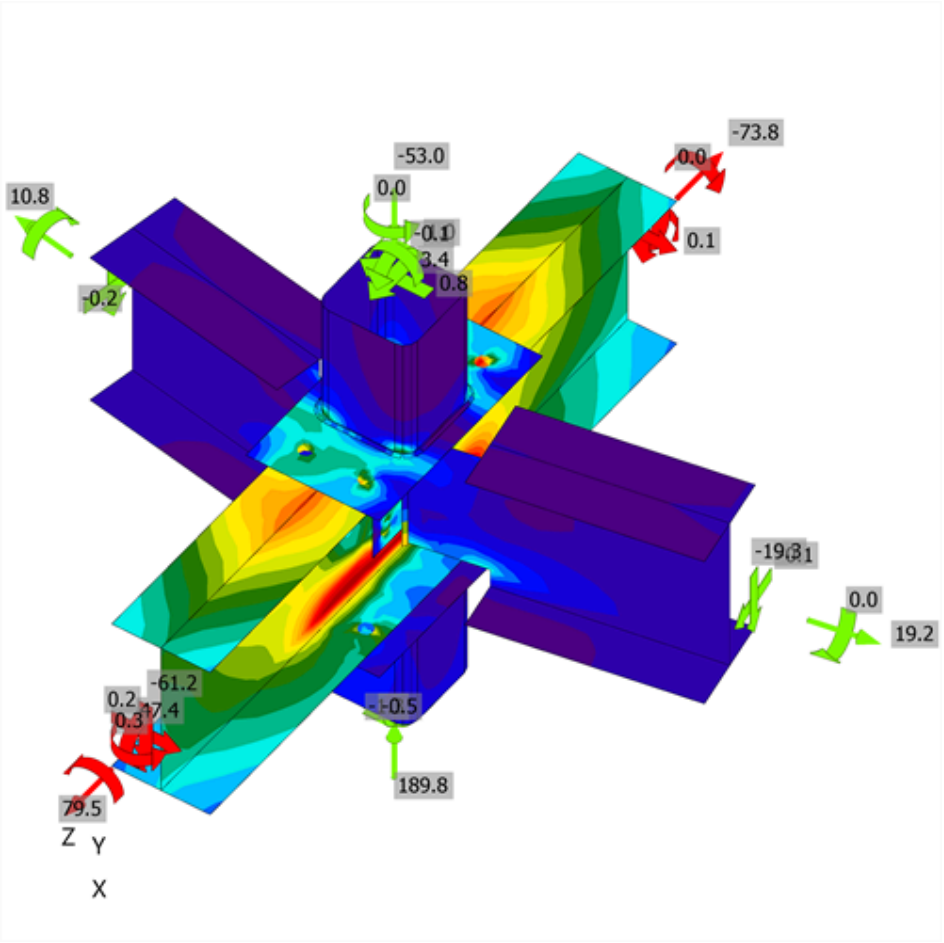
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

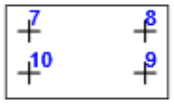
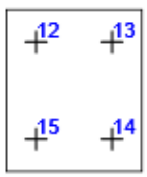


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.8	5.8	177.0	0.5	6.2	6.6	OK	OK
	B3	M20 8.8 - 1	C LC_110	13.7	2.1	123.0	9.7	2.2	9.1	OK	OK
	B4	M20 8.8 - 1	C LC_110	13.3	2.2	91.6	9.4	2.4	9.1	OK	OK
	B5	M20 8.8 - 1	C LC_124	8.5	4.1	176.2	6.0	4.3	8.7	OK	OK
	B7	M20 8.8 - 1	C LC_124	15.0	4.8	156.4	10.6	5.1	12.7	OK	OK
	B8	M20 8.8 - 1	C LC_152	24.5	3.4	88.1	17.4	3.8	16.0	OK	OK
	B9	M20 8.8 - 1	C LC_152	25.2	3.3	94.5	17.8	3.5	16.2	OK	OK
	B10	M20 8.8 - 1	C LC_151	3.5	3.2	112.2	2.5	3.4	5.2	OK	OK
	B12	M20 8.8 - 1	C LC_128	7.1	1.3	70.1	5.0	1.7	5.0	OK	OK
	B13	M20 8.8 - 1	C LC_124	7.1	1.4	68.9	5.0	2.1	5.1	OK	OK
	B14	M20 8.8 - 1	C LC_128	6.8	1.5	71.0	4.8	1.6	5.0	OK	OK
	B15	M20 8.8 - 1	C LC_128	6.6	1.4	72.9	4.7	1.5	4.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN52 column below	SLN52 column below	◢ 5.6 ◣	521	C LC_122	271.2	0.0	-131.9	-133.3	31.0	67.0	36.1	OK	OK
		◢ 5.6 ◣	520	C LC_122	278.0	0.0	-137.9	136.5	-27.8	68.7	37.7	OK	OK
End plate column above SLN85 column above	SLN85 column above	◢ 5.6 ◣	521	C LC_154	138.2	0.0	-67.8	-69.6	1.1	34.2	18.5	OK	OK
		◢ 5.6 ◣	521	C LC_154	147.2	0.0	-74.9	73.1	-1.8	36.4	19.8	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_108	293.5	0.0	37.0	-36.5	164.1	72.5	39.4	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_124	379.8	0.0	-43.8	-41.5	-213.8	93.8	61.8	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_124	241.8	0.0	-61.1	-51.5	124.8	59.7	35.9	OK	OK
		◢ 4.2 ◣	149	C LC_124	286.7	0.0	-58.0	71.9	-145.3	70.8	41.7	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_121	177.7	0.0	-33.7	-43.9	90.7	43.9	27.1	OK	OK
		◢ 4.2 ◣	149	C LC_121	157.4	0.0	-41.4	34.9	-80.4	38.9	24.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-020
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut

Name	SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

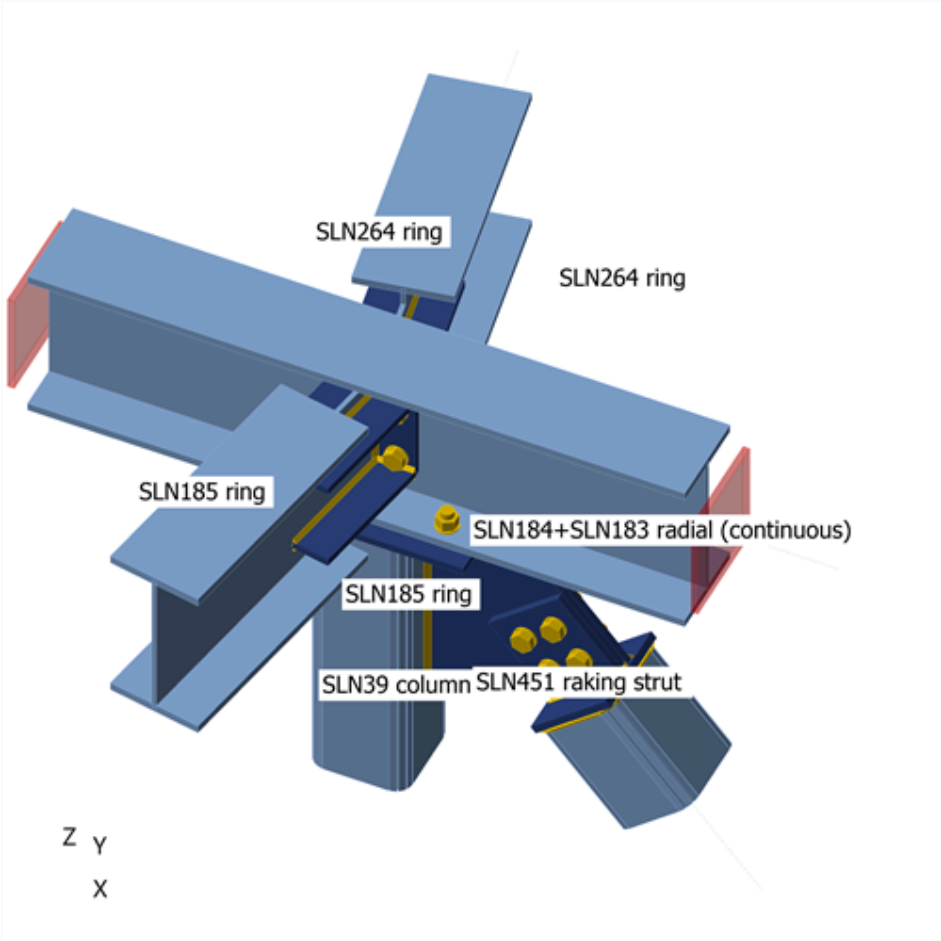
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN184+SLN183 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN39 column below	2 - SHS150x150x10	0.0	90.0	-160.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN264 ring	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN451 raking strut	2 - SHS150x150x10	20.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN184+SLN183 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN184+SLN183 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN39 column below / begin		Position	0
SLN185 ring / begin		Position	0
SLN264 ring / end		Position	0
SLN451 raking strut / end		Position	0



SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN39 column below / Begin	123.5	0.0	0.8	0.0	3.3	0.1
	SLN185 ring / Begin	-9.0	1.0	-25.6	0.0	0.0	0.7
	SLN264 ring / End	9.0	2.5	-25.5	0.0	0.0	0.3
	SLN451 raking strut / End	-13.7	0.0	-1.0	-0.1	0.0	-0.3
C LC_108	SLN39 column below / Begin	40.4	-2.6	-7.8	0.0	-5.1	0.7
	SLN185 ring / Begin	-8.0	1.6	-0.5	0.0	0.0	-0.3
	SLN264 ring / End	5.3	1.9	-0.5	0.0	0.0	1.2
	SLN451 raking strut / End	50.3	0.0	-1.2	0.1	0.0	-0.8
C LC_109	SLN39 column below / Begin	8.4	7.0	-2.8	0.0	-1.7	-1.9
	SLN185 ring / Begin	6.0	-0.8	-0.5	0.0	0.0	-0.5
	SLN264 ring / End	1.2	0.4	-0.5	0.0	0.0	0.0
	SLN451 raking strut / End	7.0	0.0	-0.9	-0.5	0.0	2.1
C LC_111	SLN39 column below / Begin	47.4	0.0	0.5	0.0	2.1	0.2
	SLN185 ring / Begin	8.0	-2.2	-16.2	0.0	0.0	1.0
	SLN264 ring / End	-7.8	-1.0	-16.2	0.0	0.0	0.0
	SLN451 raking strut / End	-50.4	-0.3	-0.5	-0.1	0.0	-0.4
C LC_112	SLN39 column below / Begin	115.4	-2.6	-7.4	0.0	-3.3	0.8
	SLN185 ring / Begin	-11.7	2.0	-16.2	0.0	0.0	0.0
	SLN264 ring / End	8.9	3.1	-16.2	0.0	0.0	1.4
	SLN451 raking strut / End	41.5	0.0	-1.5	0.1	0.0	-1.0
C LC_116	SLN39 column below / Begin	47.3	-2.6	-7.8	0.0	-4.9	0.7
	SLN185 ring / Begin	-8.8	1.7	-1.7	0.0	0.0	-0.3
	SLN264 ring / End	6.1	2.1	-1.7	0.0	0.0	1.2
	SLN451 raking strut / End	50.2	0.0	-1.5	0.1	0.0	-0.9
C LC_119	SLN39 column below / Begin	110.0	0.0	0.7	0.0	3.1	0.2
	SLN185 ring / Begin	-1.9	-0.5	-25.6	0.0	0.0	0.9
	SLN264 ring / End	2.0	1.1	-25.5	0.0	0.0	0.2
	SLN451 raking strut / End	-31.4	-0.2	-0.8	-0.2	0.0	-0.4
C LC_122	SLN39 column below / Begin	115.2	0.0	0.8	0.0	3.2	0.0
	SLN185 ring / Begin	-1.1	-0.4	-25.6	0.0	0.0	0.7
	SLN264 ring / End	0.7	0.9	-25.5	0.0	0.0	0.3
	SLN451 raking strut / End	-24.5	0.3	-0.9	-0.1	0.0	-0.1
C LC_124	SLN39 column below / Begin	126.4	-2.6	-7.3	0.0	-2.8	0.8
	SLN185 ring / Begin	-18.5	2.9	-19.4	0.0	0.0	0.5
	SLN264 ring / End	15.7	4.0	-19.4	0.0	0.0	1.0
	SLN451 raking strut / End	33.8	0.0	-1.5	0.1	0.0	-1.0
C LC_135	SLN39 column below / Begin	123.8	0.0	0.8	0.0	3.3	0.1
	SLN185 ring / Begin	-6.9	0.6	-25.6	0.0	0.0	0.7
	SLN264 ring / End	6.9	2.2	-25.5	0.0	0.0	0.4
	SLN451 raking strut / End	-13.4	0.0	-1.0	-0.2	0.0	-0.3
C LC_145	SLN39 column below / Begin	121.2	0.0	0.8	0.0	3.2	0.1
	SLN185 ring / Begin	-8.0	0.7	-25.6	0.0	0.0	0.8
	SLN264 ring / End	8.0	2.2	-25.5	0.0	0.0	0.2
	SLN451 raking strut / End	-16.8	0.0	-1.0	-0.1	0.0	-0.3
C LC_147	SLN39 column below / Begin	124.2	0.0	0.8	0.0	3.3	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN185 ring / Begin	-9.3	1.1	-25.6	0.0	0.0	0.7
	SLN264 ring / End	9.5	2.6	-25.5	0.0	0.0	0.3
	SLN451 raking strut / End	-12.7	-0.1	-1.0	-0.1	0.0	-0.3
C LC_148	SLN39 column below / Begin	122.7	0.0	0.8	0.0	3.3	0.1
	SLN185 ring / Begin	-8.7	0.9	-25.6	0.0	0.0	0.8
	SLN264 ring / End	8.5	2.3	-25.5	0.0	0.0	0.3
	SLN451 raking strut / End	-14.8	0.0	-1.0	-0.1	0.0	-0.3
C LC_152	SLN39 column below / Begin	77.0	0.0	2.2	0.0	8.8	0.1
	SLN185 ring / Begin	-11.3	1.5	-14.1	0.0	0.0	0.6
	SLN264 ring / End	11.2	2.8	-14.0	0.0	0.0	0.3
	SLN451 raking strut / End	-3.5	0.0	-1.0	-0.1	0.0	-0.3
C LC_153	SLN39 column below / Begin	123.4	0.0	0.7	0.0	3.0	0.1
	SLN185 ring / Begin	-9.0	1.0	-20.9	0.0	0.0	0.7
	SLN264 ring / End	9.0	2.5	-20.4	0.0	0.0	0.3
	SLN451 raking strut / End	-13.5	0.0	-1.0	-0.1	0.0	-0.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	26.7 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN184+SLN183 radial (continuous)-bfl 1	12.7	C LC_108	217.2	0.0	34.7	OK
SLN184+SLN183 radial (continuous)-tfl 1	12.7	C LC_153	76.1	0.0	0.0	OK
SLN184+SLN183 radial (continuous)-w 1	7.2	C LC_112	258.7	0.0	35.4	OK
SLN39 column below	10.0	C LC_111	275.2	0.1	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_124	72.9	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_124	69.3	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_124	189.1	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_112	51.4	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_112	49.4	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_112	158.5	0.0	0.0	OK
SLN451 raking strut	10.0	C LC_116	56.6	0.0	0.0	OK
End plate column below SLN39 column below	12.0	C LC_108	259.1	0.0	34.7	OK
End plate SLN185 ring	12.0	C LC_124	35.8	0.0	175.4	OK
End plate SLN264 ring	12.0	C LC_112	34.2	0.0	205.1	OK
Cap plate SLN451 raking strut	12.0	C LC_109	115.7	0.0	0.0	OK
Fin plate SLN451 raking strut	12.0	C LC_109	275.1	0.0	64.9	OK
Fin plate 2 SLN451 raking strut	12.0	C LC_109	275.1	0.0	91.4	OK
Gusset SLN451 raking strut	12.0	C LC_109	275.1	0.1	48.5	OK
Flange top a end plate SLN185 ring	12.0	C LC_119	82.1	0.0	0.0	OK
Flange top b end plate SLN185 ring	12.0	C LC_124	64.1	0.0	0.0	OK
Flange bottom a end plate SLN185 ring	12.0	C LC_124	69.7	0.0	0.0	OK
Flange bottom b end plate SLN185 ring	12.0	C LC_148	79.0	0.0	0.0	OK
Flange top a end plate SLN264 ring	12.0	C LC_112	78.8	0.0	0.0	OK
Flange top b end plate SLN264 ring	12.0	C LC_112	74.2	0.0	0.0	OK
Flange bottom a end plate SLN264 ring	12.0	C LC_112	69.3	0.0	0.0	OK
Flange bottom b end plate SLN264 ring	12.0	C LC_112	87.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

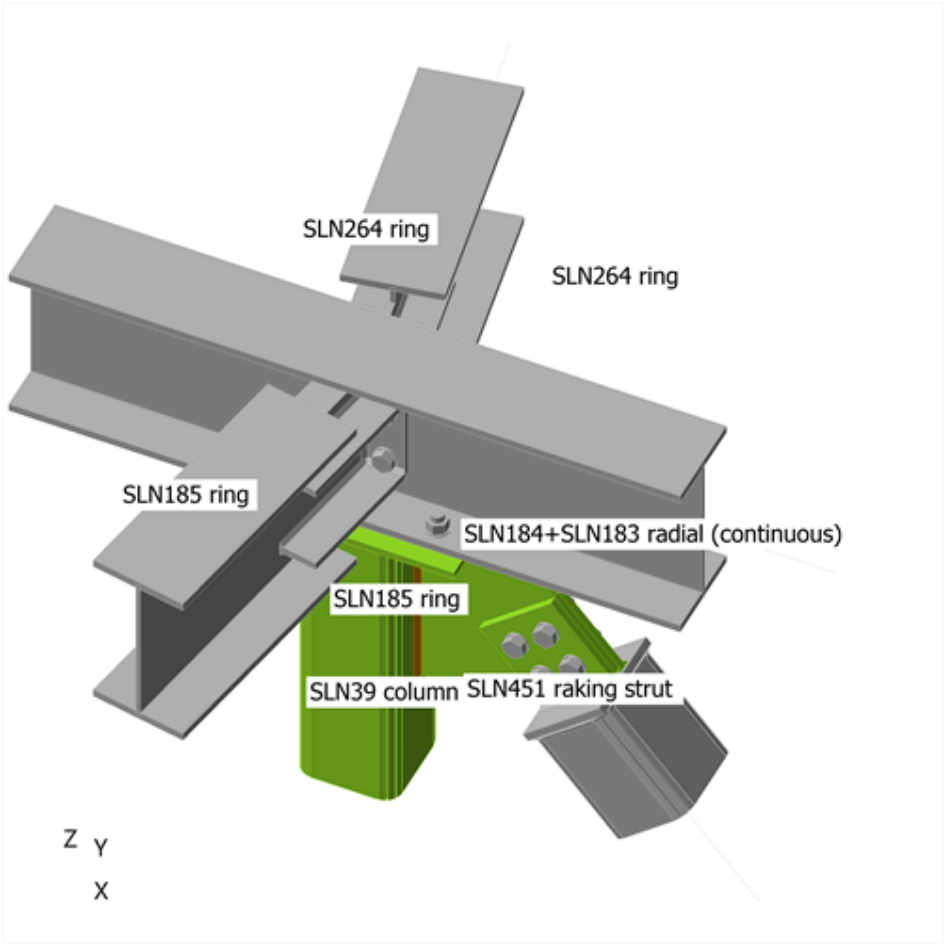
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

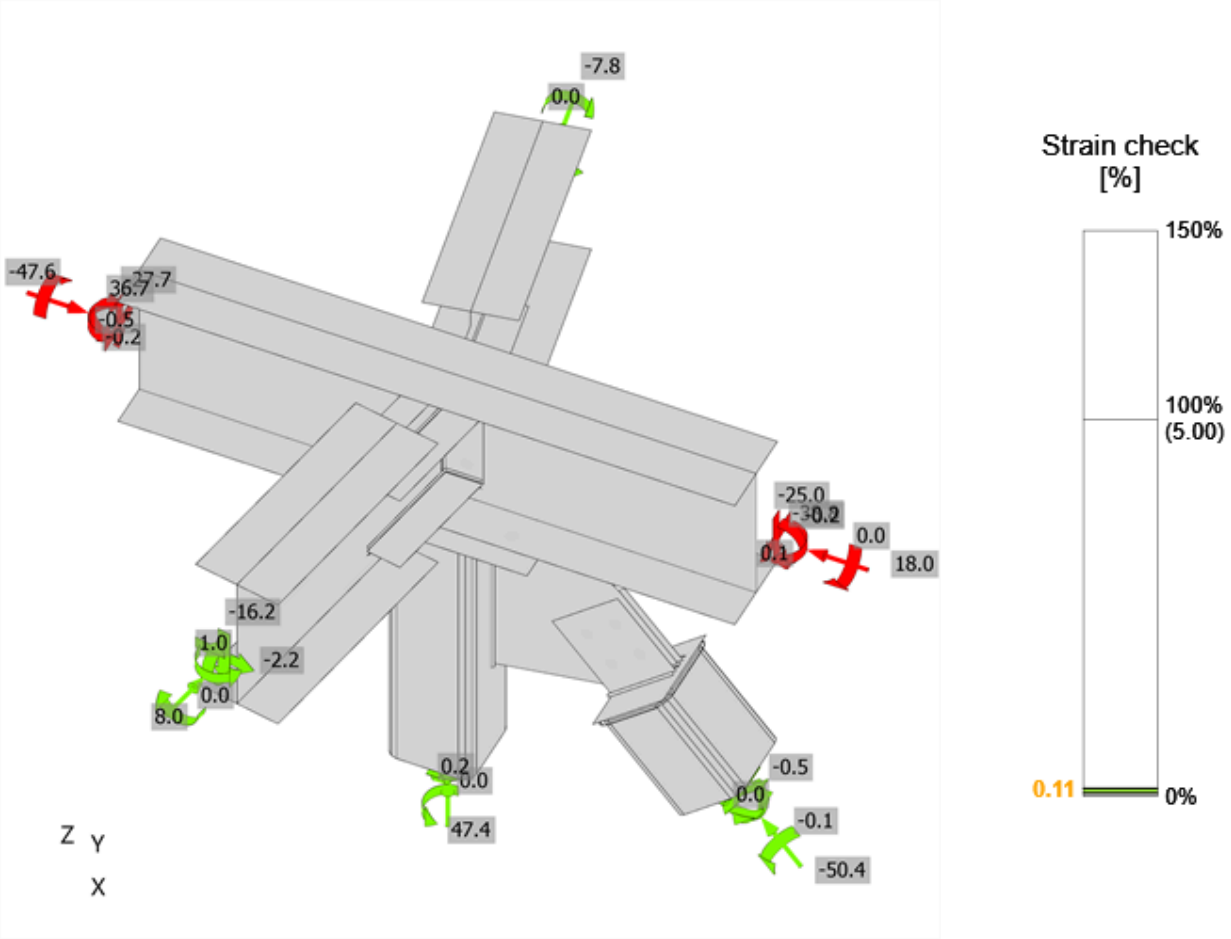
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN39 column below	150	C LC_108	0	5	0.1	OK
SLN451 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

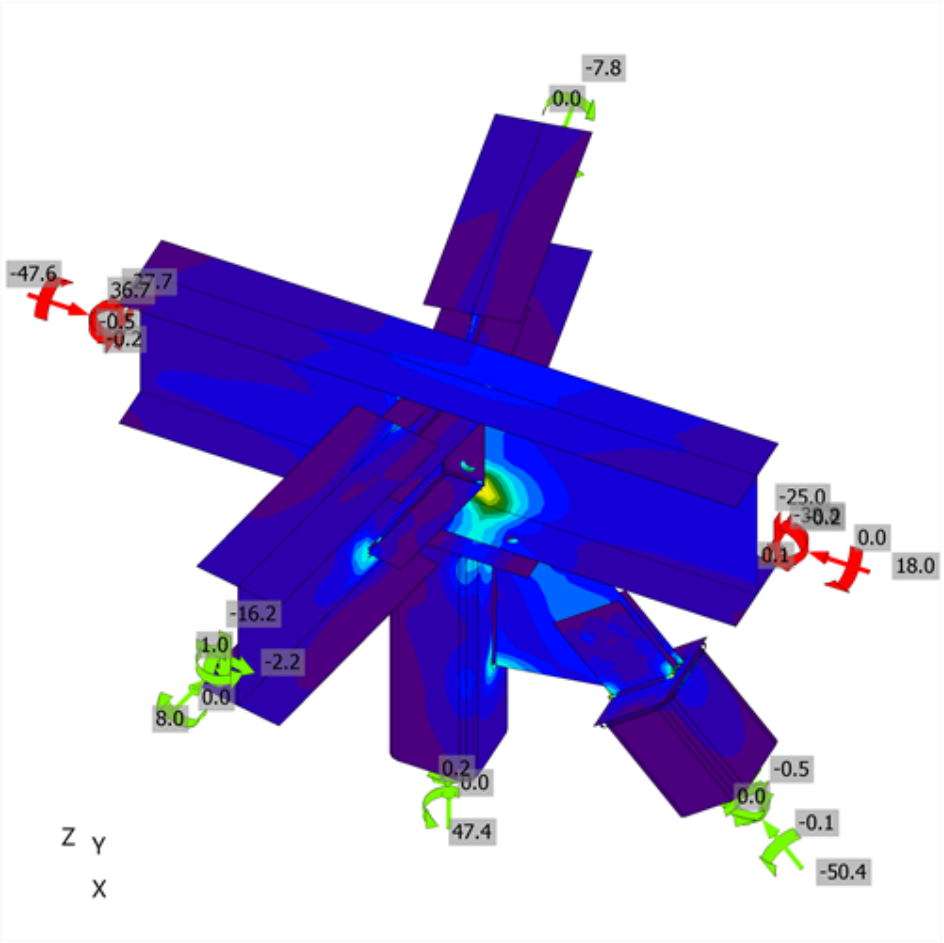
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

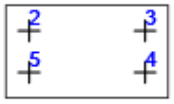
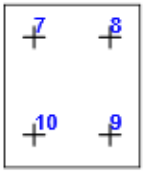
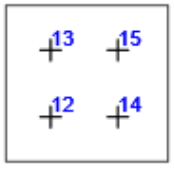


Strain check, C LC_111



Equivalent stress, C LC_111

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_111	3.2	10.0	169.8	2.2	10.6	12.2	OK	OK
	B3	M20 8.8 - 1	C LC_108	37.6	5.5	92.5	26.7	5.9	24.9	OK	OK
	B4	M20 8.8 - 1	C LC_108	36.7	5.8	111.3	26.0	6.2	24.8	OK	OK
	B5	M20 8.8 - 1	C LC_111	5.7	9.8	170.3	4.0	10.5	13.3	OK	OK
	B7	M20 8.8 - 1	C LC_111	1.9	0.4	123.8	1.3	0.7	1.4	OK	OK
	B8	M20 8.8 - 1	C LC_112	4.0	0.5	123.8	2.9	0.7	2.5	OK	OK
	B9	M20 8.8 - 1	C LC_112	5.4	0.4	123.8	3.8	0.6	3.2	OK	OK
	B10	M20 8.8 - 1	C LC_119	4.1	0.4	123.8	2.9	0.6	2.5	OK	OK
	B12	M20 8.8 - 1	C LC_109	16.8	13.1	106.0	11.9	13.9	22.4	OK	OK
	B13	M20 8.8 - 1	C LC_109	7.6	9.2	106.0	5.4	9.8	13.7	OK	OK
	B14	M20 8.8 - 1	C LC_116	1.6	10.2	106.0	1.2	14.6	11.7	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.2	7.2	115.3	0.1	11.5	7.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN451 raking strut	SLN451 raking strut	◢ 4.2	519	C LC_116	197.7	0.0	93.2	-100.6	-3.1	48.8	26.2	OK	OK
SLN39 column below-w 3	Gusset SLN451 raking strut	◢ 5.6 ◣	288	C LC_111	397.2	0.4	-128.5	-131.0	-173.0	98.2	64.2	OK	OK
		◢ 5.6 ◣	288	C LC_111	397.3	0.4	-136.0	133.6	169.0	98.2	63.9	OK	OK
SLN185 ring-w 1	Flange top b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_124	88.4	0.0	48.9	39.0	16.9	21.8	13.7	OK	OK
		◢ 4.2 ◣	197	C LC_119	41.6	0.0	-19.2	17.3	12.4	10.3	7.6	OK	OK
SLN185 ring-w 1	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_124	82.8	0.0	23.7	31.9	32.9	20.5	12.4	OK	OK
		◢ 4.2 ◣	198	C LC_148	138.0	0.0	60.0	-52.9	-48.4	34.1	22.0	OK	OK
SLN264 ring-w 1	Flange top a end plate SLN264 ring	◢ 4.2 ◣	198	C LC_112	133.7	0.0	-60.8	-53.8	42.8	33.0	16.7	OK	OK
		◢ 4.2 ◣	198	C LC_112	62.4	0.0	-18.1	25.2	-23.6	15.4	9.0	OK	OK
SLN264 ring-w 1	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	198	C LC_147	55.8	0.0	16.0	17.4	-25.5	13.8	8.6	OK	OK
		◢ 4.2 ◣	198	C LC_108	96.2	0.0	-48.2	41.6	-24.1	23.8	12.2	OK	OK
Cap plate SLN451 raking strut	Fin plate SLN451 raking strut	◢ 5.6 ◣	139	C LC_109	236.2	0.0	89.0	105.4	69.6	58.4	45.0	OK	OK
		◢ 5.6 ◣	139	C LC_116	167.9	0.0	86.1	-67.8	-48.3	41.5	26.4	OK	OK
Cap plate SLN451 raking strut	Fin plate 2 SLN451 raking strut	◢ 5.6 ◣	139	C LC_111	131.6	0.0	-53.1	-55.3	42.2	32.5	19.0	OK	OK
		◢ 5.6 ◣	139	C LC_116	262.3	0.0	99.7	-114.4	-80.8	64.8	42.4	OK	OK
		◢ 5.6 ◣	521	C LC_108	219.5	0.0	151.6	-74.2	53.8	54.2	33.4	OK	OK
End plate column below SLN39 column below	SLN39 column below	◢ 5.6 ◣	521	C LC_119	268.6	0.0	-166.3	-121.7	5.7	66.4	37.1	OK	OK
SLN184+SLN183 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_122	92.7	0.0	7.8	-3.0	53.2	22.9	13.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN184+SLN183 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_124	88.1	0.0	32.3	37.2	-29.2	21.8	13.6	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_135	91.6	0.0	-27.8	-19.9	-46.3	22.6	12.2	OK	OK
		◢ 4.2 ◣	149	C LC_145	60.4	0.0	-13.9	20.6	27.0	14.9	8.8	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_148	72.4	0.0	-14.0	-24.5	32.8	17.9	11.3	OK	OK
		◢ 4.2 ◣	149	C LC_148	57.6	0.0	-20.3	12.6	-28.4	14.2	9.0	OK	OK
SLN185 ring-w 1	Flange top a end plate SLN185 ring	◢ 4.2 ◣	199	C LC_119	144.6	0.0	-72.9	33.6	-63.8	35.7	23.2	OK	OK
		◢ 4.2 ◣	198	C LC_124	96.5	0.0	13.7	15.8	-52.8	23.9	13.9	OK	OK
SLN185 ring-w 1	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	198	C LC_124	113.5	0.0	19.7	-21.1	61.0	28.1	16.1	OK	OK
		◢ 4.2 ◣	198	C LC_124	103.1	0.0	-13.9	-4.3	58.8	25.5	15.2	OK	OK
SLN264 ring-w 1	Flange top b end plate SLN264 ring	◢ 4.2 ◣	199	C LC_112	125.3	0.0	9.5	-3.8	-72.0	31.0	18.7	OK	OK
		◢ 4.2 ◣	198	C LC_112	109.0	0.0	-18.1	-19.8	-58.8	26.9	15.1	OK	OK
SLN264 ring-w 1	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	198	C LC_108	95.1	0.0	-16.1	17.4	51.2	23.5	13.3	OK	OK
		◢ 4.2 ◣	199	C LC_112	146.2	0.0	17.7	10.9	83.1	36.1	24.6	OK	OK
End plate SLN185 ring	Flange top a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_124	103.1	0.0	-50.5	40.6	32.3	25.5	13.0	OK	OK
		◢ 4.2 ◣	58	C LC_119	92.7	0.0	-32.6	-39.6	-30.7	22.9	11.7	OK	OK
End plate SLN185 ring	Flange top b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_109	61.8	0.0	-32.7	28.4	-10.5	15.3	10.2	OK	OK
		◢ 4.2 ◣	58	C LC_109	49.8	0.0	-17.7	-22.0	15.4	12.3	8.2	OK	OK
End plate SLN185 ring	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_111	70.0	0.0	-24.2	31.8	20.6	17.3	9.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_124	103.8	0.0	-54.5	-39.8	-31.9	25.6	13.3	OK	OK
End plate SLN185 ring	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_109	48.3	0.0	-21.8	21.7	-12.2	11.9	8.4	OK	OK
		◢ 4.2 ◣	58	C LC_109	62.6	0.0	-31.1	-31.2	3.4	15.5	9.9	OK	OK
End plate SLN264 ring	Flange top a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_112	143.3	0.0	-70.6	65.0	31.0	35.4	18.1	OK	OK
		◢ 4.2 ◣	58	C LC_112	142.8	0.0	-61.3	-67.0	-32.6	35.3	17.7	OK	OK
End plate SLN264 ring	Flange top b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_108	26.7	0.0	1.6	-0.9	-15.4	6.6	6.2	OK	OK
		◢ 4.2 ◣	58	C LC_111	32.6	0.0	-18.5	-13.5	7.5	8.0	7.8	OK	OK
End plate SLN264 ring	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_112	113.2	0.0	-57.6	52.2	21.0	28.0	15.8	OK	OK
		◢ 4.2 ◣	58	C LC_112	123.5	0.0	-56.7	-62.2	-12.3	30.5	16.2	OK	OK
End plate SLN264 ring	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_148	54.7	0.0	12.0	-20.4	-23.1	13.5	7.8	OK	OK
		◢ 4.2 ◣	58	C LC_112	34.8	0.0	2.7	0.8	20.0	8.6	6.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-100
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut

Name	SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

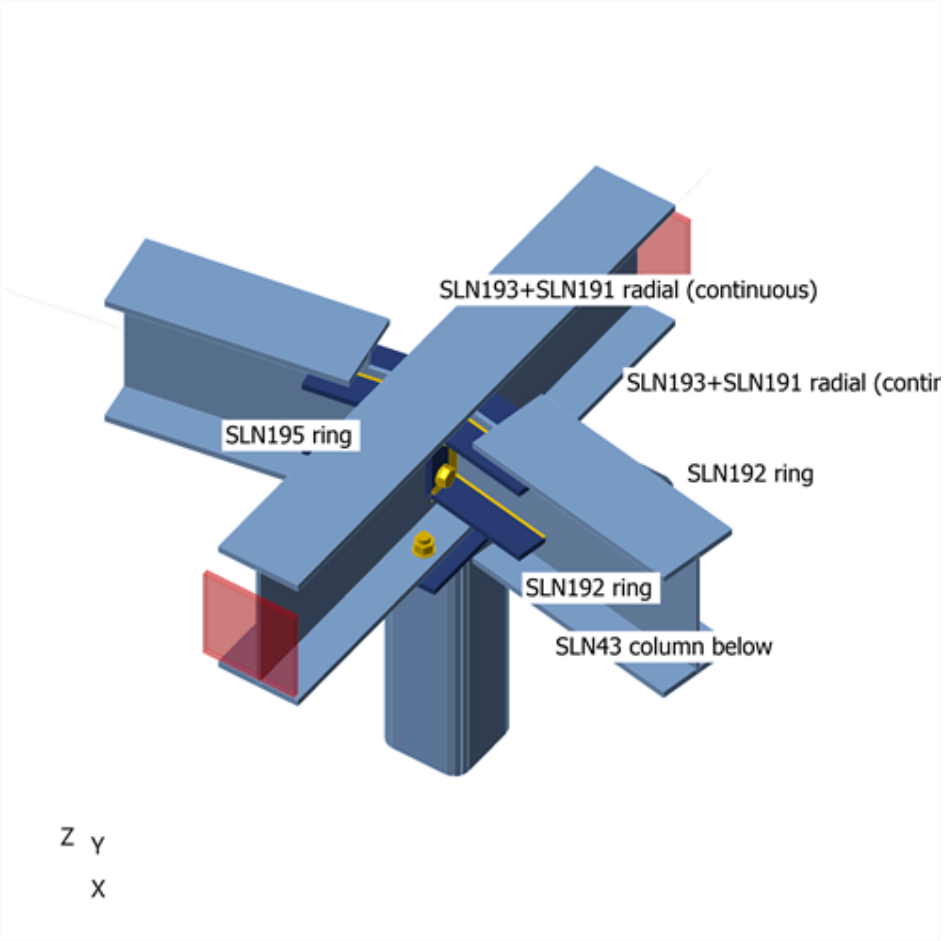
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN193+SLN191 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN43 column below	2 - SHS150x150x10	0.0	90.0	-80.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0
SLN195 ring	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN446 raking strut	2 - SHS150x150x10	100.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN193+SLN191 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN193+SLN191 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN43 column below / begin		Position	0
SLN192 ring / begin		Position	0
SLN195 ring / end		Position	0
SLN446 raking strut / end		Position	0



SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN43 column below / Begin	123.3	0.0	0.4	0.0	1.6	0.0
	SLN192 ring / Begin	-31.2	4.7	-25.6	0.0	0.0	1.2
	SLN195 ring / End	31.2	4.7	-25.7	0.0	0.0	-1.2
	SLN446 raking strut / End	-23.5	0.0	-1.0	0.0	0.0	0.0
C LC_109	SLN43 column below / Begin	39.6	-1.3	-8.3	0.0	-5.6	0.4
	SLN192 ring / Begin	-14.8	3.0	-0.5	0.0	0.0	-0.6
	SLN195 ring / End	13.4	2.7	-0.5	0.0	0.0	0.5
	SLN446 raking strut / End	48.3	0.0	-1.3	0.1	0.0	-0.4
C LC_111	SLN43 column below / Begin	79.0	7.4	-1.2	0.0	0.0	-2.0
	SLN192 ring / Begin	-8.4	1.1	-16.2	0.0	0.0	0.4
	SLN195 ring / End	15.9	2.4	-16.3	0.0	0.0	-0.8
	SLN446 raking strut / End	-14.2	0.0	-1.1	-0.5	0.0	2.2
C LC_113	SLN43 column below / Begin	114.1	-1.3	-8.1	0.0	-4.8	0.4
	SLN192 ring / Begin	-30.7	5.5	-16.2	0.0	0.0	0.0
	SLN195 ring / End	29.4	5.2	-16.3	0.0	0.0	0.0
	SLN446 raking strut / End	33.0	0.0	-1.5	0.1	0.0	-0.4
C LC_117	SLN43 column below / Begin	46.6	-1.3	-8.2	0.0	-5.4	0.4
	SLN192 ring / Begin	-17.4	3.5	-1.7	0.0	0.0	-0.5
	SLN195 ring / End	16.1	3.1	-1.8	0.0	0.0	0.4
	SLN446 raking strut / End	47.9	0.0	-1.5	0.1	0.0	-0.4
C LC_119	SLN43 column below / Begin	121.4	3.7	-0.4	0.0	1.0	-1.0
	SLN192 ring / Begin	-24.5	3.6	-25.6	0.0	0.0	1.0
	SLN195 ring / End	28.5	4.3	-25.7	0.0	0.0	-1.1
	SLN446 raking strut / End	-25.8	0.0	-1.1	-0.2	0.0	1.1
C LC_120	SLN43 column below / Begin	115.8	0.0	0.3	0.0	1.5	-0.1
	SLN192 ring / Begin	-23.8	3.3	-25.6	0.0	0.0	1.2
	SLN195 ring / End	23.3	3.4	-25.7	0.0	0.0	-1.1
	SLN446 raking strut / End	-33.3	0.4	-1.0	0.1	0.0	0.3
C LC_122	SLN43 column below / Begin	108.8	0.0	0.3	0.0	1.5	0.1
	SLN192 ring / Begin	-21.8	2.9	-25.6	0.0	0.0	1.3
	SLN195 ring / End	22.0	3.0	-25.7	0.0	0.0	-1.3
	SLN446 raking strut / End	-42.3	-0.1	-0.8	0.0	0.0	0.0
C LC_123	SLN43 column below / Begin	92.8	7.4	-1.2	0.0	0.2	-2.0
	SLN192 ring / Begin	-15.4	2.2	-19.4	0.0	0.0	0.7
	SLN195 ring / End	22.9	3.5	-19.5	0.0	0.0	-1.0
	SLN446 raking strut / End	-19.1	0.0	-1.1	-0.5	0.0	2.3
C LC_125	SLN43 column below / Begin	127.6	-1.3	-8.0	0.0	-4.5	0.4
	SLN192 ring / Begin	-38.6	6.5	-19.4	0.0	0.0	0.5
	SLN195 ring / End	37.1	6.1	-19.5	0.0	0.0	-0.6
	SLN446 raking strut / End	27.5	0.0	-1.5	0.1	0.0	-0.3
C LC_127	SLN43 column below / Begin	26.4	7.4	-1.3	0.0	-0.4	-2.0
	SLN192 ring / Begin	-2.6	0.3	-4.9	0.0	0.0	0.1
	SLN195 ring / End	10.1	1.5	-4.9	0.0	0.0	-0.5
	SLN446 raking strut / End	-2.8	0.0	-1.1	-0.5	0.0	2.2
C LC_145	SLN43 column below / Begin	123.7	0.0	0.4	0.0	1.6	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / Begin	-31.3	4.7	-25.6	0.0	0.0	1.1
	SLN195 ring / End	31.5	4.8	-25.7	0.0	0.0	-1.2
	SLN446 raking strut / End	-22.9	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN43 column below / Begin	122.8	0.0	0.4	0.0	1.6	0.0
	SLN192 ring / Begin	-31.1	4.6	-25.6	0.0	0.0	1.2
	SLN195 ring / End	30.9	4.6	-25.7	0.0	0.0	-1.1
	SLN446 raking strut / End	-24.0	0.0	-1.0	0.0	0.0	0.1
C LC_148	SLN43 column below / Begin	120.8	0.0	0.4	0.0	1.6	0.1
	SLN192 ring / Begin	-30.1	4.4	-25.6	0.0	0.0	1.2
	SLN195 ring / End	30.1	4.4	-25.7	0.0	0.0	-1.2
	SLN446 raking strut / End	-26.8	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN43 column below / Begin	73.2	0.0	2.2	0.0	8.9	0.0
	SLN192 ring / Begin	-31.1	4.8	-14.2	0.0	0.0	1.0
	SLN195 ring / End	31.1	4.8	-14.1	0.0	0.0	-1.0
	SLN446 raking strut / End	-8.5	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN43 column below / Begin	123.6	0.0	0.3	0.0	1.5	0.0
	SLN192 ring / Begin	-31.3	4.7	-21.2	0.0	0.0	1.2
	SLN195 ring / End	31.3	4.7	-21.0	0.0	0.0	-1.2
	SLN446 raking strut / End	-23.4	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	27.0 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN193+SLN191 radial (continuous)-bfl 1	12.7	C LC_109	220.5	0.0	35.3	OK
SLN193+SLN191 radial (continuous)-tfl 1	12.7	C LC_153	73.0	0.0	0.0	OK
SLN193+SLN191 radial (continuous)-w 1	7.2	C LC_122	173.7	0.0	50.2	OK
SLN43 column below	10.0	C LC_109	275.2	0.1	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_146	140.6	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_146	134.9	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_146	275.1	0.1	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_145	139.9	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_145	134.3	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_145	275.1	0.1	0.0	OK
SLN446 raking strut	10.0	C LC_123	63.8	0.0	0.0	OK
End plate column below SLN43 column below	12.0	C LC_109	259.6	0.0	35.3	OK
End plate SLN192 ring	12.0	C LC_148	141.9	0.0	0.0	OK
End plate SLN195 ring	12.0	C LC_148	120.1	0.0	24.3	OK
Cap plate SLN446 raking strut	12.0	C LC_123	141.1	0.0	0.0	OK
Fin plate SLN446 raking strut	12.0	C LC_127	275.1	0.1	130.8	OK
Fin plate 2 SLN446 raking strut	12.0	C LC_123	275.1	0.1	97.0	OK
Gusset SLN446 raking strut	12.0	C LC_123	275.2	0.1	66.3	OK
Flange top a end plate SLN192 ring	12.0	C LC_148	152.8	0.0	0.0	OK
Flange top b end plate SLN192 ring	12.0	C LC_146	141.1	0.0	0.0	OK
Flange bottom a end plate SLN192 ring	12.0	C LC_148	133.1	0.0	0.0	OK
Flange bottom b end plate SLN192 ring	12.0	C LC_146	158.0	0.0	0.0	OK
Flange top a end plate SLN195 ring	12.0	C LC_148	134.8	0.0	0.0	OK
Flange top b end plate SLN195 ring	12.0	C LC_148	149.9	0.0	0.0	OK
Flange bottom a end plate SLN195 ring	12.0	C LC_148	156.2	0.0	0.0	OK
Flange bottom b end plate SLN195 ring	12.0	C LC_148	132.7	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

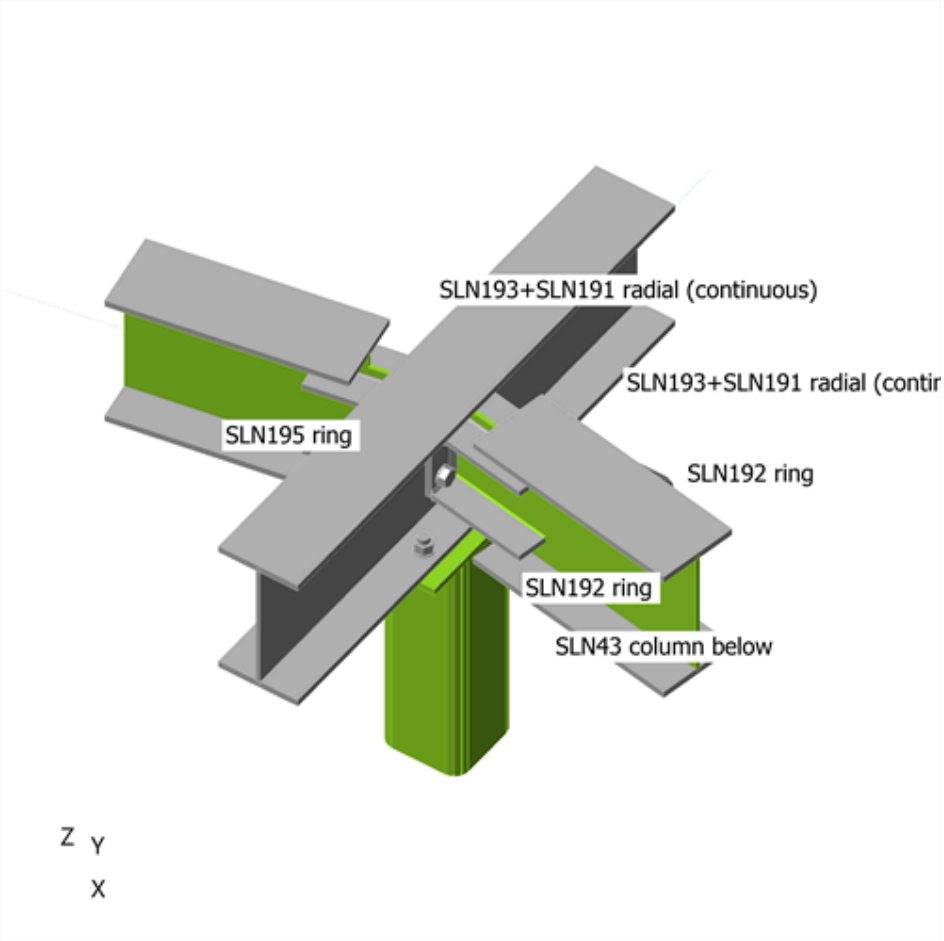
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

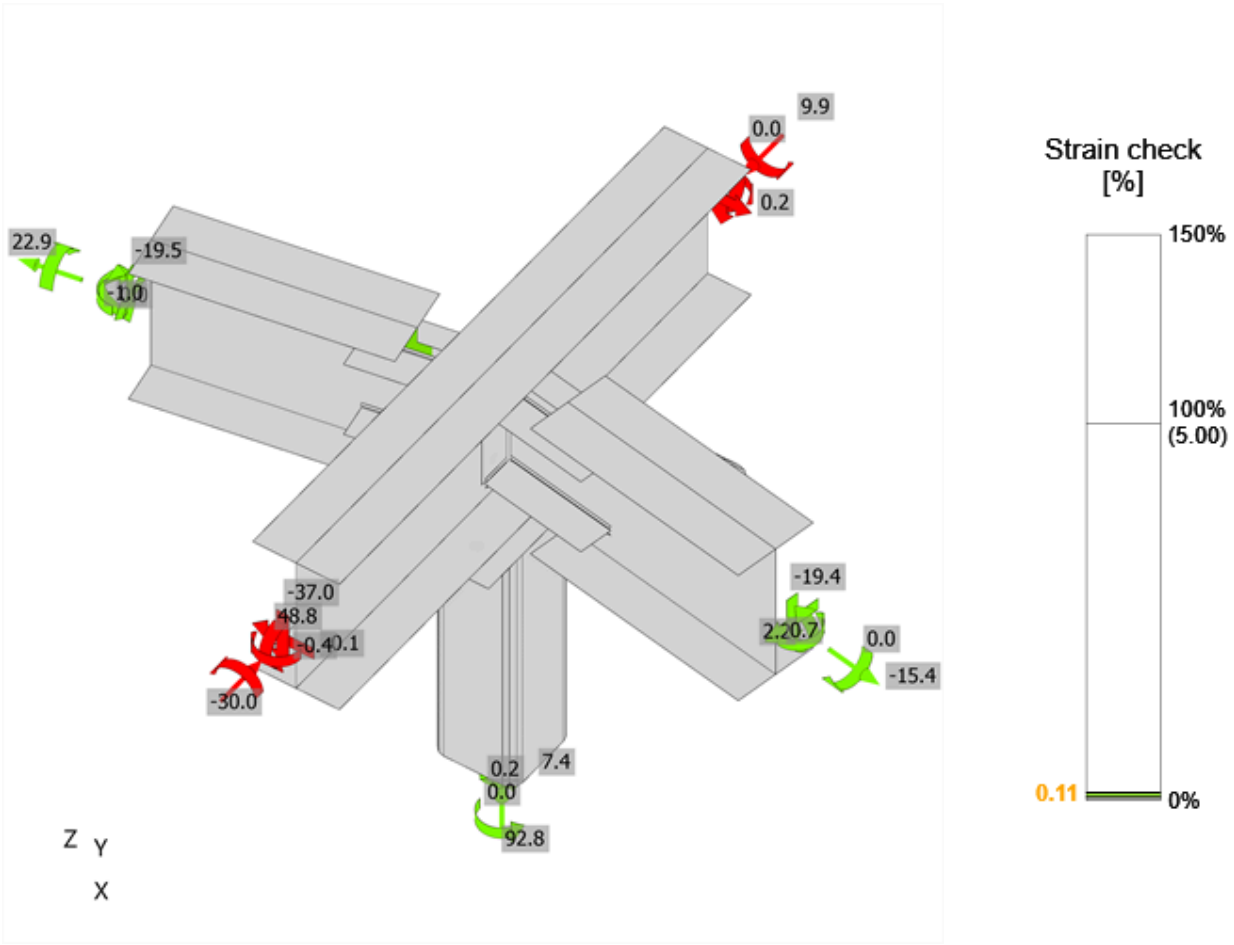
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN43 column below	150	C LC_109	0	5	0.1	OK
SLN446 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

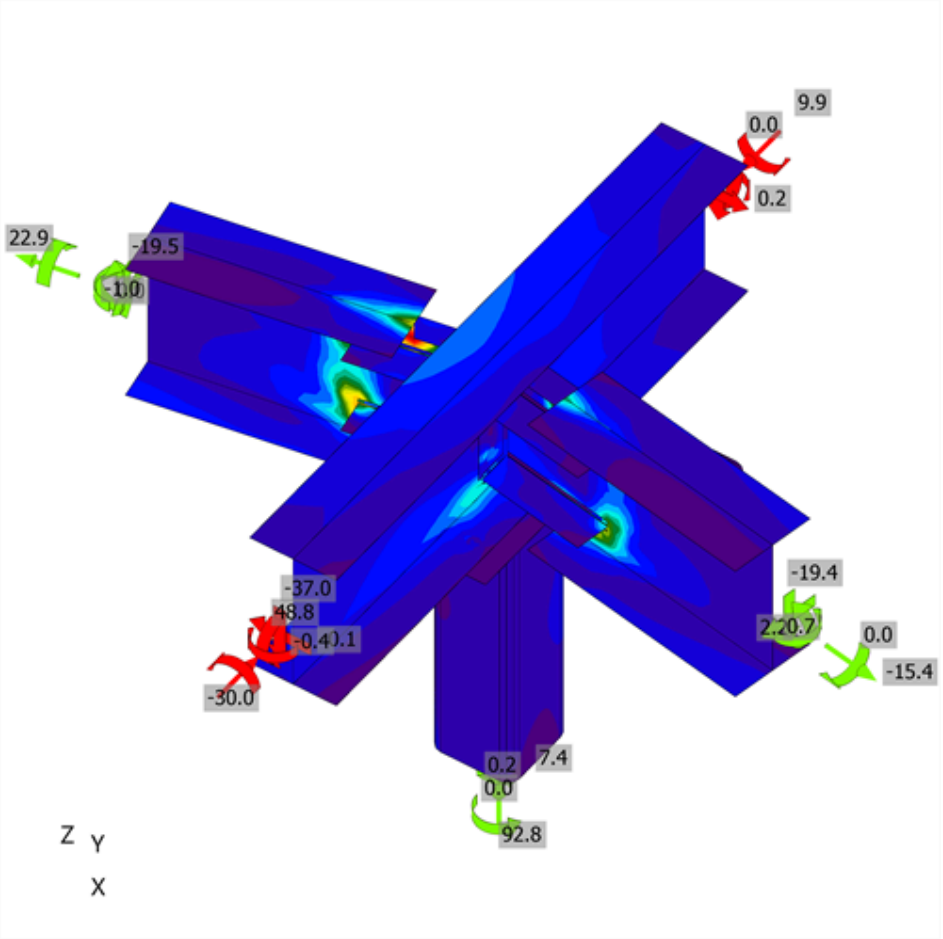
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

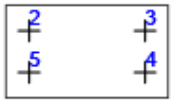
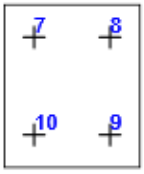
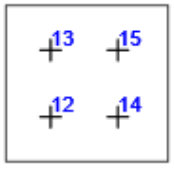


Strain check, C LC_123



Equivalent stress, C LC_123

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	16.1	2.6	107.5	11.4	2.7	10.9	OK	OK
	B3	M20 8.8 - 1	C LC_109	37.8	5.2	95.1	26.8	5.5	24.6	OK	OK
	B4	M20 8.8 - 1	C LC_109	38.0	5.3	101.7	27.0	5.7	24.9	OK	OK
	B5	M20 8.8 - 1	C LC_152	15.9	2.6	112.4	11.3	2.8	10.8	OK	OK
	B7	M20 8.8 - 1	C LC_148	19.4	1.4	123.8	13.7	1.5	11.3	OK	OK
	B8	M20 8.8 - 1	C LC_109	6.7	0.7	123.8	4.8	1.0	4.1	OK	OK
	B9	M20 8.8 - 1	C LC_109	6.6	0.7	123.8	4.7	0.9	4.0	OK	OK
	B10	M20 8.8 - 1	C LC_148	20.2	1.6	123.8	14.3	1.8	12.0	OK	OK
	B12	M20 8.8 - 1	C LC_123	17.5	15.4	115.3	12.4	16.3	25.2	OK	OK
	B13	M20 8.8 - 1	C LC_123	8.0	13.4	115.3	5.7	14.2	18.3	OK	OK
	B14	M20 8.8 - 1	C LC_117	1.0	8.6	106.0	0.7	14.0	9.7	OK	OK
	B15	M20 8.8 - 1	C LC_122	0.1	5.9	115.3	0.1	10.2	6.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN446 raking strut	SLN446 raking strut	◢ 4.2	519	C LC_123	179.2	0.0	-76.0	92.0	-17.7	44.3	25.5	OK	OK
SLN43 column below-w 3	Gusset SLN446 raking strut	◢ 5.6 ◣	288	C LC_122	397.0	0.3	-125.6	-125.6	-177.5	98.1	60.9	OK	OK
		◢ 5.6 ◣	288	C LC_109	397.0	0.3	132.7	-136.4	-167.5	98.1	62.8	OK	OK
SLN192 ring-w 1	Flange top b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_153	207.7	0.0	108.1	89.0	50.8	51.3	30.6	OK	OK
		◢ 4.2 ◣	197	C LC_148	86.0	0.0	-43.8	41.0	11.9	21.2	12.5	OK	OK
SLN192 ring-w 1	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_146	145.1	0.0	41.9	59.0	54.4	35.9	21.6	OK	OK
		◢ 4.2 ◣	198	C LC_146	271.0	0.0	124.6	-107.5	-88.0	67.0	40.6	OK	OK
SLN195 ring-w 1	Flange top a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_153	196.6	0.0	101.5	85.1	-47.0	48.6	24.4	OK	OK
		◢ 4.2 ◣	197	C LC_148	83.2	0.0	-42.7	40.4	-8.0	20.6	12.0	OK	OK
SLN195 ring-w 1	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_145	146.5	0.0	43.1	58.5	-55.8	36.2	19.1	OK	OK
		◢ 4.2 ◣	198	C LC_148	270.2	0.0	123.5	-108.0	87.1	66.8	35.0	OK	OK
Cap plate SLN446 raking strut	Fin plate SLN446 raking strut	◢ 5.6 ◣	139	C LC_127	263.7	0.0	82.8	115.0	87.5	65.1	49.1	OK	OK
		◢ 5.6 ◣	139	C LC_117	156.5	0.0	74.9	-65.2	-45.2	38.7	23.5	OK	OK
Cap plate SLN446 raking strut	Fin plate 2 SLN446 raking strut	◢ 5.6 ◣	139	C LC_123	143.9	0.0	-57.8	-73.0	21.2	35.5	21.4	OK	OK
		◢ 5.6 ◣	139	C LC_123	256.9	0.0	-118.1	103.0	-82.0	63.5	50.1	OK	OK
End plate column below SLN43 column below	SLN43 column below	◢ 5.6 ◣	521	C LC_122	291.3	0.0	-186.6	-127.8	-18.5	72.0	39.5	OK	OK
		◢ 5.6 ◣	521	C LC_109	221.6	0.0	152.8	-74.5	-55.1	54.8	34.2	OK	OK
SLN193+SLN191 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_119	89.8	0.0	-3.9	0.4	51.8	22.2	13.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN193+SLN191 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_120	93.4	0.0	-6.0	10.6	-52.7	23.1	14.4	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_148	92.1	0.0	-42.9	-1.6	-47.0	22.7	14.5	OK	OK
		◢ 4.2 ◣	149	C LC_148	84.2	0.0	-6.0	37.9	30.2	20.8	15.0	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_145	90.6	0.0	-36.9	-22.2	42.3	22.4	15.3	OK	OK
		◢ 4.2 ◣	149	C LC_148	91.8	0.0	-25.5	39.0	-32.7	22.7	16.7	OK	OK
SLN192 ring-w 1	Flange top a end plate SLN192 ring	◢ 4.2 ◣	199	C LC_148	275.9	0.0	-29.3	15.9	-157.6	68.2	47.0	OK	OK
		◢ 4.2 ◣	198	C LC_153	213.4	0.0	34.9	39.3	-115.0	52.7	30.0	OK	OK
SLN192 ring-w 1	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	198	C LC_146	234.2	0.0	40.9	-44.1	125.6	57.9	33.1	OK	OK
		◢ 4.2 ◣	198	C LC_148	242.2	0.0	-31.2	-11.5	138.2	59.8	36.6	OK	OK
SLN195 ring-w 1	Flange top b end plate SLN195 ring	◢ 4.2 ◣	199	C LC_148	276.6	0.0	-38.2	20.1	156.9	68.4	41.7	OK	OK
		◢ 4.2 ◣	198	C LC_153	211.5	0.0	35.9	39.7	113.6	52.3	28.7	OK	OK
SLN195 ring-w 1	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	198	C LC_148	236.3	0.0	40.1	-43.4	-127.3	58.4	32.8	OK	OK
		◢ 4.2 ◣	198	C LC_148	243.2	0.0	-29.7	-10.4	-139.0	60.1	34.8	OK	OK
End plate SLN192 ring	Flange top a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_148	72.2	0.0	-2.6	7.3	-41.0	17.8	11.8	OK	OK
		◢ 4.2 ◣	58	C LC_148	42.4	0.0	1.5	-16.1	-18.4	10.5	9.1	OK	OK
End plate SLN192 ring	Flange top b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_148	115.1	0.0	-1.8	-0.8	-66.4	28.4	19.5	OK	OK
		◢ 4.2 ◣	58	C LC_148	112.1	0.0	56.6	39.8	-39.2	27.7	22.7	OK	OK
End plate SLN192 ring	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_145	65.0	0.0	18.2	-10.5	34.5	16.1	11.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_148	67.3	0.0	-5.5	-9.3	37.6	16.6	11.8	OK	OK
End plate SLN192 ring	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_122	121.3	0.0	64.6	-48.0	34.7	30.0	23.1	OK	OK
		◢ 4.2 ◣	58	C LC_148	125.0	0.0	-4.6	0.2	72.1	30.9	20.7	OK	OK
End plate SLN195 ring	Flange top a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	99.7	0.0	0.8	-4.8	57.4	24.6	17.2	OK	OK
		◢ 4.2 ◣	58	C LC_148	91.2	0.0	12.5	19.8	48.3	22.5	17.6	OK	OK
End plate SLN195 ring	Flange top b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	61.6	0.0	-9.3	12.0	33.0	15.2	10.3	OK	OK
		◢ 4.2 ◣	58	C LC_148	45.5	0.0	-3.4	-17.1	19.9	11.2	8.3	OK	OK
End plate SLN195 ring	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_122	99.6	0.0	50.9	-40.0	-29.0	24.6	18.7	OK	OK
		◢ 4.2 ◣	58	C LC_148	108.8	0.0	-2.8	3.2	-62.7	26.9	18.7	OK	OK
End plate SLN195 ring	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_146	72.7	0.0	14.8	-8.8	-40.1	18.0	10.4	OK	OK
		◢ 4.2 ◣	58	C LC_148	62.2	0.0	-10.3	-12.6	-33.1	15.4	10.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-140
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut

Name	SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

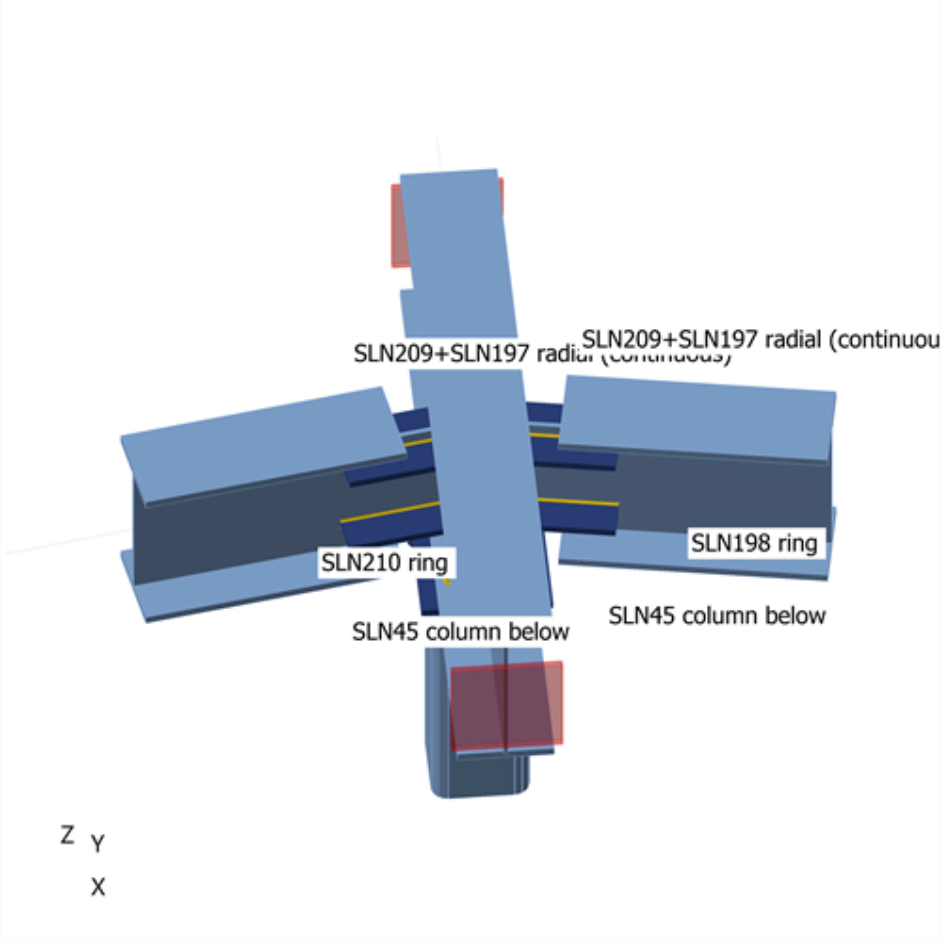
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN209+SLN197 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN45 column below	2 - SHS150x150x10	0.0	90.0	-40.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN210 ring	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN445 raking strut	2 - SHS150x150x10	140.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN209+SLN197 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN209+SLN197 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN45 column below / begin		Position	0
SLN198 ring / begin		Position	0
SLN210 ring / end		Position	0
SLN445 raking strut / end		Position	0



SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN45 column below / Begin	116.7	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-31.1	4.7	-26.0	0.0	0.0	1.2
	SLN210 ring / End	31.1	4.6	-25.6	0.0	0.0	-1.3
	SLN445 raking strut / End	-22.9	0.0	-1.0	0.0	0.0	0.1
C LC_107	SLN45 column below / Begin	31.2	4.8	-6.4	0.0	-4.1	-1.3
	SLN198 ring / Begin	-6.2	1.3	-0.5	0.0	0.0	-0.4
	SLN210 ring / End	11.0	2.2	-0.5	0.0	0.0	0.3
	SLN445 raking strut / End	38.1	0.0	-1.1	-0.3	0.0	1.5
C LC_109	SLN45 column below / Begin	23.5	-5.8	-5.3	0.0	-3.4	1.6
	SLN198 ring / Begin	-10.2	1.9	-0.5	0.0	0.0	-0.1
	SLN210 ring / End	4.4	1.0	-0.5	0.0	0.0	0.4
	SLN445 raking strut / End	27.6	0.0	-1.1	0.4	0.0	-1.7
C LC_115	SLN45 column below / Begin	37.9	4.8	-6.3	0.0	-3.9	-1.3
	SLN198 ring / Begin	-8.8	1.7	-1.8	0.0	0.0	-0.3
	SLN210 ring / End	13.6	2.6	-1.7	0.0	0.0	0.2
	SLN445 raking strut / End	37.7	0.0	-1.4	-0.3	0.0	1.5
C LC_120	SLN45 column below / Begin	105.1	0.0	0.7	0.0	3.0	-0.1
	SLN198 ring / Begin	-22.8	3.1	-26.0	0.0	0.0	1.3
	SLN210 ring / End	22.4	3.0	-25.6	0.0	0.0	-1.3
	SLN445 raking strut / End	-38.1	0.2	-0.8	0.0	0.0	0.2
C LC_122	SLN45 column below / Begin	105.9	-0.1	0.7	0.0	3.0	0.2
	SLN198 ring / Begin	-22.3	3.0	-26.0	0.0	0.0	1.2
	SLN210 ring / End	22.7	3.0	-25.6	0.0	0.0	-1.4
	SLN445 raking strut / End	-37.1	-0.3	-0.9	0.0	0.0	-0.1
C LC_123	SLN45 column below / Begin	110.8	4.8	-5.8	0.0	-2.0	-1.3
	SLN198 ring / Begin	-29.2	4.6	-19.7	0.0	0.0	0.7
	SLN210 ring / End	34.2	5.6	-19.4	0.0	0.0	-0.8
	SLN445 raking strut / End	12.9	0.0	-1.4	-0.3	0.0	1.5
C LC_135	SLN45 column below / Begin	116.1	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-31.4	4.7	-26.0	0.0	0.0	1.2
	SLN210 ring / End	31.3	4.6	-25.6	0.0	0.0	-1.3
	SLN445 raking strut / End	-23.7	0.0	-1.0	0.0	0.0	0.1
C LC_145	SLN45 column below / Begin	118.5	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-31.9	4.9	-26.0	0.0	0.0	1.1
	SLN210 ring / End	32.0	4.8	-25.6	0.0	0.0	-1.3
	SLN445 raking strut / End	-20.5	0.0	-1.0	0.0	0.0	0.1
C LC_146	SLN45 column below / Begin	114.9	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-30.4	4.5	-26.0	0.0	0.0	1.3
	SLN210 ring / End	30.2	4.3	-25.6	0.0	0.0	-1.3
	SLN445 raking strut / End	-25.3	0.0	-1.0	0.0	0.0	0.1
C LC_147	SLN45 column below / Begin	118.3	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-31.9	4.9	-26.0	0.0	0.0	1.2
	SLN210 ring / End	31.7	4.7	-25.6	0.0	0.0	-1.2
	SLN445 raking strut / End	-20.7	0.0	-1.0	0.0	0.0	0.1
C LC_148	SLN45 column below / Begin	115.1	0.0	0.8	0.0	3.1	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / Begin	-30.3	4.5	-26.0	0.0	0.0	1.2
	SLN210 ring / End	30.5	4.4	-25.6	0.0	0.0	-1.4
	SLN445 raking strut / End	-25.0	0.0	-1.0	0.0	0.0	0.1
C LC_152	SLN45 column below / Begin	71.9	0.0	2.2	0.0	8.7	0.0
	SLN198 ring / Begin	-31.0	4.8	-14.0	0.0	0.0	1.0
	SLN210 ring / End	31.0	4.7	-14.1	0.0	0.0	-1.1
	SLN445 raking strut / End	-10.0	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN45 column below / Begin	116.4	0.0	0.7	0.0	3.0	0.0
	SLN198 ring / Begin	-31.3	4.7	-21.0	0.0	0.0	1.2
	SLN210 ring / End	31.2	4.6	-20.7	0.0	0.0	-1.3
	SLN445 raking strut / End	-22.6	0.0	-1.0	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	22.0 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN209+SLN197 radial (continuous)-bfl 1	12.7	C LC_107	173.7	0.0	27.4	OK
SLN209+SLN197 radial (continuous)-tfl 1	12.7	C LC_153	69.0	0.0	0.0	OK
SLN209+SLN197 radial (continuous)-w 1	7.2	C LC_120	175.0	0.0	51.2	OK
SLN45 column below	10.0	C LC_120	275.1	0.1	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_146	141.8	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_146	136.0	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_146	275.1	0.1	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_148	145.2	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_148	141.0	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_148	275.1	0.1	0.0	OK
SLN445 raking strut	10.0	C LC_115	55.6	0.0	0.0	OK
End plate column below SLN45 column below	12.0	C LC_107	195.8	0.0	27.4	OK
End plate SLN198 ring	12.0	C LC_120	147.1	0.0	0.0	OK
End plate SLN210 ring	12.0	C LC_146	124.5	0.0	41.1	OK
Cap plate SLN445 raking strut	12.0	C LC_109	110.7	0.0	0.0	OK
Fin plate SLN445 raking strut	12.0	C LC_115	272.1	0.0	86.2	OK
Fin plate 2 SLN445 raking strut	12.0	C LC_109	275.1	0.0	103.5	OK
Gusset SLN445 raking strut	12.0	C LC_109	275.1	0.0	51.0	OK
Flange top a end plate SLN198 ring	12.0	C LC_146	157.1	0.0	0.0	OK
Flange top b end plate SLN198 ring	12.0	C LC_146	145.4	0.0	0.0	OK
Flange bottom a end plate SLN198 ring	12.0	C LC_146	133.3	0.0	0.0	OK
Flange bottom b end plate SLN198 ring	12.0	C LC_146	158.6	0.0	0.0	OK
Flange top a end plate SLN210 ring	12.0	C LC_148	148.4	0.0	0.0	OK
Flange top b end plate SLN210 ring	12.0	C LC_148	162.5	0.0	0.0	OK
Flange bottom a end plate SLN210 ring	12.0	C LC_148	163.6	0.0	0.0	OK
Flange bottom b end plate SLN210 ring	12.0	C LC_148	140.3	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

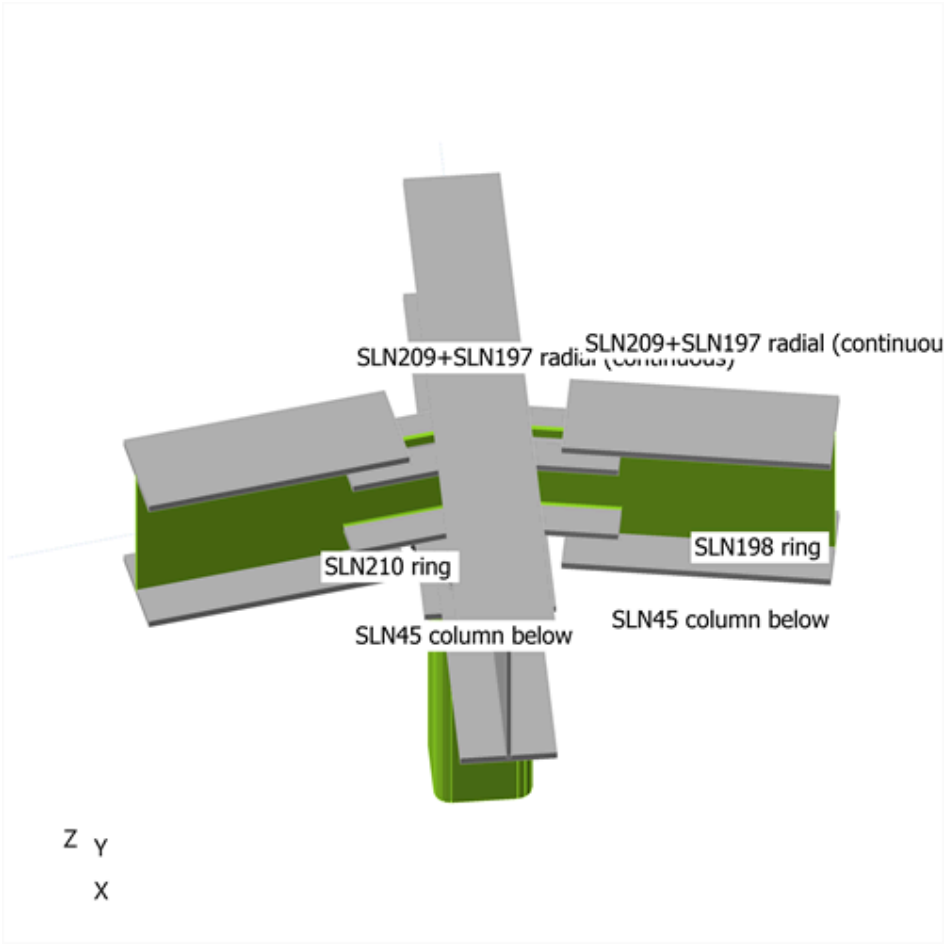
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

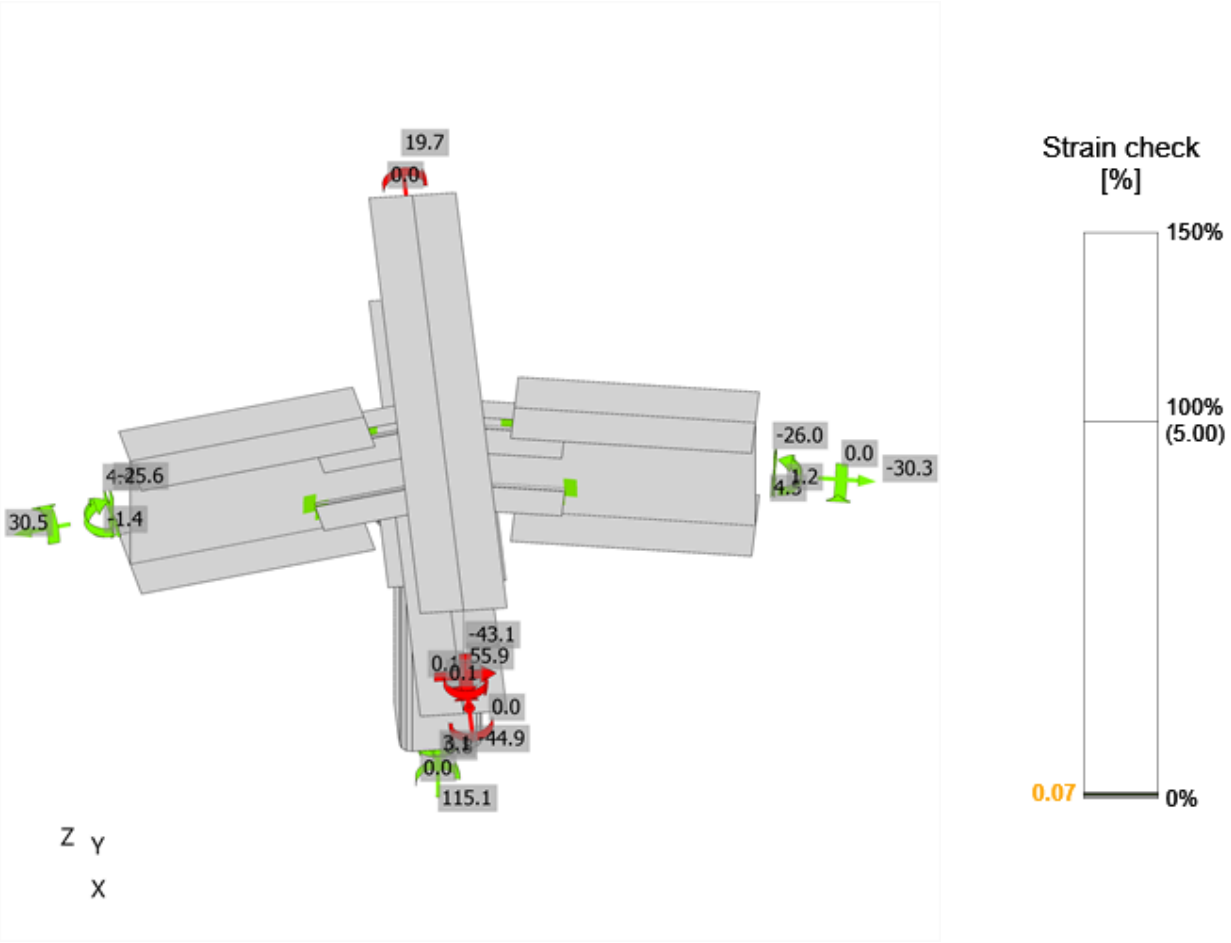
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN45 column below	150	C LC_107	0	5	0.1	OK
SLN445 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

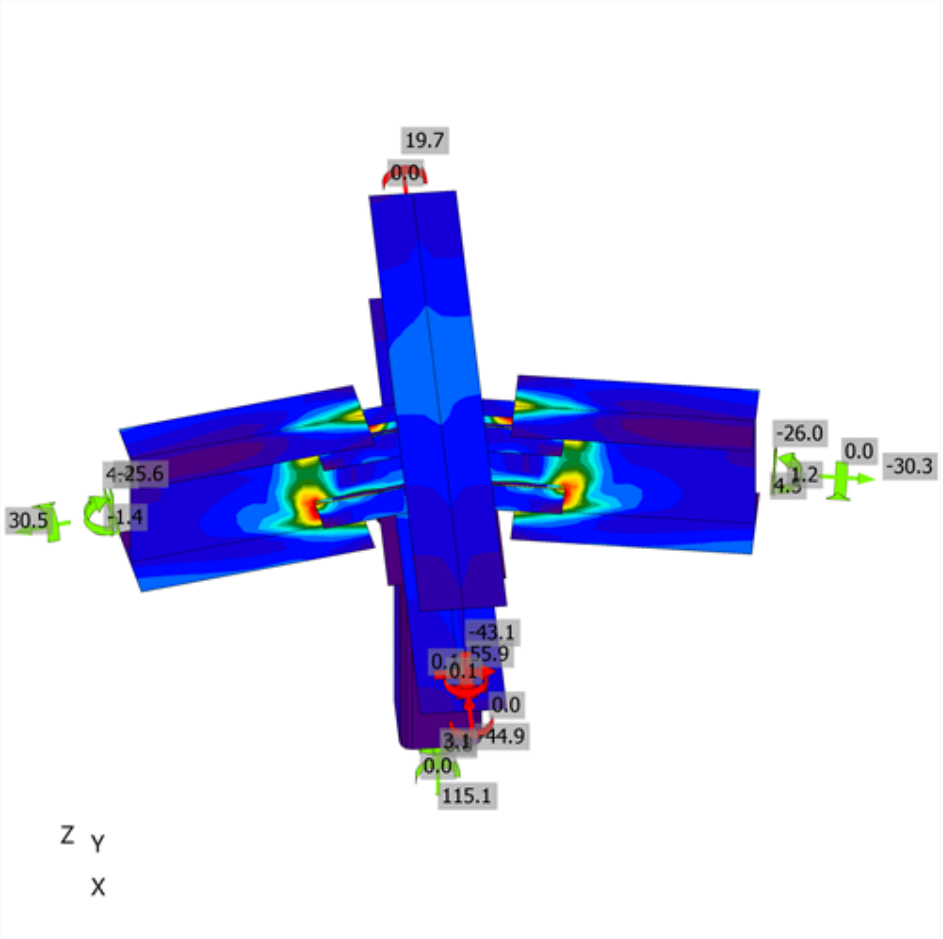
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

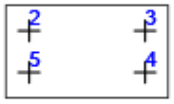
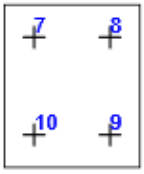
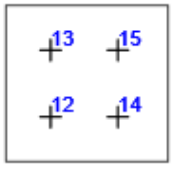


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	16.2	2.8	110.1	11.4	3.0	11.2	OK	OK
	B3	M20 8.8 - 1	C LC_107	29.1	4.6	129.5	20.6	4.9	19.6	OK	OK
	B4	M20 8.8 - 1	C LC_107	28.3	3.9	82.4	20.0	4.8	18.5	OK	OK
	B5	M20 8.8 - 1	C LC_152	15.9	2.8	114.3	11.2	3.0	11.0	OK	OK
	B7	M20 8.8 - 1	C LC_120	20.6	1.5	123.8	14.6	1.6	12.1	OK	OK
	B8	M20 8.8 - 1	C LC_107	4.2	0.5	123.8	3.0	0.7	2.7	OK	OK
	B9	M20 8.8 - 1	C LC_107	4.2	0.6	81.2	3.0	0.7	2.7	OK	OK
	B10	M20 8.8 - 1	C LC_120	21.1	1.7	123.8	14.9	1.8	12.5	OK	OK
	B12	M20 8.8 - 1	C LC_109	14.4	13.9	106.0	10.2	14.7	22.0	OK	OK
	B13	M20 8.8 - 1	C LC_109	6.7	9.5	106.0	4.8	10.1	13.5	OK	OK
	B14	M20 8.8 - 1	C LC_115	2.3	10.5	106.0	1.6	11.7	12.3	OK	OK
	B15	M20 8.8 - 1	C LC_120	0.1	5.6	115.3	0.1	9.4	6.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN445 raking strut	SLN445 raking strut	◢ 4.2	519	C LC_115	187.0	0.0	86.4	-94.6	14.7	46.2	25.6	OK	OK
SLN45 column below-w 3	Gusset SLN445 raking strut	◢ 5.6 ◣	288	C LC_120	396.9	0.2	-123.4	-121.9	-180.4	98.1	59.4	OK	OK
		◢ 5.6 ◣	288	C LC_120	396.9	0.2	-120.1	121.6	181.5	98.1	60.1	OK	OK
SLN198 ring-w 1	Flange top b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_153	215.2	0.0	112.4	92.6	51.6	53.2	31.6	OK	OK
		◢ 4.2 ◣	197	C LC_146	80.8	0.0	-41.0	38.2	12.5	20.0	12.7	OK	OK
SLN198 ring-w 1	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_135	145.2	0.0	42.0	59.0	54.4	35.9	21.6	OK	OK
		◢ 4.2 ◣	198	C LC_146	270.9	0.0	124.5	-107.5	-87.9	66.9	40.5	OK	OK
SLN210 ring-w 1	Flange top a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_153	219.3	0.0	113.1	95.2	-52.0	54.2	26.8	OK	OK
		◢ 4.2 ◣	197	C LC_148	85.9	0.0	-44.3	41.3	-9.8	21.2	12.9	OK	OK
SLN210 ring-w 1	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_145	148.6	0.0	44.2	59.8	-55.9	36.7	19.5	OK	OK
		◢ 4.2 ◣	198	C LC_148	279.8	0.0	128.4	-112.6	89.0	69.1	36.3	OK	OK
Cap plate SLN445 raking strut	Fin plate SLN445 raking strut	◢ 5.6 ◣	139	C LC_115	292.8	0.0	106.1	127.8	92.0	72.3	49.3	OK	OK
		◢ 5.6 ◣	139	C LC_109	131.3	0.0	85.7	-47.6	-32.2	32.5	21.6	OK	OK
Cap plate SLN445 raking strut	Fin plate 2 SLN445 raking strut	◢ 5.6 ◣	139	C LC_115	154.5	0.0	91.5	59.5	40.4	38.2	25.2	OK	OK
		◢ 5.6 ◣	139	C LC_109	290.2	0.0	100.4	-125.9	-94.1	71.7	50.3	OK	OK
End plate column below SLN45 column below	SLN45 column below	◢ 5.6 ◣	521	C LC_120	291.1	0.0	-183.1	-129.1	-20.2	71.9	39.7	OK	OK
		◢ 5.6 ◣	521	C LC_152	209.7	0.0	-97.0	107.2	-4.1	51.8	30.8	OK	OK
SLN209+SLN197 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_122	84.5	0.0	-1.4	-2.7	48.7	20.9	12.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN209+SLN197 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_120	87.5	0.0	-5.0	9.1	-49.6	21.6	14.0	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_148	91.5	0.0	-41.9	-3.3	-46.9	22.6	14.3	OK	OK
		◢ 4.2 ◣	149	C LC_148	84.6	0.0	-7.5	37.4	31.1	20.9	14.8	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_147	88.2	0.0	-37.1	-19.5	41.9	21.8	15.1	OK	OK
		◢ 4.2 ◣	149	C LC_146	89.7	0.0	-22.2	38.5	-32.1	22.2	16.4	OK	OK
SLN198 ring-w 1	Flange top a end plate SLN198 ring	◢ 4.2 ◣	199	C LC_146	290.3	0.0	-22.1	10.6	-166.8	71.7	49.2	OK	OK
		◢ 4.2 ◣	198	C LC_146	218.0	0.0	32.1	36.7	-119.0	53.9	30.6	OK	OK
SLN198 ring-w 1	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	198	C LC_146	234.6	0.0	43.4	-46.3	124.8	58.0	33.1	OK	OK
		◢ 4.2 ◣	198	C LC_146	241.8	0.0	-35.9	-13.7	137.4	59.7	36.5	OK	OK
SLN210 ring-w 1	Flange top b end plate SLN210 ring	◢ 4.2 ◣	199	C LC_148	302.1	0.0	-29.3	14.3	173.0	74.6	45.8	OK	OK
		◢ 4.2 ◣	198	C LC_148	228.4	0.0	34.8	39.8	124.1	56.4	31.0	OK	OK
SLN210 ring-w 1	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	198	C LC_148	248.5	0.0	42.2	-45.8	-133.8	61.4	34.1	OK	OK
		◢ 4.2 ◣	198	C LC_148	260.8	0.0	-33.2	-11.7	-148.9	64.4	36.7	OK	OK
End plate SLN198 ring	Flange top a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_146	74.2	0.0	0.0	2.3	-42.8	18.3	12.1	OK	OK
		◢ 4.2 ◣	58	C LC_146	41.0	0.0	1.4	-16.1	-17.4	10.1	9.1	OK	OK
End plate SLN198 ring	Flange top b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_146	117.5	0.0	-2.6	-0.8	-67.8	29.0	20.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	119.2	0.0	62.9	43.9	-38.5	29.4	24.2	OK	OK
End plate SLN198 ring	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_145	64.1	0.0	18.6	-11.1	33.6	15.8	11.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_146	67.2	0.0	-5.5	-9.2	37.6	16.6	11.8	OK	OK
End plate SLN198 ring	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_120	126.9	0.0	67.5	-50.0	36.7	31.4	24.6	OK	OK
		◢ 4.2 ◣	58	C LC_146	126.9	0.0	-5.6	0.2	73.2	31.3	21.0	OK	OK
End plate SLN210 ring	Flange top a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_148	106.2	0.0	0.5	-4.4	61.2	26.2	18.2	OK	OK
		◢ 4.2 ◣	58	C LC_120	95.8	0.0	12.7	19.6	51.2	23.7	18.4	OK	OK
End plate SLN210 ring	Flange top b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_146	60.2	0.0	-6.6	8.3	33.5	14.9	11.1	OK	OK
		◢ 4.2 ◣	58	C LC_120	48.5	0.0	-7.6	-19.8	19.3	12.0	8.5	OK	OK
End plate SLN210 ring	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_146	102.2	0.0	49.9	-39.4	-33.2	25.2	19.3	OK	OK
		◢ 4.2 ◣	58	C LC_148	114.4	0.0	-3.3	2.7	-66.0	28.3	19.5	OK	OK
End plate SLN210 ring	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_147	69.5	0.0	15.9	-11.4	-37.4	17.2	10.2	OK	OK
		◢ 4.2 ◣	58	C LC_146	57.8	0.0	-11.1	-12.5	-30.2	14.3	10.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-180
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut

Name	SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

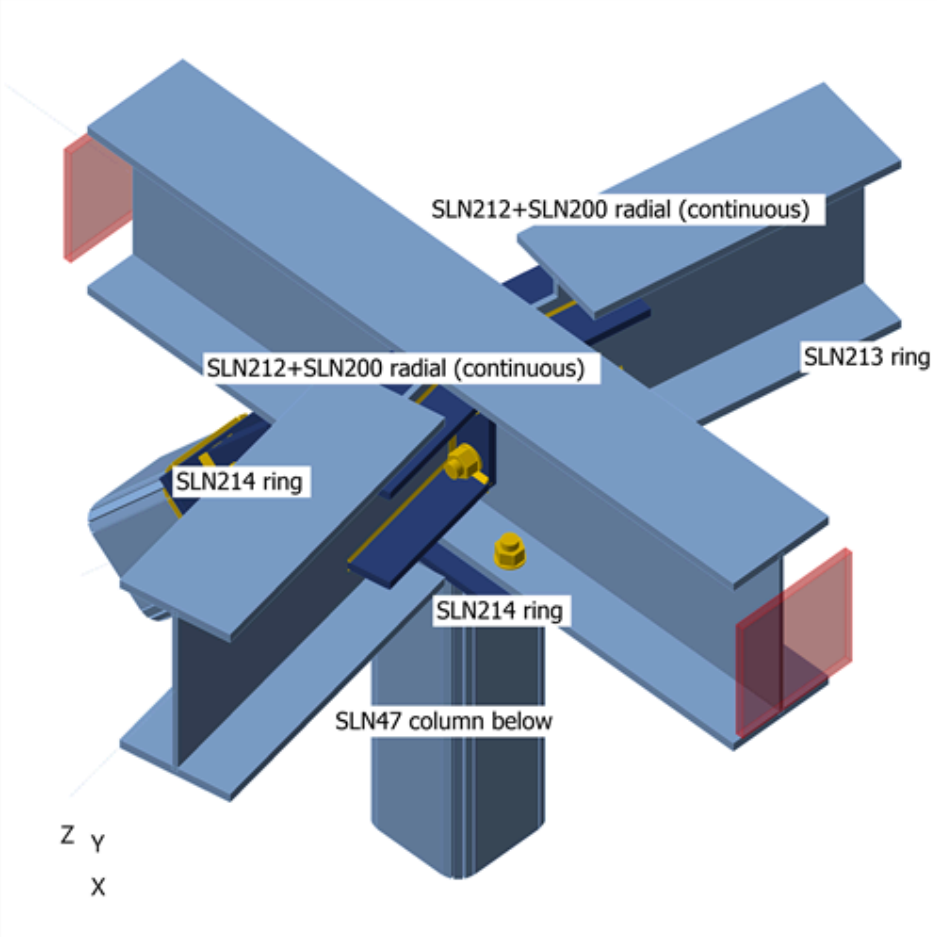
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN212+SLN200 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN47 column below	2 - SHS150x150x10	0.0	90.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0
SLN214 ring	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN434 raking strut	2 - SHS150x150x10	-179.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN212+SLN200 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN212+SLN200 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN47 column below / begin		Position	0
SLN213 ring / begin		Position	0
SLN214 ring / end		Position	0
SLN434 raking strut / end		Position	0



SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN47 column below / Begin	119.4	0.1	0.4	0.0	1.6	0.0
	SLN213 ring / Begin	-30.7	4.6	-25.6	0.0	0.0	1.2
	SLN214 ring / End	30.7	4.5	-25.6	0.0	0.0	-1.3
	SLN434 raking strut / End	-29.4	0.0	-1.0	0.0	0.0	0.1
C LC_107	SLN47 column below / Begin	42.6	0.0	-8.4	0.0	-5.7	0.0
	SLN213 ring / Begin	-13.3	2.6	-0.5	0.0	0.0	-0.4
	SLN214 ring / End	13.2	2.8	-0.5	0.0	0.0	0.7
	SLN434 raking strut / End	52.3	0.0	-1.3	0.0	0.0	0.0
C LC_114	SLN47 column below / Begin	70.6	7.5	0.2	0.0	0.9	-2.0
	SLN213 ring / Begin	-5.7	0.8	-16.2	0.0	0.0	0.2
	SLN214 ring / End	13.4	1.6	-16.2	0.0	0.0	-1.2
	SLN434 raking strut / End	-26.2	0.0	-1.0	-0.5	0.0	2.3
C LC_120	SLN47 column below / Begin	106.8	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-21.8	2.9	-25.6	0.0	0.0	1.4
	SLN214 ring / End	21.8	2.9	-25.6	0.0	0.0	-1.3
	SLN434 raking strut / End	-45.8	0.0	-0.8	0.0	0.0	0.1
C LC_121	SLN47 column below / Begin	115.3	-3.7	0.4	0.0	1.5	1.0
	SLN213 ring / Begin	-26.7	3.9	-25.6	0.0	0.0	1.3
	SLN214 ring / End	22.8	3.2	-25.6	0.0	0.0	-1.1
	SLN434 raking strut / End	-34.7	0.1	-1.0	0.3	0.0	-1.1
C LC_122	SLN47 column below / Begin	115.9	3.8	0.4	0.0	1.5	-1.0
	SLN213 ring / Begin	-22.8	3.4	-25.6	0.0	0.0	0.9
	SLN214 ring / End	26.9	3.8	-25.6	0.0	0.0	-1.4
	SLN434 raking strut / End	-34.0	0.0	-1.0	-0.2	0.0	1.2
C LC_123	SLN47 column below / Begin	123.9	0.1	-8.1	0.0	-4.6	0.0
	SLN213 ring / Begin	-36.5	5.9	-19.4	0.0	0.0	0.8
	SLN214 ring / End	36.4	6.1	-19.4	0.0	0.0	-0.5
	SLN434 raking strut / End	22.0	0.0	-1.5	0.0	0.0	0.0
C LC_125	SLN47 column below / Begin	84.3	-7.5	0.3	0.0	1.2	2.1
	SLN213 ring / Begin	-20.2	2.9	-19.4	0.0	0.0	1.1
	SLN214 ring / End	12.5	1.6	-19.4	0.0	0.0	-0.7
	SLN434 raking strut / End	-31.0	0.1	-1.0	0.5	0.0	-2.2
C LC_126	SLN47 column below / Begin	85.4	7.5	0.3	0.0	1.1	-2.0
	SLN213 ring / Begin	-12.7	1.8	-19.4	0.0	0.0	0.5
	SLN214 ring / End	20.4	2.7	-19.4	0.0	0.0	-1.4
	SLN434 raking strut / End	-29.8	0.0	-1.0	-0.5	0.0	2.3
C LC_146	SLN47 column below / Begin	117.1	0.1	0.4	0.0	1.6	0.0
	SLN213 ring / Begin	-29.6	4.3	-25.6	0.0	0.0	1.2
	SLN214 ring / End	29.6	4.2	-25.6	0.0	0.0	-1.4
	SLN434 raking strut / End	-32.5	0.0	-1.0	0.0	0.0	0.1
C LC_147	SLN47 column below / Begin	119.3	0.1	0.4	0.0	1.6	0.0
	SLN213 ring / Begin	-30.8	4.6	-25.6	0.0	0.0	1.2
	SLN214 ring / End	30.6	4.5	-25.6	0.0	0.0	-1.3
	SLN434 raking strut / End	-29.5	0.0	-1.0	0.0	0.0	0.1
C LC_148	SLN47 column below / Begin	119.5	0.1	0.4	0.0	1.6	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / Begin	-30.6	4.6	-25.6	0.0	0.0	1.2
	SLN214 ring / End	30.8	4.5	-25.6	0.0	0.0	-1.3
	SLN434 raking strut / End	-29.3	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN47 column below / Begin	77.0	0.0	-1.6	0.0	-6.7	0.0
	SLN213 ring / Begin	-15.6	2.4	-11.6	0.0	0.0	0.5
	SLN214 ring / End	15.5	2.2	-12.3	0.0	0.0	-0.7
	SLN434 raking strut / End	-16.5	0.0	-1.0	0.0	0.0	0.1
C LC_152	SLN47 column below / Begin	69.6	0.0	2.2	0.0	9.0	0.0
	SLN213 ring / Begin	-30.7	4.8	-14.2	0.0	0.0	0.9
	SLN214 ring / End	30.7	4.7	-16.2	0.0	0.0	-1.1
	SLN434 raking strut / End	-13.4	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN47 column below / Begin	119.1	0.1	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-30.8	4.6	-20.9	0.0	0.0	1.2
	SLN214 ring / End	30.9	4.5	-23.6	0.0	0.0	-1.3
	SLN434 raking strut / End	-29.2	0.0	-1.0	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	28.6 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN212+SLN200 radial (continuous)-bfl 1	12.7	C LC_107	234.6	0.0	37.8	OK
SLN212+SLN200 radial (continuous)-tfl 1	12.7	C LC_122	73.6	0.0	0.0	OK
SLN212+SLN200 radial (continuous)-w 1	7.2	C LC_126	181.9	0.0	38.1	OK
SLN47 column below	10.0	C LC_107	275.3	0.1	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_147	140.0	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_147	134.0	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_147	275.1	0.1	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_148	144.8	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_148	140.3	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_148	275.1	0.1	0.0	OK
SLN434 raking strut	10.0	C LC_126	68.7	0.0	0.0	OK
End plate column below SLN47 column below	12.0	C LC_107	269.6	0.0	37.8	OK
End plate SLN213 ring	12.0	C LC_120	147.9	0.0	0.0	OK
End plate SLN214 ring	12.0	C LC_146	124.9	0.0	61.9	OK
Cap plate SLN434 raking strut	12.0	C LC_126	149.2	0.0	0.0	OK
Fin plate SLN434 raking strut	12.0	C LC_125	275.2	0.1	96.8	OK
Fin plate 2 SLN434 raking strut	12.0	C LC_126	275.2	0.1	98.2	OK
Gusset SLN434 raking strut	12.0	C LC_126	275.2	0.1	66.6	OK
Flange top a end plate SLN213 ring	12.0	C LC_146	155.0	0.0	0.0	OK
Flange top b end plate SLN213 ring	12.0	C LC_146	143.4	0.0	0.0	OK
Flange bottom a end plate SLN213 ring	12.0	C LC_146	132.7	0.0	0.0	OK
Flange bottom b end plate SLN213 ring	12.0	C LC_146	156.9	0.0	0.0	OK
Flange top a end plate SLN214 ring	12.0	C LC_146	149.2	0.0	0.0	OK
Flange top b end plate SLN214 ring	12.0	C LC_122	163.3	0.0	0.0	OK
Flange bottom a end plate SLN214 ring	12.0	C LC_146	160.8	0.0	0.0	OK
Flange bottom b end plate SLN214 ring	12.0	C LC_122	137.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

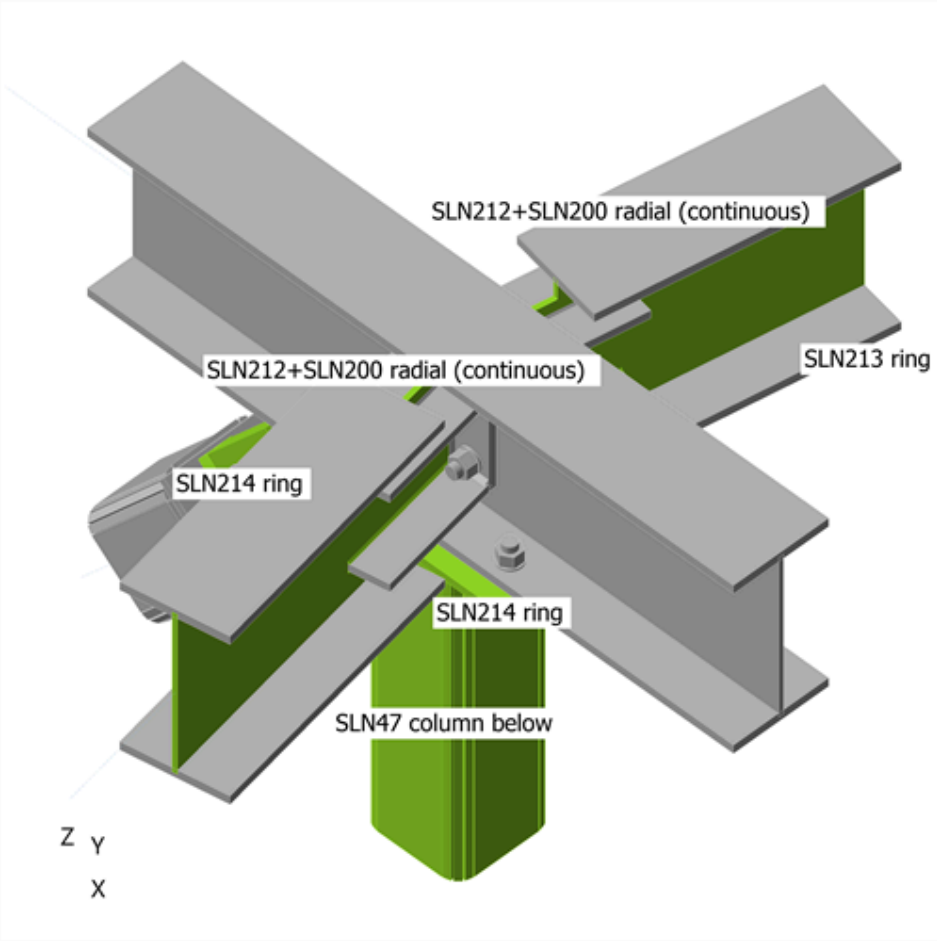
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

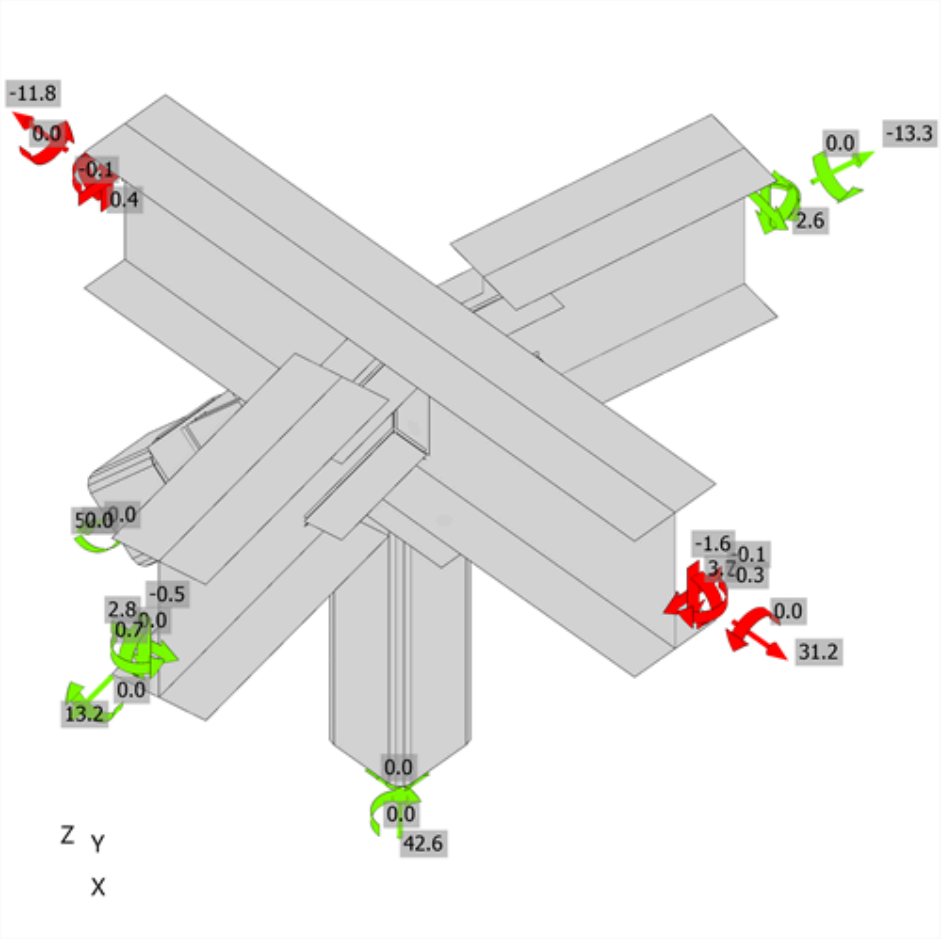
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN47 column below	150	C LC_107	0	5	0.1	OK
SLN434 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

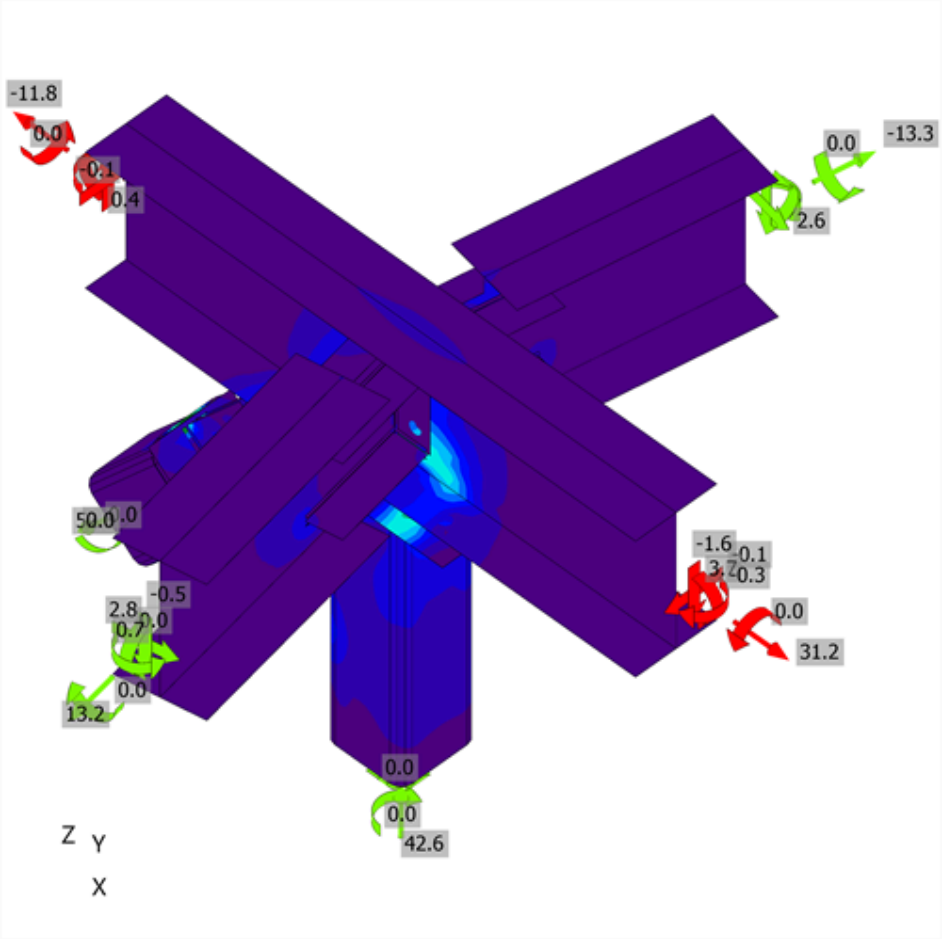
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

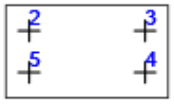
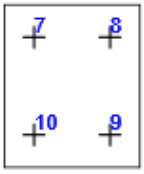
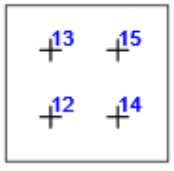


Strain check, C LC_107



Equivalent stress, C LC_107

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	18.6	3.2	109.4	13.2	3.4	12.8	OK	OK
	B3	M20 8.8 - 1	C LC_107	40.4	5.8	100.9	28.6	6.2	26.6	OK	OK
	B4	M20 8.8 - 1	C LC_107	39.5	5.8	98.6	28.0	6.2	26.2	OK	OK
	B5	M20 8.8 - 1	C LC_152	18.2	3.3	113.9	12.9	3.5	12.6	OK	OK
	B7	M20 8.8 - 1	C LC_120	20.8	1.6	123.8	14.7	1.6	12.2	OK	OK
	B8	M20 8.8 - 1	C LC_107	4.0	0.7	123.8	2.8	0.9	2.7	OK	OK
	B9	M20 8.8 - 1	C LC_107	3.9	0.6	123.8	2.8	0.9	2.6	OK	OK
	B10	M20 8.8 - 1	C LC_120	21.3	1.7	123.8	15.1	1.8	12.7	OK	OK
	B12	M20 8.8 - 1	C LC_126	17.5	16.8	115.3	12.4	17.9	26.8	OK	OK
	B13	M20 8.8 - 1	C LC_126	7.9	14.6	115.3	5.6	15.6	19.6	OK	OK
	B14	M20 8.8 - 1	C LC_107	0.8	7.7	106.0	0.6	14.5	8.6	OK	OK
	B15	M20 8.8 - 1	C LC_120	0.1	6.4	115.3	0.1	10.9	6.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN434 raking strut	SLN434 raking strut	◢ 4.2	519	C LC_126	202.0	0.0	-87.3	104.1	-14.6	49.9	28.5	OK	OK
SLN47 column below-w 3	Gusset SLN434 raking strut	◢ 5.6 ◣	288	C LC_107	397.2	0.4	137.0	137.0	166.0	98.1	61.7	OK	OK
		◢ 5.6 ◣	288	C LC_107	397.2	0.4	136.9	-136.8	-166.2	98.1	61.7	OK	OK
SLN213 ring-w 1	Flange top b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_153	213.1	0.0	111.4	91.8	50.9	52.7	31.3	OK	OK
		◢ 4.2 ◣	197	C LC_146	78.5	0.0	-40.1	37.2	11.4	19.4	12.5	OK	OK
SLN213 ring-w 1	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_147	144.0	0.0	41.9	59.1	53.3	35.6	21.4	OK	OK
		◢ 4.2 ◣	198	C LC_147	268.9	0.0	124.2	-107.0	-86.6	66.4	40.4	OK	OK
SLN214 ring-w 1	Flange top a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_153	219.2	0.0	113.9	95.6	-50.6	54.2	26.8	OK	OK
		◢ 4.2 ◣	197	C LC_146	84.2	0.0	-43.8	40.3	-9.8	20.8	12.9	OK	OK
SLN214 ring-w 1	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_148	145.0	0.0	43.1	58.2	-54.8	35.8	19.1	OK	OK
		◢ 4.2 ◣	198	C LC_146	273.4	0.0	125.0	-109.9	87.4	67.6	35.6	OK	OK
Cap plate SLN434 raking strut	Fin plate SLN434 raking strut	◢ 5.6 ◣	139	C LC_125	279.6	0.0	-125.4	-112.8	90.0	69.1	53.6	OK	OK
		◢ 5.6 ◣	139	C LC_125	171.6	0.0	-72.0	84.5	-30.7	42.4	24.4	OK	OK
Cap plate SLN434 raking strut	Fin plate 2 SLN434 raking strut	◢ 5.6 ◣	139	C LC_126	169.0	0.0	-68.5	-83.8	30.4	41.8	24.0	OK	OK
		◢ 5.6 ◣	139	C LC_126	283.2	0.0	-129.0	113.8	-90.8	70.0	54.1	OK	OK
End plate column below SLN47 column below	SLN47 column below	◢ 5.6 ◣	521	C LC_120	298.8	0.0	-191.2	-129.6	-27.9	73.8	40.4	OK	OK
		◢ 5.6 ◣	521	C LC_107	232.8	0.0	159.1	-77.4	-60.3	57.5	36.3	OK	OK
SLN212+SLN200 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_122	97.3	0.0	-5.8	6.4	55.7	24.0	15.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN212+SLN200 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_121	97.2	0.0	5.7	11.6	54.8	24.0	15.4	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_122	93.8	0.0	-30.2	-9.5	-50.4	23.2	13.9	OK	OK
		◢ 4.2 ◣	149	C LC_146	85.2	0.0	-7.3	37.6	31.4	21.1	14.7	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_147	87.9	0.0	-35.4	-19.2	42.3	21.7	15.3	OK	OK
		◢ 4.2 ◣	149	C LC_146	88.4	0.0	-21.7	36.8	-33.1	21.8	16.0	OK	OK
SLN213 ring-w 1	Flange top a end plate SLN213 ring	◢ 4.2 ◣	199	C LC_146	287.8	0.0	-21.0	9.5	-165.4	71.1	48.5	OK	OK
		◢ 4.2 ◣	198	C LC_153	216.1	0.0	30.5	35.8	-118.2	53.4	30.3	OK	OK
SLN213 ring-w 1	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	198	C LC_146	233.9	0.0	41.9	-44.8	125.1	57.8	33.1	OK	OK
		◢ 4.2 ◣	198	C LC_146	242.4	0.0	-31.9	-12.0	138.2	59.9	36.5	OK	OK
SLN214 ring-w 1	Flange top b end plate SLN214 ring	◢ 4.2 ◣	199	C LC_146	302.0	0.0	-25.1	12.4	173.3	74.6	46.1	OK	OK
		◢ 4.2 ◣	198	C LC_146	227.3	0.0	34.0	39.5	123.6	56.2	31.1	OK	OK
SLN214 ring-w 1	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	198	C LC_146	243.9	0.0	44.7	-46.6	-130.4	60.3	33.4	OK	OK
		◢ 4.2 ◣	198	C LC_146	254.3	0.0	-38.0	-14.5	-144.4	62.8	35.7	OK	OK
End plate SLN213 ring	Flange top a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	74.1	0.0	-0.7	3.7	-42.6	18.3	11.7	OK	OK
		◢ 4.2 ◣	58	C LC_121	40.8	0.0	0.3	-18.4	-14.7	10.1	9.0	OK	OK
End plate SLN213 ring	Flange top b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_146	116.9	0.0	-2.6	-0.8	-67.5	28.9	19.6	OK	OK
		◢ 4.2 ◣	58	C LC_120	120.4	0.0	63.8	44.7	-38.5	29.8	24.3	OK	OK
End plate SLN213 ring	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_148	63.2	0.0	17.9	-10.7	33.3	15.6	10.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_146	66.2	0.0	-5.4	-9.1	37.0	16.4	11.7	OK	OK
End plate SLN213 ring	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	129.0	0.0	68.9	-51.1	36.8	31.9	24.6	OK	OK
		◢ 4.2 ◣	58	C LC_146	125.9	0.0	-5.6	-0.1	72.6	31.1	20.6	OK	OK
End plate SLN214 ring	Flange top a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_146	106.5	0.0	0.4	-4.1	61.3	26.3	18.2	OK	OK
		◢ 4.2 ◣	58	C LC_120	95.6	0.0	12.4	19.8	51.0	23.6	18.6	OK	OK
End plate SLN214 ring	Flange top b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_114	125.7	0.0	-64.0	49.3	-38.3	31.0	19.3	OK	OK
		◢ 4.2 ◣	58	C LC_114	100.6	0.0	-28.9	-43.7	34.5	24.9	15.2	OK	OK
End plate SLN214 ring	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_120	105.4	0.0	53.4	-43.2	-29.8	26.0	19.4	OK	OK
		◢ 4.2 ◣	58	C LC_146	114.4	0.0	-3.8	2.2	-66.0	28.3	19.5	OK	OK
End plate SLN214 ring	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_114	87.9	0.0	-25.0	36.9	-31.7	21.7	13.4	OK	OK
		◢ 4.2 ◣	58	C LC_114	122.5	0.0	-61.4	-49.5	35.9	30.3	18.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-220
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut

Name	SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

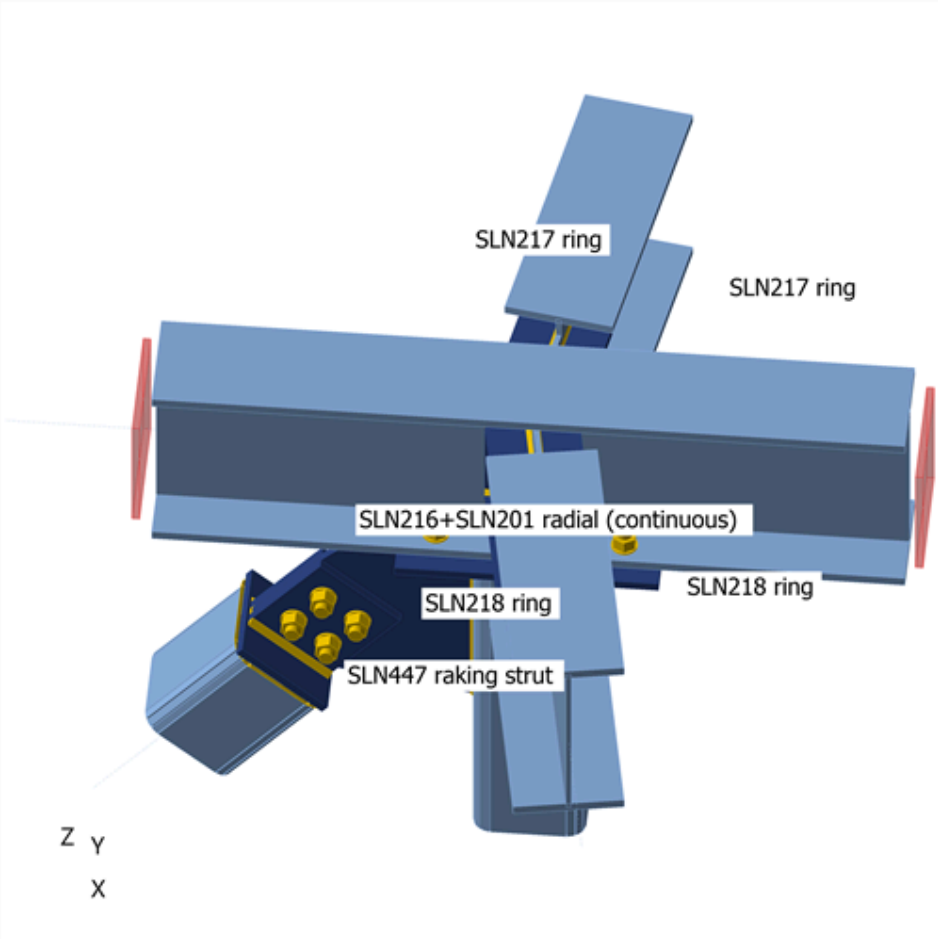
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN216+SLN201 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN49 column below	2 - SHS150x150x10	0.0	90.0	40.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN218 ring	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN447 raking strut	2 - SHS150x150x10	-139.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN216+SLN201 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN216+SLN201 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN49 column below / begin		Position	0
SLN217 ring / begin		Position	0
SLN218 ring / end		Position	0
SLN447 raking strut / end		Position	0



SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN49 column below / Begin	122.1	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-30.4	4.6	-25.6	0.0	0.0	1.1
	SLN218 ring / End	30.2	4.6	-26.1	0.0	0.0	-1.1
	SLN447 raking strut / End	-25.3	0.0	-1.0	-0.1	0.0	0.0
C LC_107	SLN49 column below / Begin	30.7	-4.8	-6.4	0.0	-4.3	1.3
	SLN217 ring / Begin	-11.0	2.0	-0.5	0.0	0.0	-0.1
	SLN218 ring / End	6.0	1.5	-0.5	0.0	0.0	0.7
	SLN447 raking strut / End	36.2	0.0	-1.1	0.3	0.0	-1.4
C LC_110	SLN49 column below / Begin	23.7	5.7	-5.4	0.0	-3.5	-1.6
	SLN217 ring / Begin	-4.4	1.2	-0.5	0.0	0.0	-0.7
	SLN218 ring / End	10.2	1.8	-0.5	0.0	0.0	-0.1
	SLN447 raking strut / End	27.2	0.0	-1.1	-0.4	0.0	1.7
C LC_115	SLN49 column below / Begin	37.5	-4.8	-6.4	0.0	-4.2	1.3
	SLN217 ring / Begin	-13.5	2.5	-1.7	0.0	0.0	0.0
	SLN218 ring / End	8.5	1.9	-1.8	0.0	0.0	0.6
	SLN447 raking strut / End	35.4	0.0	-1.4	0.3	0.0	-1.4
C LC_119	SLN49 column below / Begin	134.8	-2.4	-2.9	0.0	-1.1	0.7
	SLN217 ring / Begin	-33.9	5.3	-25.6	0.0	0.0	1.1
	SLN218 ring / End	31.1	5.1	-26.1	0.0	0.0	-0.7
	SLN447 raking strut / End	-8.3	0.0	-1.2	0.1	0.0	-0.7
C LC_120	SLN49 column below / Begin	112.0	0.0	0.3	0.0	1.1	0.1
	SLN217 ring / Begin	-21.7	2.9	-25.6	0.0	0.0	1.2
	SLN218 ring / End	21.8	3.1	-26.1	0.0	0.0	-1.1
	SLN447 raking strut / End	-38.3	-0.2	-0.8	-0.1	0.0	-0.1
C LC_121	SLN49 column below / Begin	111.9	0.0	0.3	0.0	1.1	-0.1
	SLN217 ring / Begin	-22.0	3.1	-25.6	0.0	0.0	1.1
	SLN218 ring / End	21.4	2.9	-26.1	0.0	0.0	-1.2
	SLN447 raking strut / End	-38.5	0.3	-0.9	0.0	0.0	0.2
C LC_122	SLN49 column below / Begin	131.6	2.9	-2.4	0.0	-0.7	-0.8
	SLN217 ring / Begin	-30.3	4.8	-25.6	0.0	0.0	0.8
	SLN218 ring / End	33.3	5.2	-26.1	0.0	0.0	-1.1
	SLN447 raking strut / End	-12.3	0.0	-1.2	-0.3	0.0	0.9
C LC_123	SLN49 column below / Begin	116.5	-4.8	-6.2	0.0	-3.5	1.3
	SLN217 ring / Begin	-33.8	5.4	-19.4	0.0	0.0	0.9
	SLN218 ring / End	28.6	4.9	-19.7	0.0	0.0	-0.2
	SLN447 raking strut / End	12.3	0.0	-1.4	0.3	0.0	-1.4
C LC_145	SLN49 column below / Begin	123.9	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-31.4	4.9	-25.6	0.0	0.0	1.0
	SLN218 ring / End	31.0	4.8	-26.1	0.0	0.0	-1.0
	SLN447 raking strut / End	-22.9	0.0	-1.0	-0.1	0.0	0.0
C LC_146	SLN49 column below / Begin	120.3	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-29.5	4.4	-25.6	0.0	0.0	1.1
	SLN218 ring / End	29.4	4.4	-26.1	0.0	0.0	-1.2
	SLN447 raking strut / End	-27.7	0.0	-1.0	-0.1	0.0	0.0
C LC_147	SLN49 column below / Begin	120.3	0.0	0.3	0.0	1.2	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / Begin	-29.8	4.4	-25.6	0.0	0.0	1.1
	SLN218 ring / End	29.4	4.4	-26.1	0.0	0.0	-1.2
	SLN447 raking strut / End	-27.7	0.0	-1.0	-0.1	0.0	0.0
C LC_151	SLN49 column below / Begin	76.1	0.0	-1.6	0.0	-6.7	0.0
	SLN217 ring / Begin	-15.1	2.4	-11.8	0.0	0.0	0.5
	SLN218 ring / End	14.9	2.2	-11.6	0.0	0.0	-0.7
	SLN447 raking strut / End	-17.2	0.0	-1.0	0.0	0.0	0.1
C LC_152	SLN49 column below / Begin	71.0	0.0	2.1	0.0	8.6	0.0
	SLN217 ring / Begin	-30.4	4.8	-14.1	0.0	0.0	0.9
	SLN218 ring / End	30.3	4.7	-14.0	0.0	0.0	-1.0
	SLN447 raking strut / End	-10.9	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN49 column below / Begin	121.3	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-30.5	4.6	-21.0	0.0	0.0	1.1
	SLN218 ring / End	30.3	4.6	-20.6	0.0	0.0	-1.1
	SLN447 raking strut / End	-25.0	0.0	-1.0	-0.1	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	21.8 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN216+SLN201 radial (continuous)-bfl 1	12.7	C LC_107	167.8	0.0	26.7	OK
SLN216+SLN201 radial (continuous)-tfl 1	12.7	C LC_153	72.2	0.0	0.0	OK
SLN216+SLN201 radial (continuous)-w 1	7.2	C LC_123	209.9	0.0	32.8	OK
SLN49 column below	10.0	C LC_121	275.1	0.1	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_119	141.0	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_119	134.3	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_119	275.1	0.1	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_122	140.8	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_122	134.7	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_122	275.1	0.1	0.0	OK
SLN447 raking strut	10.0	C LC_115	53.1	0.0	0.0	OK
End plate column below SLN49 column below	12.0	C LC_107	200.9	0.0	26.7	OK
End plate SLN217 ring	12.0	C LC_120	134.6	0.0	0.0	OK
End plate SLN218 ring	12.0	C LC_146	110.9	0.0	39.1	OK
Cap plate SLN447 raking strut	12.0	C LC_110	110.4	0.0	0.0	OK
Fin plate SLN447 raking strut	12.0	C LC_110	275.1	0.0	101.7	OK
Fin plate 2 SLN447 raking strut	12.0	C LC_115	270.1	0.0	85.1	OK
Gusset SLN447 raking strut	12.0	C LC_110	275.1	0.0	50.8	OK
Flange top a end plate SLN217 ring	12.0	C LC_119	149.1	0.0	0.0	OK
Flange top b end plate SLN217 ring	12.0	C LC_119	139.2	0.0	0.0	OK
Flange bottom a end plate SLN217 ring	12.0	C LC_119	128.7	0.0	0.0	OK
Flange bottom b end plate SLN217 ring	12.0	C LC_119	154.7	0.0	0.0	OK
Flange top a end plate SLN218 ring	12.0	C LC_122	130.7	0.0	0.0	OK
Flange top b end plate SLN218 ring	12.0	C LC_146	144.6	0.0	0.0	OK
Flange bottom a end plate SLN218 ring	12.0	C LC_122	154.5	0.0	0.0	OK
Flange bottom b end plate SLN218 ring	12.0	C LC_146	131.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

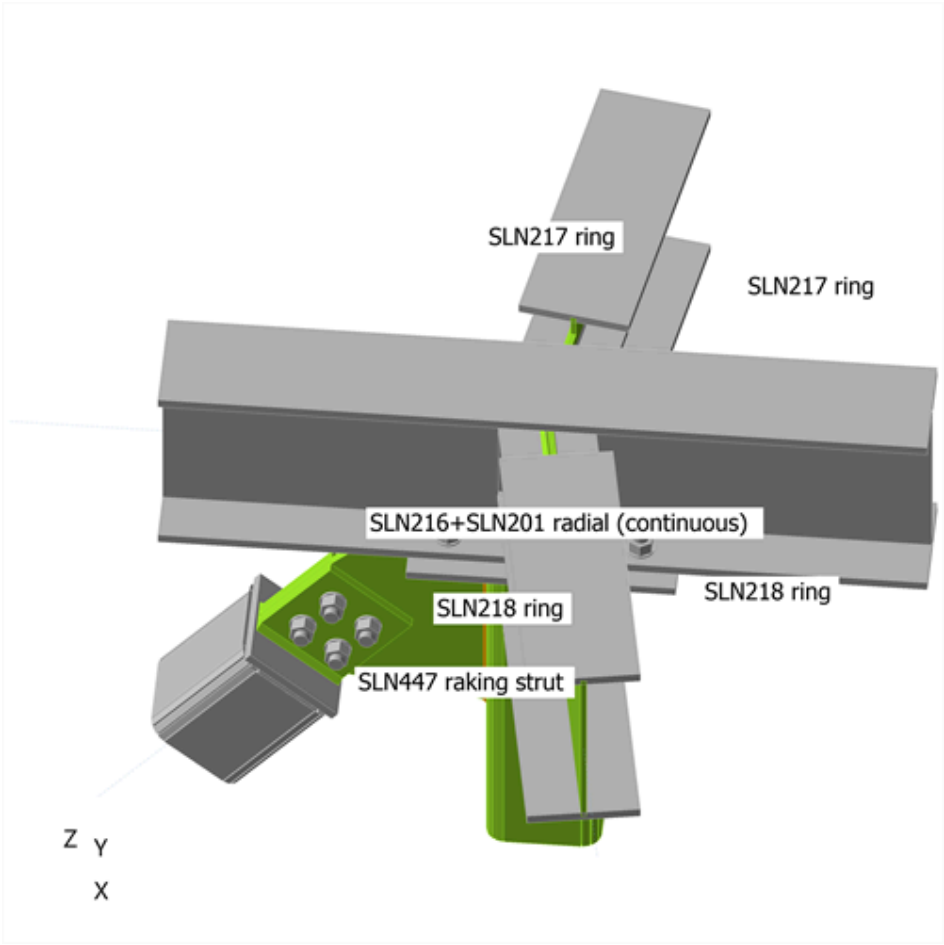
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

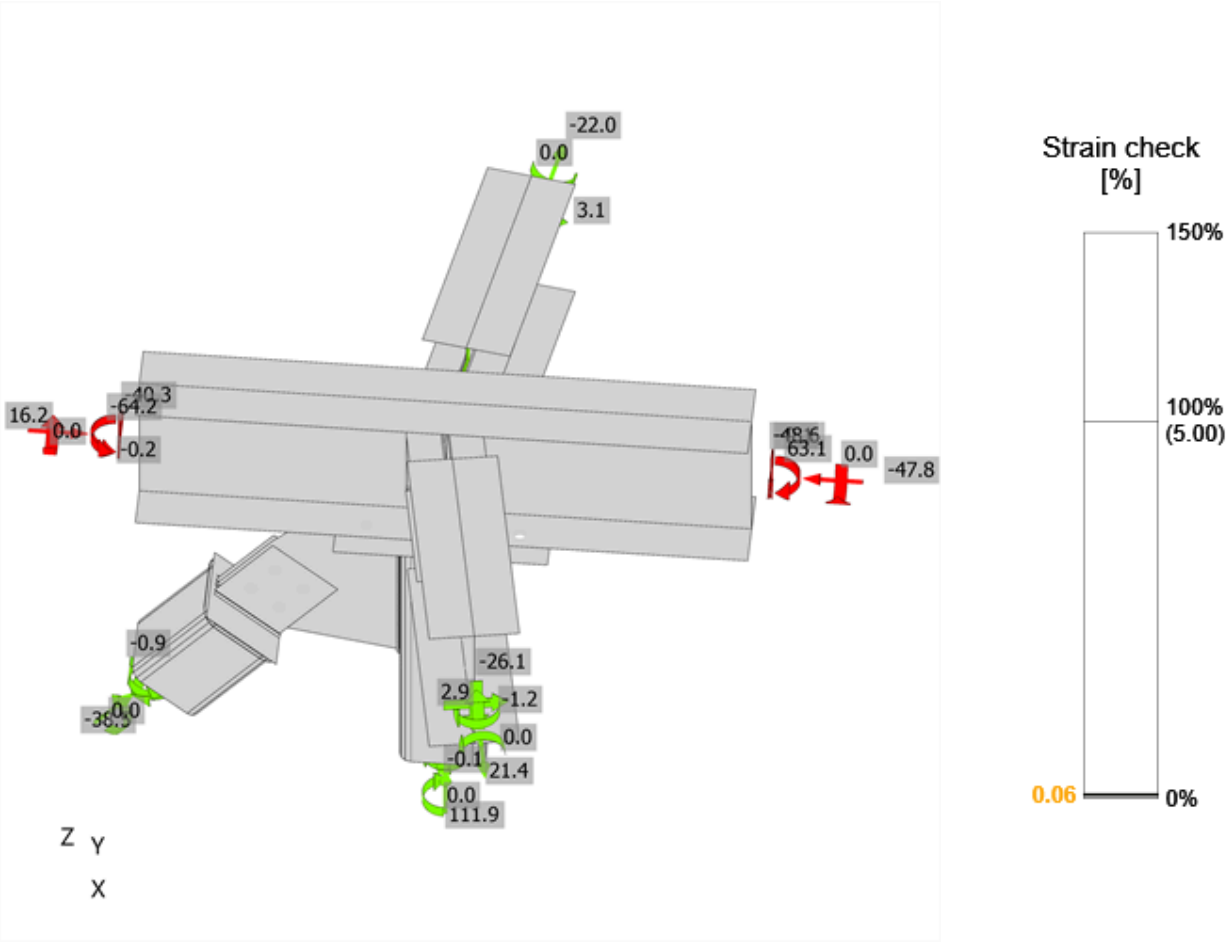
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN49 column below	150	C LC_107	0	5	0.1	OK
SLN447 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

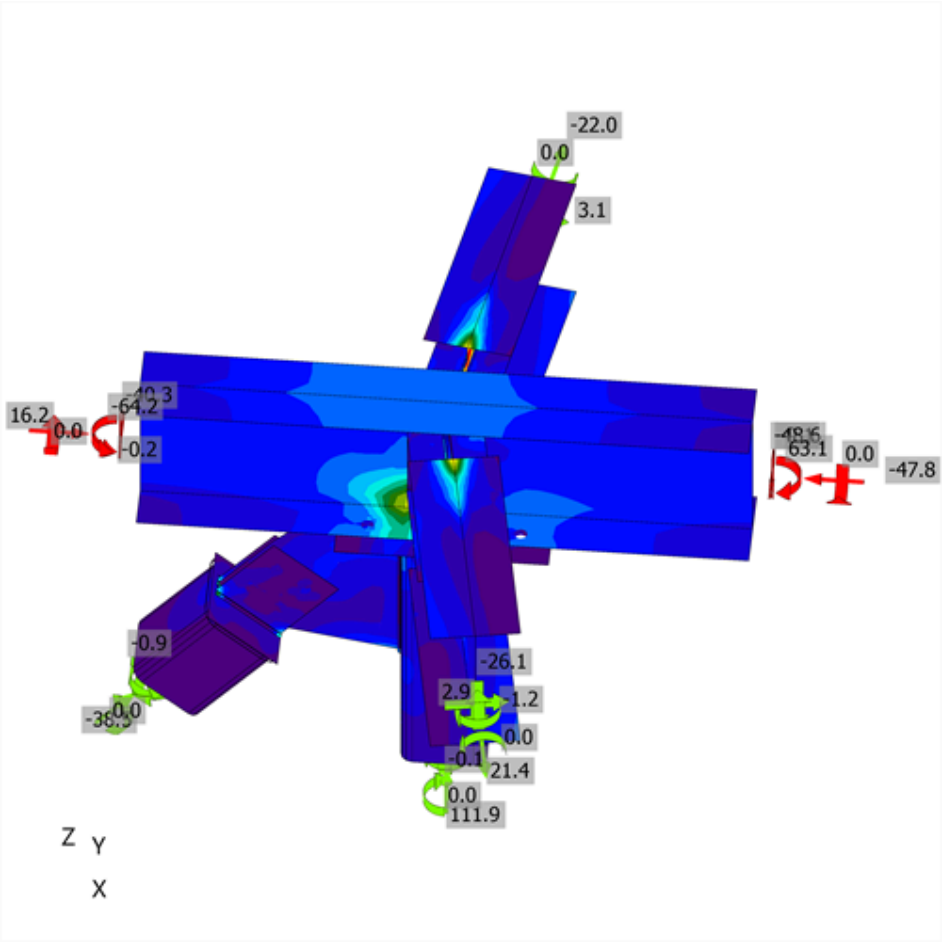
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

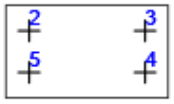
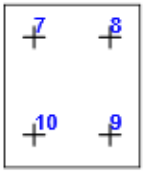
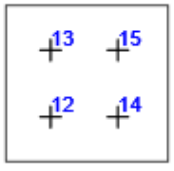


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	15.8	2.9	108.7	11.2	3.1	11.1	OK	OK
	B3	M20 8.8 - 1	C LC_107	28.7	3.7	83.2	20.3	4.4	18.4	OK	OK
	B4	M20 8.8 - 1	C LC_107	28.2	4.3	124.2	20.0	4.5	18.8	OK	OK
	B5	M20 8.8 - 1	C LC_152	16.0	3.0	119.1	11.3	3.2	11.2	OK	OK
	B7	M20 8.8 - 1	C LC_120	18.6	1.2	123.8	13.2	1.3	10.7	OK	OK
	B8	M20 8.8 - 1	C LC_110	7.9	1.0	62.9	5.6	1.0	5.1	OK	OK
	B9	M20 8.8 - 1	C LC_110	7.9	1.0	108.0	5.6	1.1	5.1	OK	OK
	B10	M20 8.8 - 1	C LC_120	19.5	1.4	123.8	13.8	1.5	11.3	OK	OK
	B12	M20 8.8 - 1	C LC_110	14.3	13.7	106.0	10.1	14.6	21.8	OK	OK
	B13	M20 8.8 - 1	C LC_110	6.7	9.4	106.0	4.7	10.0	13.4	OK	OK
	B14	M20 8.8 - 1	C LC_115	2.3	10.1	106.0	1.6	11.2	11.9	OK	OK
	B15	M20 8.8 - 1	C LC_121	0.1	5.5	115.3	0.1	9.5	6.0	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN447 raking strut	SLN447 raking strut	◢ 4.2	519	C LC_115	180.5	0.0	82.5	-92.6	3.4	44.6	25.0	OK	OK
SLN49 column below-w 3	Gusset SLN447 raking strut	◢ 5.6 ◣	288	C LC_120	396.9	0.2	-121.8	-122.3	-180.5	98.1	58.9	OK	OK
		◢ 5.6 ◣	288	C LC_121	396.9	0.2	-124.3	125.4	177.9	98.1	60.3	OK	OK
SLN217 ring-w 1	Flange top b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_119	204.7	0.0	108.4	88.6	46.9	50.6	30.1	OK	OK
		◢ 4.2 ◣	197	C LC_119	76.4	0.0	-38.9	36.2	11.4	18.9	12.4	OK	OK
SLN217 ring-w 1	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_119	146.3	0.0	42.4	59.5	54.7	36.1	21.7	OK	OK
		◢ 4.2 ◣	198	C LC_119	267.6	0.0	123.1	-105.9	-87.1	66.1	40.5	OK	OK
SLN218 ring-w 1	Flange top a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_153	190.4	0.0	97.9	82.2	-46.2	47.0	23.7	OK	OK
		◢ 4.2 ◣	197	C LC_146	79.8	0.0	-42.1	38.8	-5.2	19.7	11.6	OK	OK
SLN218 ring-w 1	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_122	149.0	0.0	43.8	59.6	-56.7	36.8	19.4	OK	OK
		◢ 4.2 ◣	198	C LC_122	271.1	0.0	123.9	-108.3	87.5	67.0	35.8	OK	OK
Cap plate SLN447 raking strut	Fin plate SLN447 raking strut	◢ 5.6 ◣	139	C LC_110	284.2	0.0	99.0	123.7	91.5	70.2	49.6	OK	OK
		◢ 5.6 ◣	139	C LC_115	147.3	0.0	87.9	-56.5	-38.3	36.4	23.9	OK	OK
Cap plate SLN447 raking strut	Fin plate 2 SLN447 raking strut	◢ 5.6 ◣	139	C LC_110	129.1	0.0	84.4	46.7	31.5	31.9	21.1	OK	OK
		◢ 5.6 ◣	139	C LC_115	281.5	0.0	102.9	-122.7	-88.4	69.5	47.6	OK	OK
End plate column below SLN49 column below	SLN49 column below	◢ 5.6 ◣	521	C LC_121	278.4	0.0	-176.9	-121.9	23.4	68.8	37.7	OK	OK
		◢ 5.6 ◣	521	C LC_152	210.2	0.0	-93.1	106.4	-22.7	51.9	30.6	OK	OK
SLN216+SLN201 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	95.0	0.0	5.5	5.2	-54.5	23.5	14.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN216+SLN201 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_122	89.8	0.0	36.6	-29.4	37.1	22.2	13.4	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	93.9	0.0	-31.0	-15.9	-48.7	23.2	13.9	OK	OK
		◢ 4.2 ◣	149	C LC_146	85.6	0.0	-10.9	36.6	32.6	21.2	14.5	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_122	83.6	0.0	-37.5	-24.2	35.7	20.6	14.0	OK	OK
		◢ 4.2 ◣	149	C LC_147	81.2	0.0	-23.4	33.2	-30.2	20.1	14.5	OK	OK
SLN217 ring-w 1	Flange top a end plate SLN217 ring	◢ 4.2 ◣	199	C LC_119	279.4	0.0	-20.0	8.8	-160.6	69.0	47.1	OK	OK
		◢ 4.2 ◣	198	C LC_119	211.7	0.0	30.2	35.2	-115.8	52.3	29.3	OK	OK
SLN217 ring-w 1	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	198	C LC_119	229.3	0.0	41.6	-43.9	122.6	56.7	32.4	OK	OK
		◢ 4.2 ◣	198	C LC_119	232.9	0.0	-31.6	-11.5	132.7	57.6	35.1	OK	OK
SLN218 ring-w 1	Flange top b end plate SLN218 ring	◢ 4.2 ◣	199	C LC_146	257.0	0.0	-36.4	20.5	145.5	63.5	39.7	OK	OK
		◢ 4.2 ◣	198	C LC_122	203.1	0.0	39.0	38.9	108.3	50.2	27.9	OK	OK
SLN218 ring-w 1	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	198	C LC_122	236.9	0.0	38.3	-41.4	-128.5	58.5	32.7	OK	OK
		◢ 4.2 ◣	199	C LC_146	241.5	0.0	-24.4	-8.1	-138.5	59.7	34.7	OK	OK
End plate SLN217 ring	Flange top a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_119	68.3	0.0	-2.6	6.0	-39.0	16.9	11.3	OK	OK
		◢ 4.2 ◣	58	C LC_110	47.0	0.0	27.4	15.4	15.8	11.6	9.9	OK	OK
End plate SLN217 ring	Flange top b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_119	112.3	0.0	-1.1	-1.2	-64.8	27.7	18.8	OK	OK
		◢ 4.2 ◣	58	C LC_120	108.5	0.0	56.6	39.7	-35.7	26.8	21.8	OK	OK
End plate SLN217 ring	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_122	66.8	0.0	18.6	-16.8	33.0	16.5	11.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_120	64.6	0.0	-18.2	-13.0	33.4	16.0	11.4	OK	OK
End plate SLN217 ring	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_120	118.7	0.0	62.8	-47.1	34.1	29.3	22.6	OK	OK
		◢ 4.2 ◣	58	C LC_119	120.6	0.0	-4.2	-0.1	69.6	29.8	19.9	OK	OK
End plate SLN218 ring	Flange top a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_146	94.1	0.0	0.8	-3.9	54.2	23.2	16.1	OK	OK
		◢ 4.2 ◣	58	C LC_120	86.3	0.0	10.5	16.9	46.5	21.3	16.2	OK	OK
End plate SLN218 ring	Flange top b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_110	69.3	0.0	-34.5	24.1	-25.0	17.1	11.2	OK	OK
		◢ 4.2 ◣	58	C LC_120	46.6	0.0	-6.2	-18.1	19.6	11.5	8.1	OK	OK
End plate SLN218 ring	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_120	90.6	0.0	52.1	-33.6	-26.5	22.4	17.3	OK	OK
		◢ 4.2 ◣	58	C LC_146	104.1	0.0	-3.3	2.6	-60.0	25.7	17.7	OK	OK
End plate SLN218 ring	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_145	69.0	0.0	17.2	-11.9	-36.7	17.0	10.0	OK	OK
		◢ 4.2 ◣	58	C LC_110	71.0	0.0	-38.0	-23.9	25.0	17.5	11.6	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-300
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut

Name	SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

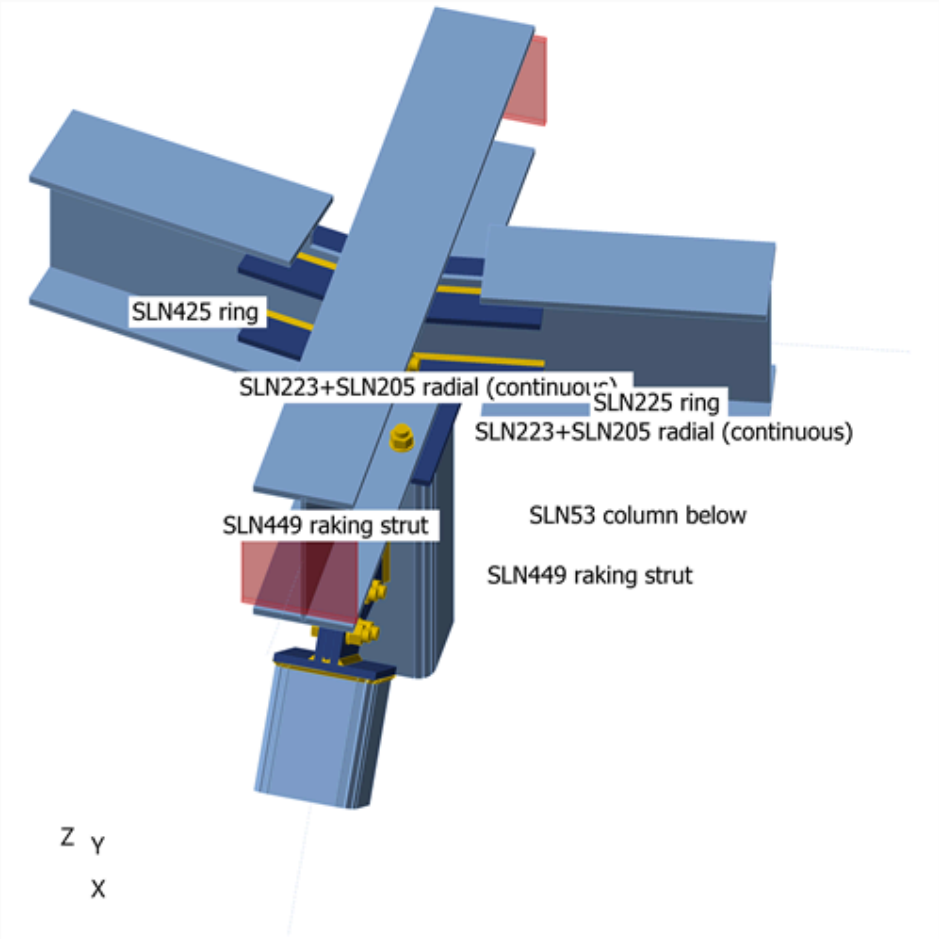
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN223+SLN205 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN53 column below	2 - SHS150x150x10	0.0	90.0	120.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN225 ring	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN449 raking strut	2 - SHS150x150x10	-60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN223+SLN205 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN223+SLN205 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN53 column below / begin		Position	0
SLN425 ring / begin		Position	0
SLN225 ring / end		Position	0
SLN449 raking strut / end		Position	0



SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN53 column below / Begin	180.3	-0.2	0.8	0.0	3.2	1.0
	SLN425 ring / Begin	-29.5	5.3	-25.7	0.0	0.0	-0.9
	SLN225 ring / End	9.7	-0.3	-25.6	0.0	0.0	-3.6
	SLN449 raking strut / End	11.0	-0.4	-1.0	-0.4	0.0	-1.5
C LC_106	SLN53 column below / Begin	188.1	-2.1	-2.9	0.0	0.4	1.5
	SLN425 ring / Begin	-30.9	5.9	-24.1	0.0	0.0	-1.2
	SLN225 ring / End	9.3	0.0	-24.0	0.0	0.0	-3.1
	SLN449 raking strut / End	35.3	-0.4	-1.2	-0.3	0.0	-2.1
C LC_110	SLN53 column below / Begin	47.1	-3.8	-7.2	0.0	-4.7	1.3
	SLN425 ring / Begin	-13.5	2.8	-0.5	0.0	0.0	-0.7
	SLN225 ring / End	5.8	1.0	-0.5	0.0	0.0	-0.1
	SLN449 raking strut / End	53.7	0.0	-1.2	0.2	0.0	-1.4
C LC_114	SLN53 column below / Begin	154.4	-3.9	-6.8	0.0	-3.0	1.9
	SLN425 ring / Begin	-28.7	5.6	-16.3	0.0	0.0	-1.3
	SLN225 ring / End	9.7	0.7	-16.2	0.0	0.0	-2.0
	SLN449 raking strut / End	58.2	-0.3	-1.5	-0.1	0.0	-2.2
C LC_118	SLN53 column below / Begin	58.4	-3.8	-7.1	0.0	-4.3	1.3
	SLN425 ring / Begin	-16.0	3.3	-1.8	0.0	0.0	-0.8
	SLN225 ring / End	6.6	1.0	-1.7	0.0	0.0	-0.4
	SLN449 raking strut / End	55.7	-0.1	-1.5	0.1	0.0	-1.5
C LC_120	SLN53 column below / Begin	188.6	3.0	-1.3	0.0	1.8	0.1
	SLN425 ring / Begin	-26.8	4.9	-25.7	0.0	0.0	-1.0
	SLN225 ring / End	10.3	0.0	-25.6	0.0	0.0	-3.5
	SLN449 raking strut / End	19.5	-0.4	-1.1	-0.6	0.0	-0.5
C LC_122	SLN53 column below / Begin	193.9	-2.1	-2.9	0.0	0.5	1.6
	SLN425 ring / Begin	-34.8	6.4	-25.7	0.0	0.0	-1.1
	SLN225 ring / End	12.4	0.4	-25.6	0.0	0.0	-3.4
	SLN449 raking strut / End	30.8	-0.4	-1.2	-0.3	0.0	-2.1
C LC_126	SLN53 column below / Begin	166.0	-4.0	-6.7	0.0	-2.8	2.0
	SLN425 ring / Begin	-36.5	6.8	-19.5	0.0	0.0	-1.0
	SLN225 ring / End	15.7	1.4	-19.4	0.0	0.0	-2.7
	SLN449 raking strut / End	49.3	-0.3	-1.5	-0.1	0.0	-2.4
C LC_130	SLN53 column below / Begin	70.0	-3.8	-7.1	0.0	-4.2	1.5
	SLN425 ring / Begin	-23.8	4.4	-4.9	0.0	0.0	-0.6
	SLN225 ring / End	12.7	1.7	-4.9	0.0	0.0	-1.0
	SLN449 raking strut / End	46.8	-0.1	-1.5	0.1	0.0	-1.7
C LC_135	SLN53 column below / Begin	186.1	-0.2	0.9	0.0	3.7	1.0
	SLN425 ring / Begin	-29.8	5.4	-25.7	0.0	0.0	-1.0
	SLN225 ring / End	7.7	-0.7	-25.6	0.0	0.0	-3.8
	SLN449 raking strut / End	13.4	-0.4	-1.0	-0.4	0.0	-1.6
C LC_147	SLN53 column below / Begin	178.4	-0.2	0.8	0.0	3.2	1.0
	SLN425 ring / Begin	-28.5	5.0	-25.7	0.0	0.0	-0.8
	SLN225 ring / End	8.9	-0.5	-25.6	0.0	0.0	-3.7
	SLN449 raking strut / End	8.4	-0.4	-1.0	-0.4	0.0	-1.5
C LC_148	SLN53 column below / Begin	182.3	-0.2	0.8	0.0	3.3	1.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN425 ring / Begin	-30.6	5.6	-25.7	0.0	0.0	-0.9
	SLN225 ring / End	10.6	0.0	-25.6	0.0	0.0	-3.6
	SLN449 raking strut / End	13.5	-0.4	-1.0	-0.4	0.0	-1.5
C LC_152	SLN53 column below / Begin	130.3	-0.2	2.4	0.0	9.9	0.9
	SLN425 ring / Begin	-29.5	5.2	-14.0	0.0	0.0	-0.6
	SLN225 ring / End	12.0	0.4	-13.9	0.0	0.0	-3.1
	SLN449 raking strut / End	29.8	-0.3	-1.0	-0.4	0.0	-1.2
C LC_153	SLN53 column below / Begin	180.0	-0.2	0.8	0.0	3.1	1.0
	SLN425 ring / Begin	-29.7	5.3	-20.7	0.0	0.0	-0.9
	SLN225 ring / End	9.7	-0.3	-20.6	0.0	0.0	-3.7
	SLN449 raking strut / End	11.4	-0.4	-1.0	-0.4	0.0	-1.5

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.7 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	29.6 < 100%	OK
Welds	98.3 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN223+SLN205 radial (continuous)-bfl 1	12.7	C LC_110	215.2	0.0	34.4	OK
SLN223+SLN205 radial (continuous)-tfl 1	12.7	C LC_135	191.5	0.0	0.0	OK
SLN223+SLN205 radial (continuous)-w 1	7.2	C LC_135	275.5	0.2	39.9	OK
SLN53 column below	10.0	C LC_110	275.3	0.1	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_122	55.3	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_122	42.6	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_126	127.9	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_135	213.0	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_135	211.5	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_135	276.5	0.7	0.0	OK
SLN449 raking strut	10.0	C LC_114	80.6	0.0	0.0	OK
End plate column below SLN53 column below	12.0	C LC_110	246.5	0.0	34.4	OK
End plate SLN425 ring	12.0	C LC_135	76.6	0.0	190.2	OK
End plate SLN225 ring	12.0	C LC_135	152.8	0.0	779.4	OK
Cap plate SLN449 raking strut	12.0	C LC_126	155.3	0.0	0.0	OK
Fin plate SLN449 raking strut	12.0	C LC_126	275.1	0.0	69.7	OK
Fin plate 2 SLN449 raking strut	12.0	C LC_126	275.2	0.1	156.3	OK
Gusset SLN449 raking strut	12.0	C LC_126	275.3	0.1	76.7	OK
Flange top a end plate SLN425 ring	12.0	C LC_135	50.8	0.0	0.0	OK
Flange top b end plate SLN425 ring	12.0	C LC_135	87.8	0.0	0.0	OK
Flange bottom a end plate SLN425 ring	12.0	C LC_135	59.2	0.0	0.0	OK
Flange bottom b end plate SLN425 ring	12.0	C LC_135	97.4	0.0	0.0	OK
Flange top a end plate SLN225 ring	12.0	C LC_135	275.1	0.0	0.0	OK
Flange top b end plate SLN225 ring	12.0	C LC_135	233.8	0.0	0.0	OK
Flange bottom a end plate SLN225 ring	12.0	C LC_135	246.8	0.1	0.0	OK
Flange bottom b end plate SLN225 ring	12.0	C LC_135	252.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

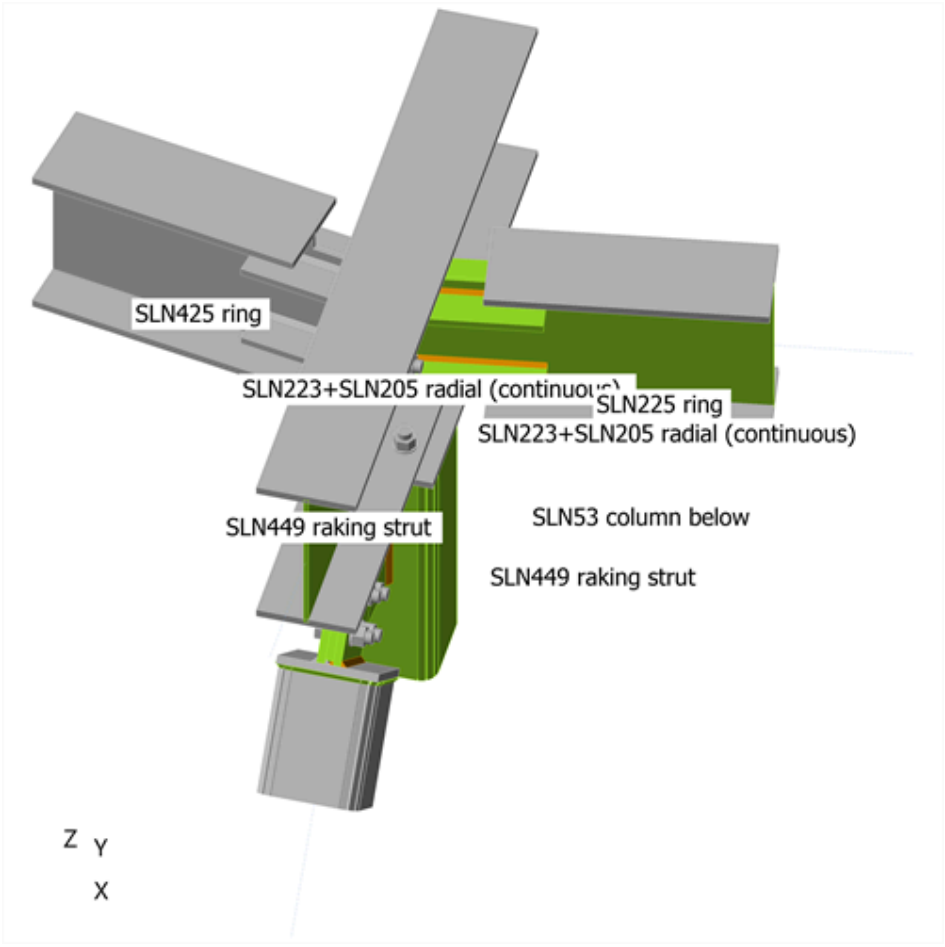
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

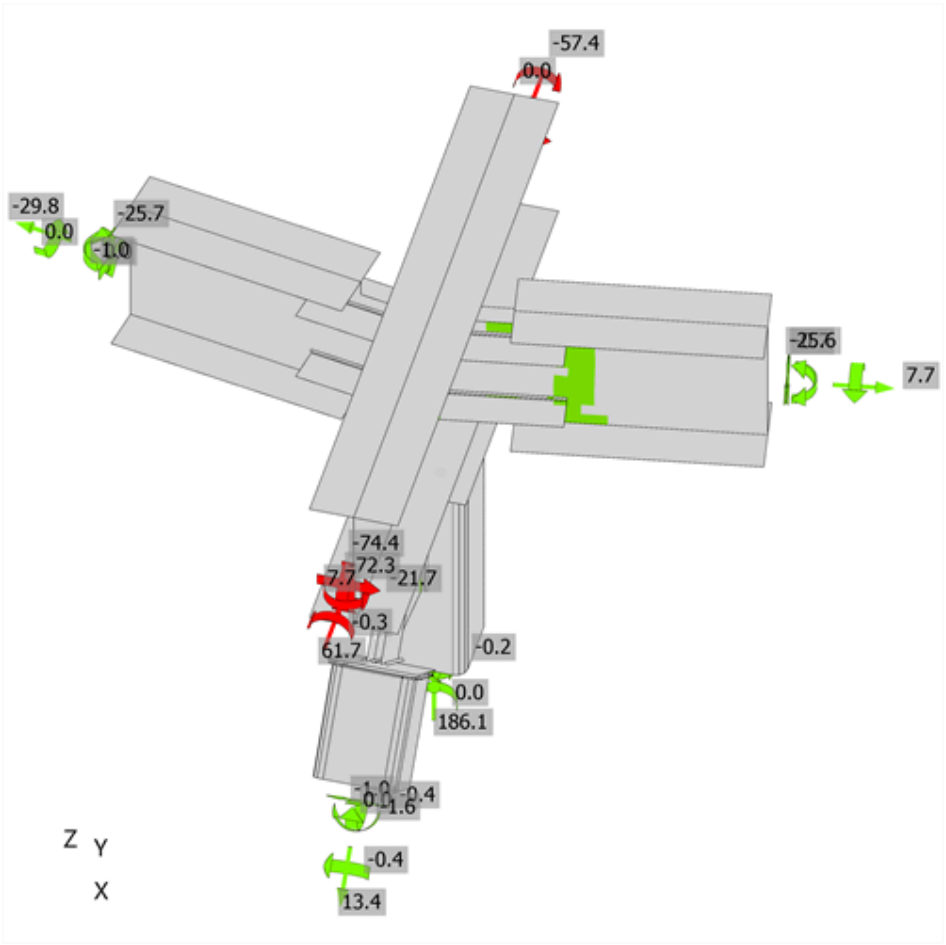
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN53 column below	150	C LC_118	0	5	0.1	OK
SLN449 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

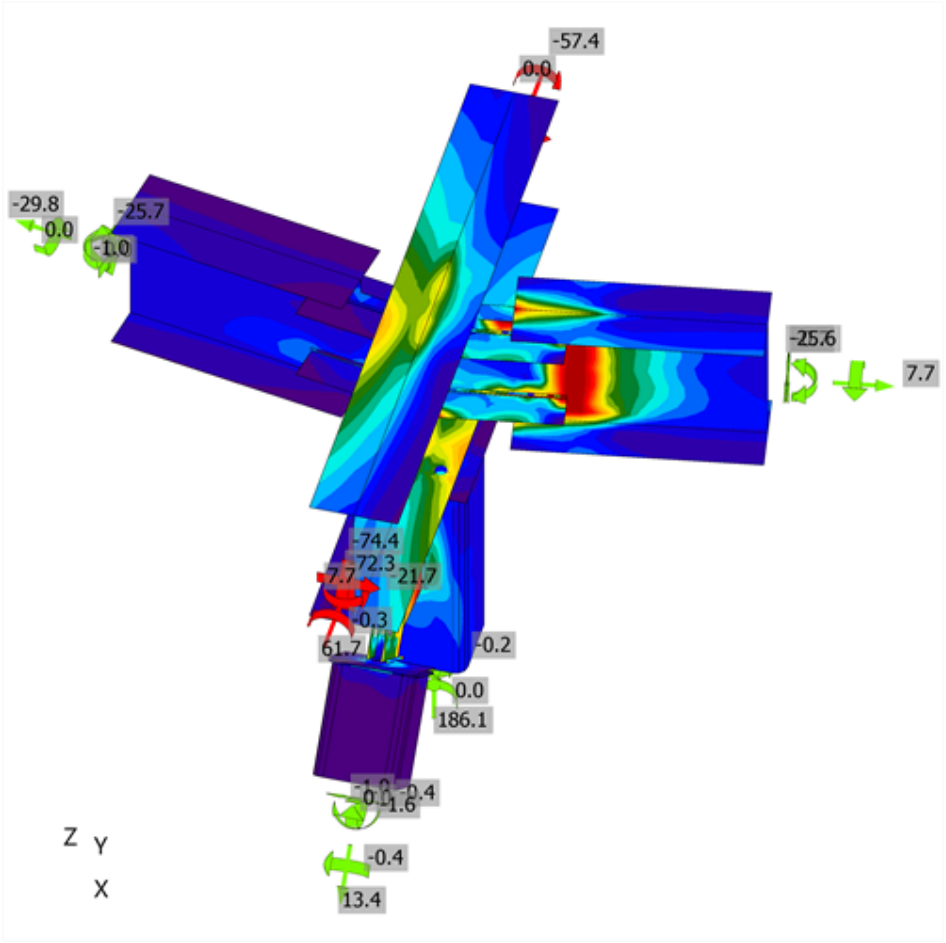
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

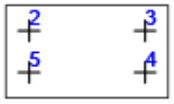
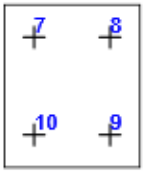
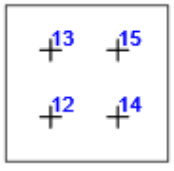


Strain check, C LC_135



Equivalent stress, C LC_135

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_118	0.7	7.4	156.4	0.5	7.9	8.3	OK	OK
	B3	M20 8.8 - 1	C LC_110	34.5	5.8	91.7	24.5	6.3	23.7	OK	OK
	B4	M20 8.8 - 1	C LC_110	36.7	7.3	124.4	26.0	7.8	26.4	OK	OK
	B5	M20 8.8 - 1	C LC_110	0.4	8.1	107.0	0.3	8.6	8.8	OK	OK
	B7	M20 8.8 - 1	C LC_135	10.3	0.4	123.8	7.3	0.6	5.7	OK	OK
	B8	M20 8.8 - 1	C LC_106	5.4	0.4	71.3	3.9	0.6	3.2	OK	OK
	B9	M20 8.8 - 1	C LC_106	7.3	0.3	123.8	5.2	0.5	4.0	OK	OK
	B10	M20 8.8 - 1	C LC_135	12.2	0.5	123.8	8.6	0.9	6.7	OK	OK
	B12	M20 8.8 - 1	C LC_126	17.6	19.5	106.0	12.5	20.7	29.6	OK	OK
	B13	M20 8.8 - 1	C LC_126	10.5	14.8	106.0	7.4	15.7	21.1	OK	OK
	B14	M20 8.8 - 1	C LC_114	3.9	14.6	106.0	2.8	16.7	17.5	OK	OK
	B15	M20 8.8 - 1	C LC_114	3.1	10.6	106.0	2.2	11.8	12.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN449 raking strut	SLN449 raking strut	◢ 4.2	519	C LC_114	271.3	0.0	123.7	-139.3	-5.4	67.0	38.0	OK	OK
SLN53 column below-w 3	Gusset SLN449 raking strut	◢ 5.6 ◣	288	C LC_118	397.2	0.3	153.5	139.3	159.1	98.1	63.7	OK	OK
		◢ 5.6 ◣	288	C LC_118	397.3	0.4	126.8	-139.5	-166.7	98.2	68.2	OK	OK
SLN425 ring-w 1	Flange top b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_153	63.6	0.0	12.5	7.5	-35.2	15.7	9.3	OK	OK
		◢ 4.2 ◣	198	C LC_122	73.1	0.0	8.5	-15.6	38.9	18.1	10.1	OK	OK
SLN425 ring-w 1	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_122	76.3	0.0	25.8	15.7	-38.3	18.8	12.3	OK	OK
		◢ 4.2 ◣	198	C LC_122	129.7	0.0	9.5	-19.6	72.0	32.0	18.6	OK	OK
SLN225 ring-w 1	Flange top a end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	397.0	0.3	212.8	181.4	-67.4	98.1	58.6	OK	OK
		◢ 4.2 ◣	197	C LC_135	200.2	0.0	-78.8	98.5	-39.8	49.5	27.8	OK	OK
SLN225 ring-w 1	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	197	C LC_135	209.9	0.0	45.7	94.8	-70.8	51.9	33.1	OK	OK
		◢ 4.2 ◣	198	C LC_135	397.3	0.4	191.1	-175.4	98.3	98.2	70.0	OK	OK
Cap plate SLN449 raking strut	Fin plate SLN449 raking strut	◢ 5.6 ◣	138	C LC_126	118.8	0.0	-59.4	-50.9	-30.6	29.4	19.9	OK	OK
		◢ 5.6 ◣	139	C LC_114	199.0	0.0	110.6	-72.9	-61.7	49.2	34.9	OK	OK
Cap plate SLN449 raking strut	Fin plate 2 SLN449 raking strut	◢ 5.6 ◣	139	C LC_114	171.8	0.0	78.9	77.0	-42.7	42.4	29.2	OK	OK
		◢ 5.6 ◣	139	C LC_114	396.6	0.0	110.1	-167.4	-142.8	98.0	68.1	OK	OK
End plate column below SLN53 column below	SLN53 column below	◢ 5.6 ◣	521	C LC_126	204.7	0.0	-94.4	-95.5	-43.3	50.6	29.0	OK	OK
		◢ 5.6 ◣	521	C LC_152	253.7	0.0	-152.2	114.7	24.2	62.7	33.5	OK	OK
SLN223+SLN205 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_135	161.1	0.0	51.2	-77.8	-41.5	39.8	24.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN223+SLN205 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_135	142.9	0.0	67.0	-53.4	49.6	35.3	25.4	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_122	117.1	0.0	-27.4	-63.4	-17.2	28.9	18.7	OK	OK
		◢ 4.2 ◣	149	C LC_147	91.0	0.0	-46.9	24.9	37.5	22.5	14.1	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_135	153.2	0.0	-79.5	-16.7	73.7	37.9	25.2	OK	OK
		◢ 4.2 ◣	149	C LC_135	129.3	0.0	-20.2	68.7	-26.7	31.9	19.6	OK	OK
SLN425 ring-w 1	Flange top a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_153	62.8	0.0	-13.1	-8.5	-34.5	15.5	8.7	OK	OK
		◢ 4.2 ◣	199	C LC_135	85.3	0.0	12.7	-6.8	48.2	21.1	11.6	OK	OK
SLN425 ring-w 1	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_135	81.8	0.0	24.6	12.5	-43.3	20.2	12.6	OK	OK
		◢ 4.2 ◣	199	C LC_135	129.2	0.0	-18.7	18.4	71.5	31.9	18.7	OK	OK
SLN225 ring-w 1	Flange top b end plate SLN225 ring	◢ 4.2 ◣	199	C LC_135	398.0	0.8	-186.4	151.4	135.3	98.3	69.7	OK	OK
		◢ 4.2 ◣	198	C LC_135	349.5	0.0	58.6	47.8	193.1	86.4	48.6	OK	OK
SLN225 ring-w 1	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	396.6	0.0	67.2	-76.6	-212.3	98.0	51.1	OK	OK
		◢ 4.2 ◣	199	C LC_135	397.4	0.5	-209.7	-147.0	-128.0	98.2	69.1	OK	OK
End plate SLN425 ring	Flange top a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_147	44.3	0.0	-7.8	20.5	-14.6	11.0	9.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	79.2	0.0	-1.6	-4.4	-45.5	19.6	14.2	OK	OK
End plate SLN425 ring	Flange top b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	232.8	0.0	-105.8	111.1	-44.6	57.5	35.7	OK	OK
		◢ 4.2 ◣	58	C LC_135	217.2	0.0	-108.9	-103.7	31.9	53.7	33.6	OK	OK
End plate SLN425 ring	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	106.4	0.0	21.9	-26.2	54.1	26.3	16.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_147	38.2	0.0	-7.2	-17.1	13.3	9.4	7.8	OK	OK
End plate SLN425 ring	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	192.8	0.0	-96.0	90.7	-33.0	47.6	32.2	OK	OK
		◢ 4.2 ◣	58	C LC_135	235.1	0.0	-106.4	-111.9	46.2	58.1	35.8	OK	OK
End plate SLN225 ring	Flange top a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	101.3	0.0	5.8	3.8	58.3	25.0	16.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	37.8	0.0	-3.3	1.6	21.7	9.3	8.7	OK	OK
End plate SLN225 ring	Flange top b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	396.9	0.2	-187.1	161.0	-122.2	98.1	59.2	OK	OK
		◢ 4.2 ◣	58	C LC_135	368.0	0.0	-109.0	-160.5	124.1	90.9	53.8	OK	OK
End plate SLN225 ring	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	71.1	0.0	11.0	-39.6	8.6	17.6	13.6	OK	OK
		◢ 4.2 ◣	58	C LC_135	116.9	0.0	-0.7	-9.2	-66.9	28.9	19.1	OK	OK
End plate SLN225 ring	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	340.8	0.0	-95.3	150.4	-114.3	84.2	49.5	OK	OK
		◢ 4.2 ◣	58	C LC_135	396.9	0.2	-189.2	-160.8	121.4	98.1	60.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-S-340
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut

Name	SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

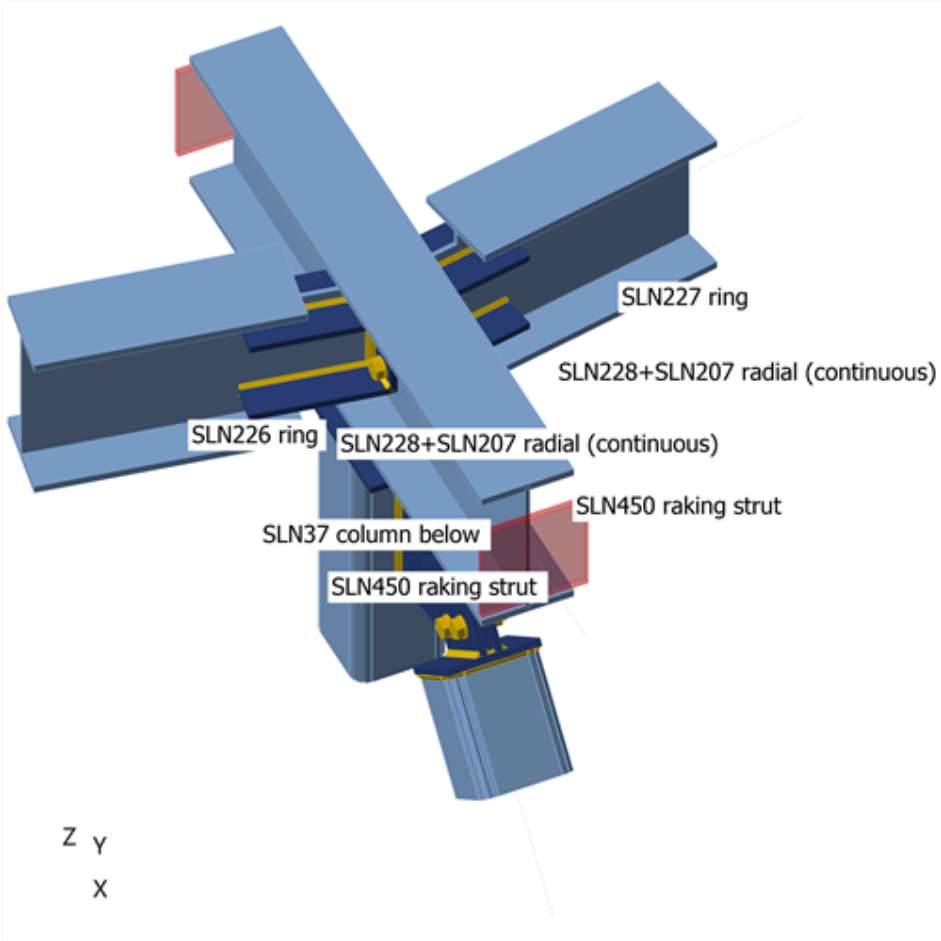
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN228+SLN207 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN37 column below	2 - SHS150x150x10	0.0	90.0	160.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN450 raking strut	2 - SHS150x150x10	-19.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN228+SLN207 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN228+SLN207 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN37 column below / begin		Position	0
SLN226 ring / begin		Position	0
SLN227 ring / end		Position	0
SLN450 raking strut / end		Position	0



SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN37 column below / Begin	124.7	0.0	0.3	0.0	1.4	0.0
	SLN226 ring / Begin	-9.0	0.8	-25.6	0.0	0.0	1.0
	SLN227 ring / End	9.0	0.8	-25.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-22.2	0.0	-1.0	0.0	0.0	0.0
C LC_108	SLN37 column below / Begin	36.8	2.6	-7.9	0.0	-5.4	-0.7
	SLN226 ring / Begin	-5.6	1.3	-0.5	0.0	0.0	-0.5
	SLN227 ring / End	8.0	1.8	-0.5	0.0	0.0	0.5
	SLN450 raking strut / End	44.6	0.0	-1.2	-0.2	0.0	0.8
C LC_110	SLN37 column below / Begin	9.7	-7.0	-2.8	0.0	-1.8	1.9
	SLN226 ring / Begin	-0.5	0.1	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-6.7	-1.0	-0.5	0.0	0.0	0.3
	SLN450 raking strut / End	7.8	0.0	-0.9	0.5	0.0	-2.1
C LC_111	SLN37 column below / Begin	49.7	0.0	0.2	0.0	0.9	-0.1
	SLN226 ring / Begin	7.8	-2.3	-16.2	0.0	0.0	1.2
	SLN227 ring / End	-8.0	-2.1	-16.3	0.0	0.0	-1.0
	SLN450 raking strut / End	-53.4	0.2	-0.5	0.0	0.0	0.2
C LC_112	SLN37 column below / Begin	113.6	2.6	-7.7	0.0	-4.7	-0.7
	SLN226 ring / Begin	-9.1	1.5	-16.2	0.0	0.0	0.1
	SLN227 ring / End	11.7	2.1	-16.3	0.0	0.0	-0.1
	SLN450 raking strut / End	32.1	0.0	-1.5	-0.2	0.0	0.8
C LC_119	SLN37 column below / Begin	113.3	0.0	0.3	0.0	1.3	0.0
	SLN226 ring / Begin	-1.9	-0.6	-25.6	0.0	0.0	1.3
	SLN227 ring / End	1.8	-0.5	-25.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-36.9	0.1	-0.8	0.0	0.0	0.1
C LC_121	SLN37 column below / Begin	116.5	0.0	0.3	0.0	1.3	0.2
	SLN226 ring / Begin	-1.2	-0.6	-25.6	0.0	0.0	1.0
	SLN227 ring / End	1.7	-0.5	-25.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-32.9	-0.3	-0.9	0.0	0.0	-0.2
C LC_122	SLN37 column below / Begin	126.9	-3.5	-1.1	0.0	0.3	1.0
	SLN226 ring / Begin	-8.6	0.8	-25.6	0.0	0.0	1.1
	SLN227 ring / End	5.0	0.2	-25.7	0.0	0.0	-0.9
	SLN450 raking strut / End	-19.1	0.0	-1.1	0.2	0.0	-1.0
C LC_124	SLN37 column below / Begin	123.6	2.6	-7.6	0.0	-4.3	-0.7
	SLN226 ring / Begin	-15.9	2.4	-19.4	0.0	0.0	0.5
	SLN227 ring / End	18.5	2.9	-19.5	0.0	0.0	-0.6
	SLN450 raking strut / End	21.8	0.0	-1.5	-0.2	0.0	0.8
C LC_126	SLN37 column below / Begin	99.2	-7.0	-2.6	0.0	-0.9	2.0
	SLN226 ring / Begin	-9.7	1.1	-19.4	0.0	0.0	1.0
	SLN227 ring / End	2.5	0.0	-19.5	0.0	0.0	-0.5
	SLN450 raking strut / End	-11.2	0.0	-1.2	0.5	0.0	-2.1
C LC_145	SLN37 column below / Begin	122.5	0.0	0.3	0.0	1.4	0.0
	SLN226 ring / Begin	-8.0	0.5	-25.6	0.0	0.0	1.1
	SLN227 ring / End	8.0	0.5	-25.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-25.1	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN37 column below / Begin	123.7	0.0	0.3	0.0	1.4	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / Begin	-8.5	0.7	-25.6	0.0	0.0	1.1
	SLN227 ring / End	8.8	0.7	-25.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-23.5	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN37 column below / Begin	125.7	0.0	0.3	0.0	1.4	0.0
	SLN226 ring / Begin	-9.4	0.9	-25.6	0.0	0.0	1.0
	SLN227 ring / End	9.3	0.9	-25.7	0.0	0.0	-1.1
	SLN450 raking strut / End	-20.8	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN37 column below / Begin	73.8	0.0	2.2	0.0	8.8	0.0
	SLN226 ring / Begin	-11.3	1.4	-14.1	0.0	0.0	0.8
	SLN227 ring / End	11.3	1.3	-14.2	0.0	0.0	-1.0
	SLN450 raking strut / End	-8.0	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN37 column below / Begin	124.4	0.0	0.3	0.0	1.2	0.0
	SLN226 ring / Begin	-9.0	0.8	-21.0	0.0	0.0	1.0
	SLN227 ring / End	9.0	0.8	-20.7	0.0	0.0	-1.2
	SLN450 raking strut / End	-21.9	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	25.4 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN228+SLN207 radial (continuous)-bfl 1	12.7	C LC_108	209.9	0.0	33.5	OK
SLN228+SLN207 radial (continuous)-tfl 1	12.7	C LC_153	72.0	0.0	0.0	OK
SLN228+SLN207 radial (continuous)-w 1	7.2	C LC_152	160.0	0.0	44.4	OK
SLN37 column below	10.0	C LC_111	275.3	0.1	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_122	84.2	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_122	81.5	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_145	232.4	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_147	91.8	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_147	89.6	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_147	249.2	0.0	0.0	OK
SLN450 raking strut	10.0	C LC_126	57.5	0.0	0.0	OK
End plate column below SLN37 column below	12.0	C LC_108	242.8	0.0	33.6	OK
End plate SLN226 ring	12.0	C LC_119	123.6	0.0	0.0	OK
End plate SLN227 ring	12.0	C LC_145	99.3	0.0	43.4	OK
Cap plate SLN450 raking strut	12.0	C LC_126	127.5	0.0	0.0	OK
Fin plate SLN450 raking strut	12.0	C LC_126	275.1	0.0	83.3	OK
Fin plate 2 SLN450 raking strut	12.0	C LC_110	275.1	0.1	124.3	OK
Gusset SLN450 raking strut	12.0	C LC_126	275.2	0.1	62.5	OK
Flange top a end plate SLN226 ring	12.0	C LC_145	107.5	0.0	0.0	OK
Flange top b end plate SLN226 ring	12.0	C LC_111	90.0	0.0	0.0	OK
Flange bottom a end plate SLN226 ring	12.0	C LC_145	83.9	0.0	0.0	OK
Flange bottom b end plate SLN226 ring	12.0	C LC_119	105.9	0.0	0.0	OK
Flange top a end plate SLN227 ring	12.0	C LC_147	92.2	0.0	0.0	OK
Flange top b end plate SLN227 ring	12.0	C LC_147	113.7	0.0	0.0	OK
Flange bottom a end plate SLN227 ring	12.0	C LC_147	111.4	0.0	0.0	OK
Flange bottom b end plate SLN227 ring	12.0	C LC_147	94.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

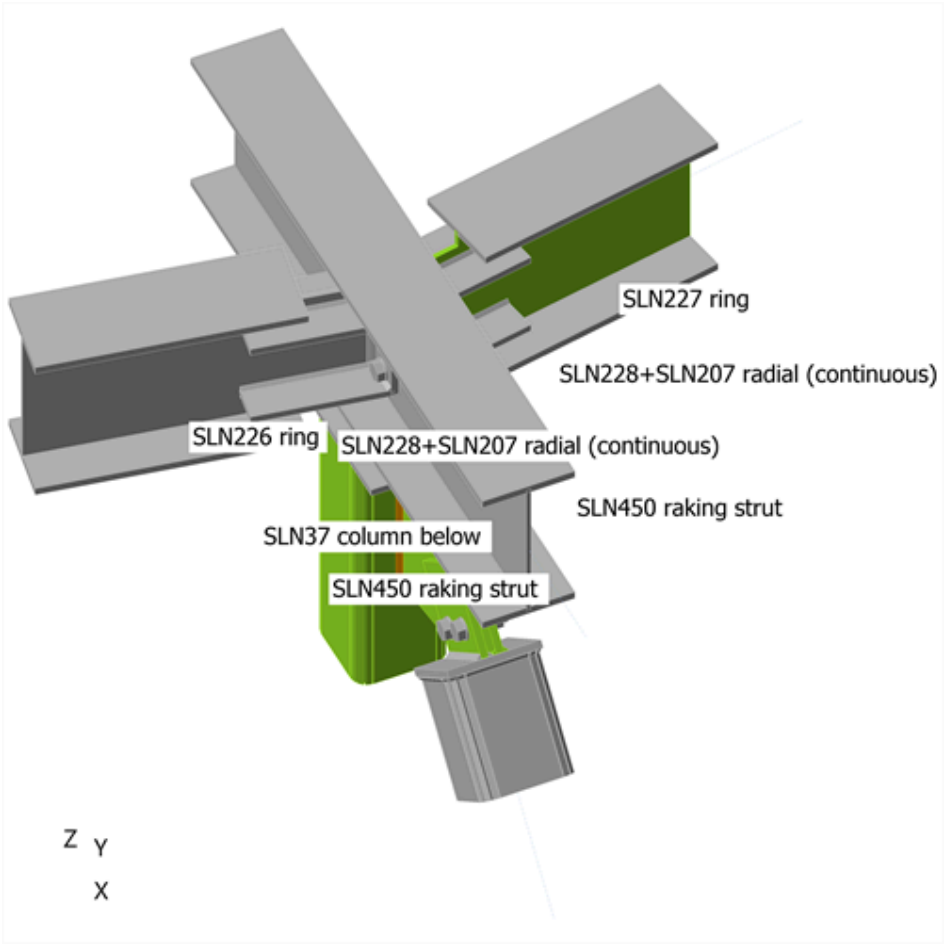
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

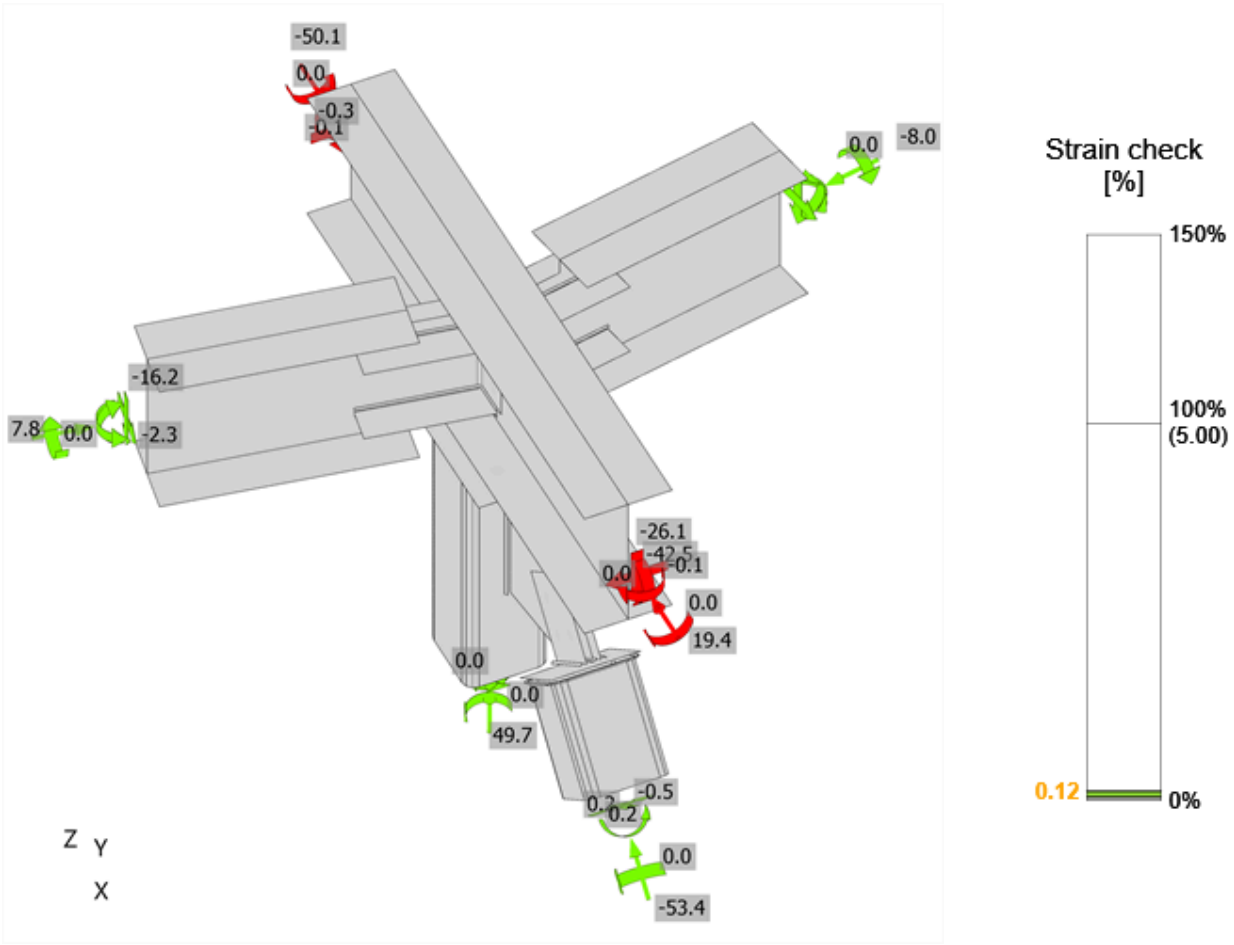
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN37 column below	150	C LC_108	0	5	0.1	OK
SLN450 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

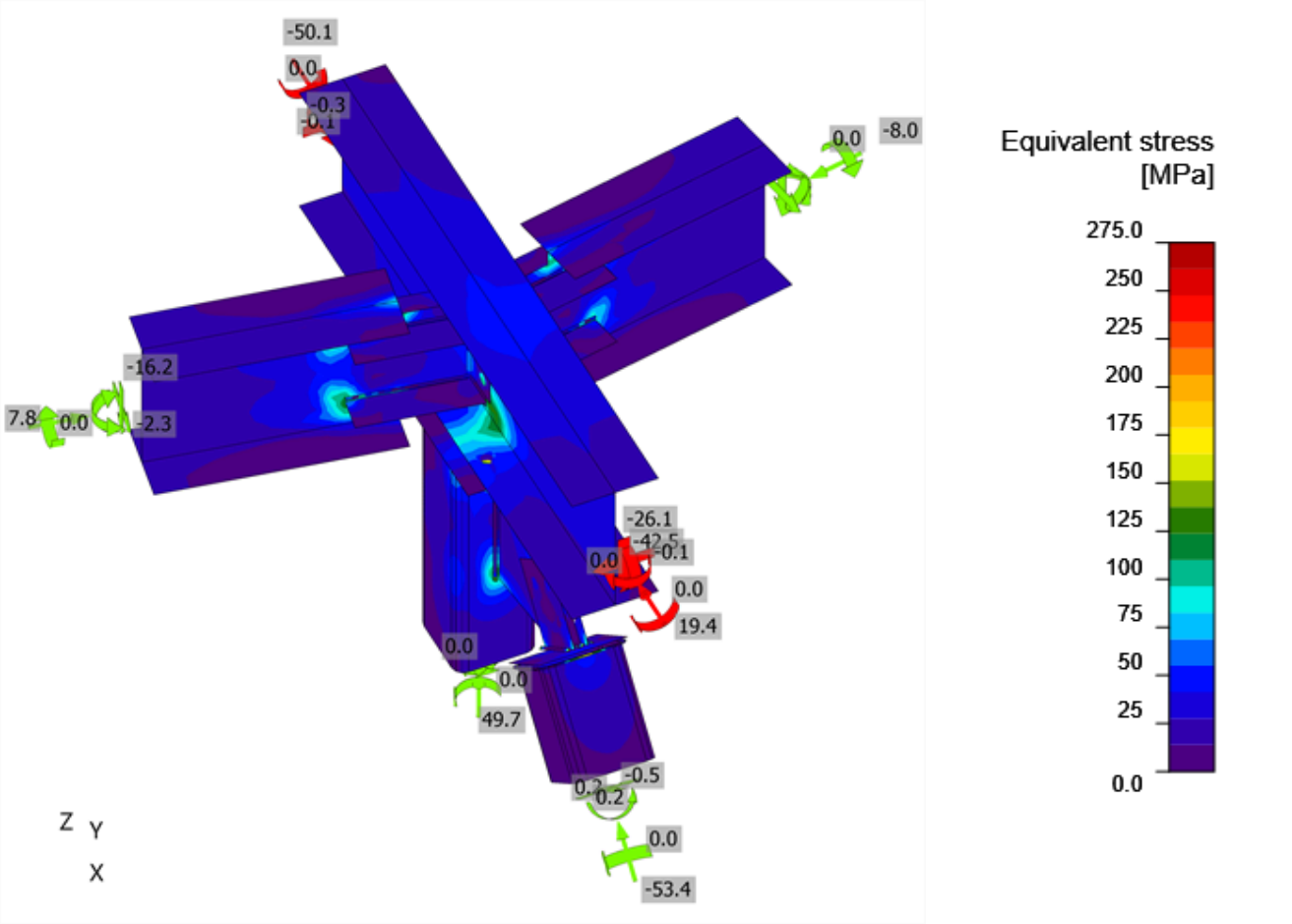
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

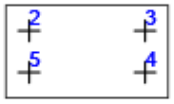
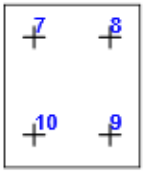
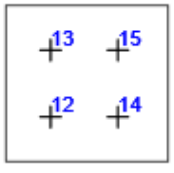


Strain check, C LC_111



Equivalent stress, C LC_111

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_111	0.7	10.6	168.9	0.5	11.2	11.6	OK	OK
	B3	M20 8.8 - 1	C LC_108	35.9	5.0	110.1	25.4	5.3	23.4	OK	OK
	B4	M20 8.8 - 1	C LC_108	35.2	4.6	87.8	24.9	5.3	22.7	OK	OK
	B5	M20 8.8 - 1	C LC_111	1.0	10.8	171.2	0.7	11.5	12.0	OK	OK
	B7	M20 8.8 - 1	C LC_119	18.1	1.2	71.5	12.8	1.2	10.4	OK	OK
	B8	M20 8.8 - 1	C LC_108	5.5	0.7	123.8	3.9	0.9	3.5	OK	OK
	B9	M20 8.8 - 1	C LC_108	5.6	0.6	123.8	3.9	0.8	3.5	OK	OK
	B10	M20 8.8 - 1	C LC_119	19.0	1.4	123.8	13.4	1.4	11.0	OK	OK
	B12	M20 8.8 - 1	C LC_110	17.0	13.3	106.0	12.0	14.2	22.8	OK	OK
	B13	M20 8.8 - 1	C LC_126	7.6	12.0	159.0	5.4	12.7	16.5	OK	OK
	B14	M20 8.8 - 1	C LC_108	1.5	9.1	106.0	1.0	12.8	10.4	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.2	7.2	115.3	0.1	12.2	7.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN450 raking strut	SLN450 raking strut	◢ 4.2	519	C LC_108	175.2	0.0	82.8	-89.1	3.1	43.3	22.8	OK	OK
SLN37 column below-w 3	Gusset SLN450 raking strut	◢ 5.6 ◣	288	C LC_111	397.3	0.4	-134.9	-133.9	-169.2	98.2	65.3	OK	OK
		◢ 5.6 ◣	288	C LC_111	397.3	0.5	-133.4	134.6	169.1	98.2	66.2	OK	OK
SLN226 ring-w 1	Flange top b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_153	113.4	0.0	62.0	50.8	20.5	28.0	17.1	OK	OK
		◢ 4.2 ◣	197	C LC_145	53.9	0.0	-26.1	24.0	12.8	13.3	8.7	OK	OK
SLN226 ring-w 1	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_148	94.4	0.0	27.9	37.2	36.5	23.3	13.9	OK	OK
		◢ 4.2 ◣	198	C LC_145	174.5	0.0	78.2	-68.9	-58.0	43.1	26.6	OK	OK
SLN227 ring-w 1	Flange top a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_153	118.9	0.0	63.5	53.4	-22.7	29.4	15.1	OK	OK
		◢ 4.2 ◣	197	C LC_147	61.8	0.0	-31.2	28.9	-10.8	15.3	9.1	OK	OK
SLN227 ring-w 1	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_147	100.4	0.0	30.3	39.6	-38.5	24.8	13.1	OK	OK
		◢ 4.2 ◣	198	C LC_147	189.7	0.0	85.3	-76.0	61.6	46.9	25.2	OK	OK
Cap plate SLN450 raking strut	Fin plate SLN450 raking strut	◢ 5.6 ◣	139	C LC_108	235.8	0.0	90.8	103.1	71.8	58.3	38.5	OK	OK
		◢ 5.6 ◣	139	C LC_111	138.9	0.0	-53.7	58.9	-44.6	34.3	20.4	OK	OK
Cap plate SLN450 raking strut	Fin plate 2 SLN450 raking strut	◢ 5.6 ◣	139	C LC_108	151.6	0.0	78.8	61.3	42.9	37.5	23.6	OK	OK
		◢ 5.6 ◣	139	C LC_110	273.8	0.0	88.7	-118.6	-91.1	67.7	49.8	OK	OK
End plate column below SLN37 column below	SLN37 column below	◢ 5.6 ◣	521	C LC_119	276.5	0.0	-174.3	-121.1	-26.4	68.3	37.7	OK	OK
		◢ 5.6 ◣	521	C LC_152	213.0	0.0	-96.8	107.2	22.2	52.6	31.0	OK	OK
SLN228+SLN207 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_121	92.2	0.0	-16.0	10.6	-51.3	22.8	13.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN228+SLN207 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_122	96.1	0.0	-4.8	15.6	53.2	23.7	15.3	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_119	89.3	0.0	-25.7	16.2	-46.6	22.1	14.6	OK	OK
		◢ 4.2 ◣	149	C LC_145	74.8	0.0	-2.3	29.5	31.5	18.5	13.4	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_122	77.6	0.0	-16.6	-10.3	42.5	19.2	12.9	OK	OK
		◢ 4.2 ◣	149	C LC_145	75.5	0.0	-16.1	28.7	-31.5	18.7	13.4	OK	OK
SLN226 ring-w 1	Flange top a end plate SLN226 ring	◢ 4.2 ◣	199	C LC_145	189.1	0.0	-15.6	9.5	-108.4	46.7	34.0	OK	OK
		◢ 4.2 ◣	198	C LC_145	128.0	0.0	18.9	21.5	-69.9	31.6	18.6	OK	OK
SLN226 ring-w 1	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	198	C LC_145	152.6	0.0	27.6	-29.2	81.6	37.7	22.4	OK	OK
		◢ 4.2 ◣	198	C LC_145	151.4	0.0	-22.0	-9.1	86.0	37.4	23.7	OK	OK
SLN227 ring-w 1	Flange top b end plate SLN227 ring	◢ 4.2 ◣	199	C LC_147	197.7	0.0	-26.9	15.7	112.0	48.8	31.7	OK	OK
		◢ 4.2 ◣	198	C LC_147	136.7	0.0	24.0	25.8	73.3	33.8	18.5	OK	OK
SLN227 ring-w 1	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	198	C LC_147	166.2	0.0	29.0	-30.8	-89.3	41.1	23.5	OK	OK
		◢ 4.2 ◣	198	C LC_147	169.9	0.0	-21.7	-8.1	-96.9	42.0	25.2	OK	OK
End plate SLN226 ring	Flange top a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_119	59.9	0.0	-1.5	5.9	-34.1	14.8	9.5	OK	OK
		◢ 4.2 ◣	58	C LC_119	35.4	0.0	-9.3	-19.1	-4.6	8.7	8.3	OK	OK
End plate SLN226 ring	Flange top b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_119	79.8	0.0	-5.0	0.8	-46.0	19.7	13.8	OK	OK
		◢ 4.2 ◣	58	C LC_119	105.8	0.0	60.4	41.7	-27.9	26.2	20.9	OK	OK
End plate SLN226 ring	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_147	45.7	0.0	9.4	-9.8	23.9	11.3	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	63.4	0.0	-18.0	-17.5	30.4	15.7	10.4	OK	OK
End plate SLN226 ring	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_119	117.3	0.0	66.8	-49.0	26.3	29.0	21.4	OK	OK
		◢ 4.2 ◣	58	C LC_119	89.7	0.0	-8.1	-1.7	51.6	22.2	15.1	OK	OK
End plate SLN227 ring	Flange top a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_145	68.6	0.0	-1.0	-2.3	39.5	17.0	12.3	OK	OK
		◢ 4.2 ◣	58	C LC_119	83.6	0.0	8.3	13.9	46.0	20.7	15.6	OK	OK
End plate SLN227 ring	Flange top b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_119	52.1	0.0	-12.4	13.9	25.7	12.9	8.2	OK	OK
		◢ 4.2 ◣	58	C LC_119	45.8	0.0	-14.9	-22.2	11.4	11.3	8.4	OK	OK
End plate SLN227 ring	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_119	91.6	0.0	50.6	-39.4	-19.9	22.6	16.5	OK	OK
		◢ 4.2 ◣	58	C LC_145	78.5	0.0	-5.1	0.8	-45.2	19.4	13.7	OK	OK
End plate SLN227 ring	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_148	52.9	0.0	7.9	-8.9	-28.9	13.1	8.0	OK	OK
		◢ 4.2 ◣	58	C LC_119	54.7	0.0	-21.1	-23.3	-17.4	13.5	8.8	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-T-060
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut

Name	SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

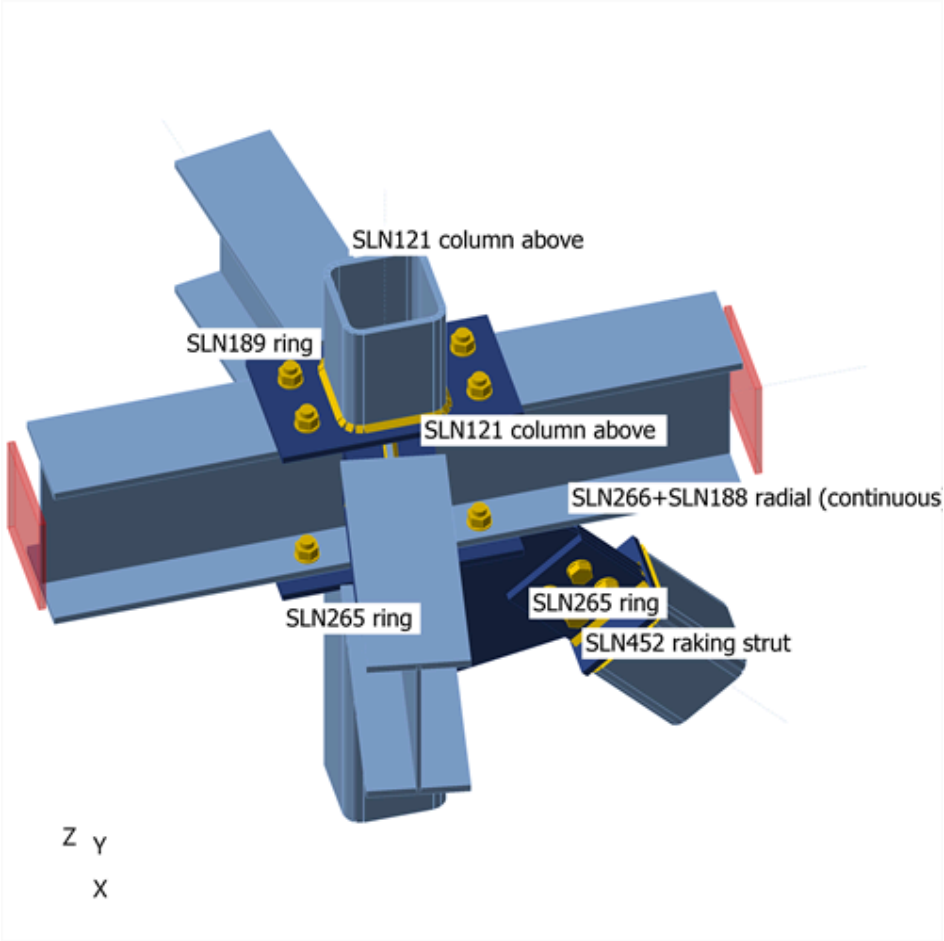
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN266+SLN188 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN41 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN189 ring	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN452 raking strut	2 - SHS150x150x10	60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN266+SLN188 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN266+SLN188 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN41 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN265 ring / begin		Position	0
SLN189 ring / end		Position	0
SLN452 raking strut / end		Position	0



SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN41 column below / Begin	173.2	0.0	0.4	0.0	1.8	-0.1
	SLN121 column above / End	-68.3	0.0	2.1	0.0	-1.9	0.3
	SLN265 ring / Begin	-31.3	3.6	-26.0	0.0	0.0	1.9
	SLN189 ring / End	31.4	4.8	-25.8	0.0	0.0	-1.3
	SLN452 raking strut / End	-47.2	-0.1	-1.0	0.0	0.0	-0.4
C LC_109	SLN41 column below / Begin	47.2	3.9	-7.3	0.0	-4.9	-1.7
	SLN121 column above / End	-5.8	0.3	-0.1	0.0	0.6	1.1
	SLN265 ring / Begin	-9.1	1.8	-0.5	0.0	0.0	-0.4
	SLN189 ring / End	13.5	2.6	-0.5	0.0	0.0	0.3
	SLN452 raking strut / End	50.5	-0.1	-1.2	-0.2	0.0	0.7
C LC_111	SLN41 column below / Begin	93.5	0.0	0.3	0.0	1.2	0.1
	SLN121 column above / End	-45.2	0.1	1.4	0.0	-1.2	0.4
	SLN265 ring / Begin	-7.7	-0.3	-16.5	0.0	0.0	1.8
	SLN189 ring / End	8.4	1.0	-16.3	0.0	0.0	-0.8
	SLN452 raking strut / End	-54.7	-0.7	-0.8	-0.1	0.0	-0.8
C LC_113	SLN41 column below / Begin	152.3	3.9	-7.1	0.0	-4.0	-1.8
	SLN121 column above / End	-45.5	0.3	1.3	0.0	-0.8	1.3
	SLN265 ring / Begin	-24.9	3.6	-16.5	0.0	0.0	0.6
	SLN189 ring / End	29.5	5.1	-16.3	0.0	0.0	-0.4
	SLN452 raking strut / End	23.4	-0.2	-1.5	-0.2	0.0	0.4
C LC_117	SLN41 column below / Begin	57.6	3.9	-7.2	0.0	-4.8	-1.7
	SLN121 column above / End	-10.4	0.3	0.0	0.0	0.5	1.1
	SLN265 ring / Begin	-11.7	2.1	-1.8	0.0	0.0	-0.3
	SLN189 ring / End	16.1	3.0	-1.8	0.0	0.0	0.2
	SLN452 raking strut / End	48.4	-0.1	-1.5	-0.2	0.0	0.6
C LC_119	SLN41 column below / Begin	165.9	0.0	0.4	0.0	1.8	-0.1
	SLN121 column above / End	-68.3	0.0	2.3	0.0	-2.2	0.4
	SLN265 ring / Begin	-23.6	2.1	-26.0	0.0	0.0	2.0
	SLN189 ring / End	24.2	3.6	-25.8	0.0	0.0	-1.2
	SLN452 raking strut / End	-56.7	-0.4	-0.9	-0.1	0.0	-0.7
C LC_120	SLN41 column below / Begin	179.6	-3.4	-1.7	0.0	0.3	1.6
	SLN121 column above / End	-68.1	-0.3	2.0	0.0	-1.6	-0.9
	SLN265 ring / Begin	-32.7	3.9	-26.0	0.0	0.0	1.8
	SLN189 ring / End	28.7	4.5	-25.8	0.0	0.0	-1.0
	SLN452 raking strut / End	-38.2	0.0	-1.1	0.1	0.0	-0.9
C LC_122	SLN41 column below / Begin	162.6	0.1	0.4	0.0	1.8	-0.2
	SLN121 column above / End	-68.4	0.0	2.4	0.0	-2.4	0.4
	SLN265 ring / Begin	-22.1	1.9	-26.0	0.0	0.0	1.9
	SLN189 ring / End	22.0	3.0	-25.8	0.0	0.0	-1.4
	SLN452 raking strut / End	-61.1	0.1	-0.8	0.0	0.0	-0.3
C LC_123	SLN41 column below / Begin	119.5	0.0	0.3	0.0	1.5	0.0
	SLN121 column above / End	-53.4	0.1	1.8	0.0	-1.7	0.5
	SLN265 ring / Begin	-14.3	0.7	-19.7	0.0	0.0	1.9
	SLN189 ring / End	15.1	2.2	-19.5	0.0	0.0	-0.9
	SLN452 raking strut / End	-53.7	-0.7	-0.8	-0.1	0.0	-0.8

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_125	SLN41 column below / Begin	161.1	3.9	-7.0	0.0	-3.8	-1.8
	SLN121 column above / End	-52.8	0.3	1.0	0.0	0.1	1.3
	SLN265 ring / Begin	-32.9	4.6	-19.7	0.0	0.0	1.0
	SLN189 ring / End	37.4	6.2	-19.5	0.0	0.0	-0.9
	SLN452 raking strut / End	2.1	-0.2	-1.5	-0.2	0.0	0.5
C LC_135	SLN41 column below / Begin	174.2	0.0	0.4	0.0	1.9	-0.1
	SLN121 column above / End	-68.3	0.0	2.1	0.0	-1.8	0.3
	SLN265 ring / Begin	-31.5	3.6	-26.0	0.0	0.0	1.9
	SLN189 ring / End	31.6	4.8	-25.8	0.0	0.0	-1.3
	SLN452 raking strut / End	-45.8	-0.1	-1.0	0.0	0.0	-0.4
C LC_145	SLN41 column below / Begin	172.1	0.0	0.4	0.0	1.9	-0.1
	SLN121 column above / End	-68.3	0.0	2.2	0.0	-1.9	0.3
	SLN265 ring / Begin	-30.7	3.4	-26.0	0.0	0.0	1.9
	SLN189 ring / End	30.9	4.7	-25.8	0.0	0.0	-1.3
	SLN452 raking strut / End	-48.7	-0.1	-1.0	0.0	0.0	-0.4
C LC_147	SLN41 column below / Begin	175.2	0.0	0.4	0.0	1.8	-0.1
	SLN121 column above / End	-68.3	0.0	2.1	0.0	-1.8	0.3
	SLN265 ring / Begin	-32.2	3.8	-26.0	0.0	0.0	1.8
	SLN189 ring / End	32.4	5.1	-25.8	0.0	0.0	-1.2
	SLN452 raking strut / End	-44.6	-0.1	-1.0	0.0	0.0	-0.4
C LC_148	SLN41 column below / Begin	171.2	0.0	0.4	0.0	1.9	-0.1
	SLN121 column above / End	-68.3	0.0	2.2	0.0	-1.9	0.3
	SLN265 ring / Begin	-30.4	3.3	-26.0	0.0	0.0	1.9
	SLN189 ring / End	30.3	4.5	-25.8	0.0	0.0	-1.3
	SLN452 raking strut / End	-49.8	-0.1	-1.0	0.0	0.0	-0.4
C LC_151	SLN41 column below / Begin	127.4	0.0	-1.3	0.0	-5.0	0.0
	SLN121 column above / End	-68.4	0.0	5.3	0.0	-10.5	0.1
	SLN265 ring / Begin	-16.9	1.3	-11.2	0.0	0.0	1.4
	SLN189 ring / End	16.8	2.4	-11.4	0.0	0.0	-1.0
	SLN452 raking strut / End	-37.6	0.0	-1.0	0.0	0.0	-0.2
C LC_152	SLN41 column below / Begin	85.2	0.0	1.9	0.0	7.7	-0.1
	SLN121 column above / End	-18.0	0.1	-2.7	0.0	8.6	0.3
	SLN265 ring / Begin	-31.0	3.9	-13.9	0.0	0.0	1.5
	SLN189 ring / End	31.1	5.0	-14.1	0.0	0.0	-1.0
	SLN452 raking strut / End	-16.5	-0.1	-1.0	0.0	0.0	-0.3
C LC_153	SLN41 column below / Begin	171.3	0.0	0.4	0.0	1.7	-0.1
	SLN121 column above / End	-67.5	0.0	2.1	0.0	-1.9	0.3
	SLN265 ring / Begin	-31.4	3.6	-20.2	0.0	0.0	1.9
	SLN189 ring / End	31.5	4.8	-20.6	0.0	0.0	-1.3
	SLN452 raking strut / End	-46.6	-0.1	-1.0	0.0	0.0	-0.4
C LC_154	SLN41 column below / Begin	127.6	0.0	-1.2	0.0	-4.9	0.0
	SLN121 column above / End	-69.2	0.0	5.3	0.0	-10.5	0.1
	SLN265 ring / Begin	-17.0	1.4	-11.2	0.0	0.0	1.4
	SLN189 ring / End	16.9	2.4	-11.4	0.0	0.0	-1.0
	SLN452 raking strut / End	-37.6	0.0	-1.0	0.0	0.0	-0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.4 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	27.2 < 100%	OK
Welds	98.4 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN266+SLN188 radial (continuous)-bfl 1	12.7	C LC_109	234.1	0.0	36.1	OK
SLN266+SLN188 radial (continuous)-tfl 1	12.7	C LC_152	239.3	0.0	27.3	OK
SLN266+SLN188 radial (continuous)-w 1	7.2	C LC_113	275.1	0.1	42.9	OK
SLN41 column below	10.0	C LC_122	275.9	0.4	0.0	OK
SLN121 column above	10.0	C LC_152	225.4	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_120	165.3	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_120	161.0	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_120	275.3	0.1	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_145	147.5	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_145	142.5	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_145	275.2	0.1	0.0	OK
SLN452 raking strut	10.0	C LC_122	54.2	0.0	0.0	OK
End plate column below SLN41 column below	12.0	C LC_109	268.1	0.0	36.1	OK
End plate column above SLN121 column above	12.0	C LC_152	213.4	0.0	27.3	OK
End plate SLN265 ring	12.0	C LC_119	212.5	0.0	0.0	OK
End plate SLN189 ring	12.0	C LC_120	147.5	0.0	74.1	OK
Cap plate SLN452 raking strut	12.0	C LC_109	72.0	0.0	0.0	OK
Fin plate SLN452 raking strut	12.0	C LC_109	209.0	0.0	38.9	OK
Fin plate 2 SLN452 raking strut	12.0	C LC_109	160.1	0.0	31.6	OK
Gusset SLN452 raking strut	12.0	C LC_122	212.8	0.0	10.3	OK
Flange top a end plate SLN265 ring	12.0	C LC_148	195.9	0.0	0.0	OK
Flange top b end plate SLN265 ring	12.0	C LC_120	187.4	0.0	0.0	OK
Flange bottom a end plate SLN265 ring	12.0	C LC_148	172.8	0.0	0.0	OK
Flange bottom b end plate SLN265 ring	12.0	C LC_120	195.9	0.0	0.0	OK
Flange top a end plate SLN189 ring	12.0	C LC_145	147.1	0.0	0.0	OK
Flange top b end plate SLN189 ring	12.0	C LC_145	160.7	0.0	0.0	OK
Flange bottom a end plate SLN189 ring	12.0	C LC_145	170.0	0.0	0.0	OK
Flange bottom b end plate SLN189 ring	12.0	C LC_145	144.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

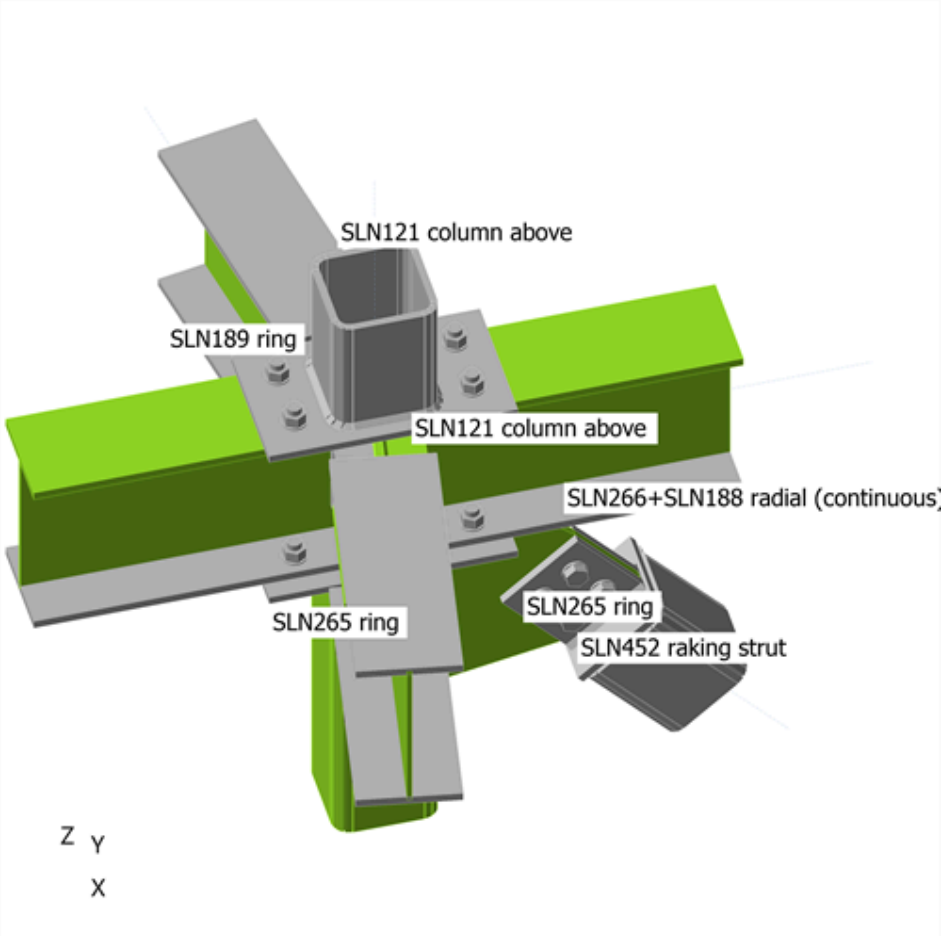
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

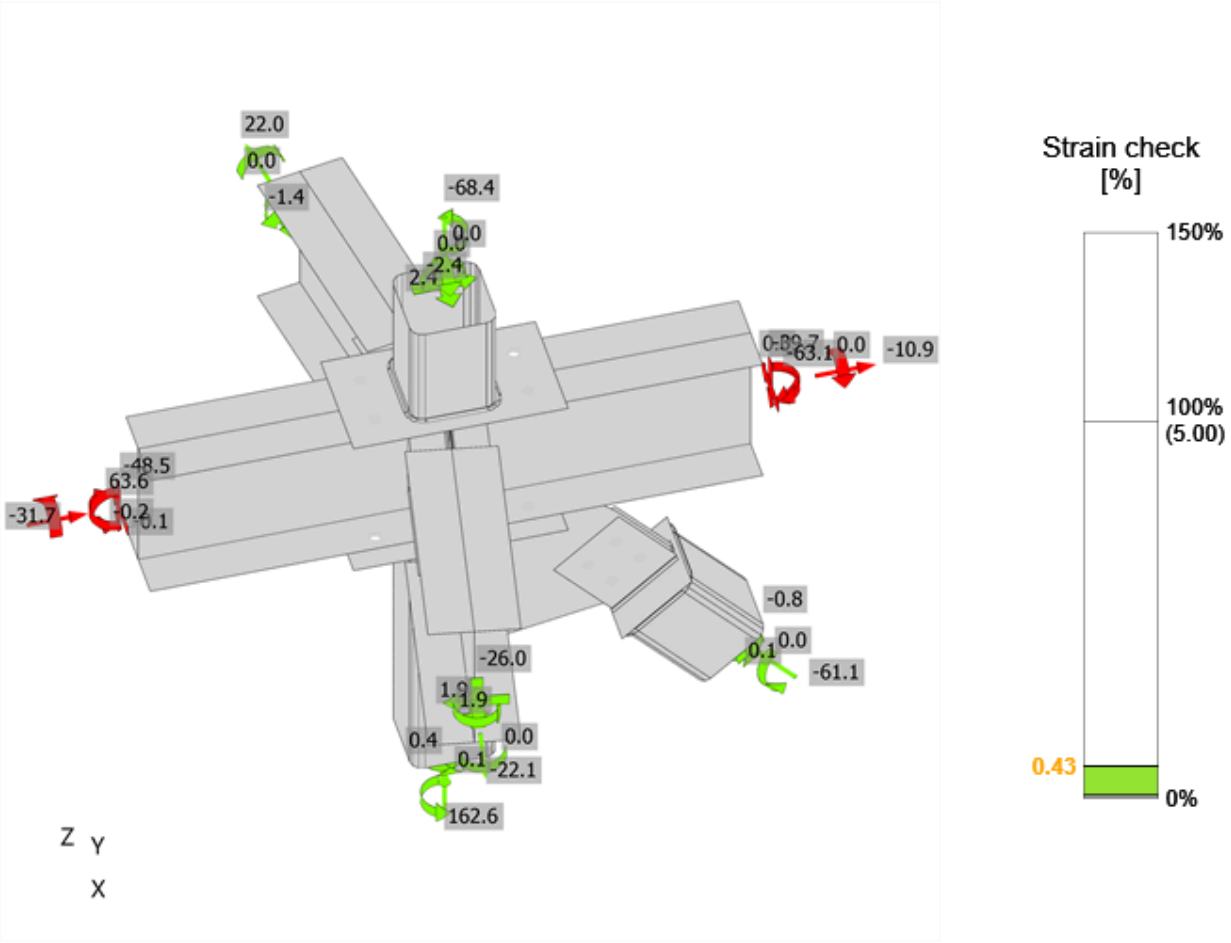
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN41 column below	150	C LC_122	0	5	0.1	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK
SLN452 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

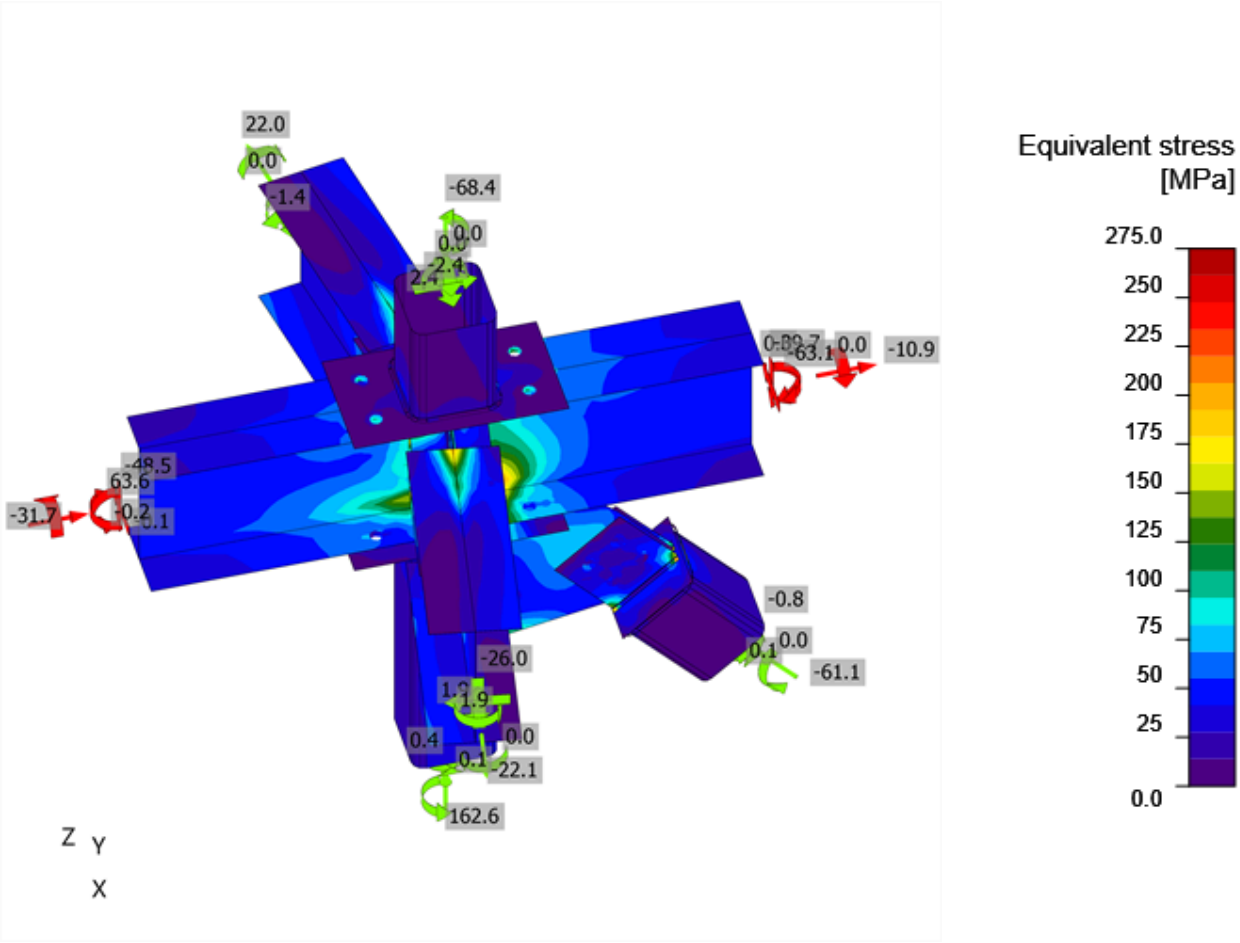
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

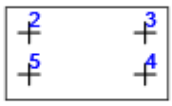
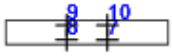
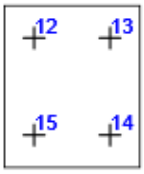
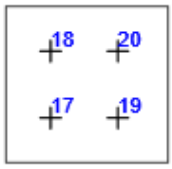


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.3	13.8	171.5	0.2	14.6	14.8	OK	OK
	B3	M20 8.8 - 1	C LC_109	31.0	6.8	105.9	22.0	7.2	22.9	OK	OK
	B4	M20 8.8 - 1	C LC_109	38.4	5.1	101.8	27.2	5.4	24.9	OK	OK
	B5	M20 8.8 - 1	C LC_122	1.4	13.0	168.6	1.0	13.8	14.5	OK	OK
	B7	M20 8.8 - 1	C LC_151	20.6	3.2	96.9	14.6	3.4	13.8	OK	OK
	B8	M20 8.8 - 1	C LC_152	29.8	3.0	86.8	21.1	3.4	18.3	OK	OK
	B9	M20 8.8 - 1	C LC_152	28.5	2.7	81.4	20.2	3.3	17.3	OK	OK
	B10	M20 8.8 - 1	C LC_151	19.0	3.1	95.0	13.4	3.3	12.9	OK	OK
	B12	M20 8.8 - 1	C LC_119	30.5	2.1	123.8	21.6	2.3	17.7	OK	OK
	B13	M20 8.8 - 1	C LC_109	4.9	0.5	123.8	3.5	0.8	3.0	OK	OK
	B14	M20 8.8 - 1	C LC_119	0.0	2.4	62.1	0.0	3.8	2.5	OK	OK
	B15	M20 8.8 - 1	C LC_119	31.2	2.4	123.8	22.1	2.5	18.3	OK	OK
	B17	M20 8.8 - 1	C LC_109	7.0	11.3	115.3	5.0	13.0	15.5	OK	OK
	B18	M20 8.8 - 1	C LC_120	2.7	9.8	115.3	1.9	10.4	11.8	OK	OK
	B19	M20 8.8 - 1	C LC_109	1.5	9.6	106.0	1.0	14.1	11.0	OK	OK
	B20	M20 8.8 - 1	C LC_122	0.3	8.8	115.3	0.2	14.2	9.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN452 raking strut	SLN452 raking strut	◢ 4.2	519	C LC_109	187.2	0.0	88.4	-95.3	0.0	46.2	24.5	OK	OK
SLN41 column below-w 3	Gusset SLN452 raking strut	◢ 5.6 ◣	287	C LC_122	398.3	1.0	-129.6	-132.1	-172.7	98.4	73.9	OK	OK
		◢ 5.6 ◣	287	C LC_122	398.3	1.0	-135.5	133.1	170.4	98.4	72.6	OK	OK
SLN265 ring-w 1	Flange top b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_153	277.8	0.0	144.2	119.0	68.1	68.6	40.2	OK	OK
		◢ 4.2 ◣	197	C LC_148	104.7	0.0	-53.4	49.9	14.6	25.9	14.8	OK	OK
SLN265 ring-w 1	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_120	170.4	0.0	48.6	70.7	62.3	42.1	25.6	OK	OK
		◢ 4.2 ◣	198	C LC_120	340.2	0.0	158.4	-136.1	-108.1	84.1	49.3	OK	OK
SLN189 ring-w 1	Flange top a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_153	213.6	0.0	112.4	90.2	-53.5	52.8	26.9	OK	OK
		◢ 4.2 ◣	197	C LC_145	93.0	0.0	-45.9	44.2	-15.0	23.0	13.4	OK	OK
SLN189 ring-w 1	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_145	151.1	0.0	42.4	63.7	-54.3	37.3	20.1	OK	OK
		◢ 4.2 ◣	198	C LC_145	287.6	0.0	134.6	-113.2	93.4	71.1	36.8	OK	OK
Cap plate SLN452 raking strut	Fin plate SLN452 raking strut	◢ 5.6 ◣	139	C LC_109	240.6	0.0	95.9	104.9	72.3	59.4	38.8	OK	OK
		◢ 5.6 ◣	139	C LC_122	170.6	0.0	-67.4	74.5	-51.5	42.2	24.4	OK	OK
Cap plate SLN452 raking strut	Fin plate 2 SLN452 raking strut	◢ 5.6 ◣	139	C LC_109	166.1	0.0	84.9	67.7	46.9	41.0	25.3	OK	OK
		◢ 5.6 ◣	139	C LC_122	152.0	0.0	-68.5	64.4	-44.6	37.6	20.8	OK	OK
End plate column below SLN41 column below	SLN41 column below	◢ 5.6 ◣	521	C LC_122	366.0	0.0	-234.6	-159.4	-30.4	90.4	48.6	OK	OK
		◢ 5.6 ◣	521	C LC_119	263.5	0.0	-74.0	145.8	8.7	65.1	34.6	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	521	C LC_154	167.0	0.0	-75.3	-85.8	-6.7	41.3	23.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_154	207.3	0.0	-107.2	100.2	21.3	51.2	31.0	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_125	251.1	0.0	8.3	3.4	144.9	62.1	34.2	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_120	210.0	0.0	-17.2	21.8	-118.9	51.9	33.6	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_125	205.7	0.0	-54.0	-24.1	-112.1	50.8	31.0	OK	OK
		◢ 4.2 ◣	149	C LC_125	149.2	0.0	-24.8	48.5	69.7	36.9	23.3	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_120	174.8	0.0	-38.9	-33.3	92.6	43.2	28.0	OK	OK
		◢ 4.2 ◣	149	C LC_120	156.2	0.0	-38.3	44.5	-75.2	38.6	29.0	OK	OK
SLN265 ring-w 1	Flange top a end plate SLN265 ring	◢ 4.2 ◣	199	C LC_120	368.5	0.0	-27.5	13.5	-211.7	91.0	62.7	OK	OK
		◢ 4.2 ◣	198	C LC_120	278.3	0.0	42.1	48.0	-151.4	68.8	38.0	OK	OK
SLN265 ring-w 1	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	198	C LC_120	295.6	0.0	53.8	-57.5	157.7	73.1	40.5	OK	OK
		◢ 4.2 ◣	199	C LC_148	314.1	0.0	-44.5	-18.5	178.5	77.6	48.3	OK	OK
SLN189 ring-w 1	Flange top b end plate SLN189 ring	◢ 4.2 ◣	199	C LC_145	286.3	0.0	-39.1	21.3	162.4	70.7	43.4	OK	OK
		◢ 4.2 ◣	198	C LC_145	219.3	0.0	39.8	43.7	116.6	54.2	30.3	OK	OK
SLN189 ring-w 1	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	198	C LC_145	243.4	0.0	43.7	-47.7	-129.8	60.1	34.2	OK	OK
		◢ 4.2 ◣	199	C LC_145	256.2	0.0	-33.9	-11.8	-146.1	63.3	36.9	OK	OK
End plate SLN265 ring	Flange top a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	106.3	0.0	-0.9	9.5	-60.6	26.3	16.6	OK	OK
		◢ 4.2 ◣	58	C LC_125	58.2	0.0	6.5	9.6	-32.0	14.4	10.4	OK	OK
End plate SLN265 ring	Flange top b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_120	161.5	0.0	-4.7	-0.9	-93.2	39.9	27.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	178.5	0.0	97.3	69.2	-51.8	44.1	34.9	OK	OK
End plate SLN265 ring	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_125	91.7	0.0	17.6	-28.3	43.5	22.6	14.8	OK	OK
		◢ 4.2 ◣	58	C LC_119	111.1	0.0	-29.0	-28.6	54.9	27.5	18.2	OK	OK
End plate SLN265 ring	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	187.6	0.0	102.4	-75.7	50.0	46.4	35.9	OK	OK
		◢ 4.2 ◣	58	C LC_120	170.3	0.0	-7.4	0.4	98.2	42.1	28.4	OK	OK
End plate SLN189 ring	Flange top a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_148	106.1	0.0	0.2	-10.3	60.4	26.2	18.7	OK	OK
		◢ 4.2 ◣	58	C LC_119	119.7	0.0	20.1	25.1	63.3	29.6	20.2	OK	OK
End plate SLN189 ring	Flange top b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_145	76.4	0.0	-7.8	14.3	41.5	18.9	11.7	OK	OK
		◢ 4.2 ◣	58	C LC_125	66.1	0.0	-24.1	-18.3	30.5	16.3	13.2	OK	OK
End plate SLN189 ring	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	119.5	0.0	68.7	-45.0	-34.1	29.5	21.7	OK	OK
		◢ 4.2 ◣	58	C LC_148	114.4	0.0	-3.3	9.7	-65.3	28.3	20.0	OK	OK
End plate SLN189 ring	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	91.4	0.0	15.5	-25.2	-45.5	22.6	13.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	90.6	0.0	-21.9	-31.6	-39.7	22.4	13.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L1-R8-T-260
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut

Name	SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

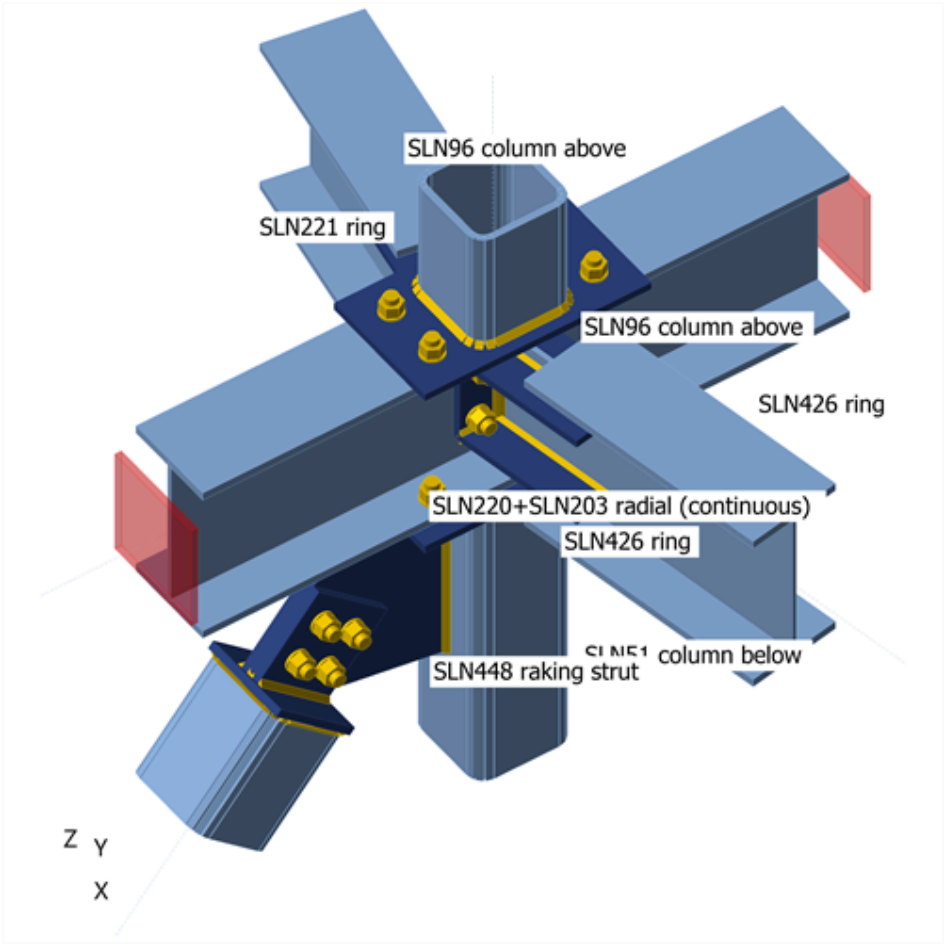
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN220+SLN203 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN51 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN96 column above	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN448 raking strut	2 - SHS150x150x10	-100.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN220+SLN203 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN220+SLN203 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN51 column below / begin		Position	0
SLN96 column above / end		Position	0
SLN221 ring / begin		Position	0
SLN426 ring / end		Position	0
SLN448 raking strut / end		Position	0



SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN51 column below / Begin	252.9	0.0	0.9	0.0	3.2	0.3
	SLN96 column above / End	-167.9	-0.1	-1.3	-1.0	3.2	-0.2
	SLN221 ring / Begin	-30.7	4.7	-25.7	0.0	0.0	0.9
	SLN426 ring / End	30.5	4.4	-25.6	0.0	0.0	-1.6
	SLN448 raking strut / End	-64.7	0.0	-1.0	0.0	0.0	-0.1
C LC_103	SLN51 column below / Begin	235.8	-3.9	0.0	0.0	2.4	2.0
	SLN96 column above / End	-155.9	-0.4	-1.3	-1.0	3.2	-1.4
	SLN221 ring / Begin	-24.6	3.6	-24.1	0.0	0.0	0.9
	SLN426 ring / End	19.8	2.8	-24.0	0.0	0.0	-1.2
	SLN448 raking strut / End	-60.2	0.1	-1.1	0.1	0.0	-0.7
C LC_105	SLN51 column below / Begin	223.2	-0.1	0.8	0.0	2.9	0.3
	SLN96 column above / End	-162.5	-0.2	-1.3	-1.0	3.2	-0.4
	SLN221 ring / Begin	-18.5	2.4	-24.1	0.0	0.0	1.0
	SLN426 ring / End	18.2	2.0	-24.0	0.0	0.0	-1.8
	SLN448 raking strut / End	-86.0	0.1	-0.8	-0.1	0.0	0.0
C LC_110	SLN51 column below / Begin	54.3	1.4	-8.2	0.0	-5.6	-0.7
	SLN96 column above / End	-13.7	0.1	-0.6	-0.1	1.7	0.6
	SLN221 ring / Begin	-13.5	2.8	-0.5	0.0	0.0	-0.6
	SLN426 ring / End	15.0	2.9	-0.5	0.0	0.0	0.3
	SLN448 raking strut / End	50.8	0.0	-1.3	0.0	0.0	0.2
C LC_113	SLN51 column below / Begin	140.8	-0.1	0.5	0.0	2.0	0.3
	SLN96 column above / End	-120.8	-0.2	-1.0	-0.7	2.5	-0.4
	SLN221 ring / Begin	-5.1	0.2	-16.3	0.0	0.0	0.9
	SLN426 ring / End	4.8	-0.2	-16.2	0.0	0.0	-1.6
	SLN448 raking strut / End	-86.5	0.1	-0.5	-0.1	0.0	0.1
C LC_119	SLN51 column below / Begin	248.4	-3.9	0.1	0.0	2.6	2.1
	SLN96 column above / End	-163.0	-0.5	-1.4	-1.0	3.5	-1.5
	SLN221 ring / Begin	-28.2	4.2	-25.7	0.0	0.0	1.1
	SLN426 ring / End	23.4	3.3	-25.6	0.0	0.0	-1.3
	SLN448 raking strut / End	-64.0	0.1	-1.1	0.1	0.0	-0.7
C LC_121	SLN51 column below / Begin	240.8	-0.1	0.8	0.0	3.1	0.4
	SLN96 column above / End	-168.3	-0.2	-1.1	-0.9	2.8	-0.4
	SLN221 ring / Begin	-21.6	3.0	-25.7	0.0	0.0	1.0
	SLN426 ring / End	21.2	2.6	-25.6	0.0	0.0	-1.8
	SLN448 raking strut / End	-81.2	0.1	-0.8	-0.1	0.0	0.1
C LC_126	SLN51 column below / Begin	226.9	1.4	-7.7	0.0	-3.3	-0.6
	SLN96 column above / End	-127.9	0.1	-1.8	-0.7	4.3	0.6
	SLN221 ring / Begin	-37.0	6.2	-19.5	0.0	0.0	0.4
	SLN426 ring / End	38.6	6.3	-19.4	0.0	0.0	-1.0
	SLN448 raking strut / End	-2.6	-0.1	-1.5	-0.1	0.0	0.0
C LC_135	SLN51 column below / Begin	254.0	0.0	0.9	0.0	3.2	0.3
	SLN96 column above / End	-168.7	-0.1	-1.3	-0.9	3.3	-0.2
	SLN221 ring / Begin	-31.0	4.7	-25.7	0.0	0.0	1.0
	SLN426 ring / End	30.7	4.4	-25.6	0.0	0.0	-1.6
	SLN448 raking strut / End	-64.4	0.0	-1.0	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_145	SLN51 column below / Begin	253.2	0.0	0.8	0.0	3.2	0.3
	SLN96 column above / End	-167.7	-0.1	-1.3	-1.0	3.2	-0.2
	SLN221 ring / Begin	-31.0	4.7	-25.7	0.0	0.0	1.0
	SLN426 ring / End	30.6	4.4	-25.6	0.0	0.0	-1.6
	SLN448 raking strut / End	-64.0	0.0	-1.0	0.0	0.0	-0.1
C LC_146	SLN51 column below / Begin	252.6	0.0	0.9	0.0	3.2	0.3
	SLN96 column above / End	-168.0	-0.1	-1.3	-1.0	3.2	-0.2
	SLN221 ring / Begin	-30.3	4.6	-25.7	0.0	0.0	0.9
	SLN426 ring / End	30.3	4.3	-25.6	0.0	0.0	-1.7
	SLN448 raking strut / End	-65.4	0.0	-1.0	0.0	0.0	-0.1
C LC_147	SLN51 column below / Begin	251.0	0.0	0.9	0.0	3.2	0.3
	SLN96 column above / End	-168.0	-0.1	-1.2	-1.0	3.1	-0.3
	SLN221 ring / Begin	-29.6	4.4	-25.7	0.0	0.0	1.0
	SLN426 ring / End	29.3	4.1	-25.6	0.0	0.0	-1.7
	SLN448 raking strut / End	-67.6	0.0	-1.0	0.0	0.0	-0.1
C LC_151	SLN51 column below / Begin	174.0	0.0	-0.9	0.0	-3.8	0.2
	SLN96 column above / End	-131.2	-0.1	2.4	-0.7	-6.2	-0.3
	SLN221 ring / Begin	-15.1	2.3	-11.7	0.0	0.0	0.4
	SLN426 ring / End	14.8	2.0	-11.9	0.0	0.0	-1.1
	SLN448 raking strut / End	-49.0	0.0	-1.0	-0.1	0.0	0.0
C LC_152	SLN51 column below / Begin	167.7	-0.1	2.0	0.0	7.9	0.3
	SLN96 column above / End	-116.7	-0.1	-4.4	-1.0	10.6	-0.3
	SLN221 ring / Begin	-30.5	4.8	-13.9	0.0	0.0	0.7
	SLN426 ring / End	30.3	4.5	-14.2	0.0	0.0	-1.5
	SLN448 raking strut / End	-39.0	0.0	-1.0	0.0	0.0	-0.1
C LC_153	SLN51 column below / Begin	252.4	0.0	0.8	0.0	3.1	0.3
	SLN96 column above / End	-167.0	-0.1	-1.2	-1.0	3.1	-0.2
	SLN221 ring / Begin	-30.8	4.7	-20.6	0.0	0.0	0.9
	SLN426 ring / End	30.6	4.4	-21.2	0.0	0.0	-1.6
	SLN448 raking strut / End	-64.6	0.0	-1.0	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	1.7 < 5.0%	OK
Loc. deformation	0.3 < 3%	OK
Bolts	25.9 < 100%	OK
Welds	99.3 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN220+SLN203 radial (continuous)-bfl 1	12.7	C LC_121	275.0	0.0	78.5	OK
SLN220+SLN203 radial (continuous)-tfl 1	12.7	C LC_119	177.3	0.0	42.9	OK
SLN220+SLN203 radial (continuous)-w 1	7.2	C LC_119	275.6	0.3	78.8	OK
SLN51 column below	10.0	C LC_105	278.6	1.7	0.0	OK
SLN96 column above	10.0	C LC_152	208.5	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_135	127.5	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_135	122.9	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_135	275.1	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_146	160.1	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_146	157.8	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_146	275.3	0.1	0.0	OK
SLN448 raking strut	10.0	C LC_105	71.2	0.0	0.0	OK
End plate column below SLN51 column below	12.0	C LC_121	275.0	0.0	71.0	OK
End plate column above SLN96 column above	12.0	C LC_152	134.9	0.0	47.3	OK
End plate SLN221 ring	12.0	C LC_147	67.8	0.0	155.8	OK
End plate SLN426 ring	12.0	C LC_147	76.3	0.0	263.4	OK
Cap plate SLN448 raking strut	12.0	C LC_105	90.9	0.0	0.0	OK
Fin plate SLN448 raking strut	12.0	C LC_119	202.3	0.0	26.7	OK
Fin plate 2 SLN448 raking strut	12.0	C LC_113	161.1	0.0	6.9	OK
Gusset SLN448 raking strut	12.0	C LC_105	236.9	0.0	6.9	OK
Flange top a end plate SLN221 ring	12.0	C LC_119	138.0	0.0	0.0	OK
Flange top b end plate SLN221 ring	12.0	C LC_135	123.2	0.0	0.0	OK
Flange bottom a end plate SLN221 ring	12.0	C LC_119	118.5	0.0	0.0	OK
Flange bottom b end plate SLN221 ring	12.0	C LC_135	139.1	0.0	0.0	OK
Flange top a end plate SLN426 ring	12.0	C LC_146	171.4	0.0	0.0	OK
Flange top b end plate SLN426 ring	12.0	C LC_146	187.6	0.0	0.0	OK
Flange bottom a end plate SLN426 ring	12.0	C LC_146	181.3	0.0	0.0	OK
Flange bottom b end plate SLN426 ring	12.0	C LC_146	161.0	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

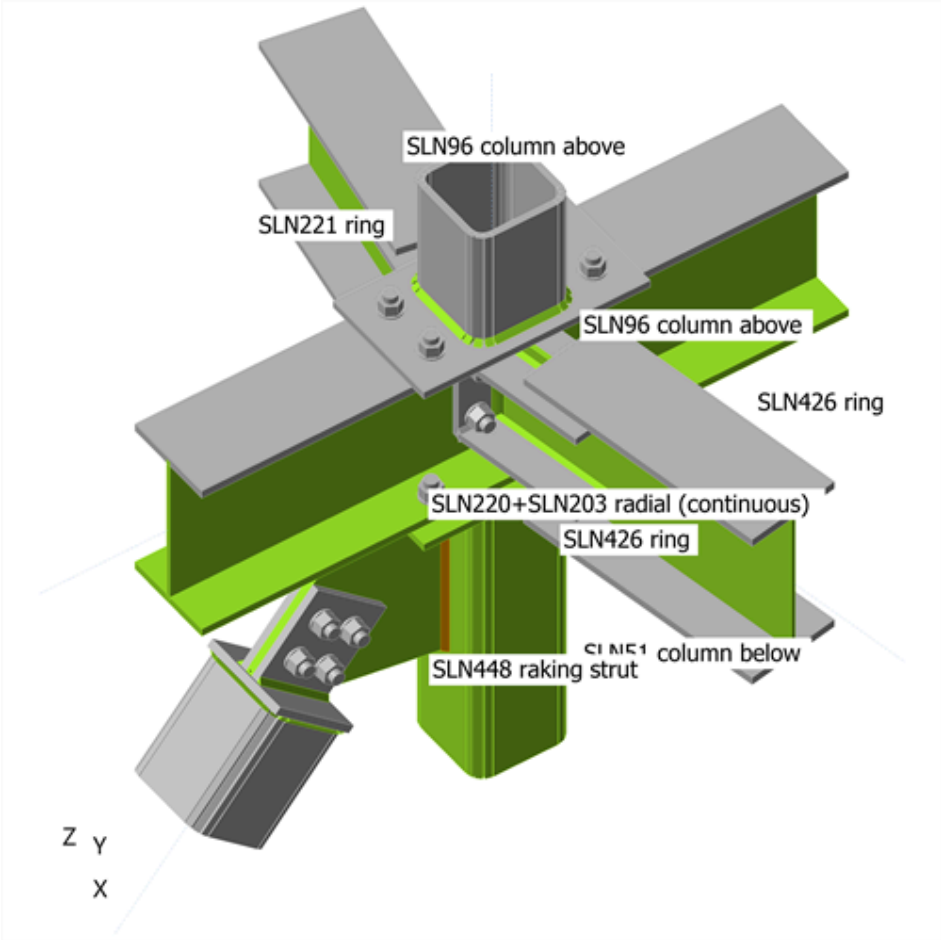
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

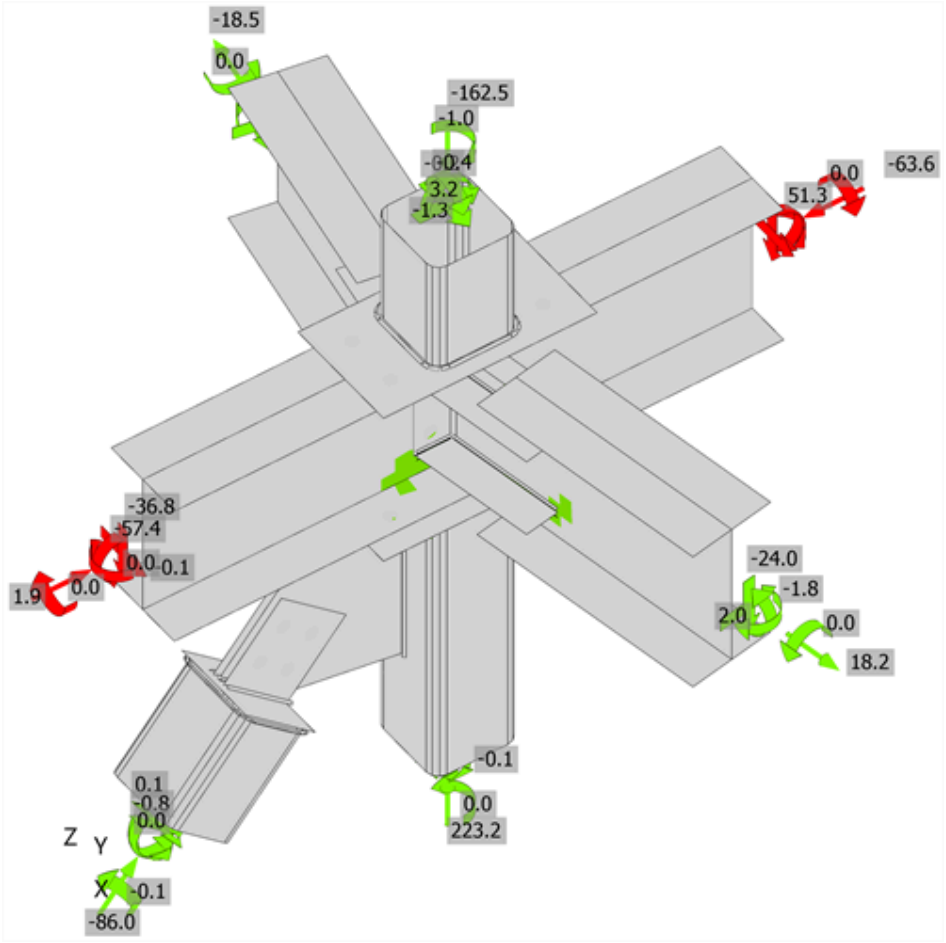
Name	d_0 [mm]	Loads	δ [mm]	$δ_{lim}$ [mm]	$δ/d_0$ [%]	Check status
SLN51 column below	150	C LC_105	0	5	0.3	OK
SLN96 column above	150	C LC_153	0	5	0.0	OK
SLN448 raking strut	150	C LC_153	0	5	0.0	OK

Symbol explanation

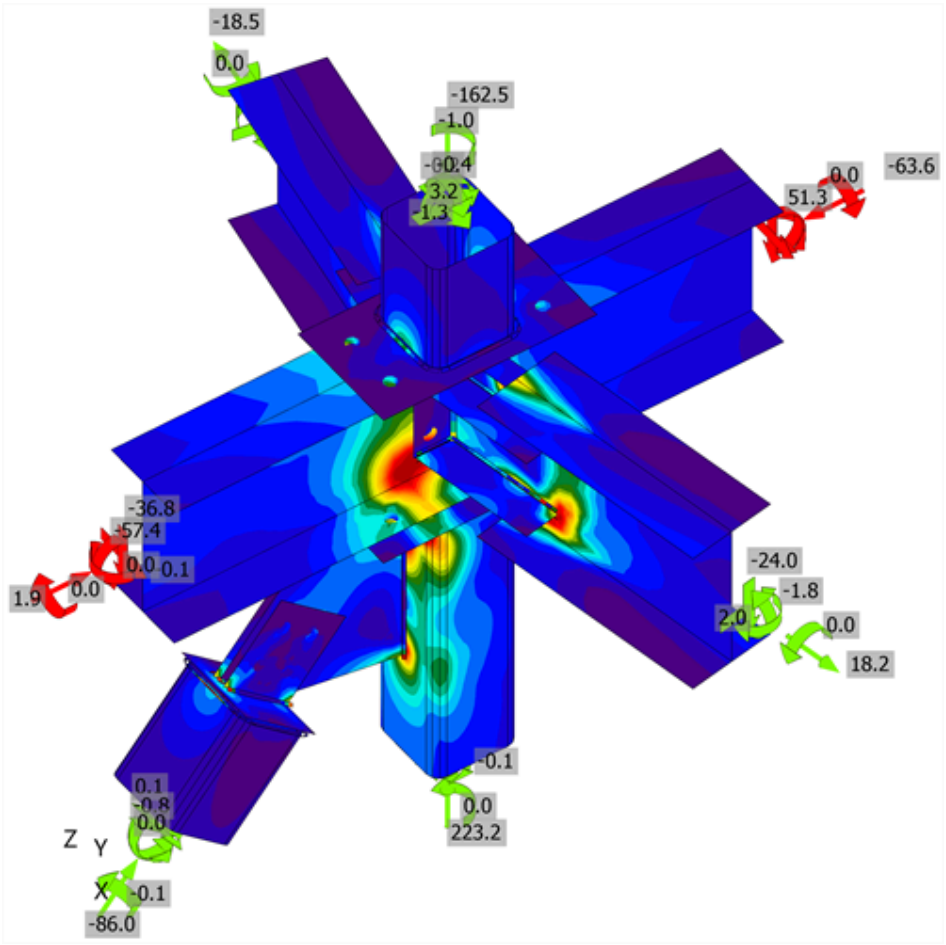
d_0	Cross-section size
δ	Local cross-section deformation
$δ_{lim}$	Allowed deformation
$δ/d_0$	Relative deformation



Overall check

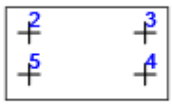
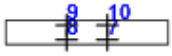
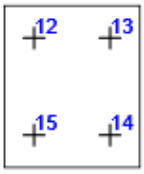
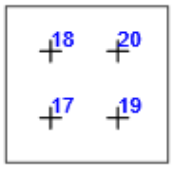


Strain check, C LC_105



Equivalent stress, C LC_105

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_105	1.3	17.3	170.0	1.0	18.4	19.1	OK	OK
	B3	M20 8.8 - 1	C LC_110	32.5	6.1	102.7	23.1	6.5	22.9	OK	OK
	B4	M20 8.8 - 1	C LC_110	36.5	5.4	103.6	25.9	5.7	24.2	OK	OK
	B5	M20 8.8 - 1	C LC_105	0.6	17.8	170.1	0.4	18.9	19.2	OK	OK
	B7	M20 8.8 - 1	C LC_151	0.9	2.4	80.3	0.6	2.9	3.0	OK	OK
	B8	M20 8.8 - 1	C LC_152	6.1	1.5	167.9	4.3	1.6	4.7	OK	OK
	B9	M20 8.8 - 1	C LC_152	6.8	3.7	79.1	4.8	4.7	7.4	OK	OK
	B10	M20 8.8 - 1	C LC_119	5.4	2.7	172.0	3.8	2.8	5.5	OK	OK
	B12	M20 8.8 - 1	C LC_152	5.5	2.0	123.8	3.9	3.3	5.0	OK	OK
	B13	M20 8.8 - 1	C LC_110	3.6	0.5	123.8	2.6	0.7	2.3	OK	OK
	B14	M20 8.8 - 1	C LC_152	1.1	2.6	123.8	0.8	4.1	3.3	OK	OK
	B15	M20 8.8 - 1	C LC_119	8.5	0.6	123.8	6.0	0.7	5.0	OK	OK
	B17	M20 8.8 - 1	C LC_119	4.0	11.6	115.3	2.9	12.4	14.4	OK	OK
	B18	M20 8.8 - 1	C LC_119	1.9	12.1	115.3	1.3	12.9	13.9	OK	OK
	B19	M20 8.8 - 1	C LC_113	0.3	10.5	115.3	0.2	17.8	11.4	OK	OK
	B20	M20 8.8 - 1	C LC_105	0.2	11.2	115.3	0.2	19.4	12.0	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN448 raking strut	SLN448 raking strut	◢ 4.2	519	C LC_105	250.8	0.0	-119.4	127.3	0.0	62.0	33.5	OK	OK
SLN51 column below-w 3	Gusset SLN448 raking strut	◢ 5.6 ◣	288	C LC_105	401.8	3.2	-133.6	-133.8	-173.1	99.3	84.8	OK	OK
		◢ 5.6 ◣	288	C LC_105	401.9	3.3	-134.3	134.6	172.4	99.3	85.6	OK	OK
SLN221 ring-w 1	Flange top b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_153	182.9	0.0	96.8	78.9	42.5	45.2	28.9	OK	OK
		◢ 4.2 ◣	197	C LC_119	70.4	0.0	-36.1	33.5	9.7	17.4	11.0	OK	OK
SLN221 ring-w 1	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_135	132.4	0.0	37.8	53.5	50.1	32.7	19.9	OK	OK
		◢ 4.2 ◣	198	C LC_135	240.8	0.0	110.6	-94.8	-79.2	59.5	38.1	OK	OK
SLN426 ring-w 1	Flange top a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_153	254.4	0.0	132.4	109.1	-61.9	62.9	32.8	OK	OK
		◢ 4.2 ◣	197	C LC_146	106.7	0.0	-54.7	51.6	-12.0	26.4	14.6	OK	OK
SLN426 ring-w 1	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_146	155.9	0.0	45.5	64.7	-56.7	38.5	21.0	OK	OK
		◢ 4.2 ◣	198	C LC_146	303.4	0.0	141.0	-121.7	96.2	75.0	41.6	OK	OK
Cap plate SLN448 raking strut	Fin plate SLN448 raking strut	◢ 5.6 ◣	139	C LC_119	250.9	0.0	-112.0	-104.4	76.8	62.0	39.6	OK	OK
		◢ 5.6 ◣	139	C LC_105	222.2	0.0	-85.3	94.2	-71.8	54.9	32.2	OK	OK
Cap plate SLN448 raking strut	Fin plate 2 SLN448 raking strut	◢ 5.6 ◣	139	C LC_105	227.7	0.0	-89.8	-97.0	72.0	56.3	32.1	OK	OK
		◢ 5.6 ◣	139	C LC_113	245.4	0.0	-109.6	102.3	-74.8	60.6	36.7	OK	OK
End plate column below SLN51 column below	SLN51 column below	◢ 5.6 ◣	521	C LC_121	396.6	0.0	-251.7	-172.8	-38.2	98.0	54.4	OK	OK
		◢ 5.6 ◣	521	C LC_135	346.2	0.0	-106.6	179.6	62.5	85.5	45.1	OK	OK
End plate column above SLN96 column above	SLN96 column above	◢ 5.6 ◣	520	C LC_152	228.7	0.0	-110.2	-115.7	-2.7	56.5	31.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_152	256.6	0.0	-129.7	127.5	8.8	63.4	36.2	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_126	206.5	0.0	15.8	5.1	-118.8	51.0	31.0	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_119	317.3	0.0	3.7	1.0	183.2	78.4	48.6	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_126	199.5	0.0	-43.5	-52.2	-99.6	49.3	28.9	OK	OK
		◢ 4.2 ◣	149	C LC_146	148.3	0.0	-41.3	43.0	70.1	36.6	22.1	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_119	281.9	0.0	-64.0	-55.2	148.6	69.6	42.8	OK	OK
		◢ 4.2 ◣	149	C LC_119	205.0	0.0	-46.7	54.0	-101.8	50.6	30.7	OK	OK
SLN221 ring-w 1	Flange top a end plate SLN221 ring	◢ 4.2 ◣	199	C LC_119	248.8	0.0	-20.6	9.4	-142.9	61.5	39.2	OK	OK
		◢ 4.2 ◣	198	C LC_135	181.2	0.0	27.1	32.6	-98.1	44.8	27.6	OK	OK
SLN221 ring-w 1	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	198	C LC_135	205.1	0.0	36.3	-39.2	109.7	50.7	30.9	OK	OK
		◢ 4.2 ◣	199	C LC_135	203.6	0.0	-25.6	-8.7	116.3	50.3	31.6	OK	OK
SLN426 ring-w 1	Flange top b end plate SLN426 ring	◢ 4.2 ◣	199	C LC_146	333.5	0.0	-36.5	20.5	190.3	82.4	50.2	OK	OK
		◢ 4.2 ◣	198	C LC_146	252.2	0.0	42.7	50.8	134.2	62.3	35.1	OK	OK
SLN426 ring-w 1	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	198	C LC_146	277.7	0.0	50.2	-53.4	-148.4	68.6	37.7	OK	OK
		◢ 4.2 ◣	198	C LC_146	294.2	0.0	-47.5	-19.2	-166.5	72.7	42.7	OK	OK
End plate SLN221 ring	Flange top a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_119	139.2	0.0	-63.1	63.8	32.6	34.4	19.0	OK	OK
		◢ 4.2 ◣	58	C LC_119	140.7	0.0	-54.5	-53.8	-52.2	34.8	20.5	OK	OK
End plate SLN221 ring	Flange top b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_119	58.8	0.0	4.4	-2.3	-33.8	14.5	11.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	85.1	0.0	-55.6	-35.7	10.5	21.0	13.9	OK	OK
End plate SLN221 ring	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_146	113.0	0.0	21.7	-40.2	49.8	27.9	19.9	OK	OK
		◢ 4.2 ◣	58	C LC_119	117.3	0.0	-49.5	-55.5	-26.3	29.0	16.9	OK	OK
End plate SLN221 ring	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_146	85.5	0.0	13.3	-47.5	-11.0	21.1	11.4	OK	OK
		◢ 4.2 ◣	58	C LC_126	82.9	0.0	-13.0	-46.8	-6.3	20.5	12.7	OK	OK
End plate SLN426 ring	Flange top a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_119	94.7	0.0	-11.3	53.7	-7.8	23.4	16.3	OK	OK
		◢ 4.2 ◣	58	C LC_119	67.1	0.0	-5.1	37.3	-10.0	16.6	12.9	OK	OK
End plate SLN426 ring	Flange top b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_146	164.9	0.0	-83.9	68.5	-45.0	40.7	26.5	OK	OK
		◢ 4.2 ◣	58	C LC_105	127.2	0.0	-38.8	-53.7	44.9	31.4	22.0	OK	OK
End plate SLN426 ring	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_119	108.8	0.0	12.6	-57.9	23.4	26.9	20.0	OK	OK
		◢ 4.2 ◣	58	C LC_119	123.6	0.0	-23.0	-68.4	15.6	30.5	20.4	OK	OK
End plate SLN426 ring	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_119	128.0	0.0	18.0	-54.0	-49.3	31.6	17.7	OK	OK
		◢ 4.2 ◣	58	C LC_121	160.4	0.0	-78.6	-67.0	45.0	39.6	27.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-000
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring

Name	SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

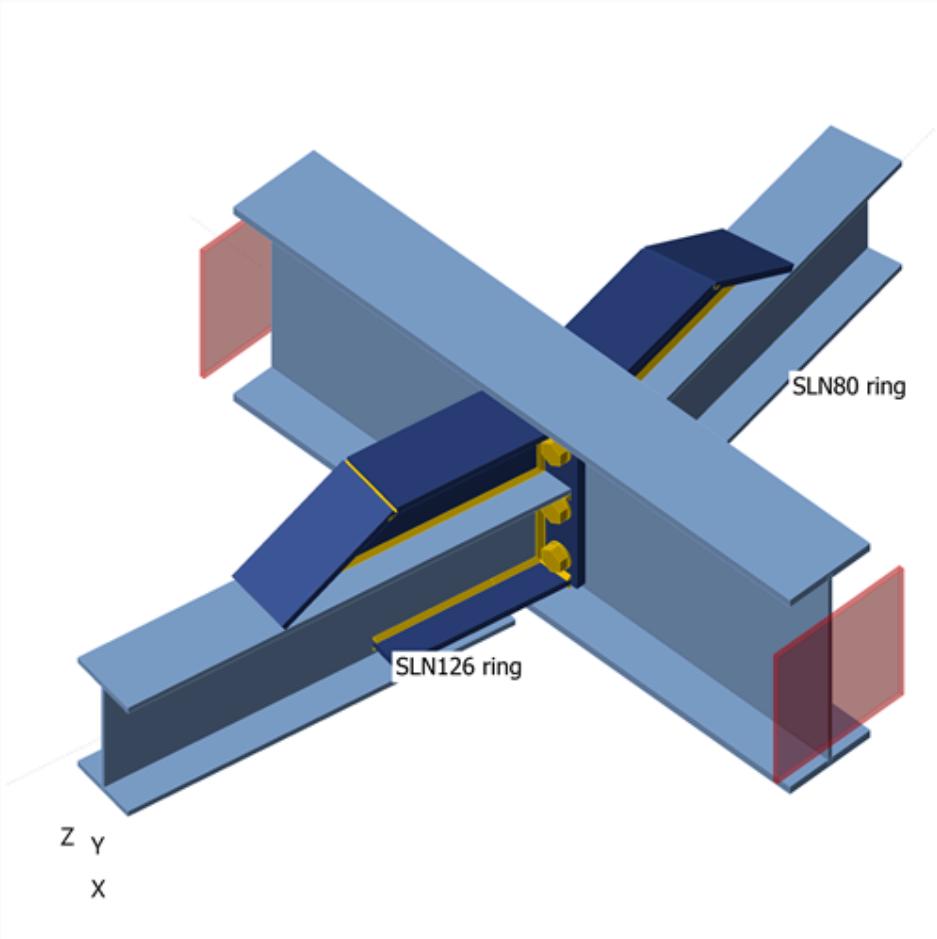
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73b+SLN73a radial (continuous)	1 - UB457x191x89	0.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN73b+SLN73a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN73b+SLN73a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN126 ring / end		Position	0



SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN80 ring / End	-16.6	-0.3	-25.6	0.0	0.0	-0.5
	SLN126 ring / End	-16.1	0.0	-25.6	0.0	0.0	-0.1
C LC_106	SLN80 ring / End	-20.2	-0.3	-24.0	0.0	0.0	-0.5
	SLN126 ring / End	-19.5	-0.2	-24.0	0.0	0.0	-0.2
C LC_114	SLN80 ring / End	-18.4	-0.4	-16.2	0.0	0.0	-0.5
	SLN126 ring / End	-17.6	-0.4	-16.2	0.0	0.0	-0.4
C LC_119	SLN80 ring / End	-17.8	-0.3	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-17.4	0.1	-25.6	0.0	0.0	0.1
C LC_122	SLN80 ring / End	-18.2	-0.3	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-17.6	-0.1	-25.6	0.0	0.0	-0.1
C LC_145	SLN80 ring / End	-17.4	-0.4	-25.6	0.0	0.0	-0.5
	SLN126 ring / End	-16.9	0.0	-25.6	0.0	0.0	0.0
C LC_148	SLN80 ring / End	-16.9	-0.3	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-16.2	0.0	-25.6	0.0	0.0	0.0
C LC_152	SLN80 ring / End	-19.9	-0.3	-14.1	0.0	0.0	-0.4
	SLN126 ring / End	-19.5	0.0	-14.2	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	19.1 < 100%	OK
Buckling	Not calculated	

Plates

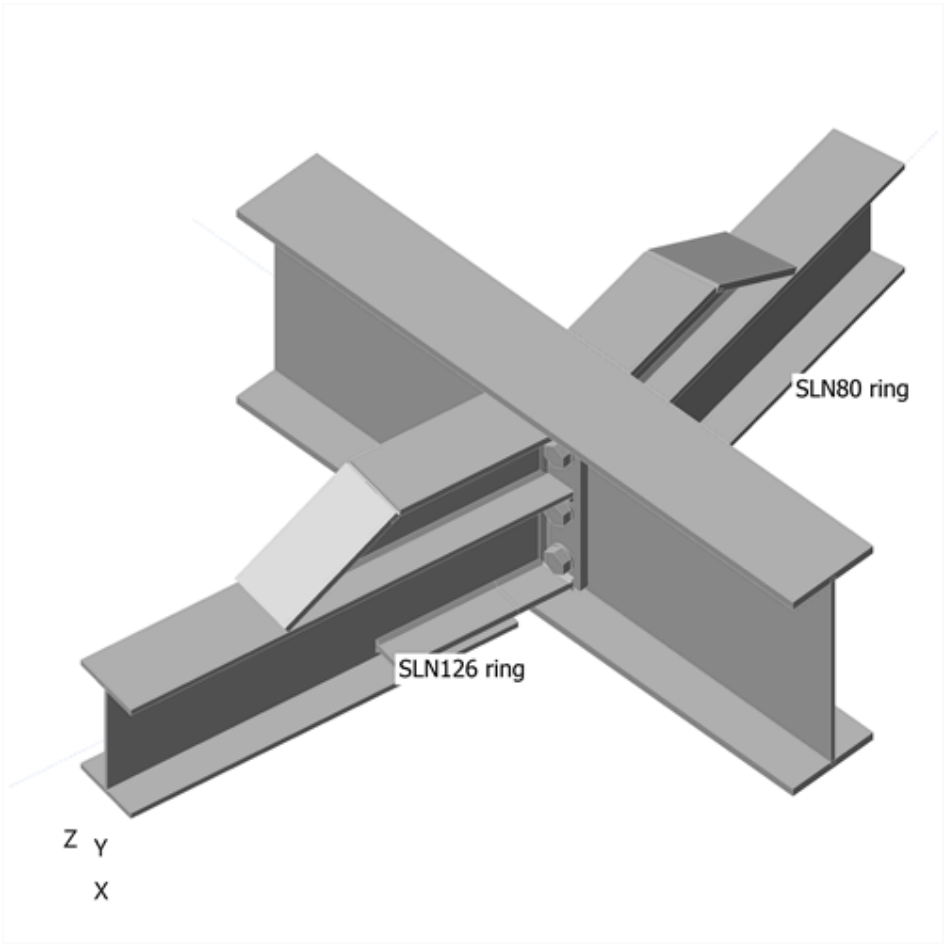
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73b+SLN73a radial (continuous)-bfl 1	17.7	C LC_148	22.6	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-tfl 1	17.7	C LC_148	18.9	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-w 1	10.5	C LC_145	27.7	0.0	1.8	OK
SLN80 ring-bfl 1	12.7	C LC_102	44.3	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_102	50.8	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_122	48.8	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_145	43.6	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_122	49.6	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_122	48.9	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_114	9.4	0.0	9.9	OK
Bottom flange plate SLN80 ring	15.0	C LC_114	22.3	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_102	8.5	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_145	8.9	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_145	33.6	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_114	6.8	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_122	15.2	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_145	6.0	0.0	7.0	OK
Bottom flange plate SLN126 ring	15.0	C LC_114	19.3	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_102	7.9	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_102	7.8	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_122	33.8	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_114	5.7	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_122	15.1	0.0	0.0	OK

Design data

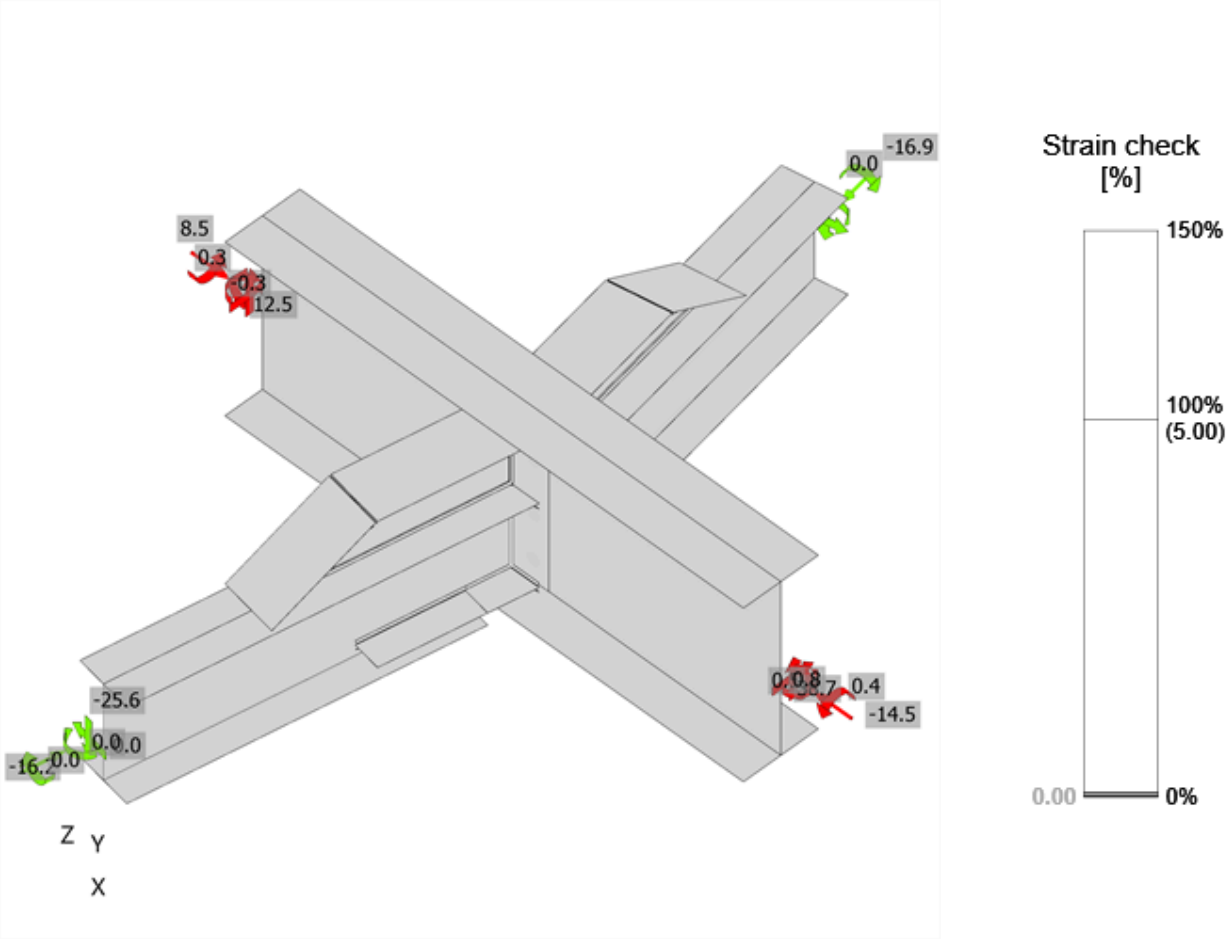
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

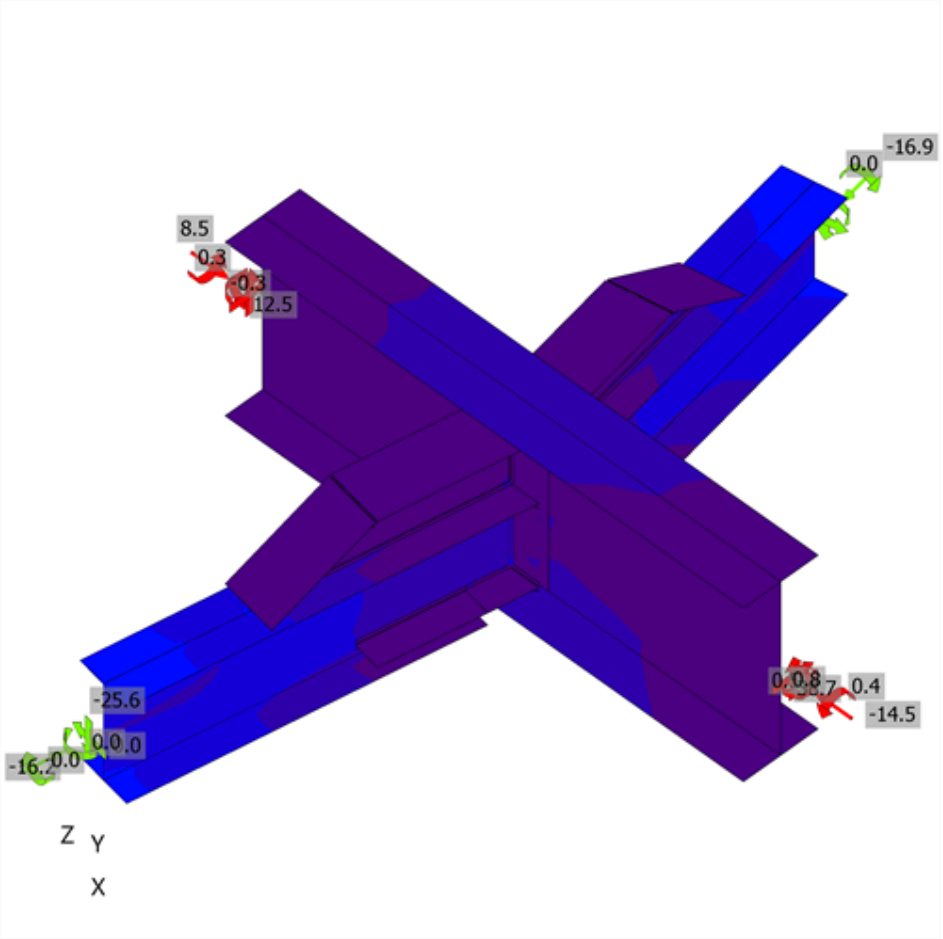
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

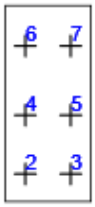


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_119	0.4	0.5	208.4	0.1	0.4	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_102	0.6	0.5	176.5	0.2	0.5	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_102	0.2	0.4	213.9	0.1	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_145	2.4	0.8	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_102	1.6	0.9	229.5	0.5	0.8	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 10.5	149	C LC_145	15.4	0.0	-6.3	6.1	-5.4	3.8	3.8	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 10.5	149	C LC_119	13.9	0.0	-7.3	6.6	1.8	3.7	3.7	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_119	9.9	0.0	0.1	-1.4	5.5	2.4	2.4	OK	OK
		◢ 6.0 ◣	208	C LC_119	7.9	0.0	-1.6	0.6	-4.4	2.0	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_119	17.8	0.0	0.6	-0.5	10.3	4.4	4.4	OK	OK
		◢ 6.0 ◣	100	C LC_145	21.8	0.0	1.7	-3.3	-12.1	5.4	5.4	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_102	8.6	0.0	-3.5	4.2	-1.7	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_102	14.2	0.0	6.8	1.0	-7.2	3.5	3.5	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_102	13.4	0.0	2.5	3.0	-7.0	3.3	3.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	16.7	0.0	4.3	-3.7	8.5	4.1	4.1	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_145	77.2	0.0	-17.4	-19.0	39.0	19.1	10.8	OK	OK
		◢ 6.0 ◣	572	C LC_122	73.0	0.0	-19.5	18.2	-36.3	18.0	10.2	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_148	8.1	0.0	-1.4	-0.5	4.6	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_119	10.1	0.0	0.0	1.4	-5.7	2.5	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_119	18.7	0.0	1.1	0.6	10.7	4.6	4.6	OK	OK
		◢ 6.0 ◣	100	C LC_102	20.2	0.0	1.2	-2.1	-11.5	5.0	5.0	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_119	9.2	0.0	2.2	2.0	4.8	2.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_102	13.7	0.0	5.8	1.5	-7.0	3.4	3.4	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_102	11.9	0.0	2.2	2.7	-6.2	2.9	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_102	14.4	0.0	3.6	-3.2	7.4	3.6	3.6	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_122	76.6	0.0	-19.3	-19.5	38.1	18.9	10.7	OK	OK
		◢ 6.0 ◣	572	C LC_119	73.2	0.0	-17.4	17.5	-37.2	18.1	10.3	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_114	27.1	0.0	-7.8	-11.1	10.0	6.7	5.8	OK	OK
		◢ 6.0 ◣	149	C LC_114	32.6	0.0	-14.1	10.7	-13.2	8.1	6.2	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_119	6.6	0.0	-0.1	-0.1	3.8	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_102	10.7	0.0	1.5	-1.4	-6.0	2.6	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_114	13.4	0.0	-6.2	-4.7	5.0	3.3	3.3	OK	OK
		◢ 6.0 ◣	149	C LC_114	10.9	0.0	-3.3	4.9	-3.5	2.7	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_145	4.7	0.0	1.7	1.6	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_145	4.3	0.0	-0.2	0.3	2.5	1.1	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_119	60.2	0.0	-10.4	-8.7	33.1	14.9	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	59.6	0.0	-10.4	12.4	-31.5	14.7	9.0	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	23.1	0.0	-9.8	-11.7	2.8	5.7	5.5	OK	OK
		◢ 6.0 ◣	149	C LC_114	29.4	0.0	-15.6	13.6	-4.7	7.3	6.0	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_119	9.9	0.0	1.2	1.0	5.6	2.4	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_145	7.9	0.0	0.2	-0.3	-4.6	2.0	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	10.0	0.0	-5.4	-4.0	2.8	2.5	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	6.9	0.0	-1.9	3.2	-2.0	1.7	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_114	3.8	0.0	1.2	1.1	1.8	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_102	3.9	0.0	0.0	0.1	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_122	61.8	0.0	-11.2	-10.7	33.4	15.3	9.3	OK	OK
		◢ 6.0 ◣	271	C LC_145	57.5	0.0	-9.5	9.8	-31.2	14.2	8.8	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_119	38.1	0.0	1.9	-4.6	-21.5	9.4	8.9	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_145	37.3	0.0	-4.2	1.9	-21.3	9.2	8.9	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_145	7.4	0.0	-5.5	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_145	3.4	0.0	-0.8	1.9	0.4	0.9	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_145	7.2	0.0	-5.0	3.0	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.0	0.0	-0.5	1.7	-0.2	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-020
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring

Name	SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

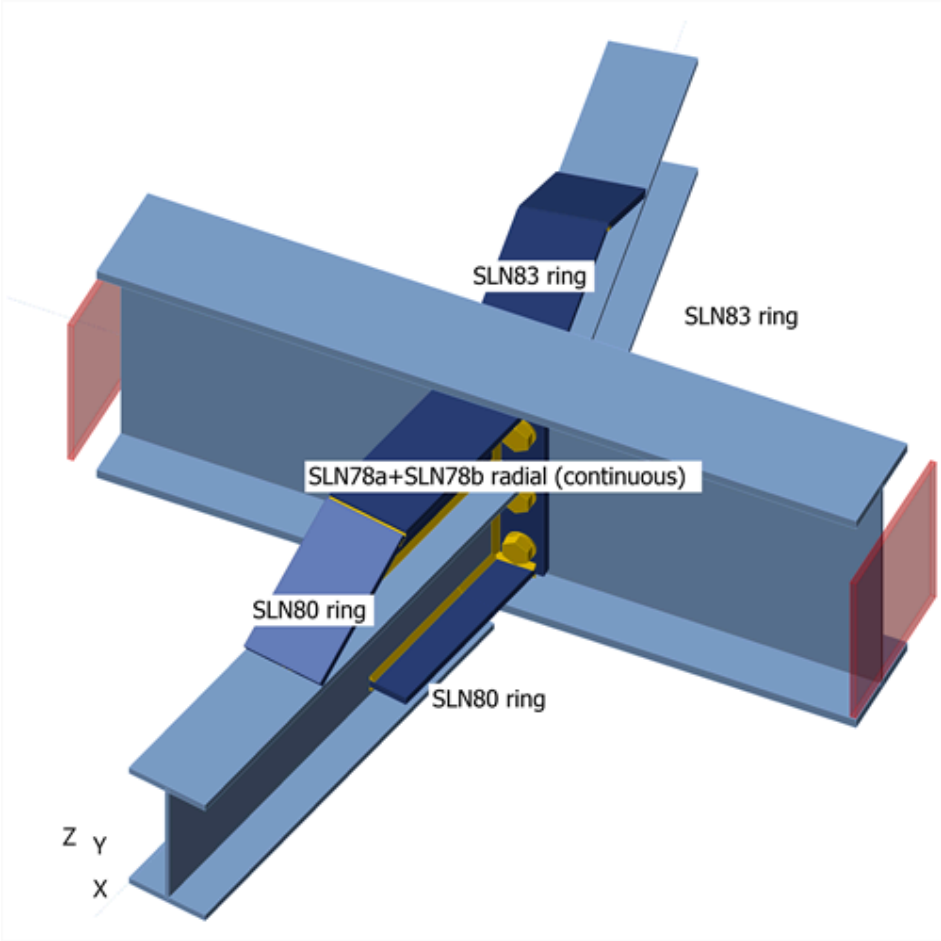
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN78a+SLN78b radial (continuous)	1 - UB457x191x89	20.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102
SLN83 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN78a+SLN78b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN78a+SLN78b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN83 ring / end		Position	0



SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN80 ring / End	-16.6	-0.3	-25.6	0.0	0.0	-0.5
	SLN83 ring / End	-17.3	-0.2	-25.6	0.0	0.0	-0.2
C LC_106	SLN80 ring / End	-20.4	-0.3	-24.0	0.0	0.0	-0.5
	SLN83 ring / End	-21.6	-0.3	-24.0	0.0	0.0	-0.4
C LC_114	SLN80 ring / End	-18.5	-0.4	-16.2	0.0	0.0	-0.6
	SLN83 ring / End	-20.1	-0.4	-16.2	0.0	0.0	-0.6
C LC_120	SLN80 ring / End	-18.0	-0.3	-25.6	0.0	0.0	-0.3
	SLN83 ring / End	-18.6	0.0	-25.6	0.0	0.0	0.0
C LC_145	SLN80 ring / End	-17.4	-0.2	-25.6	0.0	0.0	-0.4
	SLN83 ring / End	-18.0	-0.3	-25.6	0.0	0.0	-0.3
C LC_146	SLN80 ring / End	-15.9	-0.4	-25.6	0.0	0.0	-0.5
	SLN83 ring / End	-16.7	-0.1	-25.6	0.0	0.0	-0.2
C LC_148	SLN80 ring / End	-17.1	-0.3	-25.6	0.0	0.0	-0.4
	SLN83 ring / End	-18.0	-0.2	-25.6	0.0	0.0	-0.2
C LC_152	SLN80 ring / End	-19.9	-0.3	-14.2	0.0	0.0	-0.4
	SLN83 ring / End	-20.5	-0.1	-14.0	0.0	0.0	-0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	18.9 < 100%	OK
Buckling	Not calculated	

Plates

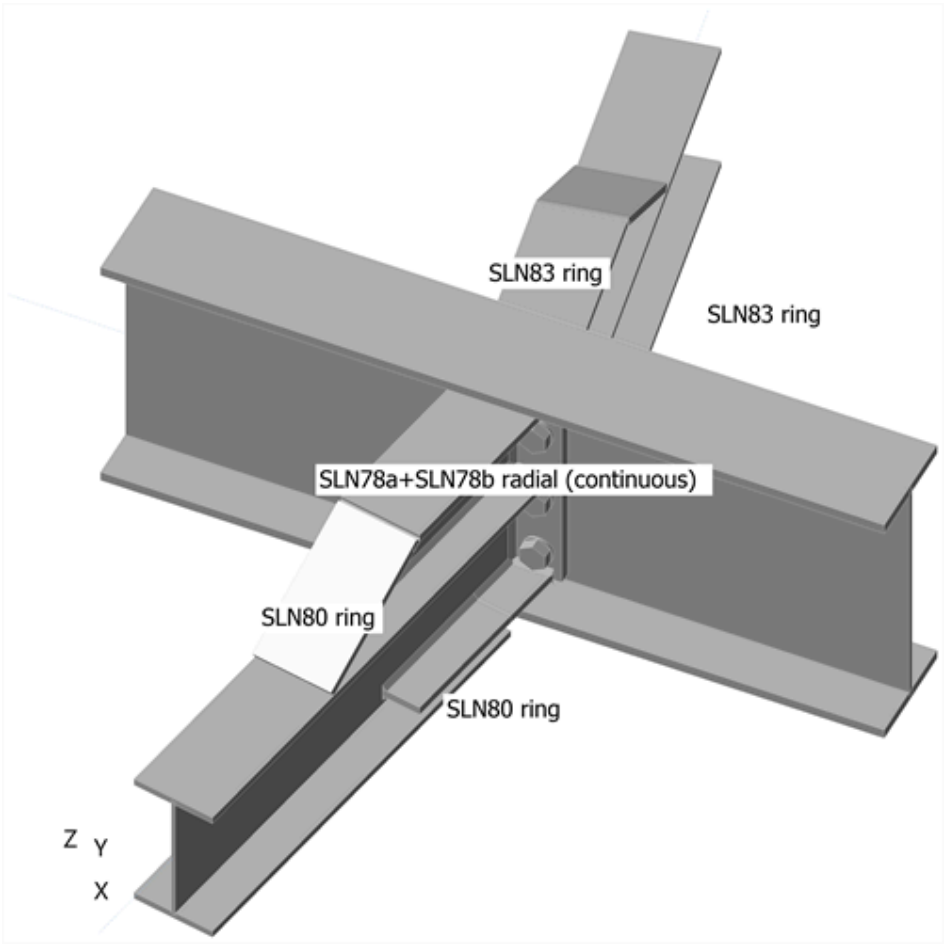
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN78a+SLN78b radial (continuous)-bfl 1	17.7	C LC_106	23.0	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-tfl 1	17.7	C LC_148	19.4	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-w 1	10.5	C LC_145	27.2	0.0	1.8	OK
SLN80 ring-bfl 1	12.7	C LC_145	44.7	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_145	51.4	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_120	48.8	0.0	0.0	OK
SLN83 ring-bfl 1	12.7	C LC_146	43.8	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_120	49.7	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_120	48.8	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_114	11.4	0.0	12.2	OK
Bottom flange plate SLN83 ring	15.0	C LC_114	27.0	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_114	8.7	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_145	7.7	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_145	33.4	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_114	9.6	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_120	15.2	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_146	7.0	0.0	9.1	OK
Bottom flange plate SLN80 ring	15.0	C LC_114	22.5	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_146	9.1	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_146	9.2	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_145	33.5	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_114	7.1	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_120	15.1	0.0	0.0	OK

Design data

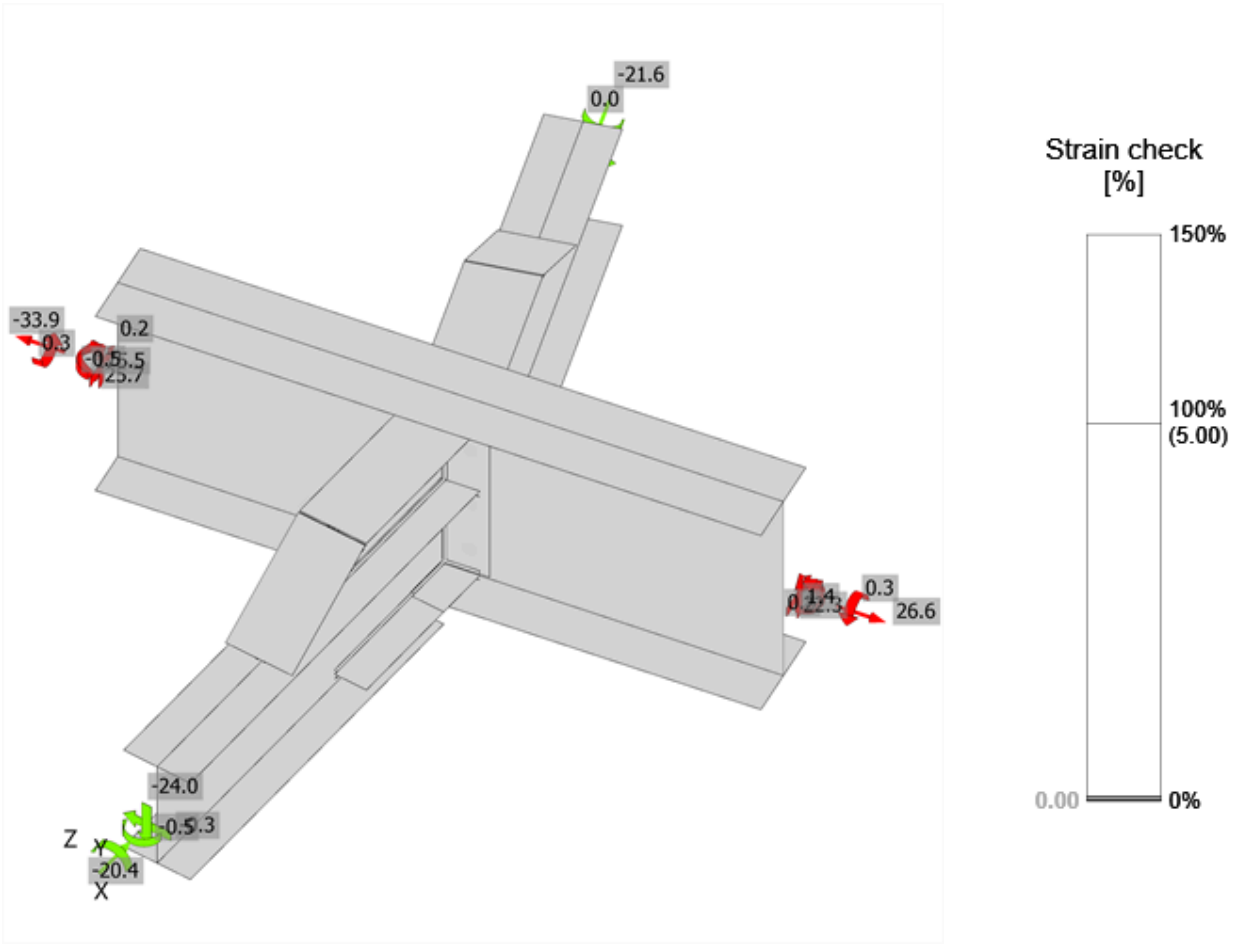
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

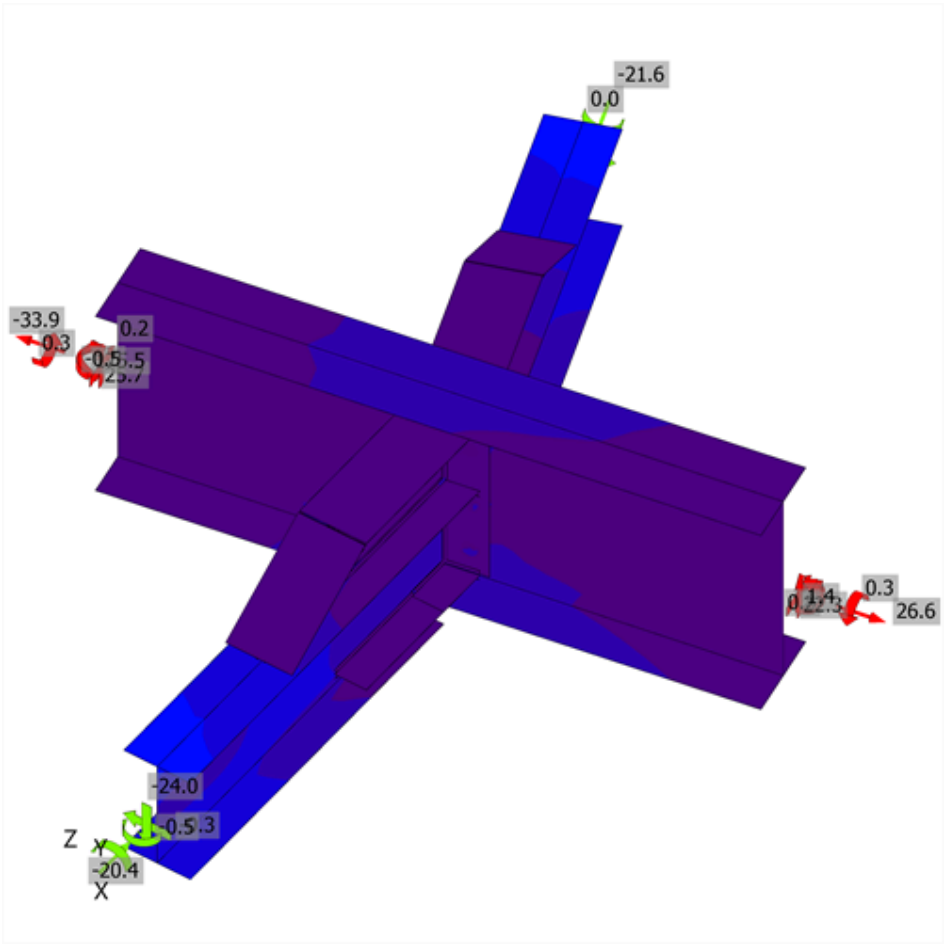
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

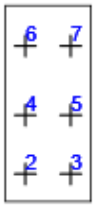


Strain check, C LC_106



Equivalent stress, C LC_106

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_145	0.5	0.4	208.4	0.1	0.4	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_146	1.0	0.5	176.5	0.3	0.5	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_120	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_146	0.2	0.5	213.9	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_145	2.1	0.8	229.5	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_146	2.3	0.9	229.5	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 10.5	149	C LC_146	15.0	0.0	-7.6	7.0	-2.6	3.7	3.7	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-tfl 1	◢ 10.5	149	C LC_120	11.7	0.0	-6.8	5.5	0.2	3.5	3.5	OK	OK
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	208	C LC_120	10.3	0.0	-0.1	-1.7	5.7	2.5	2.5	OK	OK
		◢ 6.0 ◣	208	C LC_120	7.3	0.0	-1.4	0.2	-4.1	1.8	0.0	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_120	18.3	0.0	0.8	0.3	10.6	4.5	4.5	OK	OK
		◢ 6.0 ◣	100	C LC_145	21.3	0.0	1.3	-2.6	-12.0	5.3	5.3	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_145	8.3	0.0	-3.5	3.9	-1.9	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	13.9	0.0	5.9	1.5	-7.1	3.4	3.4	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_145	12.3	0.0	2.3	2.7	-6.4	3.0	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_145	14.7	0.0	3.6	-3.2	7.6	3.6	3.6	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_145	75.9	0.0	-17.9	-18.6	38.3	18.8	10.7	OK	OK
		◢ 6.0 ◣	572	C LC_120	75.0	0.0	-18.6	18.6	-37.6	18.5	10.5	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_146	7.7	0.0	-1.1	-0.1	4.4	1.9	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_120	10.2	0.0	0.0	1.5	-5.7	2.5	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_146	17.6	0.0	0.5	-1.2	10.1	4.4	4.4	OK	OK
		◢ 6.0 ◣	100	C LC_146	22.3	0.0	1.6	-3.4	-12.4	5.5	5.5	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_146	8.9	0.0	-3.7	4.3	-1.8	2.2	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	14.7	0.0	7.2	0.8	-7.3	3.6	3.6	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_146	14.3	0.0	2.7	3.2	-7.5	3.5	3.5	OK	OK
		◢ 6.0 ◣	243	C LC_146	17.5	0.0	4.4	-3.9	9.0	4.3	4.3	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_145	76.4	0.0	-17.2	-18.6	38.8	18.9	10.7	OK	OK
		◢ 6.0 ◣	572	C LC_120	73.8	0.0	-19.3	18.4	-36.8	18.2	10.3	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	33.1	0.0	-9.5	-13.7	12.2	8.2	6.2	OK	OK
		◢ 6.0 ◣	149	C LC_114	39.6	0.0	-17.2	13.0	-16.0	9.8	6.7	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_120	6.4	0.0	0.3	0.2	3.7	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_145	10.9	0.0	1.3	-1.4	-6.1	2.7	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	17.3	0.0	-8.4	-5.7	6.6	4.3	4.3	OK	OK
		◢ 6.0 ◣	149	C LC_114	12.9	0.0	-3.5	6.1	-3.8	3.2	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_114	4.7	0.0	1.5	1.2	2.2	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_145	4.2	0.0	0.0	0.1	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_120	59.8	0.0	-10.3	-10.3	32.4	14.8	9.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_148	60.0	0.0	-10.5	11.2	-32.2	14.8	9.1	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_114	25.9	0.0	-10.7	-13.2	3.4	6.4	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_114	33.6	0.0	-18.0	15.5	-5.4	8.3	6.2	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_120	10.0	0.0	1.0	0.9	5.7	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_145	7.6	0.0	0.6	-0.6	-4.3	1.9	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_114	13.6	0.0	-7.0	-5.6	3.7	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_114	10.5	0.0	-3.4	4.8	-3.2	2.6	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_146	4.8	0.0	1.8	1.6	2.1	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_146	4.2	0.0	-0.3	0.3	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_120	59.5	0.0	-10.1	-8.8	32.7	14.7	9.1	OK	OK
		◢ 6.0 ◣	271	C LC_145	60.2	0.0	-10.7	12.8	-31.7	14.9	9.0	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_120	38.4	0.0	1.7	-4.5	-21.7	9.5	9.0	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_120	37.8	0.0	1.2	4.1	21.4	9.3	9.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_145	7.5	0.0	-5.4	3.0	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.8	0.0	-0.7	2.1	-0.2	0.9	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_145	7.5	0.0	-5.5	2.9	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	4.0	0.0	-0.7	2.2	-0.5	1.0	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-040
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN81b+SLN81a radial (continuous) - SLN83 ring - SLN84 ring

Name	SLN81b+SLN81a radial (continuous) - SLN83 ring - SLN84 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

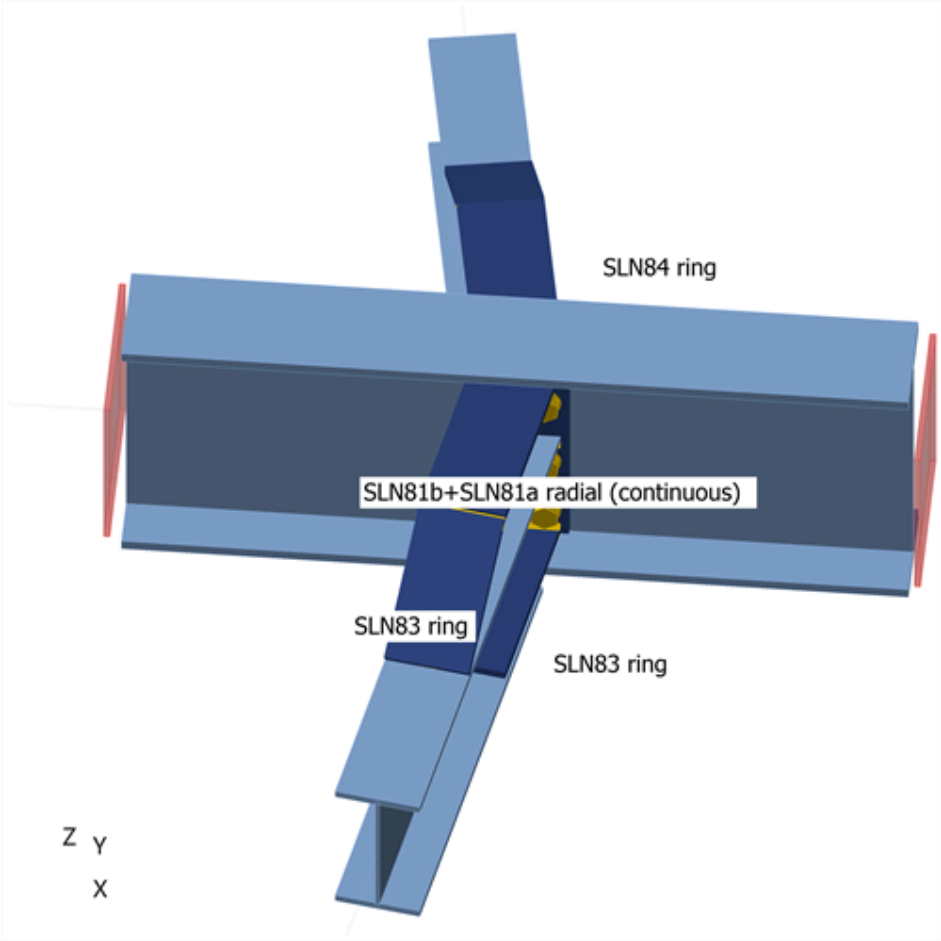
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN81b+SLN81a radial (continuous)	1 - UB457x191x89	40.0	0.0	0.0	0	0	0
SLN83 ring	2 - UB254x146x43	-60.0	0.0	0.0	0	0	-102
SLN84 ring	2 - UB254x146x43	140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN81b+SLN81a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN81b+SLN81a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN83 ring / end		Position	0
SLN84 ring / end		Position	0



SLN81b+SLN81a radial (continuous) - SLN83 ring - SLN84 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN83 ring / End	-17.3	-0.2	-25.6	0.0	0.0	-0.3
	SLN84 ring / End	-18.6	-0.4	-8.9	0.0	0.0	-0.6
C LC_106	SLN83 ring / End	-21.8	-0.2	-24.0	0.0	0.0	-0.3
	SLN84 ring / End	-23.1	-0.4	-8.4	0.0	0.0	-0.6
C LC_114	SLN83 ring / End	-20.2	-0.3	-16.2	0.0	0.0	-0.5
	SLN84 ring / End	-21.6	-0.4	-5.8	0.0	0.0	-0.7
C LC_122	SLN83 ring / End	-19.3	-0.1	-25.6	0.0	0.0	-0.2
	SLN84 ring / End	-20.6	-0.4	-8.9	0.0	0.0	-0.6
C LC_145	SLN83 ring / End	-17.9	-0.1	-25.6	0.0	0.0	-0.3
	SLN84 ring / End	-19.1	-0.5	-8.9	0.0	0.0	-0.7
C LC_146	SLN83 ring / End	-16.7	-0.2	-25.6	0.0	0.0	-0.3
	SLN84 ring / End	-18.2	-0.4	-8.9	0.0	0.0	-0.6
C LC_147	SLN83 ring / End	-16.6	-0.2	-25.6	0.0	0.0	-0.4
	SLN84 ring / End	-17.8	-0.4	-8.9	0.0	0.0	-0.7
C LC_148	SLN83 ring / End	-18.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN84 ring / End	-19.5	-0.4	-8.9	0.0	0.0	-0.6
C LC_154	SLN83 ring / End	-15.6	-0.2	-11.6	0.0	0.0	-0.4
	SLN84 ring / End	-16.9	-0.4	-8.9	0.0	0.0	-0.6

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.7 < 100%	OK
Welds	18.9 < 100%	OK
Buckling	Not calculated	

Plates

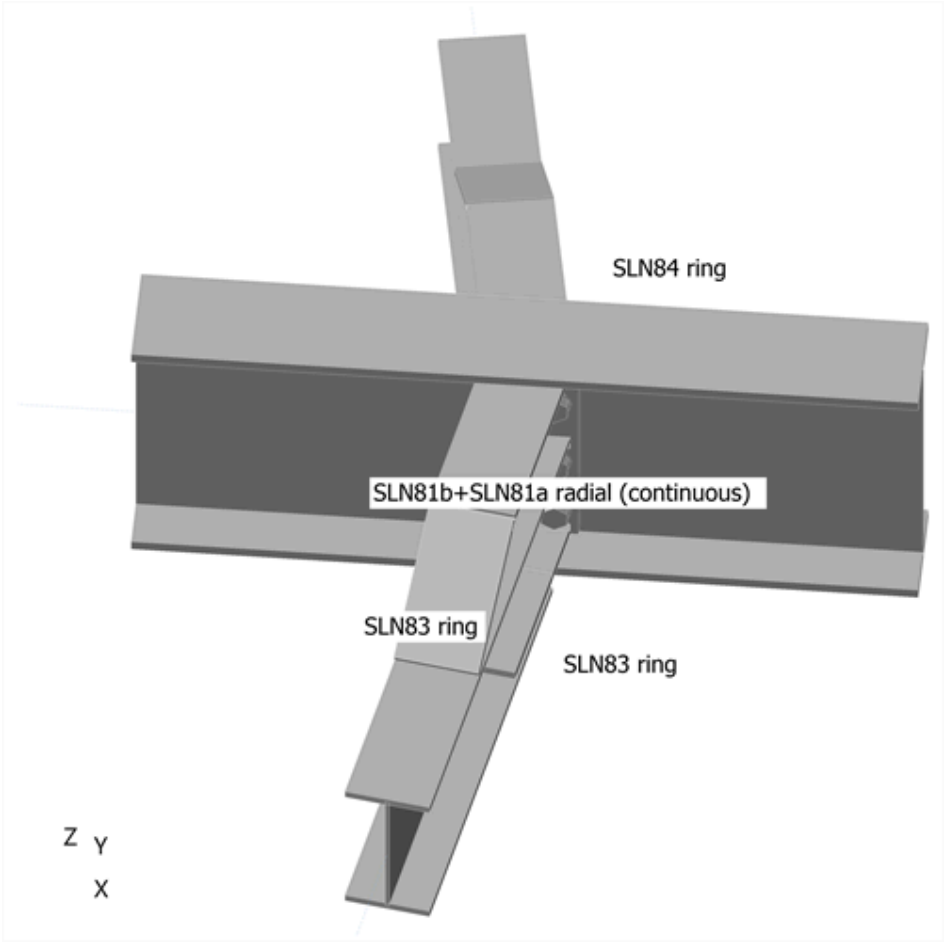
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN81b+SLN81a radial (continuous)-bfl 1	17.7	C LC_146	19.1	0.0	0.0	OK
SLN81b+SLN81a radial (continuous)-tfl 1	17.7	C LC_146	15.0	0.0	0.0	OK
SLN81b+SLN81a radial (continuous)-w 1	10.5	C LC_114	21.8	0.0	2.7	OK
SLN83 ring-bfl 1	12.7	C LC_145	44.4	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_145	50.9	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_122	49.0	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_147	14.9	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_145	22.1	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_122	19.5	0.0	0.0	OK
End plate SLN84 ring	20.0	C LC_114	11.5	0.0	12.4	OK
Bottom flange plate SLN84 ring	15.0	C LC_114	26.3	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_114	10.2	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_114	8.0	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_145	15.1	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_114	10.1	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_122	6.4	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_114	6.0	0.0	13.8	OK
Bottom flange plate SLN83 ring	15.0	C LC_114	22.3	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_147	8.7	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_147	8.8	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_122	33.6	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_114	7.7	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_122	15.2	0.0	0.0	OK

Design data

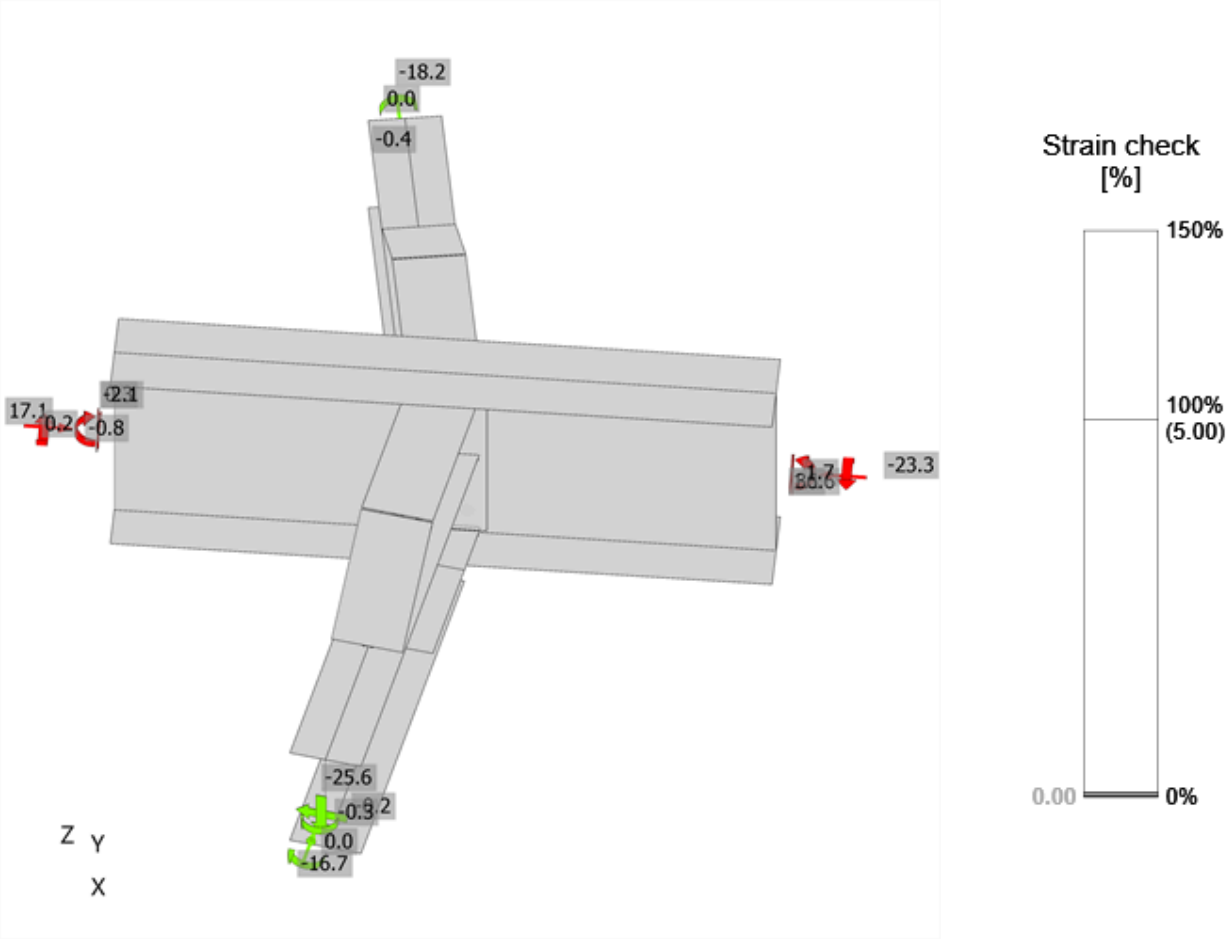
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

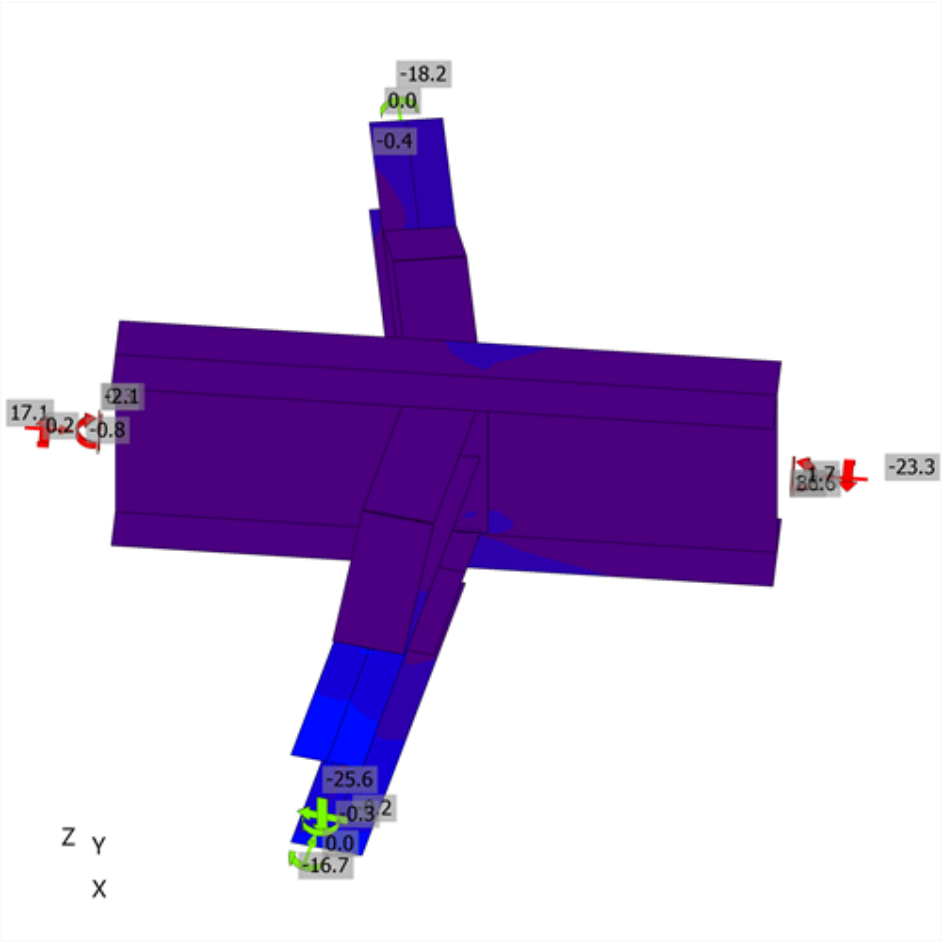
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

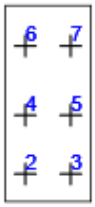


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_145	0.6	0.3	208.4	0.2	0.3	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_147	0.8	0.3	176.5	0.3	0.3	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_122	0.3	0.3	213.9	0.1	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.3	0.3	213.9	0.1	0.3	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_147	2.1	0.5	229.5	0.7	0.4	0.7	OK	OK
	B7	M30 8.8 - 1	C LC_147	1.6	0.6	270.9	0.5	0.4	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN81b+SLN81a radial (continuous)-w 1	SLN83 ring-tfl 1	◢ 10.5	149	C LC_122	10.1	0.0	-5.5	4.7	1.3	2.9	2.9	OK	OK
SLN81b+SLN81a radial (continuous)-w 1	SLN84 ring-tfl 1	◢ 10.5	149	C LC_145	13.2	0.0	-5.0	5.0	-5.0	3.3	3.3	OK	OK
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	208	C LC_106	8.5	0.0	-3.0	2.3	4.0	2.1	2.1	OK	OK
		◢ 6.0 ◣	208	C LC_122	4.8	0.0	-1.4	0.9	-2.5	1.2	0.0	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_145	7.5	0.0	-1.3	-3.7	2.2	1.8	0.0	OK	OK
		◢ 6.0 ◣	100	C LC_147	10.8	0.0	0.5	-2.9	-5.5	2.7	2.7	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_114	11.0	0.0	-2.3	-2.9	-5.5	2.7	2.7	OK	OK
		◢ 6.0 ◣	243	C LC_114	15.1	0.0	-4.6	4.0	7.3	3.7	3.7	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	243	C LC_114	5.7	0.0	0.2	-2.1	2.6	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_114	9.8	0.0	-5.0	2.8	-4.0	2.4	0.0	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_145	33.6	0.0	-5.3	-7.6	17.6	8.3	6.6	OK	OK
		◢ 6.0 ◣	572	C LC_122	29.4	0.0	-9.6	7.6	-14.1	7.3	6.2	OK	OK
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	208	C LC_147	5.6	0.0	-0.8	-0.5	3.2	1.4	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	7.0	0.0	-0.1	0.8	-3.9	1.7	0.0	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_146	18.0	0.0	0.6	-0.8	10.3	4.4	4.4	OK	OK
		◢ 6.0 ◣	100	C LC_147	22.4	0.0	1.4	-3.0	-12.5	5.5	5.5	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_147	8.9	0.0	-3.8	4.2	-2.0	2.2	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_147	14.2	0.0	7.0	1.0	-7.1	3.5	3.5	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_147	13.8	0.0	2.6	3.1	-7.2	3.4	3.4	OK	OK
		◢ 6.0 ◣	243	C LC_147	16.7	0.0	4.2	-3.7	8.6	4.1	4.1	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_122	76.6	0.0	-18.0	-18.7	38.7	18.9	10.7	OK	OK
		◢ 6.0 ◣	572	C LC_122	74.7	0.0	-19.5	18.8	-37.1	18.5	10.4	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	149	C LC_114	35.3	0.0	-10.6	-14.5	13.0	8.7	6.4	OK	OK
		◢ 6.0 ◣	149	C LC_114	40.1	0.0	-17.2	13.3	-16.1	9.9	6.7	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_147	3.2	0.0	-1.0	-0.7	-1.7	0.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	6.0	0.0	1.3	-1.1	-3.2	1.5	0.0	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_114	18.4	0.0	-8.8	-6.4	6.8	4.6	4.6	OK	OK
		◢ 6.0 ◣	149	C LC_114	14.4	0.0	-4.3	6.6	-4.3	3.6	3.6	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_114	3.8	0.0	1.1	0.8	1.9	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_114	3.1	0.0	-0.3	0.5	1.7	0.8	0.0	OK	OK
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_145	24.8	0.0	-3.5	-0.1	14.2	6.1	6.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	26.3	0.0	-4.9	7.7	-12.8	6.5	6.1	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	26.1	0.0	-10.8	-13.3	3.2	6.4	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_114	33.5	0.0	-18.0	15.4	-5.3	8.3	6.2	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_122	8.2	0.0	0.9	0.7	4.7	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	7.5	0.0	0.5	-0.5	-4.3	1.8	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	14.0	0.0	-7.4	-5.7	3.8	3.5	3.5	OK	OK
		◢ 6.0 ◣	149	C LC_114	10.2	0.0	-3.1	4.8	-2.9	2.5	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_147	5.1	0.0	1.7	1.5	2.3	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_147	4.3	0.0	-0.1	0.2	2.5	1.1	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_122	59.9	0.0	-10.2	-9.0	32.9	14.8	9.1	OK	OK
		◢ 6.0 ◣	271	C LC_122	60.5	0.0	-10.7	11.9	-32.3	15.0	9.1	OK	OK
SLN81b+SLN81a radial (continuous)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_106	26.2	0.0	0.8	-10.0	-11.3	6.5	6.4	OK	OK
SLN81b+SLN81a radial (continuous)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_122	32.3	0.0	2.4	5.0	17.9	8.0	8.0	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_114	5.0	0.0	-4.5	-1.2	0.0	1.5	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.6	0.0	-0.7	2.0	-0.5	0.9	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_147	7.9	0.0	-5.8	3.1	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.8	0.0	-0.7	2.1	-0.4	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-100
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring

Name	SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

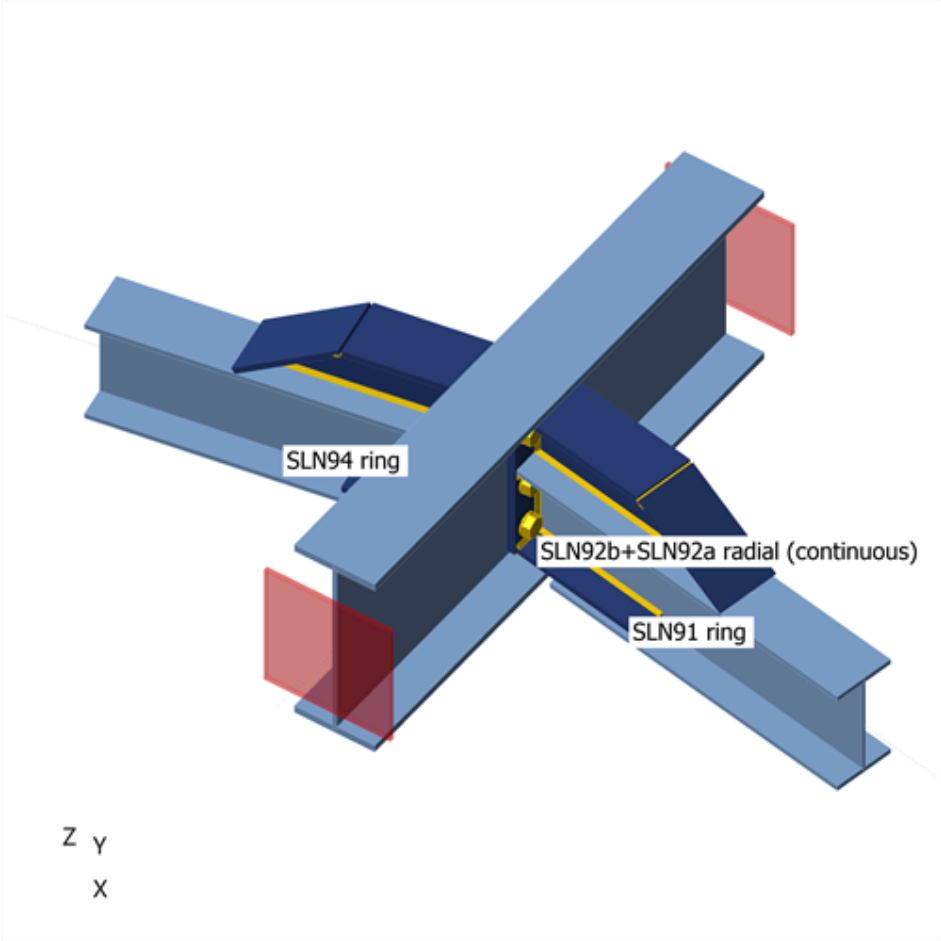
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN92b+SLN92a radial (continuous)	1 - UB457x191x89	100.0	0.0	0.0	0	0	0
SLN91 ring	2 - UB254x146x43	0.0	0.0	0.0	0	0	-102
SLN94 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN92b+SLN92a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN92b+SLN92a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN91 ring / end		Position	0
SLN94 ring / end		Position	0



SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN91 ring / End	33.4	1.3	-25.6	0.0	0.0	1.0
	SLN94 ring / End	33.2	-0.1	-25.7	0.0	0.0	-0.3
C LC_106	SLN91 ring / End	29.4	1.3	-24.0	0.0	0.0	0.9
	SLN94 ring / End	28.9	-0.4	-24.1	0.0	0.0	-0.6
C LC_120	SLN91 ring / End	32.1	1.4	-25.6	0.0	0.0	1.2
	SLN94 ring / End	31.9	-0.1	-25.7	0.0	0.0	-0.2
C LC_121	SLN91 ring / End	31.3	1.2	-25.6	0.0	0.0	1.0
	SLN94 ring / End	31.3	0.0	-25.7	0.0	0.0	-0.1
C LC_122	SLN91 ring / End	32.2	1.3	-25.6	0.0	0.0	0.9
	SLN94 ring / End	31.7	-0.4	-25.7	0.0	0.0	-0.6
C LC_126	SLN91 ring / End	27.2	1.0	-19.4	0.0	0.0	0.6
	SLN94 ring / End	26.5	-0.4	-19.5	0.0	0.0	-0.7
C LC_135	SLN91 ring / End	31.3	1.1	-25.6	0.0	0.0	0.9
	SLN94 ring / End	31.1	-0.1	-25.7	0.0	0.0	-0.2
C LC_146	SLN91 ring / End	33.5	1.3	-25.6	0.0	0.0	1.1
	SLN94 ring / End	33.1	-0.1	-25.7	0.0	0.0	-0.3
C LC_147	SLN91 ring / End	34.4	1.2	-25.6	0.0	0.0	1.0
	SLN94 ring / End	34.2	0.0	-25.7	0.0	0.0	-0.2
C LC_148	SLN91 ring / End	32.5	1.4	-25.6	0.0	0.0	1.1
	SLN94 ring / End	32.3	-0.2	-25.7	0.0	0.0	-0.4
C LC_152	SLN91 ring / End	34.5	1.4	-14.1	0.0	0.0	1.1
	SLN94 ring / End	34.4	-0.2	-14.0	0.0	0.0	-0.4

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	2.1 < 100%	OK
Welds	29.4 < 100%	OK
Buckling	Not calculated	

Plates

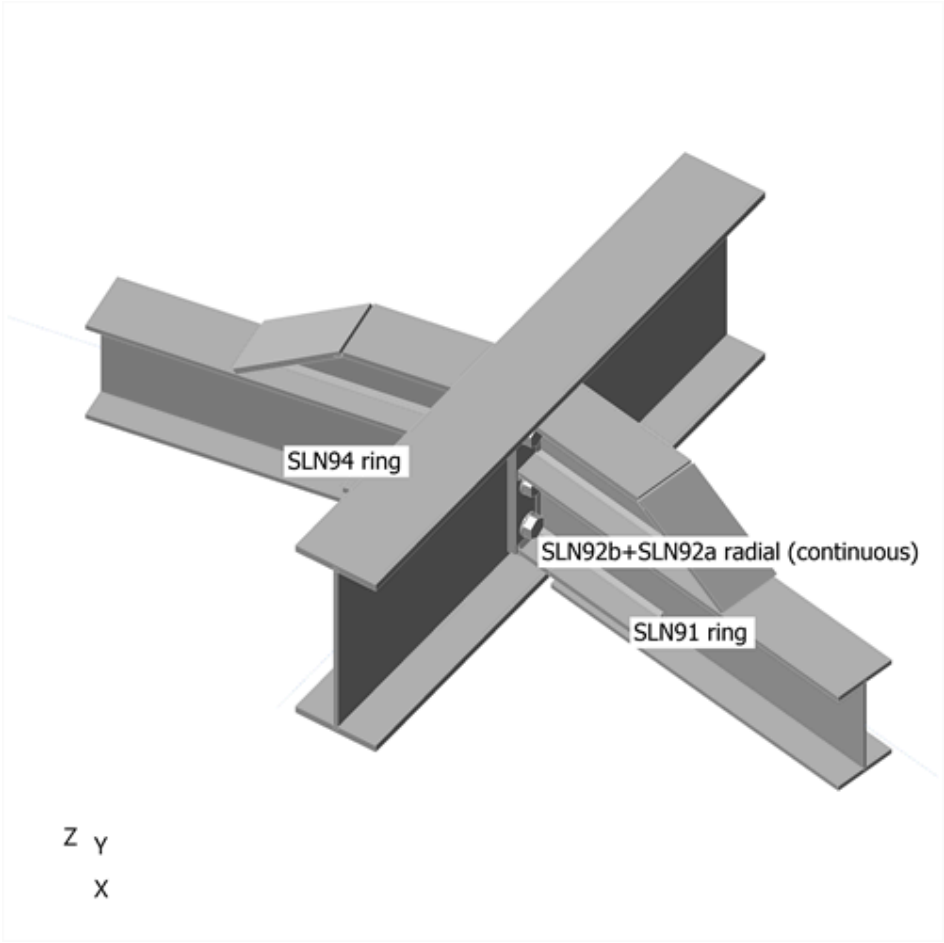
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN92b+SLN92a radial (continuous)-bfl 1	17.7	C LC_120	23.0	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-tfl 1	17.7	C LC_120	20.1	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-w 1	10.5	C LC_147	30.6	0.0	1.5	OK
SLN91 ring-bfl 1	12.7	C LC_122	55.4	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_122	41.9	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	51.4	0.0	0.0	OK
SLN94 ring-bfl 1	12.7	C LC_147	53.8	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_122	43.3	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	51.8	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_120	20.4	0.0	8.3	OK
Bottom flange plate SLN94 ring	15.0	C LC_122	30.5	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	25.0	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	24.7	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_120	29.1	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_120	7.9	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_122	12.0	0.0	0.0	OK
End plate SLN91 ring	20.0	C LC_120	23.0	0.0	11.9	OK
Bottom flange plate SLN91 ring	15.0	C LC_120	35.9	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_120	25.3	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_147	25.8	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_120	27.9	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_120	10.3	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_120	12.0	0.0	0.0	OK

Design data

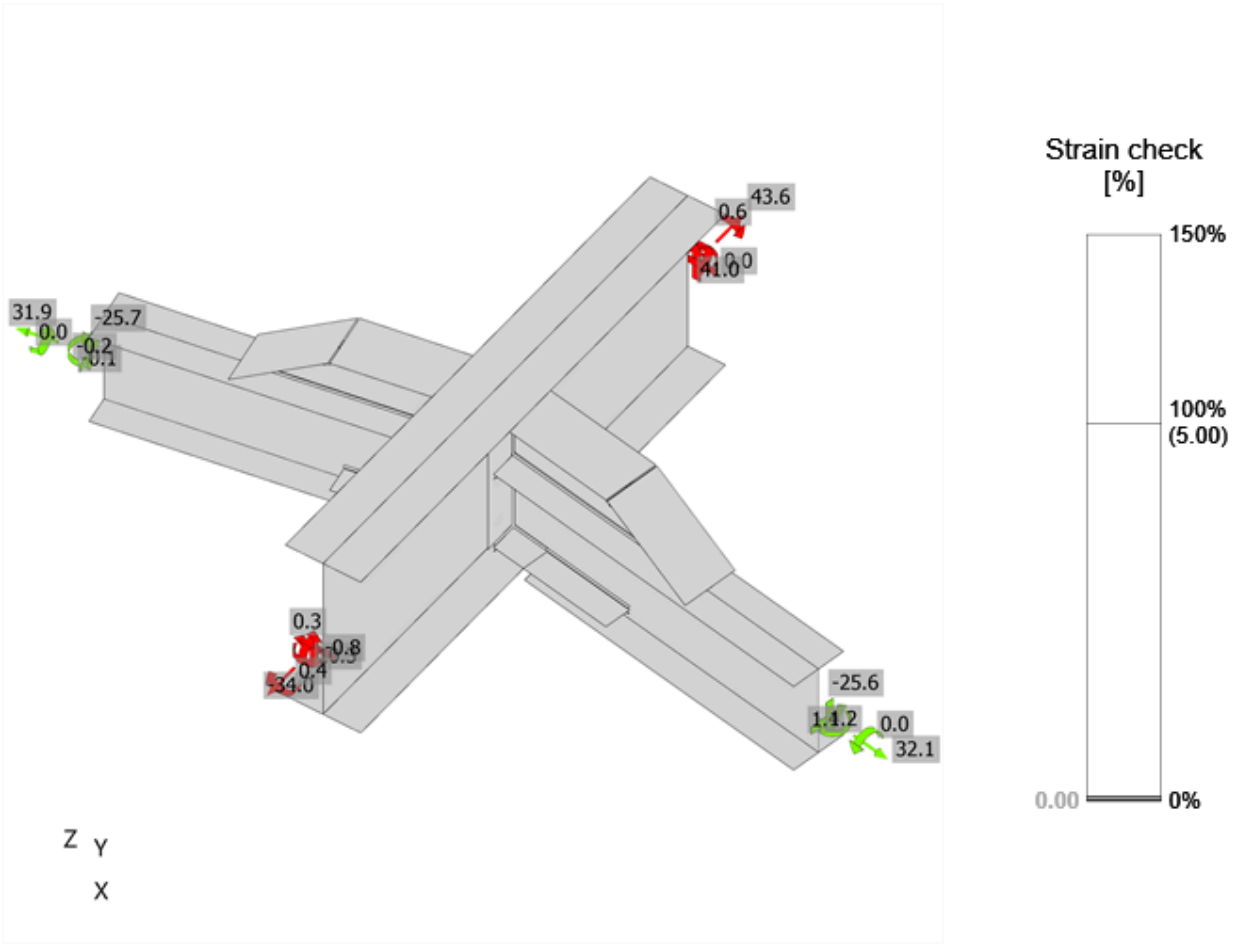
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

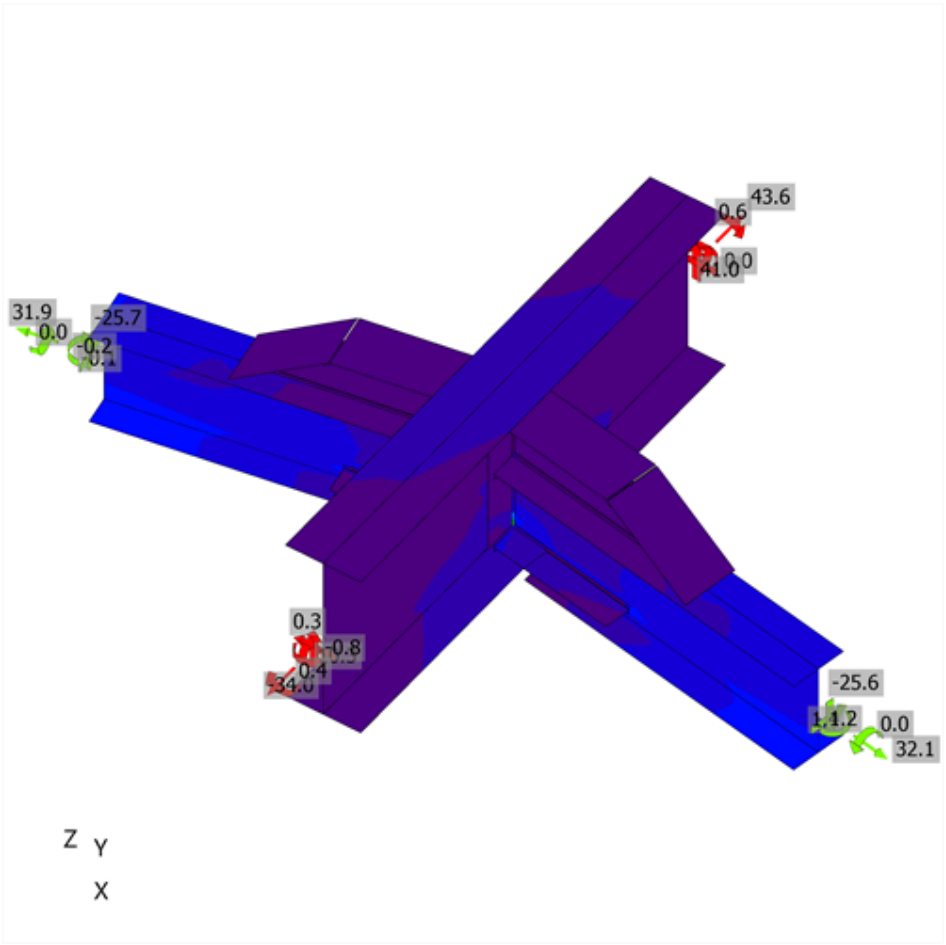
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

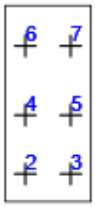


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_120	6.8	0.7	176.5	2.1	0.6	1.8	OK	OK
	B3	M30 8.8 - 1	C LC_147	1.0	0.7	208.4	0.3	0.6	0.5	OK	OK
	B4	M30 8.8 - 1	C LC_122	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.4	0.5	213.9	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_120	2.9	0.8	229.5	0.9	0.7	1.0	OK	OK
	B7	M30 8.8 - 1	C LC_120	1.9	0.9	229.5	0.6	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 10.5	149	C LC_152	18.0	0.0	8.6	-9.1	0.8	4.4	4.4	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 10.5	149	C LC_152	12.9	0.0	6.3	-6.2	2.1	3.2	3.2	OK	OK
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_147	38.9	0.0	-17.0	-19.0	-6.9	9.6	7.2	OK	OK
		◢ 6.0 ◣	208	C LC_147	31.9	0.0	-16.7	15.5	-2.0	7.9	6.5	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_120	30.7	0.0	10.2	15.4	6.6	7.6	7.6	OK	OK
		◢ 6.0 ◣	100	C LC_120	34.8	0.0	-0.2	3.2	-19.8	8.6	8.1	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	30.4	0.0	8.4	7.4	15.1	7.5	7.0	OK	OK
		◢ 6.0 ◣	243	C LC_122	32.5	0.0	12.5	1.1	-17.3	8.0	6.3	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_122	31.5	0.0	6.5	7.9	-15.9	7.8	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_122	38.8	0.0	10.8	-9.4	19.4	9.6	7.8	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	67.0	0.0	-16.1	-19.1	32.3	16.6	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_121	52.2	0.0	-12.4	11.3	-27.0	12.9	8.2	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	35.5	0.0	-16.5	-18.1	-0.9	8.8	6.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	46.6	0.0	-22.1	20.2	12.4	11.5	7.9	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_120	39.4	0.0	-1.7	-6.0	22.0	9.7	8.5	OK	OK
		◢ 6.0 ◣	100	C LC_122	28.2	0.0	7.0	-14.9	-5.2	7.0	7.0	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_120	34.7	0.0	7.0	8.6	17.6	8.6	7.0	OK	OK
		◢ 6.0 ◣	243	C LC_120	44.5	0.0	12.6	-10.9	-22.1	11.0	8.9	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	243	C LC_122	26.1	0.0	7.9	6.3	-12.9	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_147	33.7	0.0	13.2	1.0	17.9	8.3	6.7	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_135	50.2	0.0	-15.2	-11.4	25.2	12.4	7.9	OK	OK
		◢ 6.0 ◣	572	C LC_120	68.2	0.0	-13.7	18.9	-33.6	16.8	9.6	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	148	C LC_122	23.1	0.0	14.3	10.5	-0.4	5.7	5.7	OK	OK
		◢ 6.0 ◣	148	C LC_122	23.5	0.0	-3.2	-1.3	13.4	5.8	5.8	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	9.0	0.0	1.9	1.9	4.7	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	9.4	0.0	2.7	-2.6	-4.5	2.3	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_120	13.2	0.0	-7.3	-5.0	-4.0	3.3	3.3	OK	OK
		◢ 6.0 ◣	149	C LC_120	8.3	0.0	-1.9	4.2	2.0	2.1	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_122	4.4	0.0	1.8	1.6	1.7	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	4.3	0.0	1.2	-1.2	-2.1	1.1	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_122	55.7	0.0	-11.5	-7.2	30.6	13.8	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_135	40.4	0.0	-5.8	7.8	-21.8	10.0	7.2	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	149	C LC_120	44.2	0.0	-15.4	-13.6	-19.7	10.9	8.6	OK	OK
		◢ 6.0 ◣	149	C LC_120	46.2	0.0	-15.3	17.1	18.5	11.4	8.2	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	11.8	0.0	2.9	3.3	5.7	2.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	6.1	0.0	0.3	-0.6	3.5	1.5	0.0	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_120	16.9	0.0	-7.9	-6.0	-6.2	4.2	4.2	OK	OK
		◢ 6.0 ◣	149	C LC_120	14.1	0.0	-4.3	6.2	4.7	3.5	3.5	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_120	5.6	0.0	-0.3	-0.4	-3.2	1.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	7.4	0.0	2.5	-2.1	-3.5	1.8	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_121	42.9	0.0	-6.7	-12.8	20.9	10.6	7.5	OK	OK
		◢ 6.0 ◣	271	C LC_120	54.7	0.0	-10.6	3.5	-30.8	13.5	8.7	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_147	119.1	0.0	52.2	57.0	-24.1	29.4	19.7	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_147	115.5	0.0	61.1	-56.3	6.1	28.5	19.1	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_120	7.3	0.0	-5.3	2.8	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.2	0.0	-0.8	1.3	1.2	0.8	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_120	8.6	0.0	-7.8	-2.1	-0.1	2.5	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	5.6	0.0	-1.3	3.1	-0.2	1.4	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-120
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring

Name	SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

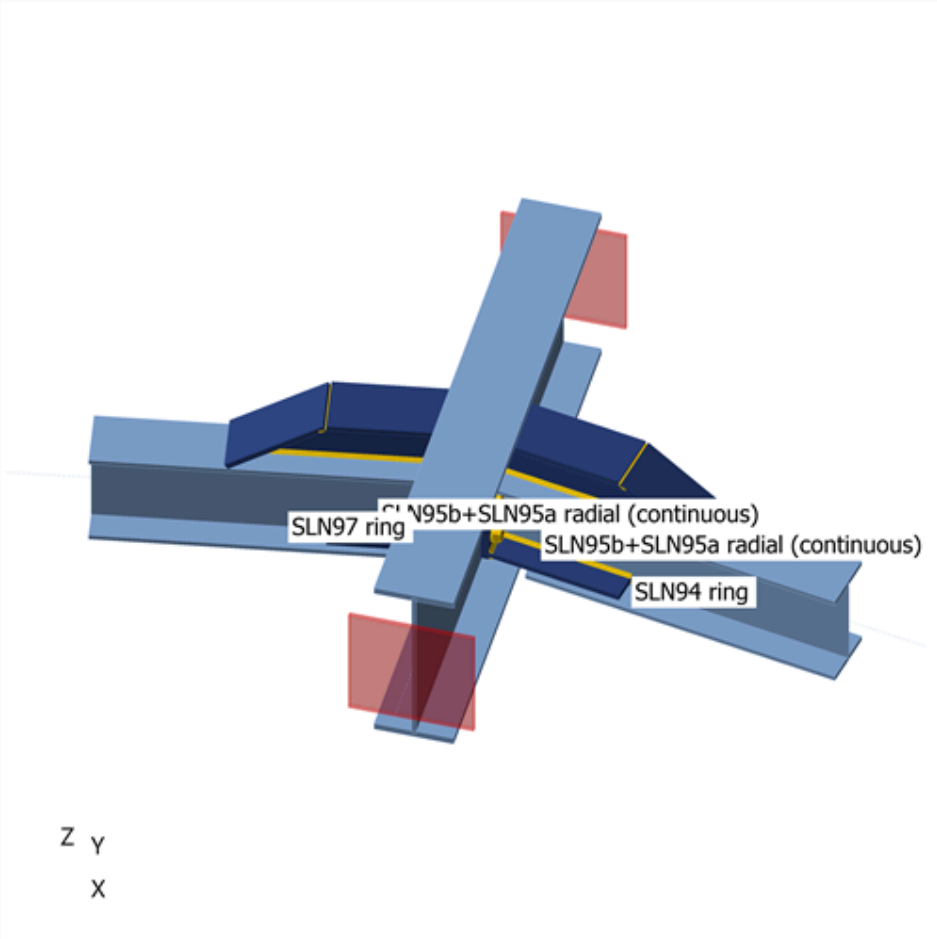
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN95b+SLN95a radial (continuous)	1 - UB457x191x89	120.0	0.0	0.0	0	0	0
SLN94 ring	2 - UB254x146x43	20.0	0.0	0.0	0	0	-102
SLN97 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN95b+SLN95a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN95b+SLN95a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN94 ring / end		Position	0
SLN97 ring / end		Position	0



SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN94 ring / End	33.2	-0.1	-25.7	0.0	0.0	-0.1
	SLN97 ring / End	33.0	-0.2	-26.0	0.0	0.0	-0.3
C LC_106	SLN94 ring / End	28.9	-0.2	-24.1	0.0	0.0	-0.3
	SLN97 ring / End	28.4	-0.4	-24.4	0.0	0.0	-0.5
C LC_119	SLN94 ring / End	31.1	-0.2	-25.7	0.0	0.0	-0.2
	SLN97 ring / End	31.0	-0.1	-26.0	0.0	0.0	-0.2
C LC_120	SLN94 ring / End	31.7	0.0	-25.7	0.0	0.0	0.0
	SLN97 ring / End	31.2	-0.2	-26.0	0.0	0.0	-0.2
C LC_122	SLN94 ring / End	31.7	-0.2	-25.7	0.0	0.0	-0.3
	SLN97 ring / End	31.3	-0.4	-26.0	0.0	0.0	-0.5
C LC_126	SLN94 ring / End	26.5	-0.3	-19.5	0.0	0.0	-0.4
	SLN97 ring / End	25.9	-0.5	-19.7	0.0	0.0	-0.6
C LC_145	SLN94 ring / End	33.5	-0.2	-25.7	0.0	0.0	-0.1
	SLN97 ring / End	33.4	-0.1	-26.0	0.0	0.0	-0.3
C LC_146	SLN94 ring / End	33.0	-0.1	-25.7	0.0	0.0	0.0
	SLN97 ring / End	32.5	-0.2	-26.0	0.0	0.0	-0.3
C LC_147	SLN94 ring / End	34.1	-0.2	-25.7	0.0	0.0	-0.1
	SLN97 ring / End	33.8	-0.1	-26.0	0.0	0.0	-0.2
C LC_152	SLN94 ring / End	34.4	-0.2	-14.1	0.0	0.0	-0.2
	SLN97 ring / End	34.0	-0.2	-14.0	0.0	0.0	-0.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.2 < 100%	OK
Welds	29.8 < 100%	OK
Buckling	Not calculated	

Plates

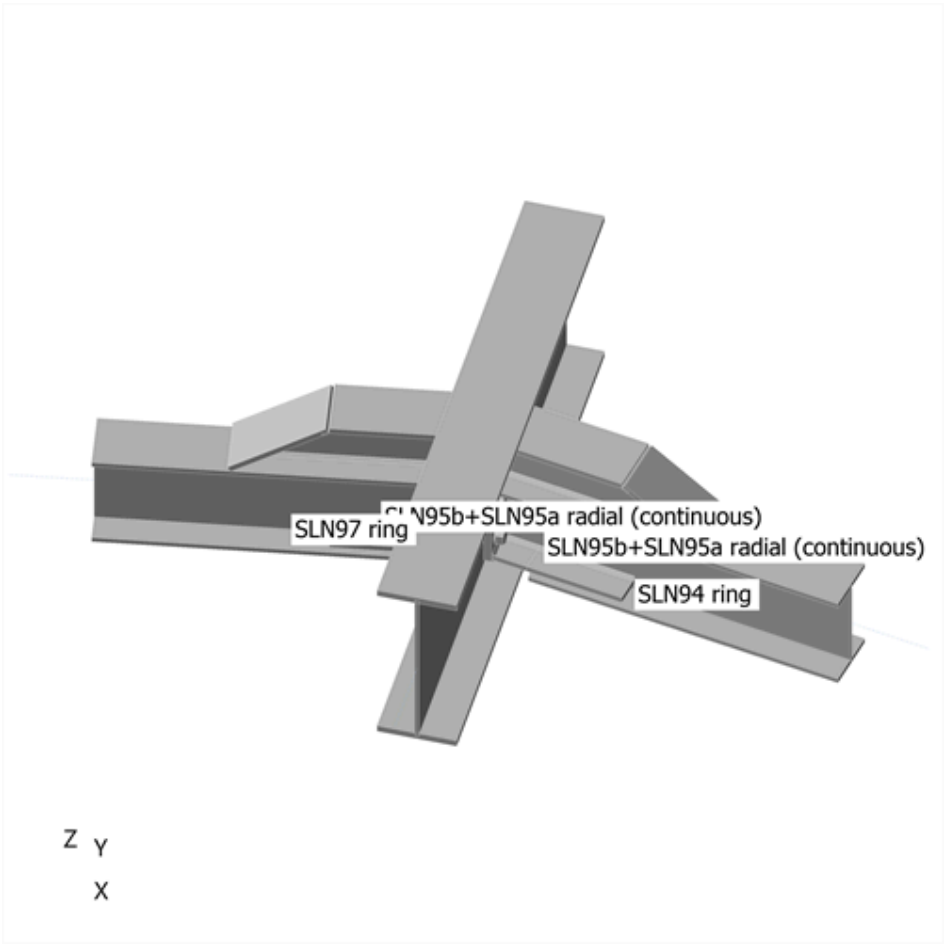
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN95b+SLN95a radial (continuous)-bfl 1	17.7	C LC_120	22.4	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-tfl 1	17.7	C LC_122	19.6	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-w 1	10.5	C LC_122	32.1	0.0	1.4	OK
SLN94 ring-bfl 1	12.7	C LC_147	53.3	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_122	41.7	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	51.7	0.0	0.0	OK
SLN97 ring-bfl 1	12.7	C LC_145	53.6	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	42.8	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_147	51.9	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_122	13.8	0.0	8.1	OK
Bottom flange plate SLN97 ring	15.0	C LC_147	25.6	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_147	25.2	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	24.6	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_122	26.4	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_126	9.1	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_122	11.8	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_147	14.0	0.0	2.7	OK
Bottom flange plate SLN94 ring	15.0	C LC_147	25.5	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	25.0	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	25.0	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_119	26.9	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_126	8.3	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_119	11.7	0.0	0.0	OK

Design data

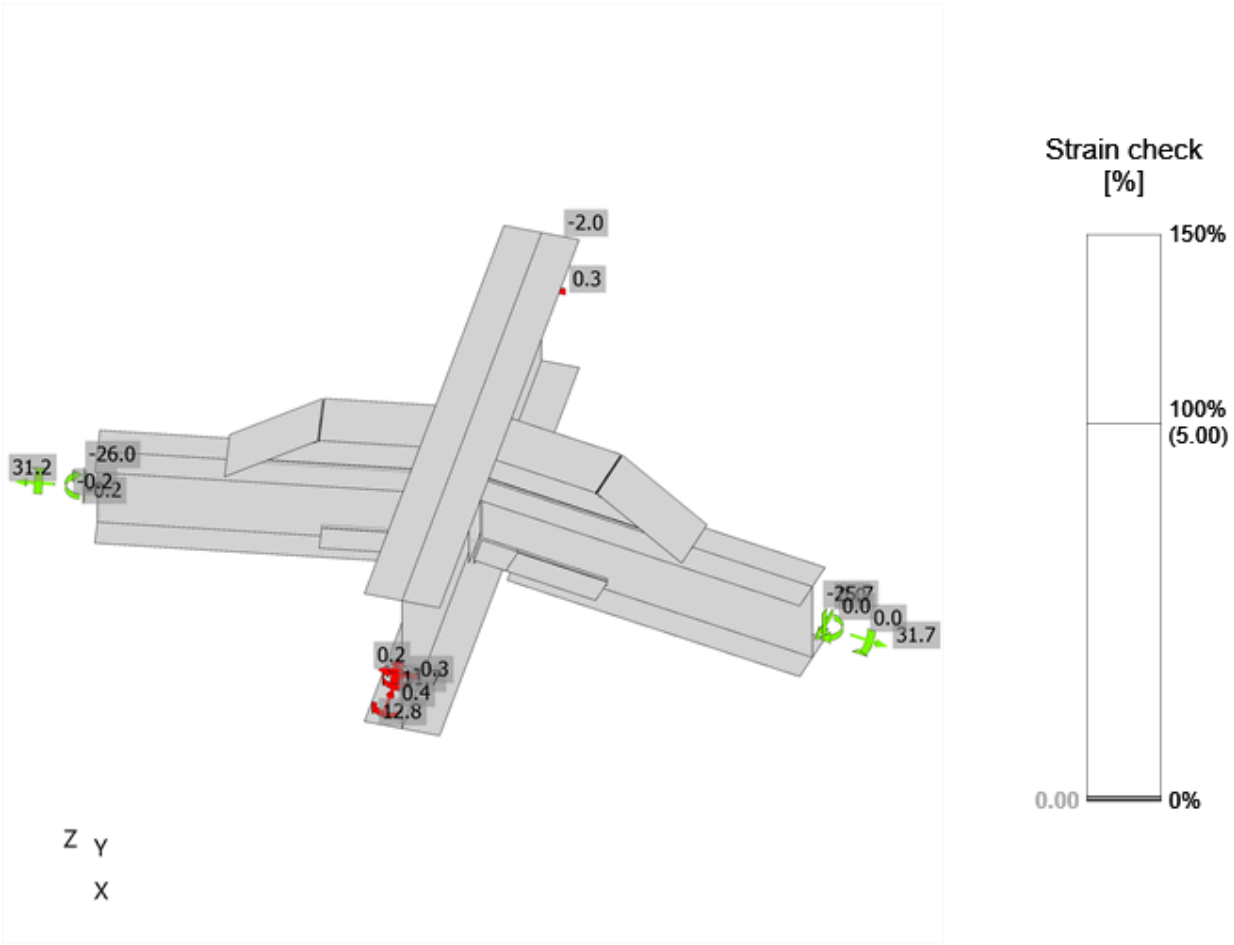
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

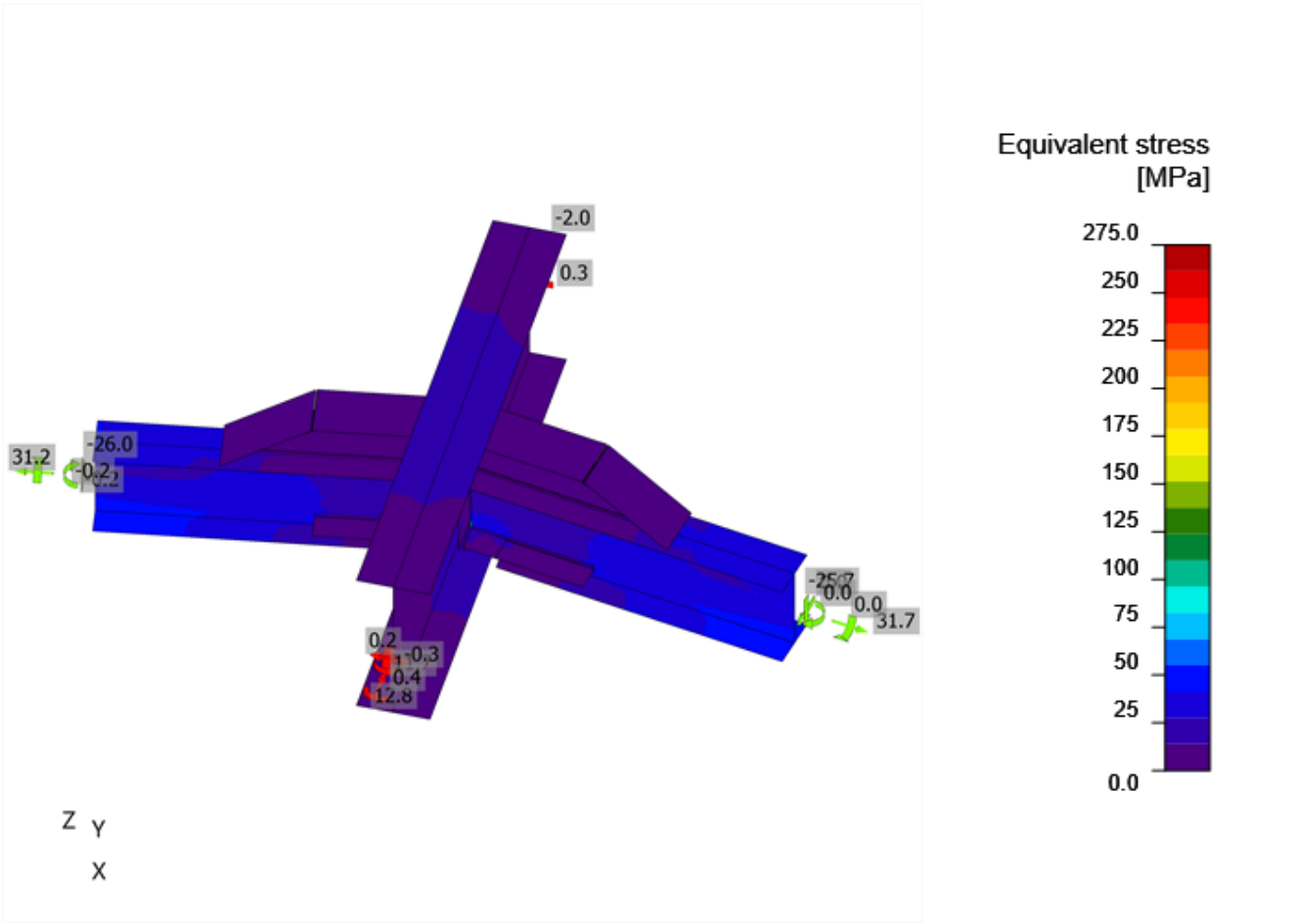
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

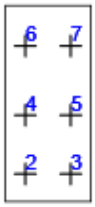


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	3.5	0.5	176.5	1.1	0.5	1.0	OK	OK
	B3	M30 8.8 - 1	C LC_147	3.7	0.7	176.5	1.1	0.8	1.2	OK	OK
	B4	M30 8.8 - 1	C LC_120	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.2	0.5	213.9	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.5	0.8	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_122	1.9	0.9	229.5	0.6	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 10.5	149	C LC_122	11.7	0.0	-5.1	-6.1	0.3	2.9	0.0	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 10.5	149	C LC_122	11.7	0.0	-5.1	-6.0	1.1	2.9	2.9	OK	OK
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	39.7	0.0	-16.9	-19.5	-7.1	9.8	7.2	OK	OK
		◢ 6.0 ◣	208	C LC_147	31.7	0.0	-17.2	15.3	1.2	7.8	6.5	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_147	26.5	0.0	-2.3	-1.5	15.2	6.6	6.6	OK	OK
		◢ 6.0 ◣	100	C LC_122	32.8	0.0	-1.6	2.7	-18.7	8.1	8.0	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	29.8	0.0	7.7	7.4	14.9	7.4	6.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	33.9	0.0	10.8	2.2	-18.4	8.4	6.7	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_122	32.1	0.0	7.3	7.9	-16.2	7.9	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_122	35.7	0.0	9.2	-8.6	17.9	8.8	7.3	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	62.8	0.0	-13.6	-16.1	31.5	15.5	9.1	OK	OK
		◢ 6.0 ◣	572	C LC_119	56.7	0.0	-15.7	14.3	-28.0	14.0	8.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_147	32.7	0.0	-18.9	-15.3	-2.0	8.1	6.6	OK	OK
		◢ 6.0 ◣	208	C LC_147	40.8	0.0	-15.5	20.8	6.5	10.1	7.3	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_122	31.2	0.0	-2.0	-2.3	17.8	7.7	7.7	OK	OK
		◢ 6.0 ◣	100	C LC_122	28.7	0.0	3.0	-4.5	-15.9	7.1	7.1	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	30.3	0.0	8.1	7.5	15.1	7.5	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_147	32.6	0.0	10.3	2.5	-17.7	8.1	6.6	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_122	31.4	0.0	7.0	7.8	-15.9	7.8	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_122	35.9	0.0	9.5	-8.7	18.0	8.9	7.4	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	62.9	0.0	-14.8	-16.8	31.0	15.5	9.0	OK	OK
		◢ 6.0 ◣	572	C LC_120	54.5	0.0	-13.6	12.8	-27.6	13.5	8.3	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	26.5	0.0	-7.0	-9.6	11.2	6.6	6.2	OK	OK
		◢ 6.0 ◣	149	C LC_126	35.4	0.0	-14.1	11.5	-14.8	8.7	6.4	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_120	6.3	0.0	1.7	1.5	3.2	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.5	0.0	2.8	-3.1	-5.7	2.9	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	17.1	0.0	-8.2	-5.9	6.4	4.2	4.2	OK	OK
		◢ 6.0 ◣	149	C LC_126	13.5	0.0	-3.9	6.2	-4.1	3.3	3.3	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_122	5.9	0.0	2.0	1.7	2.7	1.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.6	0.0	0.0	0.2	2.6	1.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_122	49.5	0.0	-8.8	-5.3	27.7	12.2	8.1	OK	OK
		◢ 6.0 ◣	271	C LC_122	47.5	0.0	-8.5	12.0	-24.1	11.7	7.8	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	148	C LC_147	23.0	0.0	11.3	6.6	9.5	5.7	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	32.1	0.0	-16.0	14.5	-6.9	7.9	6.0	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	9.4	0.0	2.3	2.5	4.6	2.3	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	9.1	0.0	2.0	-2.0	-4.7	2.2	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_126	13.6	0.0	-7.5	-5.4	3.7	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_126	8.8	0.0	-2.2	4.3	-2.4	2.2	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_122	5.2	0.0	1.7	1.5	2.4	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.6	0.0	0.0	0.1	2.6	1.1	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_119	51.5	0.0	-10.0	-7.9	28.1	12.7	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_122	44.1	0.0	-7.3	10.2	-22.9	10.9	7.5	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_147	114.5	0.0	49.8	57.4	-15.9	28.3	19.1	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_147	120.8	0.0	48.6	-62.1	15.0	29.8	20.1	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_122	7.5	0.0	-6.0	2.6	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.3	0.0	-0.9	2.4	-0.1	1.1	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.7	2.6	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.7	0.0	-0.7	2.1	-0.3	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-140
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring

Name	SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

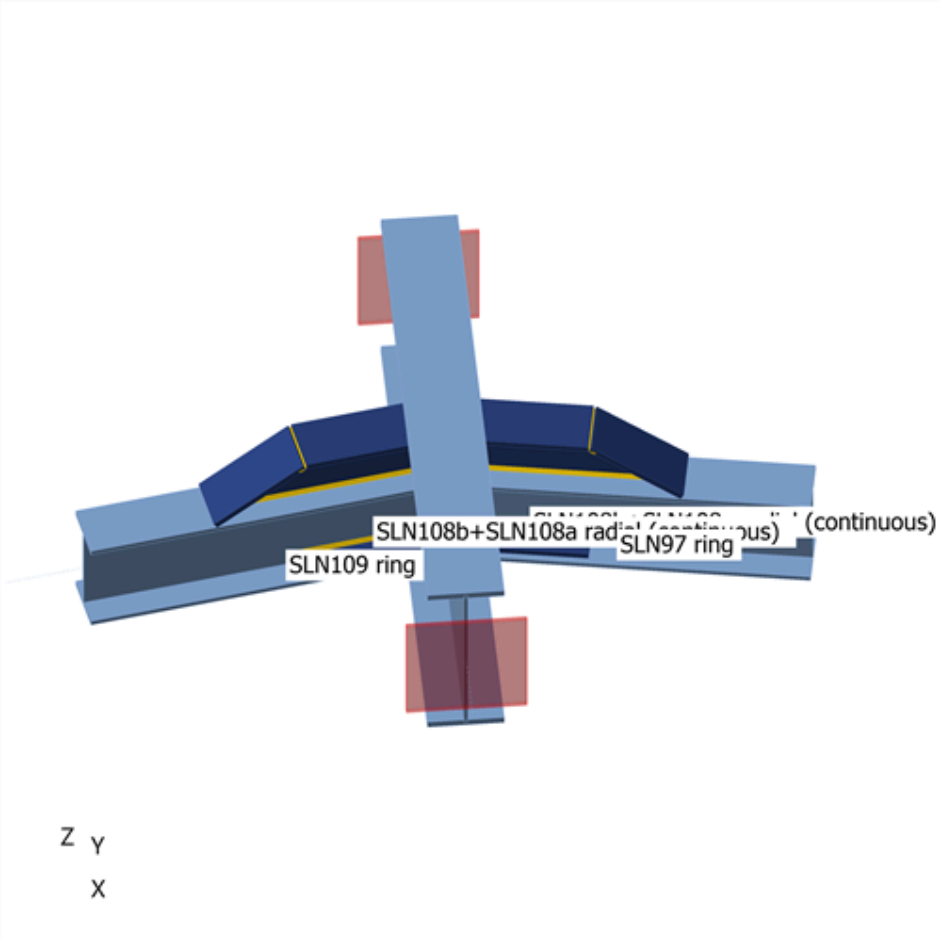
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN108b+SLN108a radial (continuous)	1 - UB457x191x89	140.0	0.0	0.0	0	0	0
SLN97 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN109 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN108b+SLN108a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN108b+SLN108a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN97 ring / end		Position	0
SLN109 ring / end		Position	0



SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN97 ring / End	33.0	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	32.8	0.1	-25.6	0.0	0.0	0.1
C LC_119	SLN97 ring / End	31.1	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	30.9	0.1	-25.6	0.0	0.0	0.1
C LC_121	SLN97 ring / End	30.9	-0.1	-26.0	0.0	0.0	-0.1
	SLN109 ring / End	30.9	0.2	-25.6	0.0	0.0	0.3
C LC_122	SLN97 ring / End	31.4	-0.3	-26.0	0.0	0.0	-0.5
	SLN109 ring / End	30.9	-0.2	-25.6	0.0	0.0	-0.2
C LC_126	SLN97 ring / End	26.0	-0.4	-19.7	0.0	0.0	-0.6
	SLN109 ring / End	25.1	-0.3	-19.4	0.0	0.0	-0.4
C LC_145	SLN97 ring / End	33.6	-0.3	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	33.5	0.1	-25.6	0.0	0.0	0.1
C LC_147	SLN97 ring / End	33.7	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	33.3	0.2	-25.6	0.0	0.0	0.2
C LC_151	SLN97 ring / End	26.4	-0.2	-11.7	0.0	0.0	-0.3
	SLN109 ring / End	26.0	0.0	-11.5	0.0	0.0	-0.1
C LC_152	SLN97 ring / End	34.0	-0.2	-14.0	0.0	0.0	-0.3
	SLN109 ring / End	33.7	0.0	-14.0	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.2 < 100%	OK
Welds	29.2 < 100%	OK
Buckling	Not calculated	

Plates

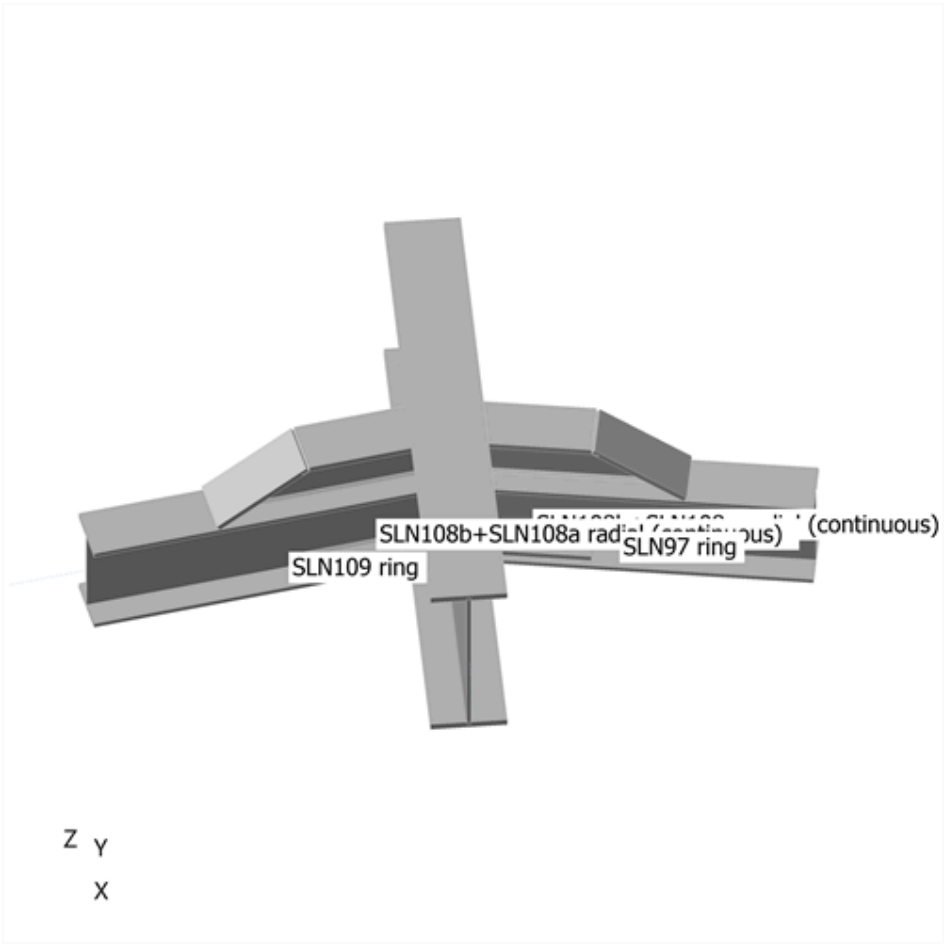
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN108b+SLN108a radial (continuous)-bfl 1	17.7	C LC_122	22.3	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-tfl 1	17.7	C LC_122	19.5	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-w 1	10.5	C LC_122	31.4	0.0	1.2	OK
SLN97 ring-bfl 1	12.7	C LC_122	54.1	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	43.7	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_145	51.9	0.0	0.0	OK
SLN109 ring-bfl 1	12.7	C LC_147	52.4	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_121	41.4	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_145	51.2	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_122	13.2	0.0	6.6	OK
Bottom flange plate SLN109 ring	15.0	C LC_145	25.3	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_145	24.7	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_147	24.7	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_122	25.8	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_126	9.4	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_122	11.6	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_145	14.4	0.0	1.7	OK
Bottom flange plate SLN97 ring	15.0	C LC_147	25.5	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_145	25.1	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	24.9	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_122	26.5	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_126	7.9	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_122	11.8	0.0	0.0	OK

Design data

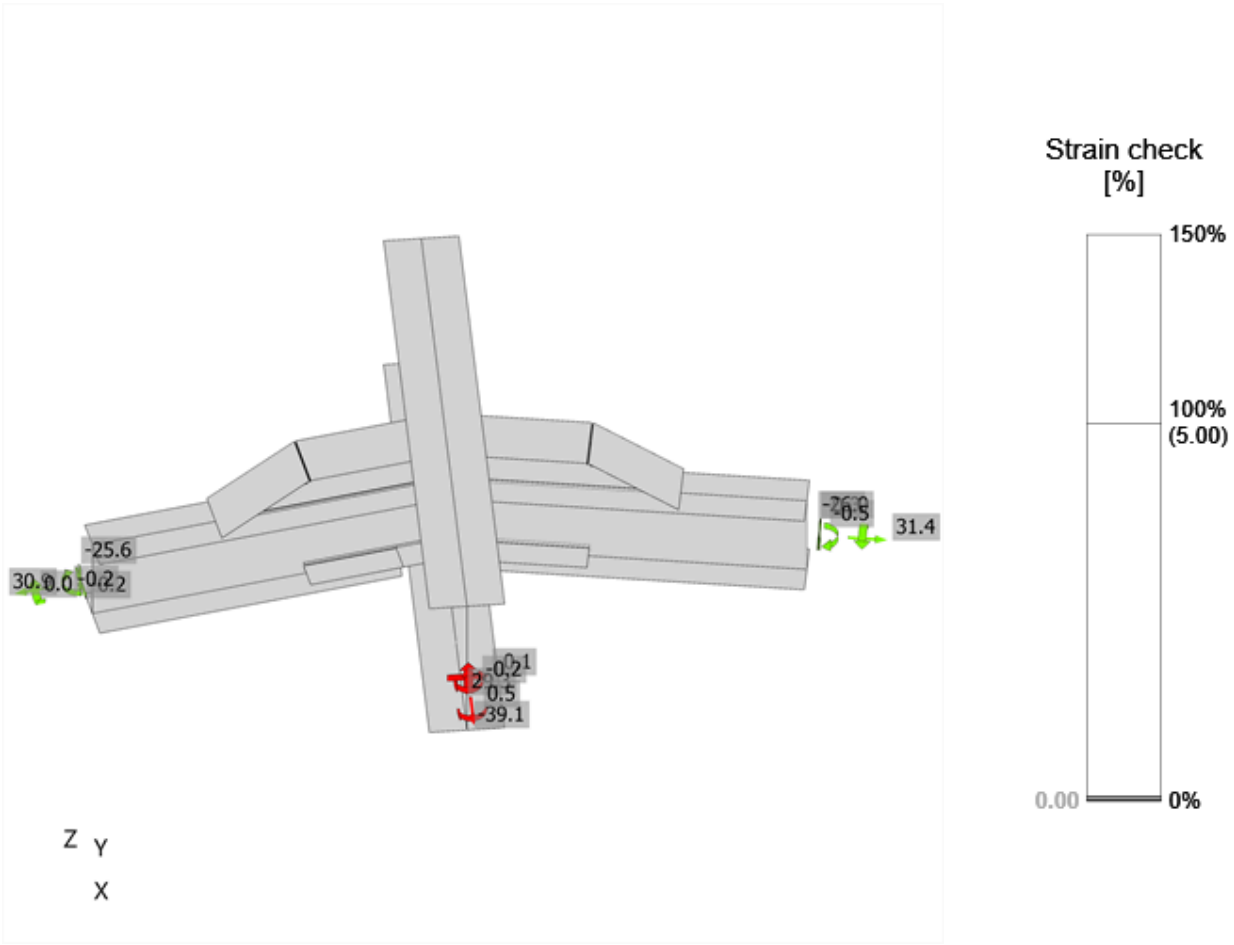
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

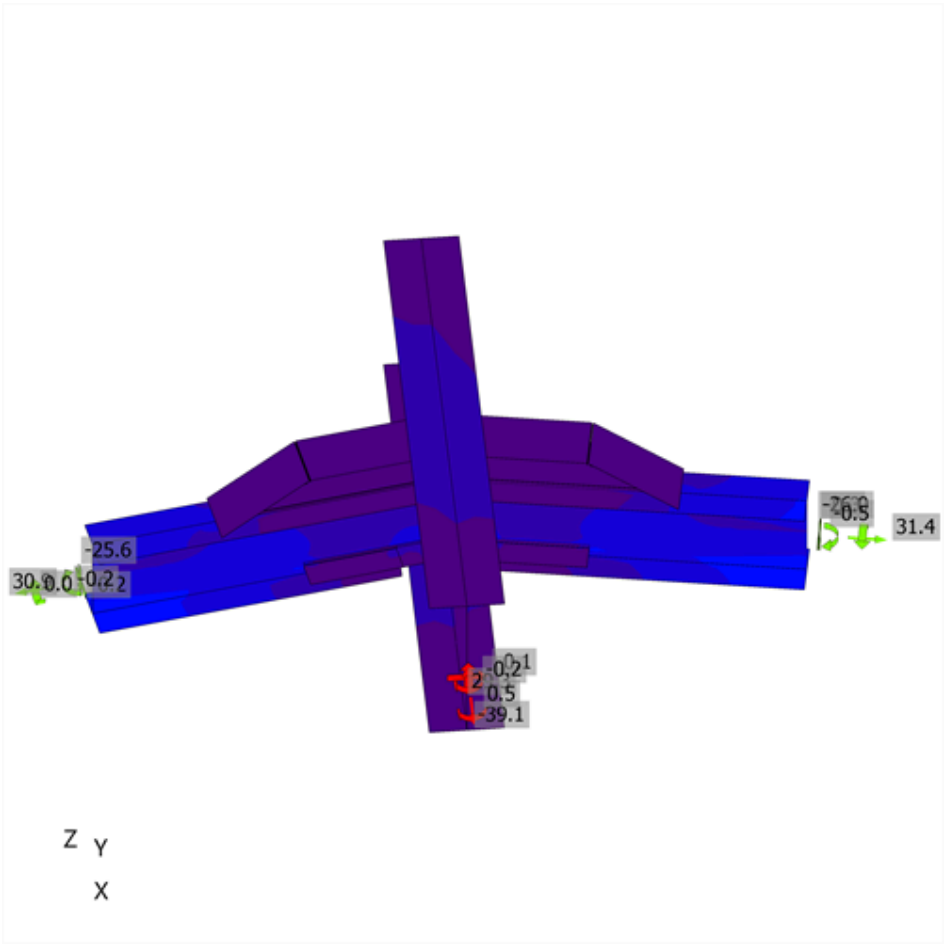
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

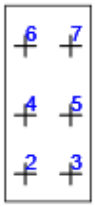


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	3.2	0.5	176.5	1.0	0.5	0.9	OK	OK
	B3	M30 8.8 - 1	C LC_145	3.8	0.7	176.5	1.2	0.8	1.2	OK	OK
	B4	M30 8.8 - 1	C LC_147	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	213.9	0.0	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.1	0.9	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.3	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 10.5	149	C LC_122	11.7	0.0	-5.1	-6.1	0.3	2.9	2.9	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 10.5	149	C LC_122	11.9	0.0	-5.0	-6.1	1.0	2.9	0.0	OK	OK
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_145	38.7	0.0	-15.9	-19.4	-6.2	9.6	7.1	OK	OK
		◢ 6.0 ◣	208	C LC_147	31.0	0.0	-17.2	14.8	1.8	7.7	6.4	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_147	26.8	0.0	3.0	3.2	15.1	6.6	6.6	OK	OK
		◢ 6.0 ◣	100	C LC_122	31.6	0.0	-1.6	2.7	-18.0	7.8	7.8	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	30.9	0.0	7.7	7.6	15.5	7.6	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_145	32.9	0.0	9.5	2.8	-18.0	8.1	6.9	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_122	30.6	0.0	7.1	7.6	-15.4	7.6	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_122	33.6	0.0	8.6	-8.1	16.9	8.3	7.2	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	60.8	0.0	-14.4	-15.8	30.2	15.0	8.9	OK	OK
		◢ 6.0 ◣	572	C LC_121	58.8	0.0	-13.6	14.8	-29.5	14.5	8.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	32.0	0.0	-18.4	-14.9	-2.5	7.9	6.5	OK	OK
		◢ 6.0 ◣	208	C LC_145	39.7	0.0	-15.2	20.4	5.7	9.8	7.2	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_122	30.4	0.0	-1.8	-2.2	17.4	7.5	7.5	OK	OK
		◢ 6.0 ◣	100	C LC_122	29.4	0.0	3.2	-4.8	-16.2	7.3	7.3	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	30.0	0.0	7.8	7.4	15.0	7.4	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_145	33.2	0.0	10.8	2.3	-18.0	8.2	6.6	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_122	32.0	0.0	7.3	7.9	-16.1	7.9	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_122	35.6	0.0	9.2	-8.6	17.9	8.8	7.3	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	63.1	0.0	-13.7	-16.2	31.6	15.6	9.1	OK	OK
		◢ 6.0 ◣	572	C LC_121	57.2	0.0	-14.9	14.2	-28.6	14.1	8.6	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	24.2	0.0	-6.1	-9.0	10.1	6.0	6.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	33.5	0.0	-13.6	10.6	-14.1	8.3	6.3	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	7.4	0.0	2.1	1.8	3.7	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.3	0.0	2.4	-3.0	-5.6	2.8	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	16.3	0.0	-8.1	-5.2	6.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	11.6	0.0	-2.8	5.7	-3.2	2.9	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_126	5.0	0.0	1.5	1.3	2.4	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.4	0.0	0.1	0.0	2.5	1.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_122	48.9	0.0	-9.0	-7.0	26.9	12.1	8.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	46.5	0.0	-8.1	6.5	-25.7	11.5	7.8	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	148	C LC_145	23.2	0.0	10.9	6.5	9.9	5.7	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	31.6	0.0	-15.8	14.4	-6.7	7.8	6.0	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	2.1	2.6	5.2	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.9	0.0	2.3	-2.1	-4.5	2.2	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	14.6	0.0	-7.6	-6.1	3.8	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	11.1	0.0	-3.6	5.1	-3.2	2.8	0.0	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_122	5.7	0.0	2.0	1.7	2.6	1.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.4	0.0	-0.1	0.2	2.5	1.1	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_119	50.0	0.0	-9.2	-6.7	27.6	12.3	8.1	OK	OK
		◢ 6.0 ◣	271	C LC_122	47.0	0.0	-8.3	12.0	-23.9	11.6	7.8	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_147	113.2	0.0	47.8	57.2	-15.4	28.0	18.8	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_145	118.1	0.0	47.4	-60.3	16.1	29.2	19.6	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.5	2.6	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.5	0.0	-0.8	2.0	-0.1	0.9	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_126	6.8	0.0	-6.3	-1.4	0.0	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.5	0.0	-0.9	2.5	-0.3	1.1	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-160
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring

Name	SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

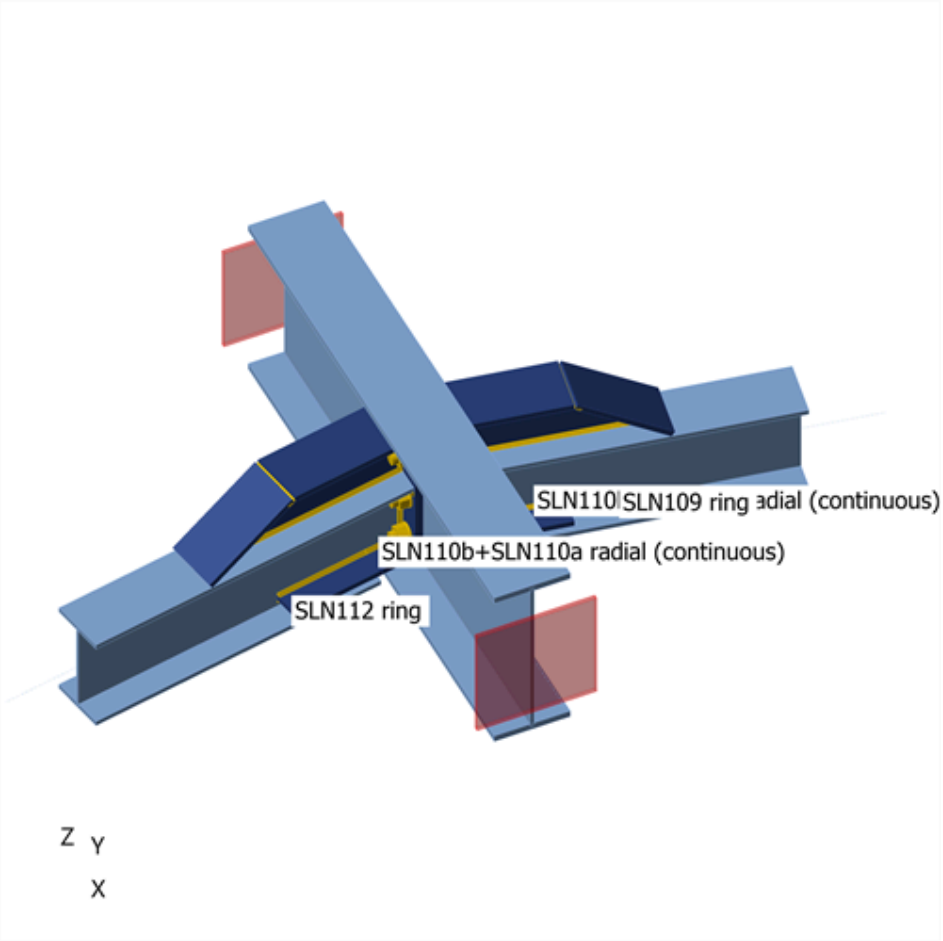
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN110b+SLN110a radial (continuous)	1 - UB457x191x89	160.0	0.0	0.0	0	0	0
SLN109 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102
SLN112 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN110b+SLN110a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN110b+SLN110a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN109 ring / end		Position	0
SLN112 ring / end		Position	0



SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN109 ring / End	32.8	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	32.7	-0.2	-25.5	0.0	0.0	-0.2
C LC_106	SLN109 ring / End	28.2	-0.1	-24.0	0.0	0.0	-0.2
	SLN112 ring / End	27.9	-0.3	-24.0	0.0	0.0	-0.4
C LC_120	SLN109 ring / End	30.7	0.2	-25.6	0.0	0.0	0.2
	SLN112 ring / End	30.5	-0.2	-25.5	0.0	0.0	-0.2
C LC_121	SLN109 ring / End	30.8	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	30.7	0.0	-25.5	0.0	0.0	0.0
C LC_122	SLN109 ring / End	31.0	-0.1	-25.6	0.0	0.0	-0.1
	SLN112 ring / End	30.8	-0.3	-25.5	0.0	0.0	-0.5
C LC_126	SLN109 ring / End	25.3	-0.2	-19.4	0.0	0.0	-0.3
	SLN112 ring / End	24.8	-0.4	-19.4	0.0	0.0	-0.6
C LC_146	SLN109 ring / End	32.0	0.2	-25.6	0.0	0.0	0.2
	SLN112 ring / End	31.8	-0.2	-25.5	0.0	0.0	-0.3
C LC_147	SLN109 ring / End	33.2	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	32.9	-0.1	-25.5	0.0	0.0	-0.2
C LC_148	SLN109 ring / End	32.4	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	32.5	-0.2	-25.5	0.0	0.0	-0.3
C LC_152	SLN109 ring / End	33.7	0.0	-14.0	0.0	0.0	0.1
	SLN112 ring / End	33.6	-0.1	-14.1	0.0	0.0	-0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.1 < 100%	OK
Welds	28.3 < 100%	OK
Buckling	Not calculated	

Plates

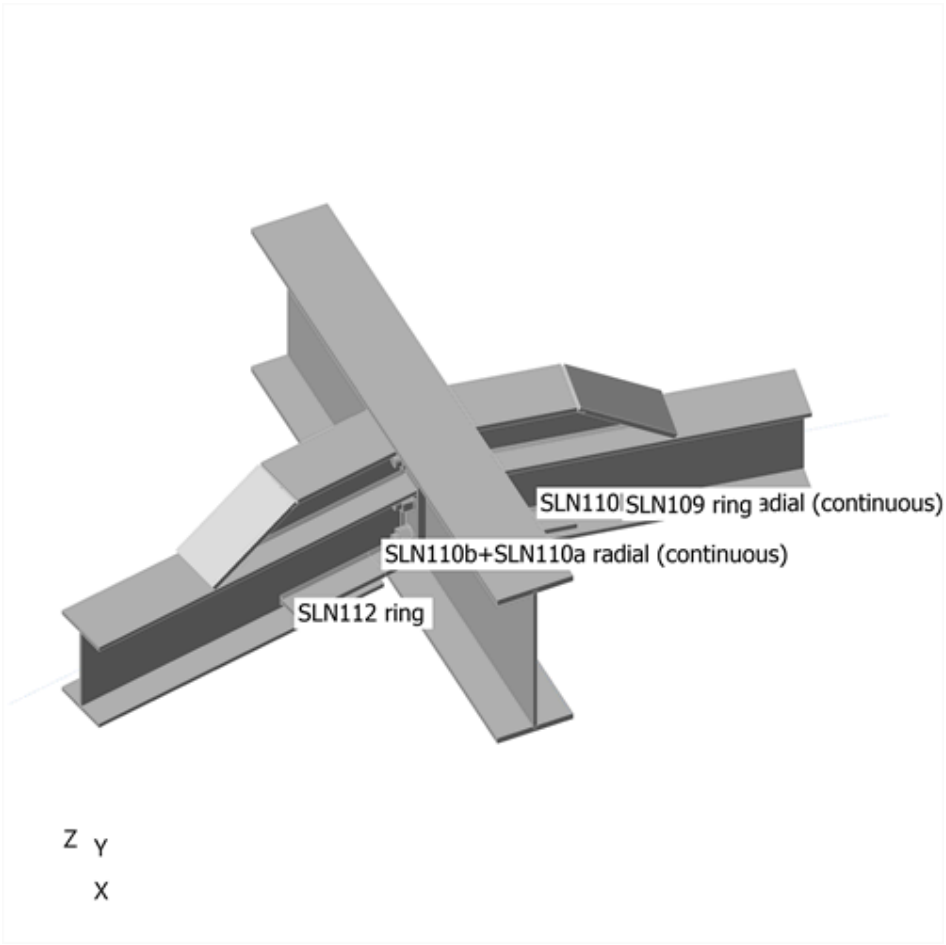
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN110b+SLN110a radial (continuous)-bfl 1	17.7	C LC_148	22.0	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-tfl 1	17.7	C LC_122	18.8	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-w 1	10.5	C LC_122	30.8	0.0	1.0	OK
SLN109 ring-bfl 1	12.7	C LC_147	52.7	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_121	41.6	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_147	51.1	0.0	0.0	OK
SLN112 ring-bfl 1	12.7	C LC_122	52.6	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	42.3	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_147	50.9	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_148	13.8	0.0	1.5	OK
Bottom flange plate SLN112 ring	15.0	C LC_147	25.1	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_147	24.7	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_147	24.4	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	26.0	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_126	8.6	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_122	11.5	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_147	12.8	0.0	0.8	OK
Bottom flange plate SLN109 ring	15.0	C LC_147	25.1	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_147	24.6	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_147	24.9	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_122	25.8	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_126	7.8	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_122	11.6	0.0	0.0	OK

Design data

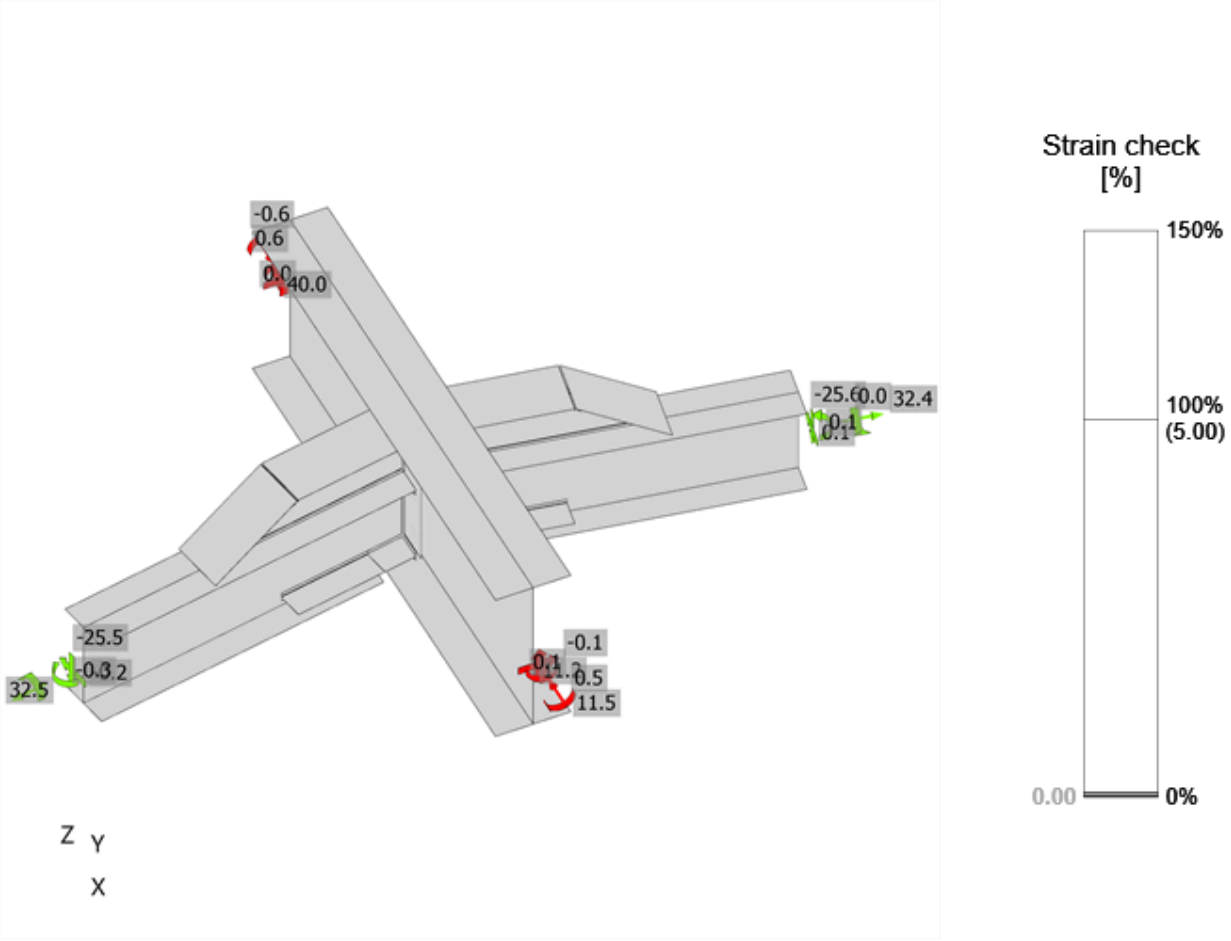
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

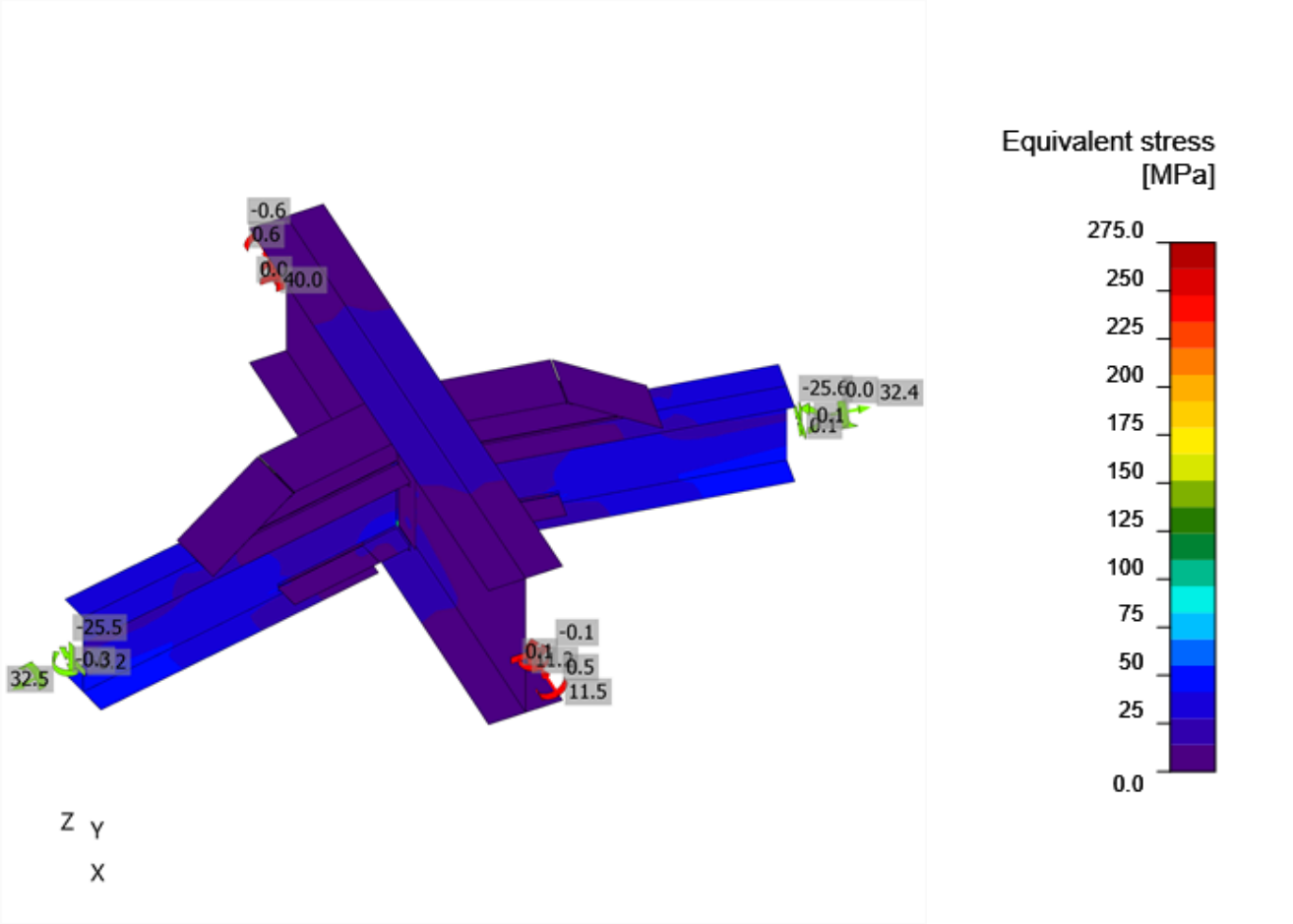
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

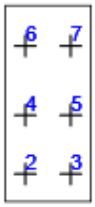


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	3.6	0.5	176.5	1.1	0.5	1.0	OK	OK
	B3	M30 8.8 - 1	C LC_122	3.1	0.7	176.5	1.0	0.8	1.0	OK	OK
	B4	M30 8.8 - 1	C LC_120	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.5	0.8	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_122	1.7	0.9	229.5	0.5	0.8	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 10.5	149	C LC_122	13.2	0.0	-5.9	-6.8	0.9	3.3	3.3	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 10.5	149	C LC_122	12.9	0.0	-6.0	-6.6	0.4	3.2	3.2	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_148	38.7	0.0	-16.3	-18.9	-7.3	9.6	7.1	OK	OK
		◢ 6.0 ◣	208	C LC_147	30.8	0.0	-17.0	14.8	0.9	7.6	6.4	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.9	0.0	2.7	2.7	14.6	6.4	6.4	OK	OK
		◢ 6.0 ◣	100	C LC_122	32.0	0.0	-1.6	3.0	-18.2	7.9	7.9	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_147	29.6	0.0	7.5	7.3	14.8	7.3	6.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	33.2	0.0	10.0	2.5	-18.1	8.2	6.8	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_122	31.3	0.0	7.2	7.7	-15.8	7.7	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_122	34.4	0.0	8.8	-8.3	17.3	8.5	7.2	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	61.2	0.0	-13.3	-15.5	30.8	15.1	8.9	OK	OK
		◢ 6.0 ◣	572	C LC_121	58.2	0.0	-14.7	14.9	-28.9	14.4	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_147	31.3	0.0	-17.5	-14.9	-1.4	7.7	6.5	OK	OK
		◢ 6.0 ◣	208	C LC_147	39.3	0.0	-15.8	19.7	6.7	9.7	7.2	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_147	31.1	0.0	-1.7	-2.5	17.7	7.7	7.7	OK	OK
		◢ 6.0 ◣	100	C LC_122	27.7	0.0	2.8	-3.7	-15.5	6.8	6.8	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	30.7	0.0	7.7	7.6	15.4	7.6	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_147	32.4	0.0	9.1	3.1	-17.7	8.0	7.0	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_122	30.3	0.0	7.1	7.5	-15.3	7.5	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_147	33.5	0.0	9.9	2.5	18.3	8.3	7.1	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	60.3	0.0	-14.5	-15.6	30.0	14.9	8.8	OK	OK
		◢ 6.0 ◣	572	C LC_121	58.4	0.0	-13.6	14.5	-29.4	14.4	8.7	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	24.8	0.0	-6.6	-8.8	10.6	6.1	6.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	32.9	0.0	-13.0	10.8	-13.7	8.1	6.3	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	7.6	0.0	1.7	1.6	3.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.1	0.0	2.7	-3.0	-5.4	2.7	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	16.5	0.0	-7.8	-5.7	6.1	4.1	4.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	13.3	0.0	-4.0	6.1	-4.1	3.3	3.3	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_126	5.6	0.0	1.8	1.5	2.7	1.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.3	0.0	-0.1	0.3	2.5	1.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_122	48.2	0.0	-8.4	-5.2	26.9	11.9	8.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	47.4	0.0	-8.6	8.4	-25.6	11.7	7.8	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	148	C LC_147	20.8	0.0	11.1	6.6	7.7	5.1	5.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	28.2	0.0	-14.3	12.5	-6.4	7.0	5.7	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	2.5	2.7	5.2	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.5	0.0	1.9	-1.9	-4.4	2.1	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	13.0	0.0	-7.2	-5.2	3.5	3.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	8.5	0.0	-2.2	4.2	-2.3	2.1	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_126	4.7	0.0	1.5	1.2	2.3	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.3	0.0	1.5	-1.4	-1.8	1.1	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_122	48.7	0.0	-8.9	-7.3	26.7	12.0	8.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	46.2	0.0	-7.9	6.7	-25.4	11.4	7.8	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_147	112.4	0.0	48.2	56.4	-15.9	27.8	18.7	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_122	114.5	0.0	46.4	-57.5	18.6	28.3	19.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.8	2.5	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.2	0.0	-0.9	2.4	-0.1	1.0	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.4	2.7	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.5	0.0	-0.7	1.9	-0.3	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-180
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring

Name	SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

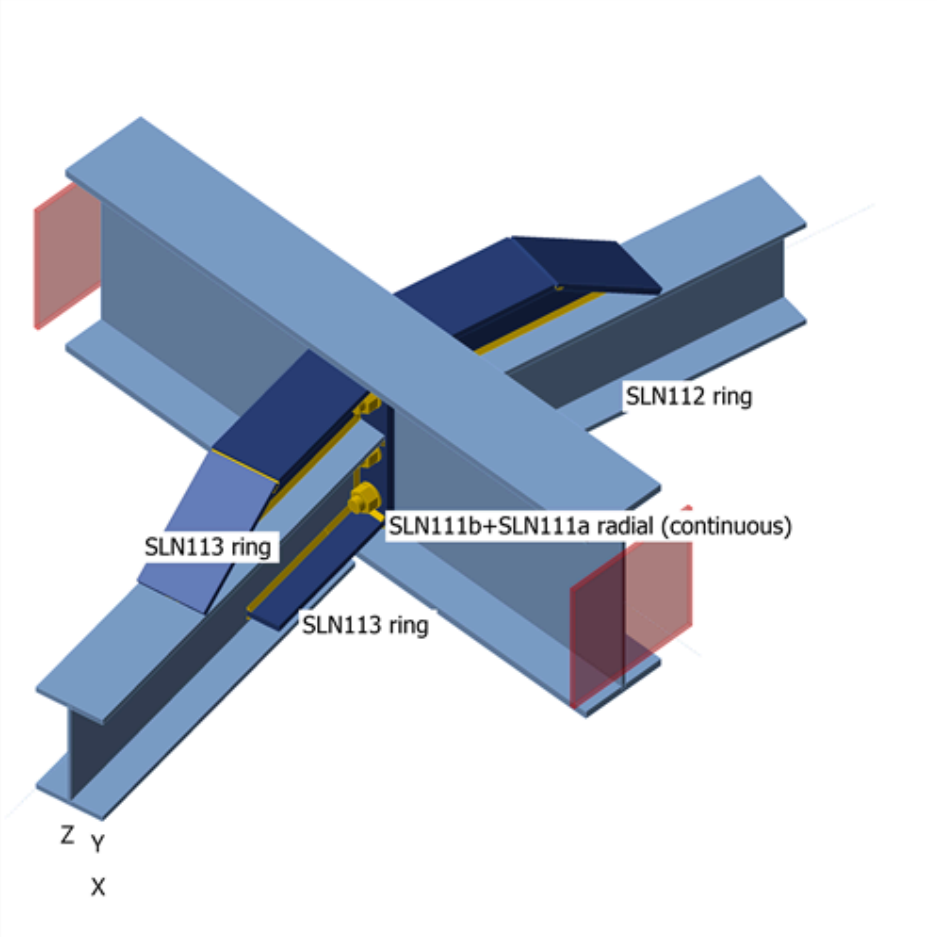
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN111b+SLN111a radial (continuous)	1 - UB457x191x89	-180.0	0.0	0.0	0	0	0
SLN112 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102
SLN113 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN111b+SLN111a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN111b+SLN111a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN112 ring / end		Position	0
SLN113 ring / end		Position	0



SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN112 ring / End	32.7	-0.2	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	32.8	0.1	-25.6	0.0	0.0	0.2
C LC_121	SLN112 ring / End	30.6	0.0	-25.6	0.0	0.0	0.0
	SLN113 ring / End	30.9	0.2	-25.6	0.0	0.0	0.4
C LC_122	SLN112 ring / End	31.0	-0.3	-25.6	0.0	0.0	-0.5
	SLN113 ring / End	30.7	-0.1	-25.6	0.0	0.0	-0.1
C LC_126	SLN112 ring / End	24.9	-0.4	-19.4	0.0	0.0	-0.7
	SLN113 ring / End	24.1	-0.2	-19.4	0.0	0.0	-0.3
C LC_135	SLN112 ring / End	30.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	30.9	0.1	-25.6	0.0	0.0	0.2
C LC_145	SLN112 ring / End	33.6	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	33.7	0.2	-25.6	0.0	0.0	0.3
C LC_147	SLN112 ring / End	32.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	32.7	0.2	-25.6	0.0	0.0	0.3
C LC_148	SLN112 ring / End	32.6	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	32.9	0.1	-25.6	0.0	0.0	0.2
C LC_152	SLN112 ring / End	33.6	-0.1	-14.2	0.0	0.0	-0.2
	SLN113 ring / End	33.6	0.1	-14.1	0.0	0.0	0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.1 < 100%	OK
Welds	28.2 < 100%	OK
Buckling	Not calculated	

Plates

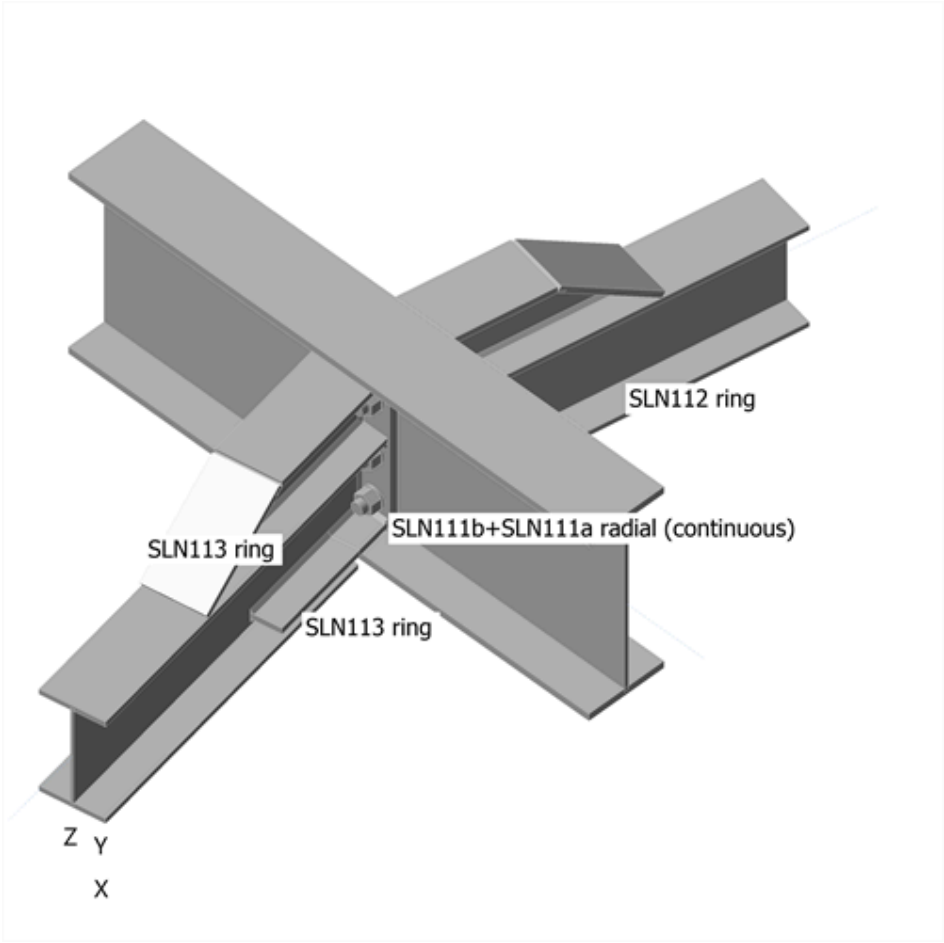
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN111b+SLN111a radial (continuous)-bfl 1	17.7	C LC_148	22.1	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-tfl 1	17.7	C LC_122	18.9	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-w 1	10.5	C LC_122	30.7	0.0	1.0	OK
SLN112 ring-bfl 1	12.7	C LC_122	52.9	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	42.7	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_145	51.1	0.0	0.0	OK
SLN113 ring-bfl 1	12.7	C LC_147	52.8	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_121	42.3	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_145	51.4	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_122	13.5	0.0	6.0	OK
Bottom flange plate SLN113 ring	15.0	C LC_145	25.4	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_145	24.7	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_145	24.8	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_122	26.9	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_126	9.1	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_122	11.6	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_145	13.3	0.0	1.2	OK
Bottom flange plate SLN112 ring	15.0	C LC_145	25.2	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_145	24.8	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_145	24.8	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	26.0	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_126	7.9	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_122	11.6	0.0	0.0	OK

Design data

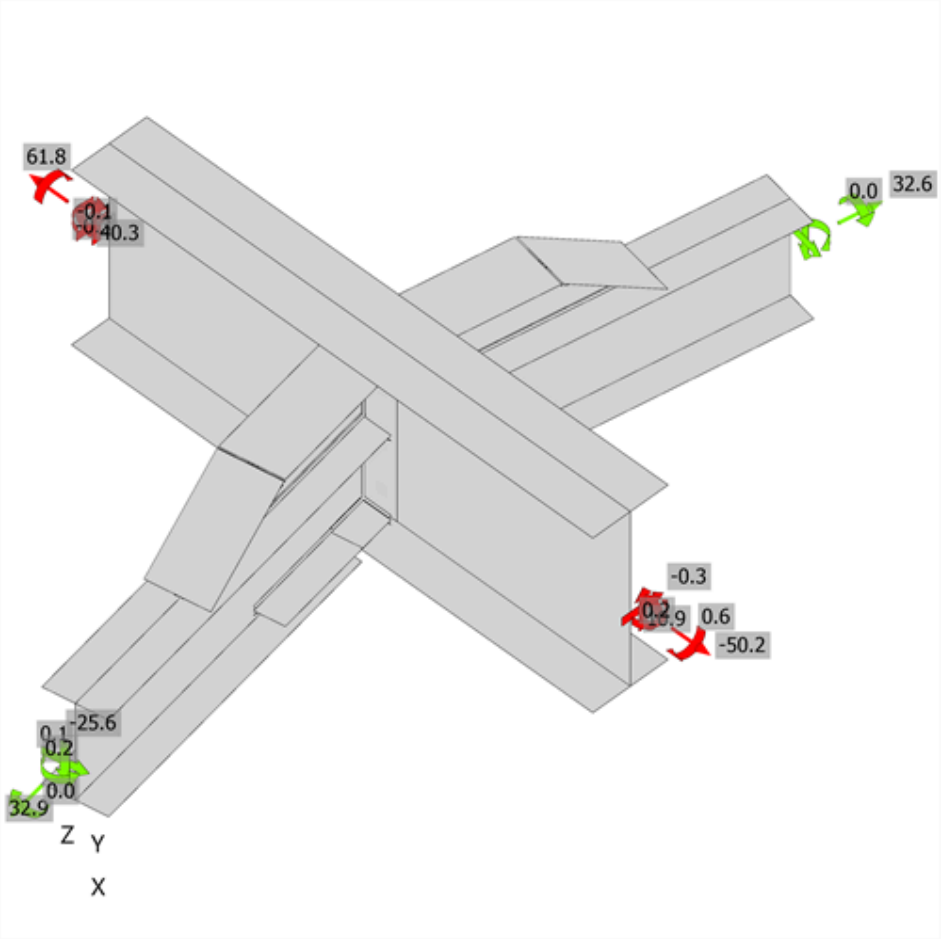
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

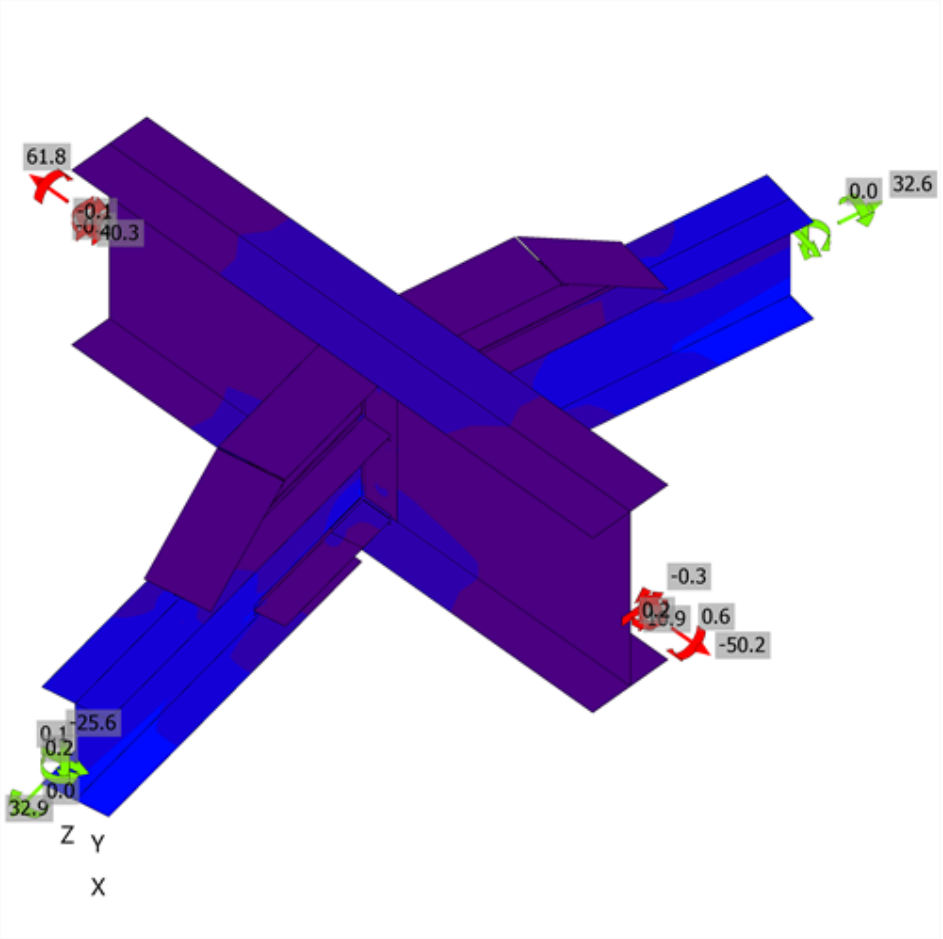
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

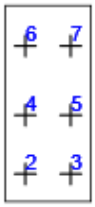


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	3.5	0.5	176.5	1.1	0.5	1.0	OK	OK
	B3	M30 8.8 - 1	C LC_122	3.5	0.7	176.5	1.1	0.8	1.1	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	213.9	0.0	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.8	0.9	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_121	2.3	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 10.5	149	C LC_121	15.2	0.0	-6.9	-7.7	-1.2	3.7	3.7	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 10.5	149	C LC_121	15.9	0.0	-6.5	-8.4	-0.3	3.9	3.9	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_145	38.0	0.0	-15.3	-18.9	-6.8	9.4	7.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	30.3	0.0	-16.8	14.5	1.8	7.5	6.4	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_121	27.6	0.0	3.0	3.5	15.5	6.8	6.8	OK	OK
		◢ 6.0 ◣	100	C LC_122	30.8	0.0	-1.6	2.8	-17.5	7.6	7.6	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_145	31.5	0.0	8.1	7.8	15.8	7.8	7.0	OK	OK
		◢ 6.0 ◣	243	C LC_145	32.5	0.0	9.4	2.9	-17.8	8.0	6.9	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_122	29.9	0.0	6.7	7.4	-15.1	7.4	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_122	33.9	0.0	8.9	-8.2	17.0	8.4	7.3	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	62.4	0.0	-15.6	-16.9	30.5	15.4	9.0	OK	OK
		◢ 6.0 ◣	572	C LC_121	57.6	0.0	-12.1	13.6	-29.5	14.2	8.7	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_145	30.8	0.0	-17.5	-14.5	-1.9	7.6	6.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	38.6	0.0	-15.2	19.7	5.6	9.5	7.1	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_147	30.2	0.0	-1.7	-2.4	17.2	7.5	7.5	OK	OK
		◢ 6.0 ◣	100	C LC_122	28.5	0.0	3.1	-4.4	-15.8	7.1	7.1	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_145	30.0	0.0	7.7	7.4	15.0	7.4	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_145	32.7	0.0	10.0	2.6	-17.8	8.1	6.8	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_122	31.2	0.0	7.2	7.7	-15.7	7.7	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_122	34.3	0.0	8.8	-8.3	17.3	8.5	7.2	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	61.8	0.0	-13.3	-15.9	31.0	15.3	9.0	OK	OK
		◢ 6.0 ◣	572	C LC_121	57.9	0.0	-14.4	14.6	-28.9	14.3	8.6	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	149	C LC_126	25.0	0.0	-6.5	-9.2	10.5	6.2	6.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	33.7	0.0	-13.5	10.8	-14.2	8.3	6.3	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	8.0	0.0	2.3	2.1	3.9	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.2	0.0	2.3	-2.8	-5.7	2.8	2.8	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_126	14.9	0.0	-7.6	-4.5	5.9	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	10.0	0.0	-2.0	5.0	-2.5	2.5	0.0	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_126	4.8	0.0	1.5	1.3	2.3	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	5.2	0.0	1.8	-1.6	-2.3	1.3	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_122	51.5	0.0	-10.1	-8.1	28.0	12.7	8.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	44.1	0.0	-6.9	4.9	-24.6	10.9	7.7	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	148	C LC_145	21.8	0.0	10.7	6.4	8.9	5.4	5.4	OK	OK
		◢ 6.0 ◣	149	C LC_126	31.4	0.0	-15.7	14.2	-6.6	7.8	6.0	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	10.7	0.0	2.2	2.8	5.4	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.8	0.0	2.3	-2.1	-4.4	2.2	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	14.0	0.0	-7.2	-6.0	3.5	3.5	3.5	OK	OK
		◢ 6.0 ◣	149	C LC_126	11.3	0.0	-3.9	5.1	-3.3	2.8	0.0	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_126	5.7	0.0	1.9	1.5	2.7	1.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.3	0.0	-0.2	0.3	2.5	1.1	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_122	48.8	0.0	-8.6	-5.1	27.2	12.0	8.1	OK	OK
		◢ 6.0 ◣	271	C LC_121	46.7	0.0	-8.2	8.0	-25.3	11.5	7.8	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_145	98.1	0.0	54.2	-45.0	14.2	24.2	16.4	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_145	114.1	0.0	47.2	-57.7	16.1	28.2	18.9	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.3	2.7	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.6	0.0	-0.8	2.0	-0.1	0.9	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_126	6.9	0.0	-6.4	-1.5	0.0	2.1	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.5	0.0	-0.9	2.5	-0.3	1.1	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-200
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring

Name	SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

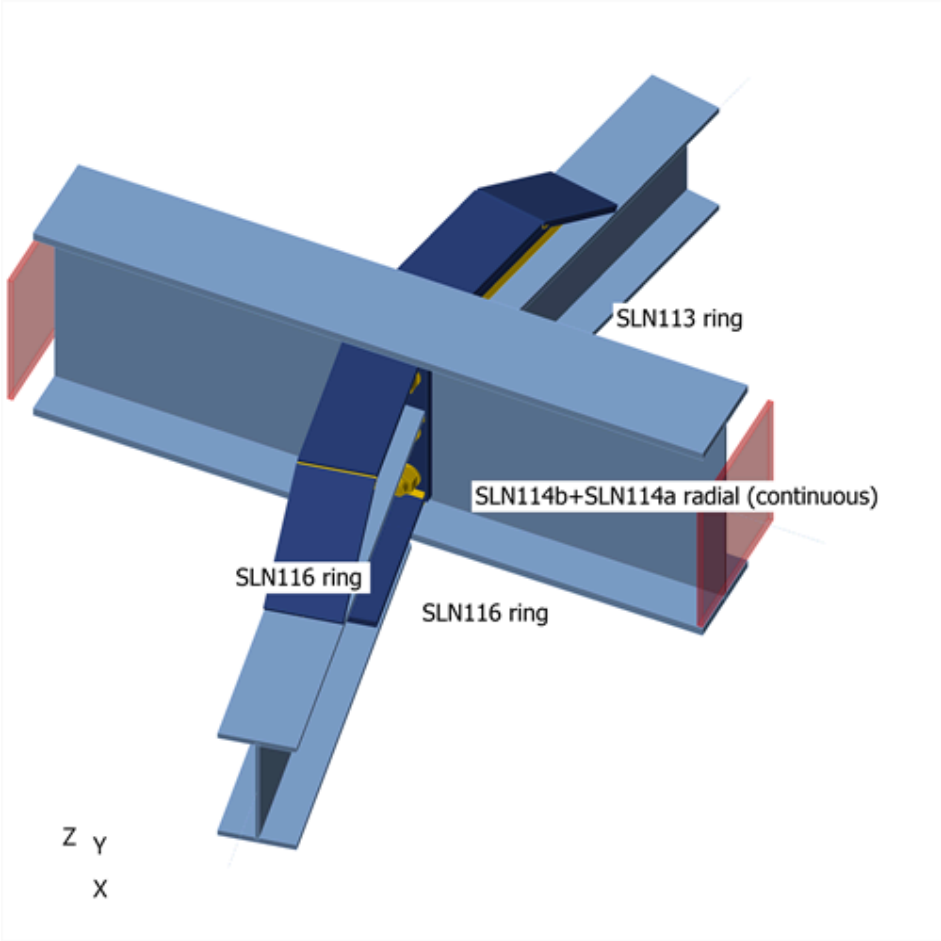
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN114b+SLN114a radial (continuous)	1 - UB457x191x89	-160.0	0.0	0.0	0	0	0
SLN113 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN116 ring	2 - UB254x146x43	-60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN114b+SLN114a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN114b+SLN114a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN113 ring / end		Position	0
SLN116 ring / end		Position	0



SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN113 ring / End	32.8	0.1	-25.6	0.0	0.0	0.2
	SLN116 ring / End	32.9	-0.1	-25.6	0.0	0.0	-0.2
C LC_119	SLN113 ring / End	30.9	0.0	-25.6	0.0	0.0	0.1
	SLN116 ring / End	30.8	0.0	-25.6	0.0	0.0	-0.1
C LC_120	SLN113 ring / End	30.5	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	30.6	-0.2	-25.6	0.0	0.0	-0.2
C LC_121	SLN113 ring / End	30.7	0.3	-25.6	0.0	0.0	0.4
	SLN116 ring / End	30.9	0.0	-25.6	0.0	0.0	0.0
C LC_122	SLN113 ring / End	30.8	-0.1	-25.6	0.0	0.0	-0.1
	SLN116 ring / End	30.9	-0.2	-25.6	0.0	0.0	-0.4
C LC_146	SLN113 ring / End	31.9	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	32.1	-0.2	-25.6	0.0	0.0	-0.3
C LC_147	SLN113 ring / End	32.5	0.2	-25.6	0.0	0.0	0.3
	SLN116 ring / End	32.4	-0.2	-25.6	0.0	0.0	-0.2
C LC_148	SLN113 ring / End	33.0	0.1	-25.6	0.0	0.0	0.1
	SLN116 ring / End	33.3	-0.1	-25.6	0.0	0.0	-0.2
C LC_152	SLN113 ring / End	33.6	0.1	-14.2	0.0	0.0	0.2
	SLN116 ring / End	33.7	-0.1	-14.0	0.0	0.0	-0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.2 < 100%	OK
Welds	23.7 < 100%	OK
Buckling	Not calculated	

Plates

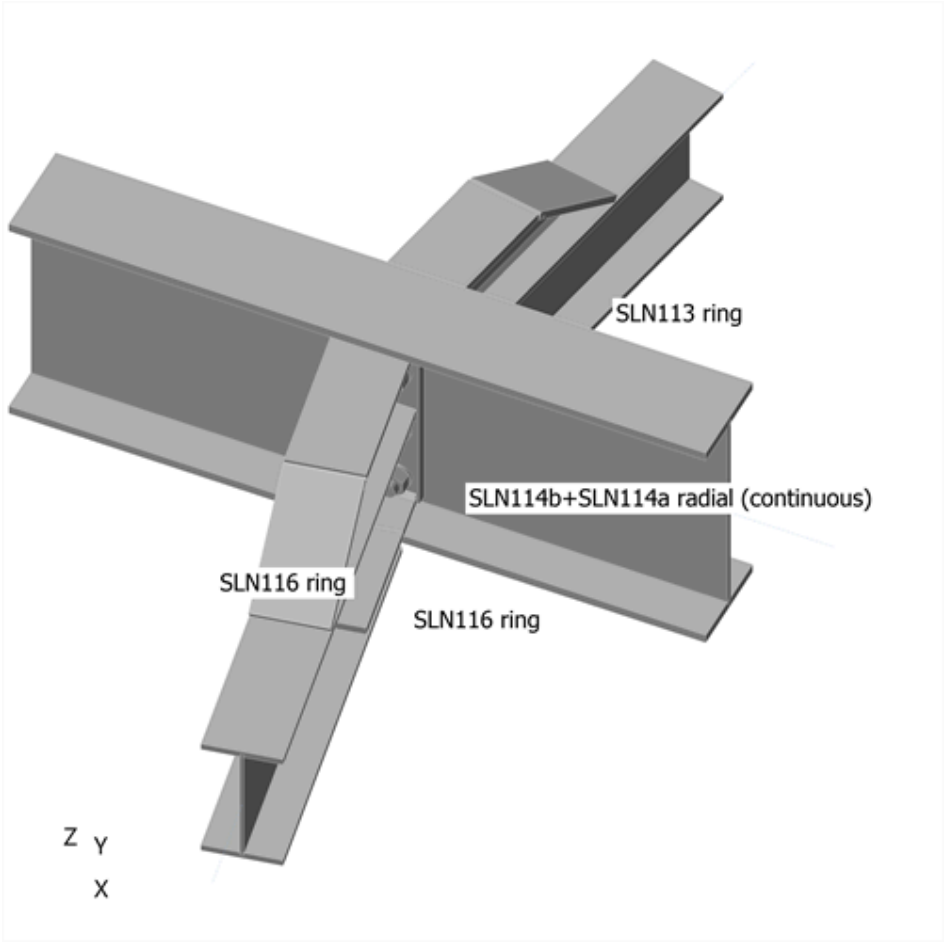
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN114b+SLN114a radial (continuous)-bfl 1	17.7	C LC_122	22.1	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-tfl 1	17.7	C LC_122	18.7	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-w 1	10.5	C LC_122	30.1	0.0	0.8	OK
SLN113 ring-bfl 1	12.7	C LC_148	52.6	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_121	41.9	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_148	51.3	0.0	0.0	OK
SLN116 ring-bfl 1	12.7	C LC_148	52.7	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	42.2	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_148	51.1	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_148	12.8	0.0	0.6	OK
Bottom flange plate SLN116 ring	15.0	C LC_148	25.2	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_148	24.8	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_148	24.5	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_122	25.7	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_122	6.3	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_121	11.5	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_122	13.8	0.0	5.8	OK
Bottom flange plate SLN113 ring	15.0	C LC_148	25.0	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_148	24.4	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_148	24.7	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_122	26.9	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_121	6.1	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_122	11.7	0.0	0.0	OK

Design data

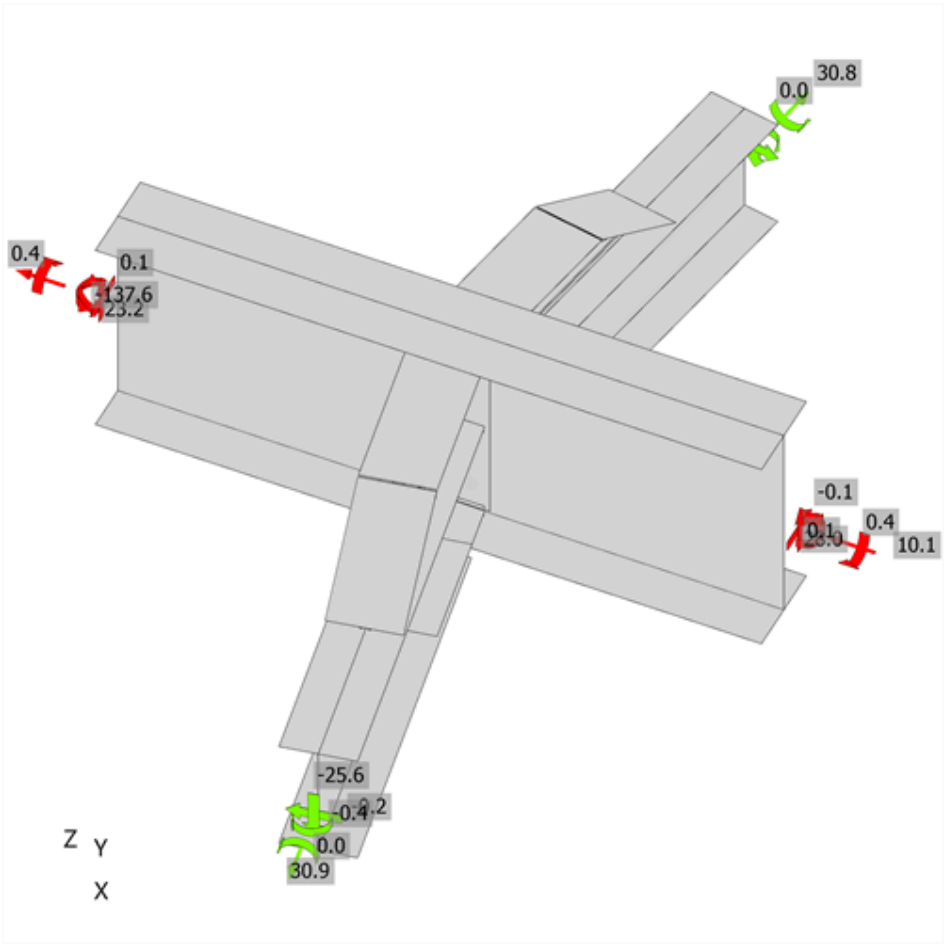
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

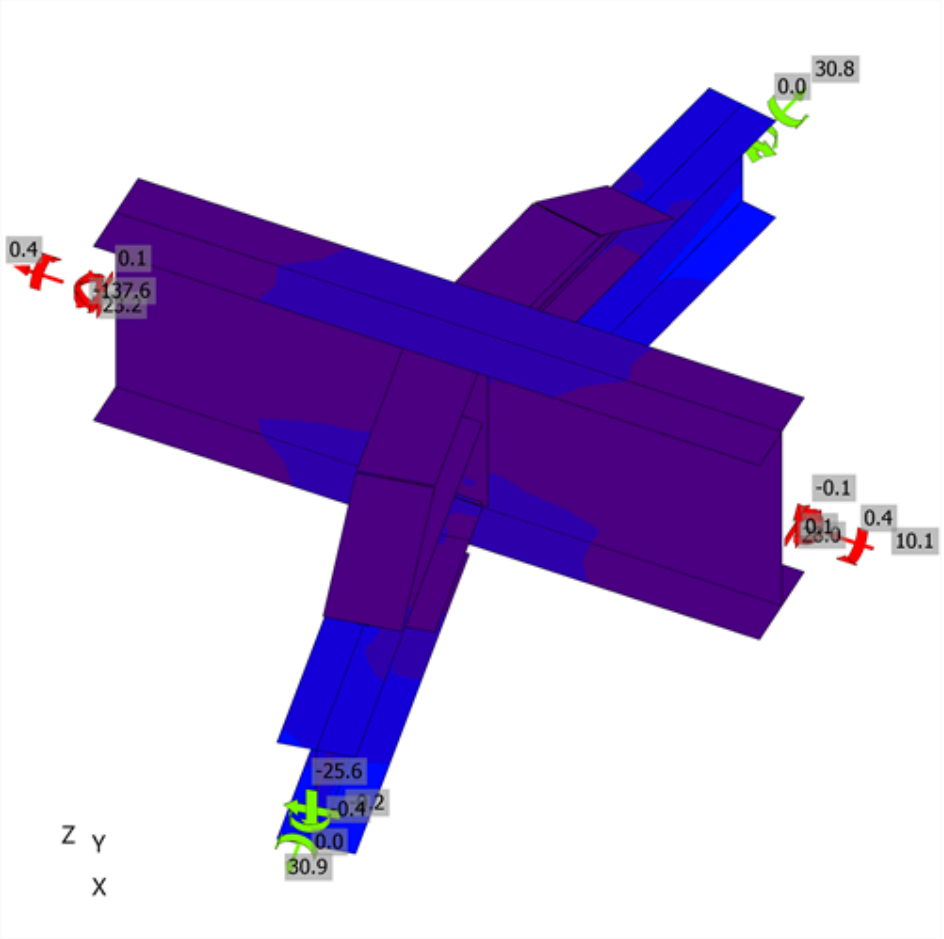
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

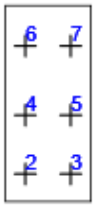


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	3.4	0.5	176.5	1.1	0.5	1.0	OK	OK
	B3	M30 8.8 - 1	C LC_122	3.8	0.7	176.5	1.2	0.8	1.2	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_102	0.2	0.5	213.9	0.1	0.5	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_121	2.4	0.8	229.5	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.0	1.0	229.5	0.3	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 10.5	149	C LC_122	13.7	0.0	-6.0	-7.1	0.6	3.4	3.4	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 10.5	149	C LC_121	13.6	0.0	-6.1	-7.0	-0.7	3.4	3.4	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_148	37.3	0.0	-14.9	-18.5	-7.0	9.2	7.0	OK	OK
		◢ 6.0 ◣	208	C LC_148	29.5	0.0	-16.4	14.1	1.6	7.3	6.3	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	26.6	0.0	2.8	3.2	14.9	6.6	6.6	OK	OK
		◢ 6.0 ◣	100	C LC_148	30.2	0.0	-1.8	2.6	-17.2	7.5	7.5	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_148	29.7	0.0	7.4	7.4	14.9	7.3	6.8	OK	OK
		◢ 6.0 ◣	243	C LC_148	33.3	0.0	9.9	2.6	-18.2	8.2	6.9	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_122	30.4	0.0	7.3	7.5	-15.3	7.5	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_148	32.5	0.0	9.2	3.0	17.8	8.0	7.0	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	59.7	0.0	-13.0	-14.8	30.2	14.7	8.8	OK	OK
		◢ 6.0 ◣	572	C LC_121	59.2	0.0	-15.0	15.4	-29.2	14.6	8.7	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_102	28.7	0.0	-16.4	-13.5	-1.5	7.1	6.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	37.1	0.0	-14.2	18.9	5.7	9.2	7.0	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_147	29.8	0.0	-1.7	-2.3	17.0	7.4	7.4	OK	OK
		◢ 6.0 ◣	100	C LC_122	27.2	0.0	2.9	-3.8	-15.2	6.7	6.7	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_147	30.9	0.0	7.9	7.6	15.5	7.6	6.9	OK	OK
		◢ 6.0 ◣	243	C LC_148	32.0	0.0	9.6	2.8	-17.4	7.9	6.7	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_122	29.8	0.0	6.7	7.4	-15.0	7.4	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_148	33.6	0.0	9.5	2.8	18.4	8.3	7.2	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	62.3	0.0	-15.7	-17.0	30.4	15.4	9.0	OK	OK
		◢ 6.0 ◣	572	C LC_121	57.5	0.0	-12.3	13.6	-29.5	14.2	8.6	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	148	C LC_148	20.4	0.0	11.0	6.6	-7.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	149	C LC_122	14.7	0.0	-3.0	-1.2	8.2	3.6	3.6	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	7.8	0.0	1.7	1.7	4.1	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.8	0.0	2.7	-2.8	-5.3	2.7	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_122	7.7	0.0	-3.4	-3.0	2.7	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_122	7.5	0.0	-2.7	3.1	-2.6	1.9	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	5.1	0.0	1.8	1.6	2.3	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.1	0.0	-0.1	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_122	46.7	0.0	-7.9	-5.4	26.0	11.5	7.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	48.4	0.0	-9.0	8.4	-26.2	12.0	7.9	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	148	C LC_122	21.0	0.0	10.4	6.3	8.4	5.2	5.2	OK	OK
		◢ 6.0 ◣	149	C LC_122	17.7	0.0	-7.8	7.7	-5.0	4.4	4.4	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	2.6	2.8	5.1	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.3	0.0	1.8	-1.7	-4.4	2.0	0.0	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_121	7.3	0.0	-3.1	-2.9	-2.5	1.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_121	7.6	0.0	-2.9	3.1	2.7	1.9	0.0	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_122	4.2	0.0	1.4	1.3	1.9	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	5.0	0.0	1.7	-1.5	-2.2	1.2	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_122	51.6	0.0	-10.1	-8.3	28.0	12.8	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_121	44.1	0.0	-6.9	5.2	-24.6	10.9	7.7	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_148	95.9	0.0	52.6	-44.2	13.5	23.7	15.8	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_148	95.5	0.0	53.6	42.5	-16.7	23.6	15.8	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.6	2.6	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.5	0.0	-0.8	1.9	0.3	0.9	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_122	7.0	0.0	-5.2	2.7	0.4	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.4	0.0	-0.8	1.9	-0.2	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-220
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring

Name	SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

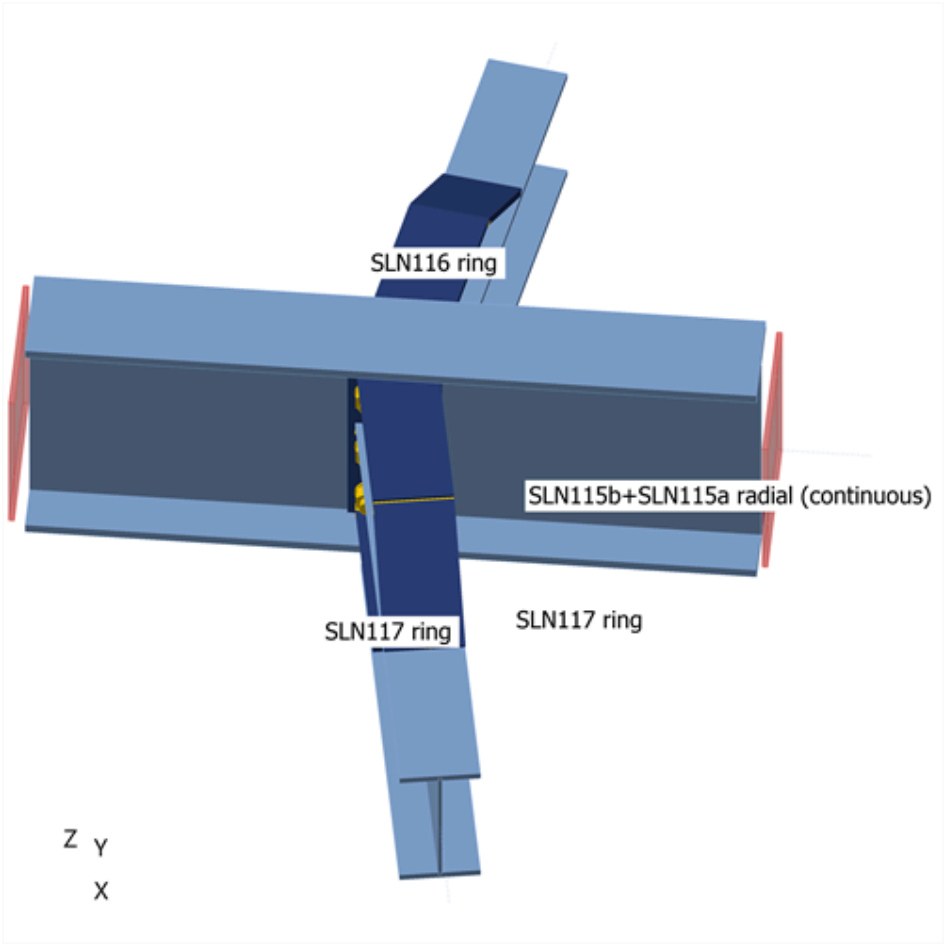
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN115b+SLN115a radial (continuous)	1 - UB457x191x89	-140.0	0.0	0.0	0	0	0
SLN116 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102
SLN117 ring	2 - UB254x146x43	-40.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN115b+SLN115a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN115b+SLN115a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN116 ring / end		Position	0
SLN117 ring / end		Position	0



SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN116 ring / End	32.9	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	33.0	0.3	-26.0	0.0	0.0	0.4
C LC_121	SLN116 ring / End	30.7	0.0	-25.6	0.0	0.0	0.0
	SLN117 ring / End	31.1	0.3	-26.0	0.0	0.0	0.5
C LC_122	SLN116 ring / End	31.0	-0.3	-25.6	0.0	0.0	-0.5
	SLN117 ring / End	30.9	0.2	-26.0	0.0	0.0	0.1
C LC_145	SLN116 ring / End	33.6	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	33.6	0.4	-26.0	0.0	0.0	0.4
C LC_146	SLN116 ring / End	32.2	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	32.5	0.2	-26.0	0.0	0.0	0.3
C LC_148	SLN116 ring / End	33.5	-0.2	-25.6	0.0	0.0	-0.3
	SLN117 ring / End	33.8	0.4	-26.0	0.0	0.0	0.4
C LC_152	SLN116 ring / End	33.7	-0.1	-14.0	0.0	0.0	-0.2
	SLN117 ring / End	33.9	0.3	-14.0	0.0	0.0	0.4

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.2 < 100%	OK
Welds	23.9 < 100%	OK
Buckling	Not calculated	

Plates

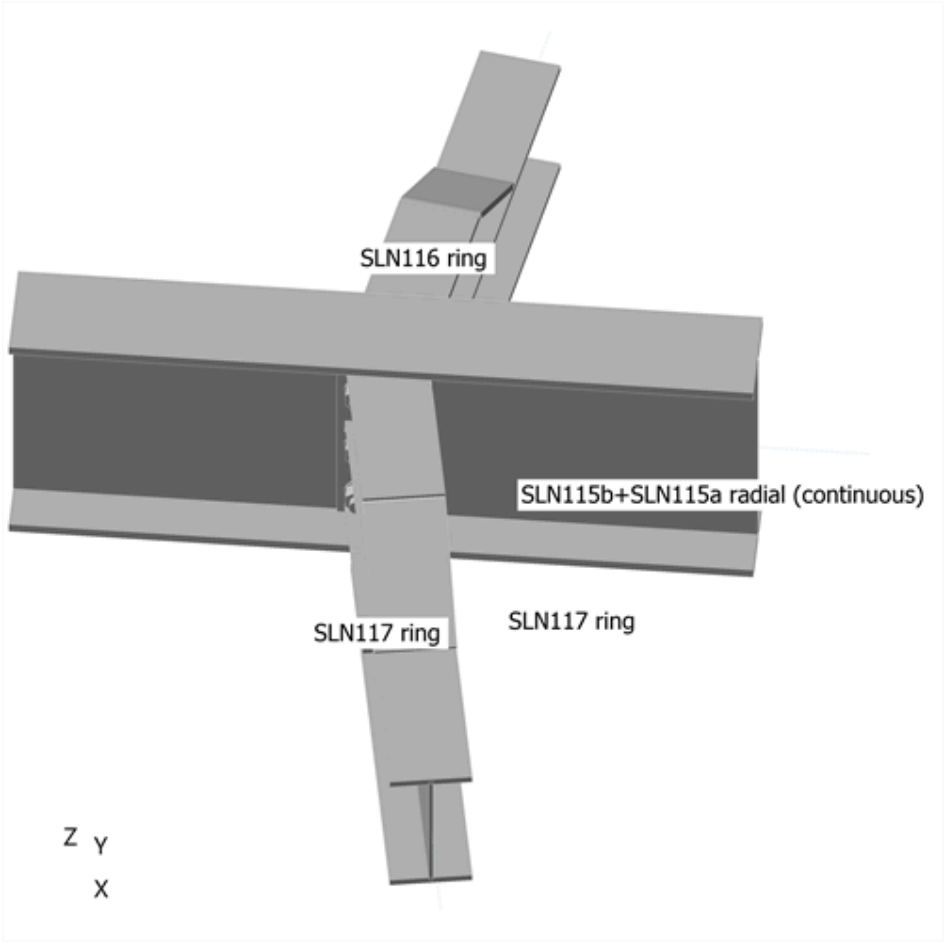
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN115b+SLN115a radial (continuous)-bfl 1	17.7	C LC_146	22.2	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-tfl 1	17.7	C LC_121	19.1	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-w 1	10.5	C LC_121	30.2	0.0	0.7	OK
SLN116 ring-bfl 1	12.7	C LC_146	52.8	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	42.4	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_145	51.2	0.0	0.0	OK
SLN117 ring-bfl 1	12.7	C LC_121	53.4	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	43.0	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	52.0	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_145	14.4	0.0	2.5	OK
Bottom flange plate SLN117 ring	15.0	C LC_148	25.6	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	24.9	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_148	25.0	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_121	26.8	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_121	7.1	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_122	11.7	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_122	13.7	0.0	3.3	OK
Bottom flange plate SLN116 ring	15.0	C LC_148	25.2	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_148	24.7	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_145	24.6	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_122	25.9	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_122	6.6	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_121	11.6	0.0	0.0	OK

Design data

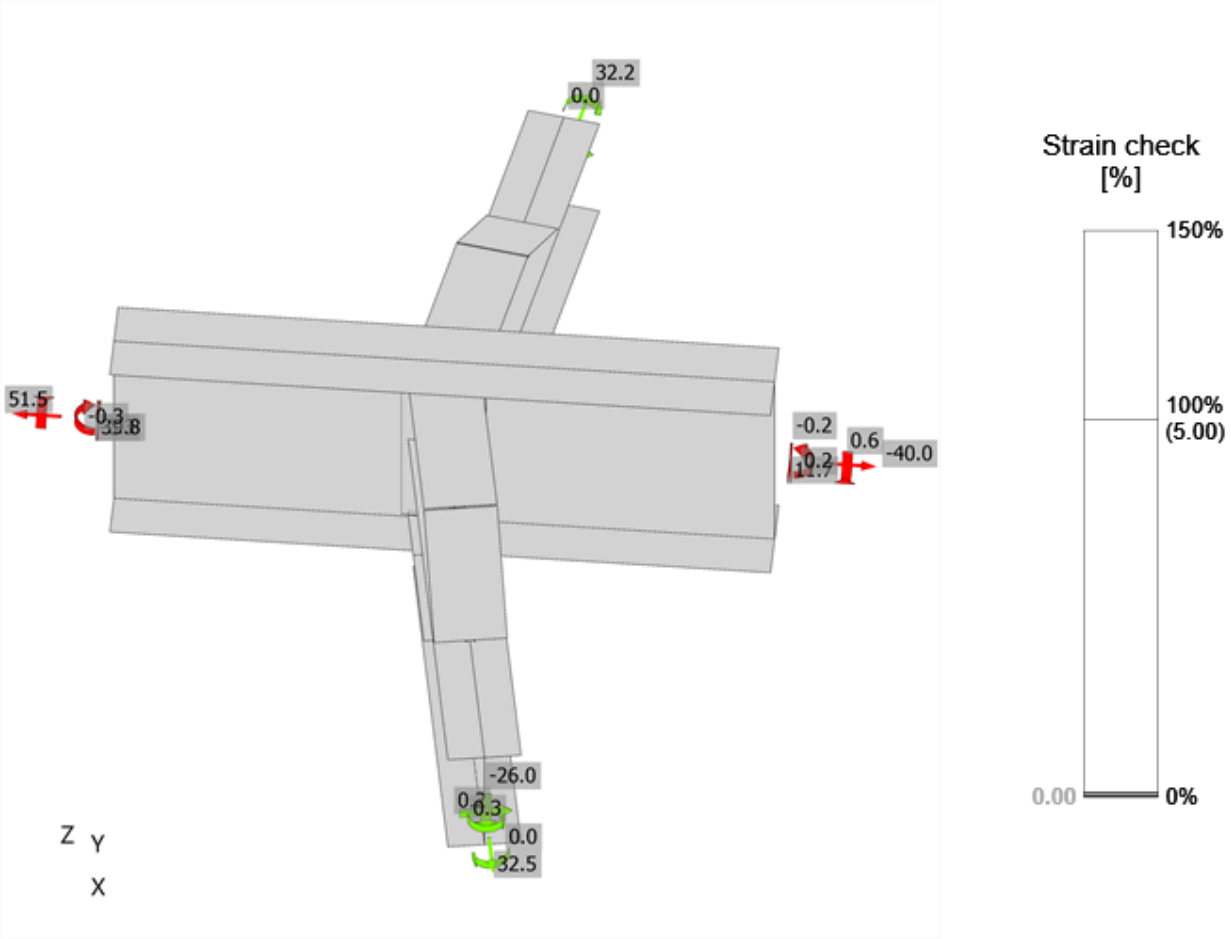
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

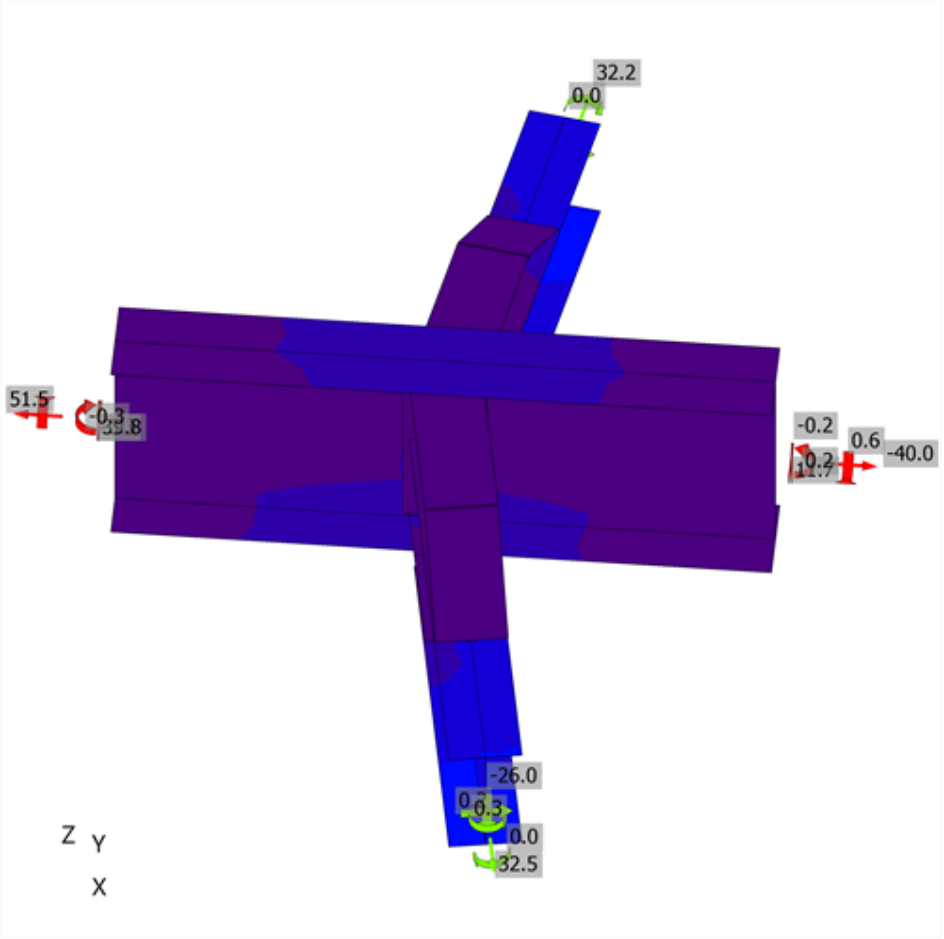
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

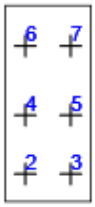


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	3.4	0.5	176.5	1.1	0.5	1.0	OK	OK
	B3	M30 8.8 - 1	C LC_145	3.9	0.8	176.5	1.2	0.8	1.2	OK	OK
	B4	M30 8.8 - 1	C LC_148	0.3	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.2	0.5	213.9	0.1	0.5	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.8	0.9	229.5	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_121	2.4	0.9	229.5	0.7	0.8	1.0	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 10.5	149	C LC_121	13.8	0.0	-5.4	-7.3	-1.1	3.4	3.4	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 10.5	149	C LC_145	13.6	0.0	-5.3	-7.2	0.3	3.4	3.4	OK	OK
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_148	37.8	0.0	-14.9	-19.1	-6.3	9.3	7.0	OK	OK
		◢ 6.0 ◣	208	C LC_148	29.8	0.0	-16.9	14.1	1.9	7.4	6.4	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_121	28.4	0.0	3.1	3.9	15.8	7.0	7.0	OK	OK
		◢ 6.0 ◣	100	C LC_148	29.1	0.0	-1.9	2.1	-16.7	7.2	7.2	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_145	31.8	0.0	7.9	7.8	15.9	7.8	7.0	OK	OK
		◢ 6.0 ◣	243	C LC_148	33.1	0.0	9.2	3.0	-18.1	8.2	7.1	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_148	29.1	0.0	7.2	7.2	-14.6	7.2	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_148	33.6	0.0	10.2	2.5	18.3	8.3	7.0	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	61.0	0.0	-16.2	-16.3	29.8	15.1	8.9	OK	OK
		◢ 6.0 ◣	572	C LC_121	59.8	0.0	-12.4	14.4	-30.5	14.8	8.8	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_145	29.2	0.0	-16.1	-14.0	-1.5	7.2	6.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	36.7	0.0	-14.8	18.5	6.0	9.1	6.9	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	29.8	0.0	-1.4	-2.4	17.0	7.4	7.4	OK	OK
		◢ 6.0 ◣	100	C LC_122	27.7	0.0	3.1	-4.0	-15.4	6.9	6.9	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_121	29.9	0.0	7.1	7.4	15.0	7.4	6.7	OK	OK
		◢ 6.0 ◣	243	C LC_148	33.0	0.0	10.1	2.6	-18.0	8.2	6.8	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_148	30.6	0.0	7.5	7.6	-15.4	7.6	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	32.9	0.0	9.3	2.9	18.0	8.1	7.0	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	60.8	0.0	-13.2	-15.4	30.6	15.0	8.9	OK	OK
		◢ 6.0 ◣	572	C LC_121	59.2	0.0	-14.9	15.3	-29.3	14.6	8.7	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	148	C LC_145	21.9	0.0	10.7	6.4	-9.0	5.4	5.4	OK	OK
		◢ 6.0 ◣	149	C LC_121	15.7	0.0	-7.5	6.6	4.5	3.9	3.9	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_121	8.3	0.0	2.4	2.2	4.1	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.8	0.0	2.1	-2.7	-5.5	2.7	0.0	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_121	8.3	0.0	-3.8	-3.7	-2.1	2.0	2.0	OK	OK
		◢ 6.0 ◣	149	C LC_121	8.3	0.0	-3.3	3.5	2.6	2.0	0.0	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_121	4.0	0.0	-0.2	-0.3	-2.3	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	5.6	0.0	2.0	-1.7	-2.5	1.4	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_122	50.9	0.0	-9.8	-9.5	27.2	12.6	8.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	45.9	0.0	-7.3	4.6	-25.7	11.3	7.9	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	148	C LC_148	21.1	0.0	10.7	6.4	8.3	5.2	5.2	OK	OK
		◢ 6.0 ◣	149	C LC_121	15.9	0.0	-5.4	4.5	7.4	3.9	3.9	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	10.9	0.0	2.2	2.8	5.5	2.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	2.3	-2.0	-3.9	2.0	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_121	8.4	0.0	-4.3	-2.1	-3.6	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_122	5.8	0.0	-2.5	2.2	-2.0	1.4	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	5.2	0.0	1.9	1.6	2.3	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.1	0.0	1.3	-1.3	-1.8	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_122	47.8	0.0	-8.3	-5.3	26.6	11.8	8.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	48.2	0.0	-8.8	8.2	-26.1	11.9	7.9	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_148	96.8	0.0	54.7	-43.7	14.6	23.9	16.2	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_148	96.0	0.0	52.6	43.6	-15.8	23.7	16.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_121	7.4	0.0	-5.9	2.5	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	4.0	0.0	-0.9	2.3	0.0	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_121	7.1	0.0	-5.2	2.7	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.6	0.0	-0.8	2.0	0.3	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-240
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring

Name	SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

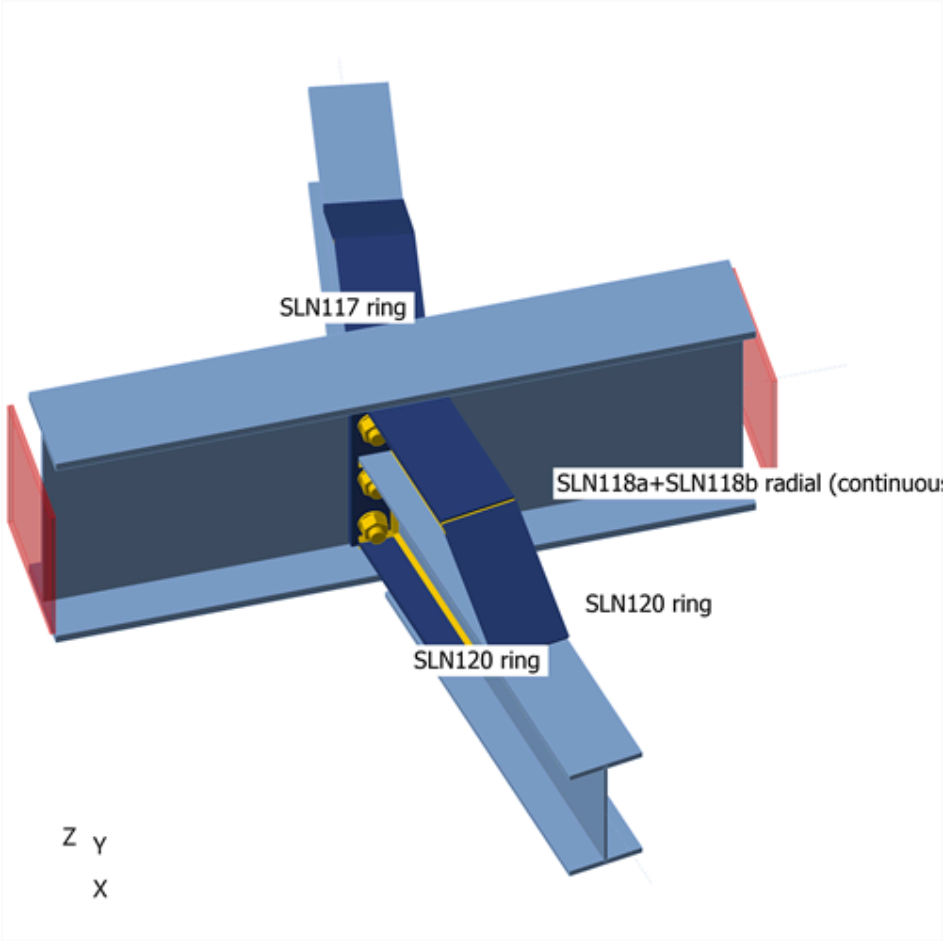
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN118a+SLN118b radial (continuous)	1 - UB457x191x89	-120.0	0.0	0.0	0	0	0
SLN117 ring	2 - UB254x146x43	140.0	0.0	0.0	0	0	-102
SLN120 ring	2 - UB254x146x43	-20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN118a+SLN118b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN118a+SLN118b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN117 ring / end		Position	0
SLN120 ring / end		Position	0



SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN117 ring / End	33.0	0.3	-26.0	0.0	0.0	0.6
	SLN120 ring / End	33.2	-1.7	-25.7	0.0	0.0	-1.5
C LC_105	SLN117 ring / End	28.8	0.5	-24.4	0.0	0.0	0.7
	SLN120 ring / End	29.2	-1.7	-24.1	0.0	0.0	-1.4
C LC_120	SLN117 ring / End	31.1	0.4	-26.0	0.0	0.0	0.6
	SLN120 ring / End	31.4	-1.7	-25.7	0.0	0.0	-1.5
C LC_121	SLN117 ring / End	31.0	0.5	-26.0	0.0	0.0	0.7
	SLN120 ring / End	31.4	-1.7	-25.7	0.0	0.0	-1.4
C LC_122	SLN117 ring / End	31.0	0.1	-26.0	0.0	0.0	0.3
	SLN120 ring / End	31.1	-1.6	-25.7	0.0	0.0	-1.5
C LC_145	SLN117 ring / End	33.5	0.3	-26.0	0.0	0.0	0.6
	SLN120 ring / End	33.4	-1.6	-25.7	0.0	0.0	-1.5
C LC_146	SLN117 ring / End	32.6	0.4	-26.0	0.0	0.0	0.6
	SLN120 ring / End	33.0	-1.7	-25.7	0.0	0.0	-1.6
C LC_147	SLN117 ring / End	32.2	0.4	-26.0	0.0	0.0	0.6
	SLN120 ring / End	32.3	-1.7	-25.7	0.0	0.0	-1.5
C LC_148	SLN117 ring / End	33.9	0.2	-26.0	0.0	0.0	0.5
	SLN120 ring / End	34.1	-1.6	-25.7	0.0	0.0	-1.5
C LC_152	SLN117 ring / End	33.9	0.3	-14.0	0.0	0.0	0.6
	SLN120 ring / End	34.1	-1.8	-13.8	0.0	0.0	-1.6
C LC_153	SLN117 ring / End	33.1	0.3	-20.9	0.0	0.0	0.6
	SLN120 ring / End	33.2	-1.7	-20.7	0.0	0.0	-1.5

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	2.3 < 100%	OK
Welds	28.5 < 100%	OK
Buckling	Not calculated	

Plates

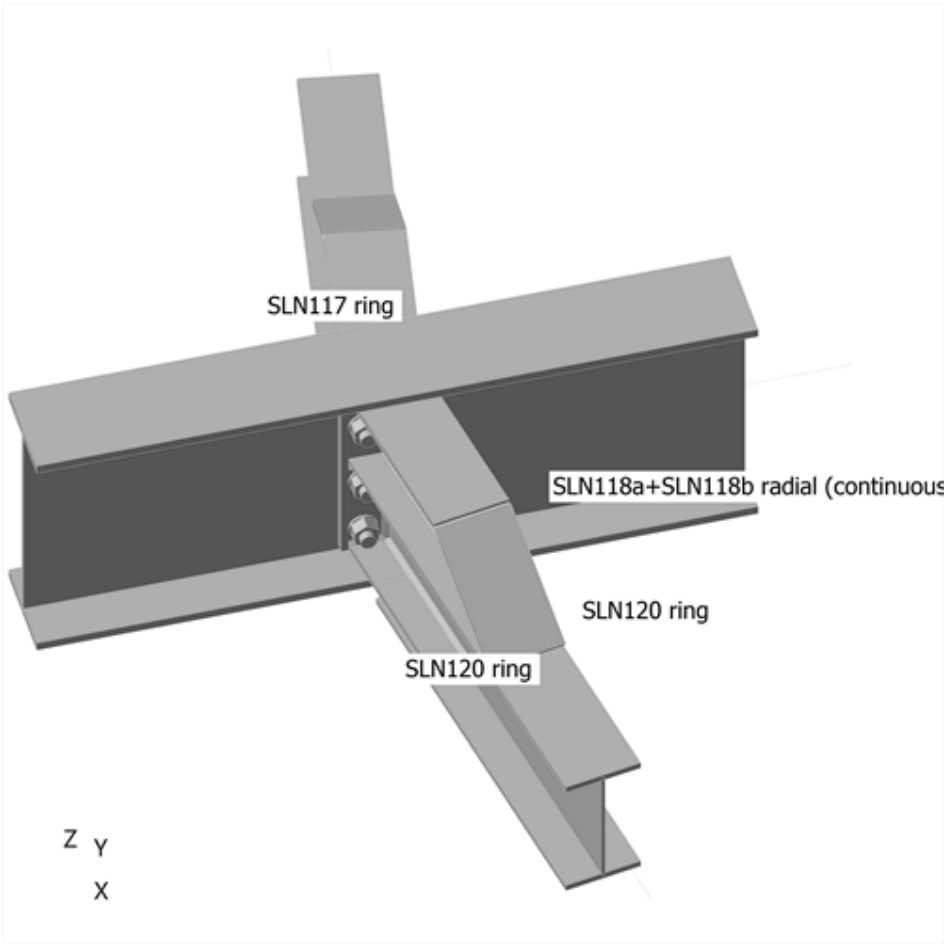
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN118a+SLN118b radial (continuous)-bfl 1	17.7	C LC_122	23.7	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-tfl 1	17.7	C LC_122	20.5	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-w 1	10.5	C LC_122	33.2	0.0	2.6	OK
SLN117 ring-bfl 1	12.7	C LC_145	54.9	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	44.4	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	51.9	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_121	54.5	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_122	42.3	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	51.5	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_146	24.8	0.0	13.1	OK
Bottom flange plate SLN120 ring	15.0	C LC_146	38.6	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	26.6	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_146	26.5	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_120	28.4	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_122	13.3	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_120	12.4	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_146	21.3	0.0	8.8	OK
Bottom flange plate SLN117 ring	15.0	C LC_122	29.0	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	24.4	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_145	24.8	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_121	26.7	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_122	10.4	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_120	11.8	0.0	0.0	OK

Design data

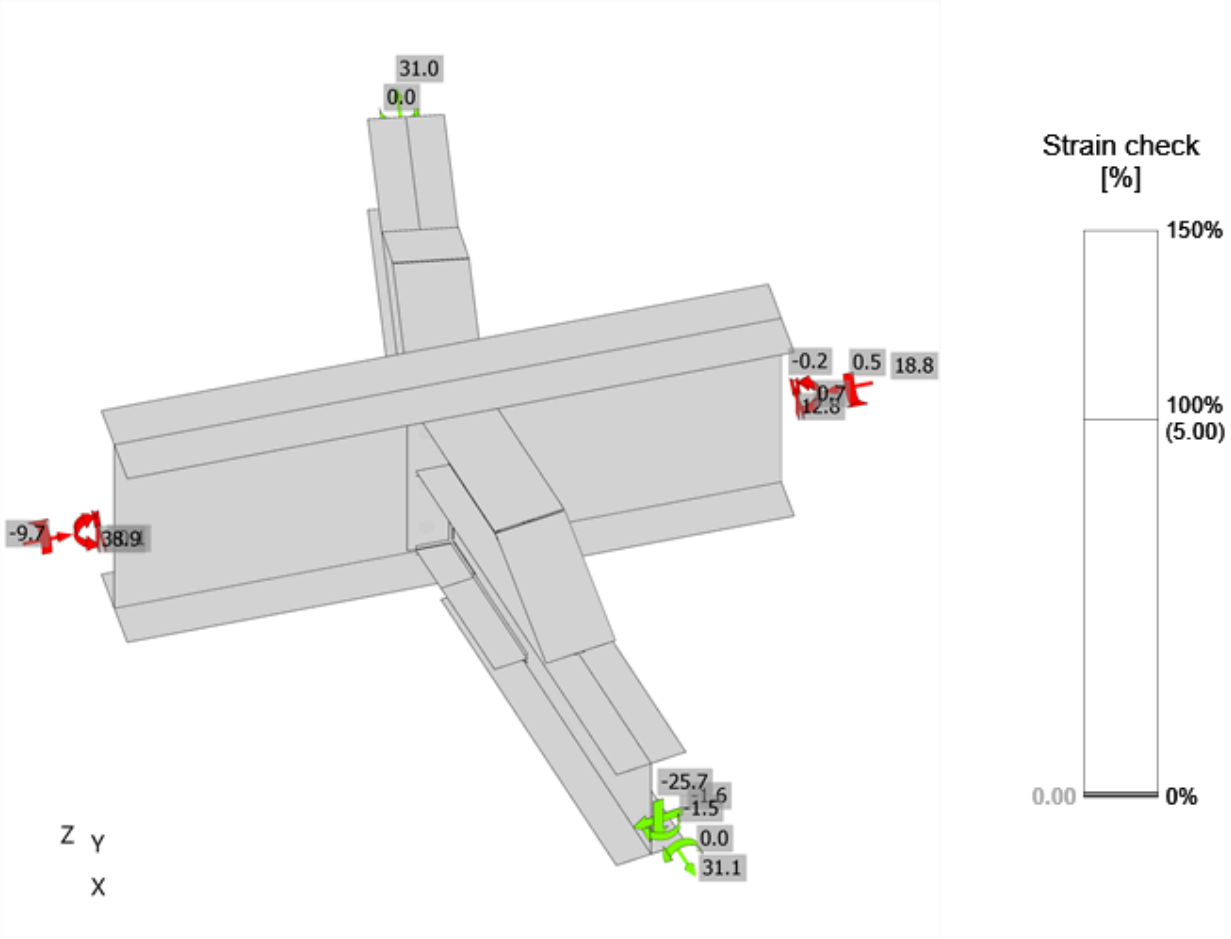
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

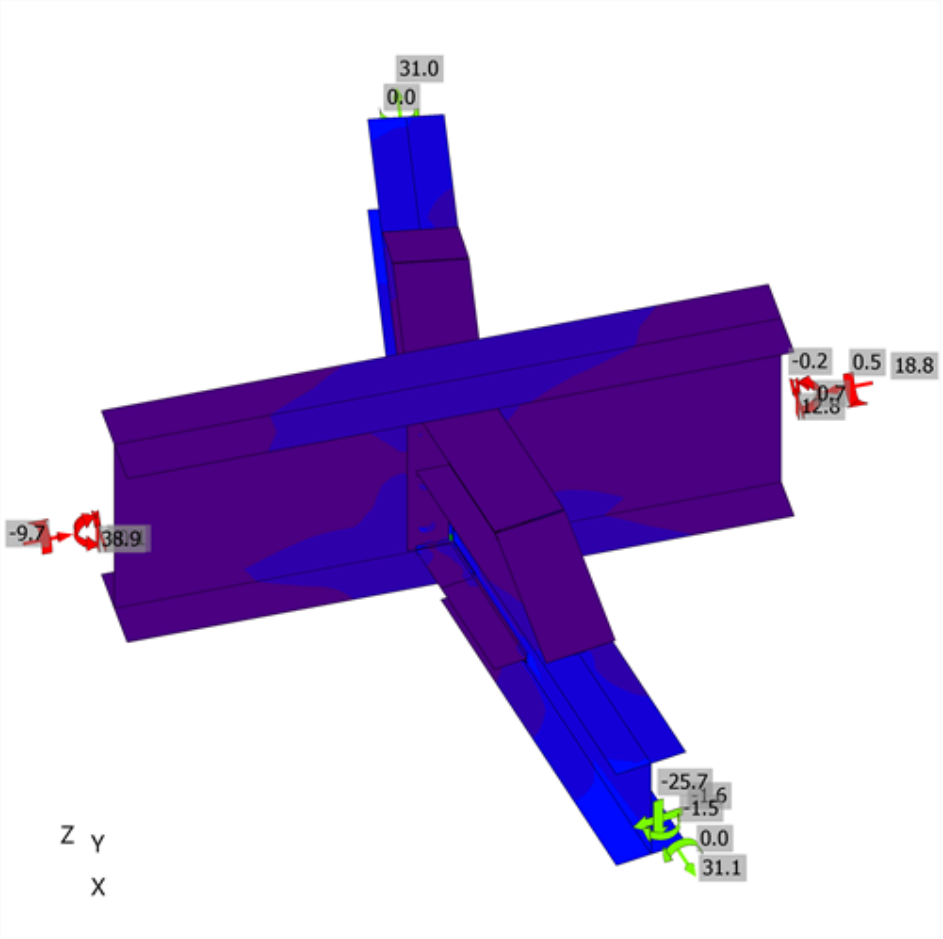
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

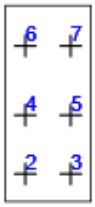


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	7.5	0.7	176.5	2.3	0.7	2.0	OK	OK
	B3	M30 8.8 - 1	C LC_122	1.6	0.7	208.4	0.5	0.6	0.7	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.4	0.4	213.9	0.1	0.3	0.3	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.5	0.5	213.9	0.2	0.5	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_122	3.5	0.8	229.5	1.1	0.7	1.2	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.4	1.0	229.5	0.4	0.8	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 10.5	149	C LC_152	21.2	0.0	10.9	-10.4	-1.2	5.2	5.2	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 10.5	149	C LC_152	27.5	0.0	13.4	-13.8	-1.2	6.8	6.8	OK	OK
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	43.4	0.0	-20.3	-17.9	-13.0	10.7	7.6	OK	OK
		◢ 6.0 ◣	208	C LC_148	36.9	0.0	-16.7	19.0	0.2	9.1	7.0	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_146	28.9	0.0	7.4	15.6	4.0	7.1	7.1	OK	OK
		◢ 6.0 ◣	100	C LC_148	39.6	0.0	-1.5	5.5	-22.2	9.8	8.6	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_145	24.3	0.0	7.7	6.0	11.9	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_148	34.4	0.0	14.8	0.2	-17.9	8.5	6.7	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	36.1	0.0	7.1	9.0	-18.4	8.9	6.6	OK	OK
		◢ 6.0 ◣	243	C LC_146	46.7	0.0	13.3	-11.3	23.3	11.5	8.3	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_120	70.7	0.0	-12.7	-19.4	35.2	17.5	9.8	OK	OK
		◢ 6.0 ◣	572	C LC_122	47.1	0.0	-17.6	10.8	-22.8	11.6	7.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_148	31.6	0.0	-17.0	-15.2	2.4	7.8	6.5	OK	OK
		◢ 6.0 ◣	208	C LC_148	38.4	0.0	-15.5	18.6	8.2	9.5	7.1	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_148	35.2	0.0	-0.8	-3.2	20.1	8.7	8.2	OK	OK
		◢ 6.0 ◣	100	C LC_146	28.1	0.0	9.4	-14.0	-6.1	6.9	6.9	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_121	31.2	0.0	7.2	7.7	15.7	7.7	6.8	OK	OK
		◢ 6.0 ◣	243	C LC_121	34.9	0.0	9.1	-8.5	-17.5	8.6	7.5	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_122	29.7	0.0	7.5	7.3	-14.9	7.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	33.0	0.0	11.0	2.2	17.9	8.2	6.8	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	57.0	0.0	-15.1	-14.1	28.4	14.1	8.6	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.9	0.0	-13.9	16.7	-31.9	15.8	9.1	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_122	51.8	0.0	-17.4	-17.0	22.5	12.8	8.5	OK	OK
		◢ 6.0 ◣	149	C LC_122	56.0	0.0	-19.9	20.1	-22.5	13.8	8.0	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_122	6.6	0.0	-0.1	-0.4	-3.8	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	12.0	0.0	3.4	-3.5	-5.7	3.0	0.0	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_122	22.4	0.0	-10.1	-8.6	7.7	5.5	5.5	OK	OK
		◢ 6.0 ◣	149	C LC_122	20.3	0.0	-7.1	8.5	-7.0	5.0	5.0	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_122	9.1	0.0	3.1	2.4	4.3	2.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	6.5	0.0	-0.9	1.3	3.4	1.6	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_120	56.2	0.0	-10.5	-1.3	31.9	13.9	8.9	OK	OK
		◢ 6.0 ◣	271	C LC_122	45.1	0.0	-7.0	16.4	-19.8	11.1	7.8	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	148	C LC_146	23.2	0.0	14.8	10.2	-1.9	5.7	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_122	30.4	0.0	-16.4	11.9	-8.7	7.5	6.0	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_121	8.7	0.0	3.0	2.7	3.9	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.1	0.0	1.3	-2.1	-5.4	2.5	2.5	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_122	14.5	0.0	-8.6	-5.1	4.4	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_152	6.8	0.0	-3.8	1.5	2.9	1.7	1.7	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_122	4.2	0.0	1.1	1.1	2.1	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.9	0.0	2.0	-1.8	-1.9	1.2	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_122	46.7	0.0	-8.1	-9.5	24.8	11.5	7.7	OK	OK
		◢ 6.0 ◣	271	C LC_121	50.8	0.0	-9.2	5.3	-28.3	12.5	8.2	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	115.4	0.0	52.7	-58.6	9.0	28.5	18.9	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_148	106.3	0.0	54.9	48.2	-21.0	26.3	17.6	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_122	10.8	0.0	-9.8	-2.6	0.0	3.2	3.2	OK	OK
		◢ 6.0 ◣	147	C LC_122	7.3	0.0	-1.7	4.1	0.0	1.8	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.3	2.8	0.4	2.1	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.7	0.0	-1.0	1.6	-1.2	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-300
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring

Name	SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

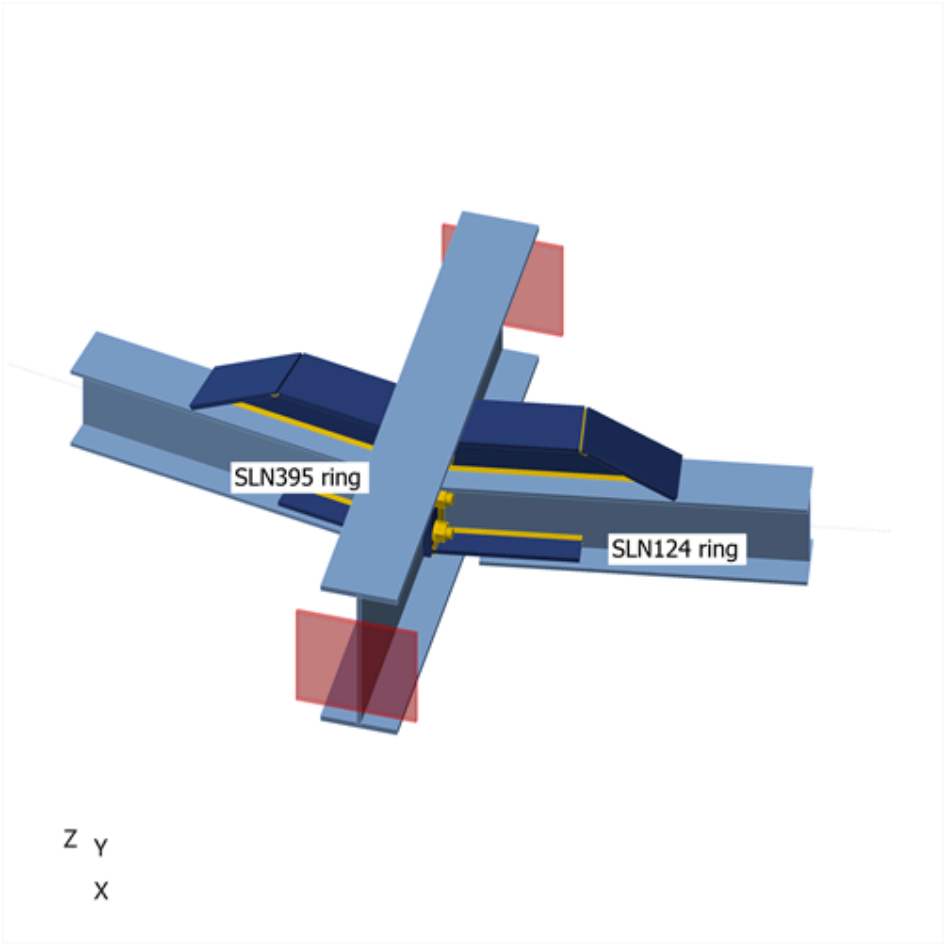
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN122a+SLN122b radial (continuous)	1 - UB457x191x89	-60.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN395 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN122a+SLN122b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN122a+SLN122b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN395 ring / end		Position	0



SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN124 ring / End	-15.5	0.0	-25.6	0.0	0.0	0.0
	SLN395 ring / End	-15.5	0.1	-8.7	0.0	0.0	0.0
C LC_104	SLN124 ring / End	-18.2	0.2	-24.0	0.0	0.0	0.3
	SLN395 ring / End	-19.2	0.2	-8.2	0.0	0.0	0.3
C LC_105	SLN124 ring / End	-18.8	0.1	-24.0	0.0	0.0	0.2
	SLN395 ring / End	-19.6	0.2	-8.2	0.0	0.0	0.2
C LC_112	SLN124 ring / End	-15.7	0.3	-16.2	0.0	0.0	0.4
	SLN395 ring / End	-17.4	0.3	-5.8	0.0	0.0	0.5
C LC_113	SLN124 ring / End	-15.8	0.3	-16.2	0.0	0.0	0.5
	SLN395 ring / End	-17.1	0.2	-5.8	0.0	0.0	0.3
C LC_114	SLN124 ring / End	-14.5	-0.3	-16.2	0.0	0.0	-0.5
	SLN395 ring / End	-13.4	0.0	-5.8	0.0	0.0	-0.2
C LC_119	SLN124 ring / End	-16.6	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-16.6	0.2	-8.7	0.0	0.0	0.2
C LC_120	SLN124 ring / End	-17.9	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-18.1	0.1	-8.7	0.0	0.0	0.0
C LC_121	SLN124 ring / End	-17.3	0.0	-25.6	0.0	0.0	0.0
	SLN395 ring / End	-17.5	0.1	-8.7	0.0	0.0	0.0
C LC_145	SLN124 ring / End	-15.8	0.0	-25.6	0.0	0.0	0.0
	SLN395 ring / End	-15.6	0.1	-8.7	0.0	0.0	0.0
C LC_146	SLN124 ring / End	-15.2	0.1	-25.6	0.0	0.0	0.0
	SLN395 ring / End	-15.3	0.0	-8.7	0.0	0.0	-0.1
C LC_148	SLN124 ring / End	-14.8	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-14.8	0.1	-8.7	0.0	0.0	0.0
C LC_154	SLN124 ring / End	-13.2	-0.1	-11.6	0.0	0.0	-0.1
	SLN395 ring / End	-12.9	0.0	-8.7	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.6 < 100%	OK
Welds	19.2 < 100%	OK
Buckling	Not calculated	

Plates

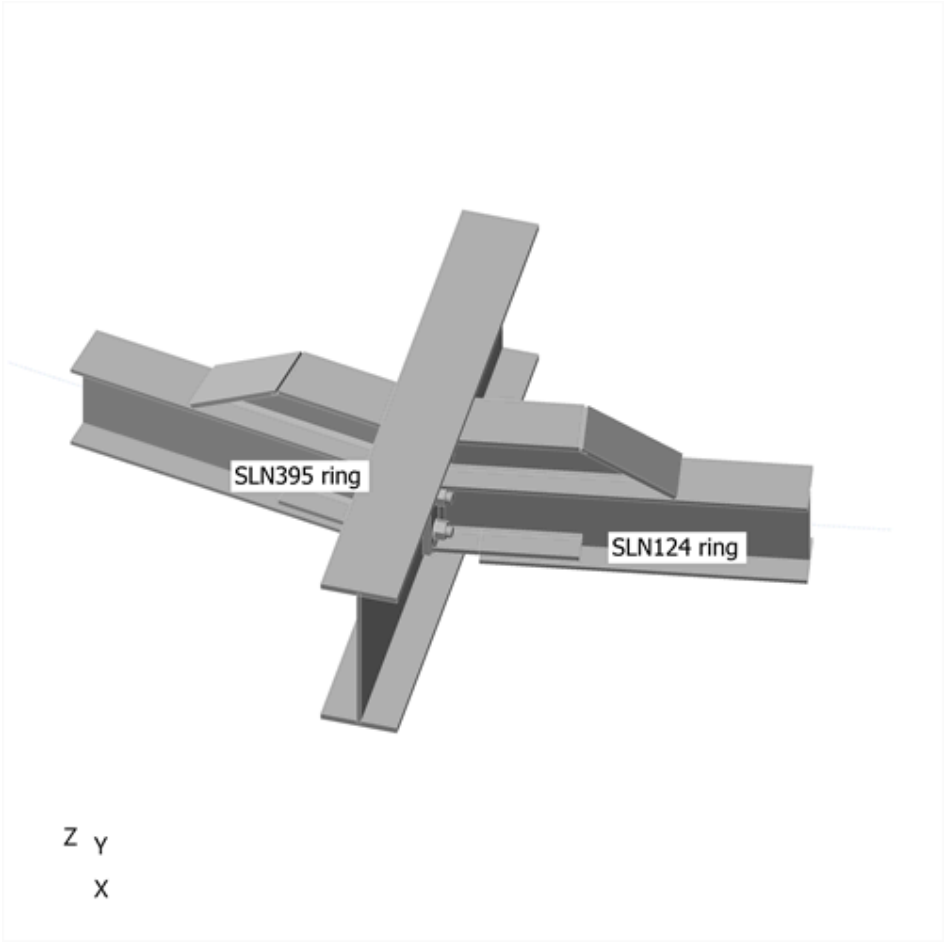
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN122a+SLN122b radial (continuous)-bfl 1	17.7	C LC_104	15.9	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-tfl 1	17.7	C LC_104	13.1	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-w 1	10.5	C LC_121	18.5	0.0	1.5	OK
SLN124 ring-bfl 1	12.7	C LC_119	43.8	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_119	49.8	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_120	49.1	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_154	14.5	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_121	19.7	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_120	18.6	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_114	7.7	0.0	8.6	OK
Bottom flange plate SLN124 ring	15.0	C LC_114	19.7	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_114	7.7	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_114	8.1	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_120	34.6	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_119	5.7	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_120	15.1	0.0	0.0	OK
End plate SLN395 ring	20.0	C LC_112	9.2	0.0	10.0	OK
Bottom flange plate SLN395 ring	15.0	C LC_112	21.4	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_112	6.1	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_112	8.5	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_120	13.7	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_112	8.5	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_120	6.1	0.0	0.0	OK

Design data

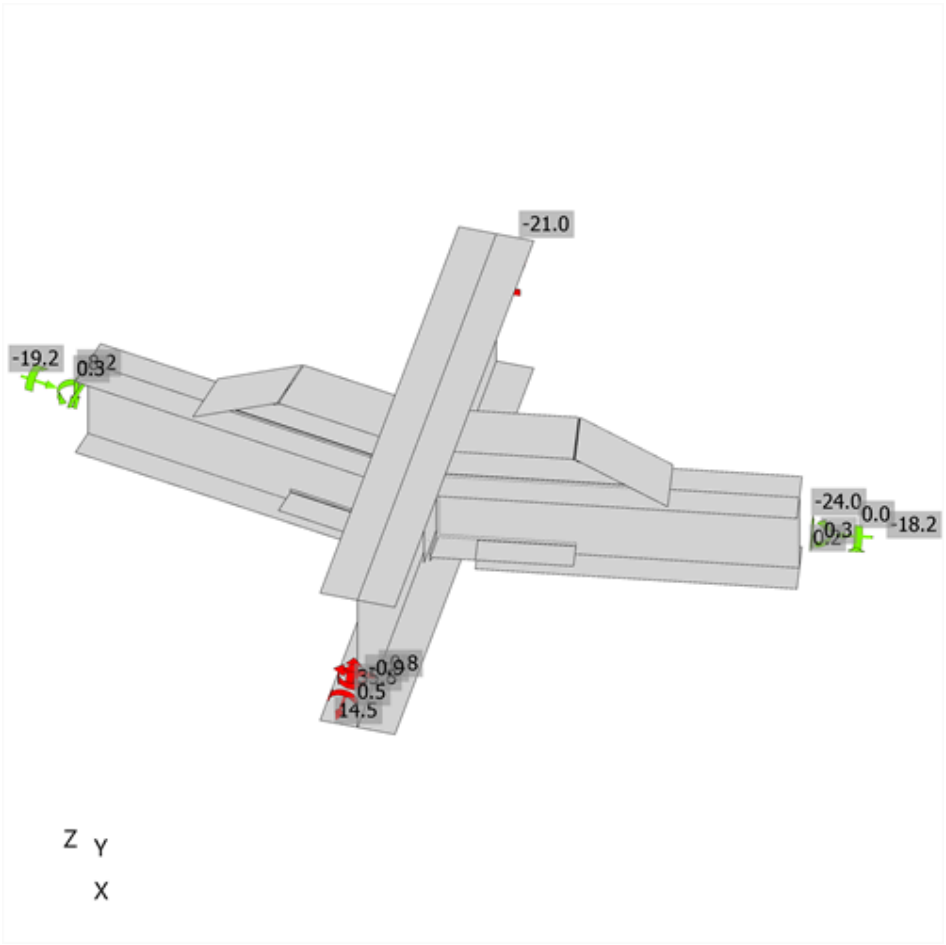
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

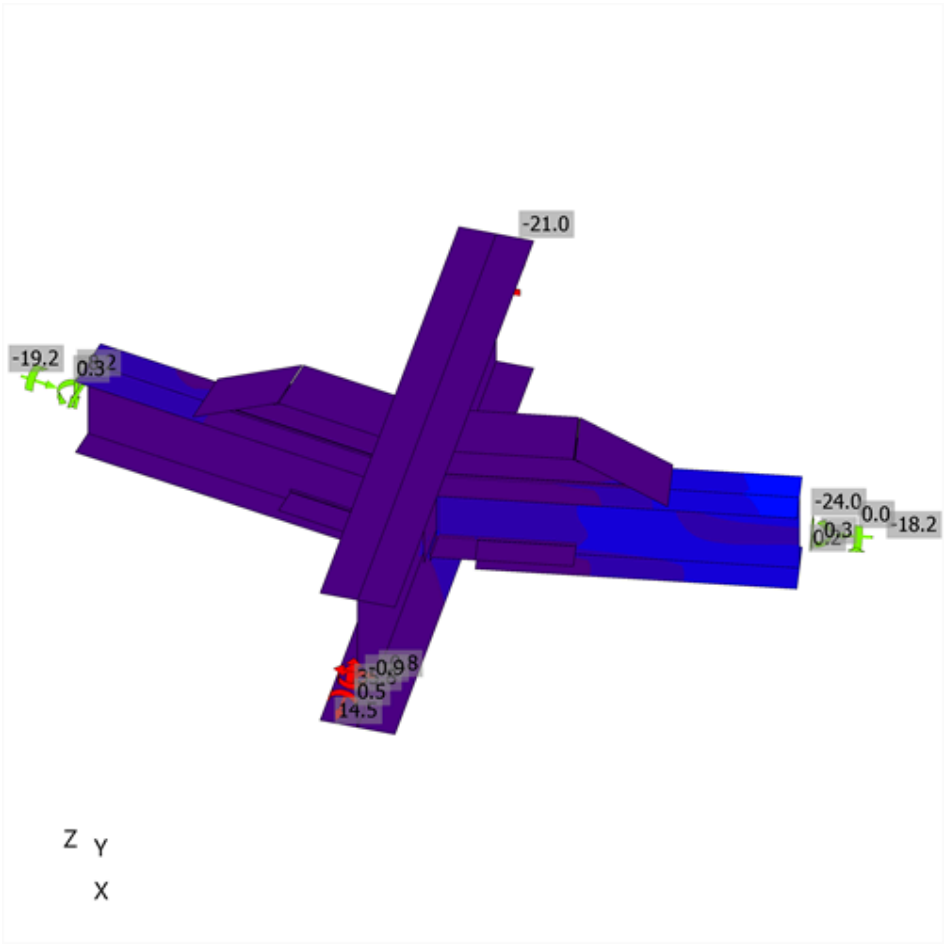
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

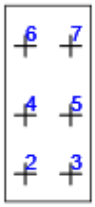


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	0.2	0.4	208.4	0.1	0.3	0.2	OK	OK
	B3	M30 8.8 - 1	C LC_148	0.2	0.3	176.5	0.1	0.3	0.2	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.3	213.9	0.0	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.0	0.3	213.9	0.0	0.3	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_119	1.3	0.6	229.5	0.4	0.5	0.6	OK	OK
	B7	M30 8.8 - 1	C LC_104	1.5	0.5	229.5	0.5	0.4	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 10.5	149	C LC_120	11.1	0.0	-6.3	5.2	-0.5	3.2	3.2	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-tfl 1	◢ 10.5	149	C LC_120	9.0	0.0	-7.7	-2.7	0.5	2.5	2.5	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_146	7.1	0.0	-0.1	-1.3	3.9	1.7	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_148	5.4	0.0	-0.8	-0.1	-3.1	1.3	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_119	20.1	0.0	1.1	1.1	11.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_146	19.5	0.0	1.4	-2.0	-11.0	4.8	4.8	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_119	11.2	0.0	2.8	2.4	5.8	2.8	2.8	OK	OK
		◢ 6.0 ◣	243	C LC_146	13.3	0.0	5.3	1.8	-6.8	3.3	3.3	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_146	10.9	0.0	1.9	2.4	-5.7	2.7	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_114	14.5	0.0	4.3	-3.2	7.3	3.6	3.6	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_120	77.9	0.0	-20.2	-20.5	38.3	19.2	10.8	OK	OK
		◢ 6.0 ◣	572	C LC_120	71.7	0.0	-16.8	16.6	-36.7	17.7	10.2	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	208	C LC_146	4.9	0.0	-1.5	-0.6	2.6	1.2	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	7.4	0.0	-0.6	1.8	-3.8	1.8	0.0	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_112	7.8	0.0	0.1	2.2	3.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	100	C LC_154	6.8	0.0	0.2	-0.7	-3.9	1.7	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_112	4.5	0.0	0.1	-1.5	-2.1	1.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_112	7.5	0.0	-4.0	2.4	2.8	1.8	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_112	8.6	0.0	-1.7	-2.3	4.3	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_112	12.2	0.0	-3.7	3.2	-5.9	3.0	3.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_120	30.9	0.0	-7.4	-7.8	15.4	7.6	6.3	OK	OK
		◢ 6.0 ◣	572	C LC_104	29.1	0.0	-6.1	6.6	-15.0	7.2	6.3	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_114	25.8	0.0	-8.1	-10.3	9.7	6.4	5.9	OK	OK
		◢ 6.0 ◣	149	C LC_114	29.8	0.0	-12.5	10.3	-11.7	7.4	6.1	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_119	6.1	0.0	0.3	0.3	3.5	1.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	9.3	0.0	0.6	-0.6	-5.3	2.3	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_112	11.3	0.0	-5.7	-4.8	-3.0	2.8	2.8	OK	OK
		◢ 6.0 ◣	149	C LC_112	9.3	0.0	-3.3	4.3	2.7	2.3	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_114	4.1	0.0	1.4	1.2	1.9	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_104	4.3	0.0	1.5	-1.4	-1.8	1.1	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_119	63.9	0.0	-12.2	-12.1	34.1	15.8	9.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	55.2	0.0	-8.5	9.0	-30.2	13.6	8.6	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	28.3	0.0	-8.4	-11.6	-10.4	7.0	6.3	OK	OK
		◢ 6.0 ◣	149	C LC_112	32.3	0.0	-14.0	10.8	12.9	8.0	6.6	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_119	4.7	0.0	0.3	0.3	2.7	1.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	3.2	0.0	-0.3	0.3	-1.8	0.8	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	15.0	0.0	-7.3	-5.0	-5.7	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_112	11.1	0.0	-3.0	5.3	3.2	2.8	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_112	2.3	0.0	-0.2	-0.4	-1.3	0.6	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_112	3.0	0.0	0.9	-0.7	-1.5	0.7	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_120	24.7	0.0	-4.5	-3.9	13.5	6.1	5.9	OK	OK
		◢ 6.0 ◣	271	C LC_121	22.6	0.0	-3.7	4.3	-12.1	5.6	5.6	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_119	29.7	0.0	-4.5	-1.9	16.9	7.3	7.3	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_120	19.2	0.0	-7.7	0.4	-10.2	4.7	4.7	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_119	7.5	0.0	-5.3	3.1	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_113	3.4	0.0	-0.6	1.9	0.3	0.8	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_112	3.8	0.0	-3.5	-0.9	0.1	1.1	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_112	2.8	0.0	-0.5	1.5	0.4	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-320
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring

Name	SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

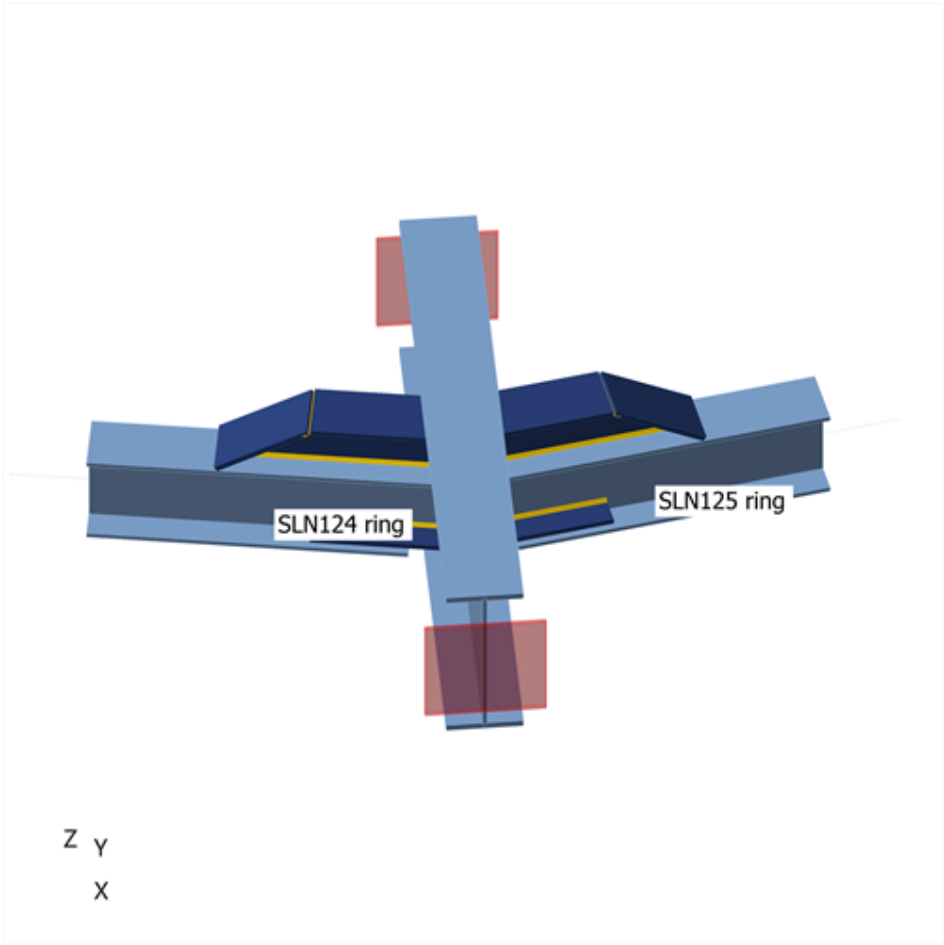
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN123b+SLN123a radial (continuous)	1 - UB457x191x89	-40.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102
SLN125 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN123b+SLN123a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN123b+SLN123a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN125 ring / end		Position	0



SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN124 ring / End	-15.5	0.0	-25.6	0.0	0.0	0.0
	SLN125 ring / End	-15.7	-0.2	-25.6	0.0	0.0	-0.4
C LC_105	SLN124 ring / End	-18.7	0.2	-24.0	0.0	0.0	0.3
	SLN125 ring / End	-18.2	-0.1	-24.0	0.0	0.0	-0.1
C LC_106	SLN124 ring / End	-17.4	-0.2	-24.0	0.0	0.0	-0.2
	SLN125 ring / End	-17.9	-0.2	-24.0	0.0	0.0	-0.3
C LC_119	SLN124 ring / End	-16.7	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-16.9	-0.2	-25.6	0.0	0.0	-0.2
C LC_120	SLN124 ring / End	-17.7	0.0	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-17.6	-0.1	-25.6	0.0	0.0	-0.2
C LC_121	SLN124 ring / End	-17.2	0.1	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-17.0	-0.2	-25.6	0.0	0.0	-0.3
C LC_122	SLN124 ring / End	-16.2	-0.1	-25.6	0.0	0.0	-0.1
	SLN125 ring / End	-16.7	-0.2	-25.6	0.0	0.0	-0.3
C LC_145	SLN124 ring / End	-16.0	0.1	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-16.3	-0.3	-25.6	0.0	0.0	-0.4
C LC_147	SLN124 ring / End	-16.1	0.1	-25.6	0.0	0.0	0.0
	SLN125 ring / End	-16.1	-0.3	-25.6	0.0	0.0	-0.4
C LC_148	SLN124 ring / End	-14.9	0.0	-25.6	0.0	0.0	0.0
	SLN125 ring / End	-15.3	-0.2	-25.6	0.0	0.0	-0.3
C LC_151	SLN124 ring / End	-13.3	-0.1	-11.6	0.0	0.0	-0.1
	SLN125 ring / End	-13.6	-0.3	-11.5	0.0	0.0	-0.4
C LC_152	SLN124 ring / End	-19.1	0.0	-13.9	0.0	0.0	0.1
	SLN125 ring / End	-19.2	-0.2	-14.0	0.0	0.0	-0.3
C LC_153	SLN124 ring / End	-15.7	0.0	-20.6	0.0	0.0	0.0
	SLN125 ring / End	-15.9	-0.2	-20.6	0.0	0.0	-0.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.0 < 100%	OK
Welds	18.8 < 100%	OK
Buckling	Not calculated	

Plates

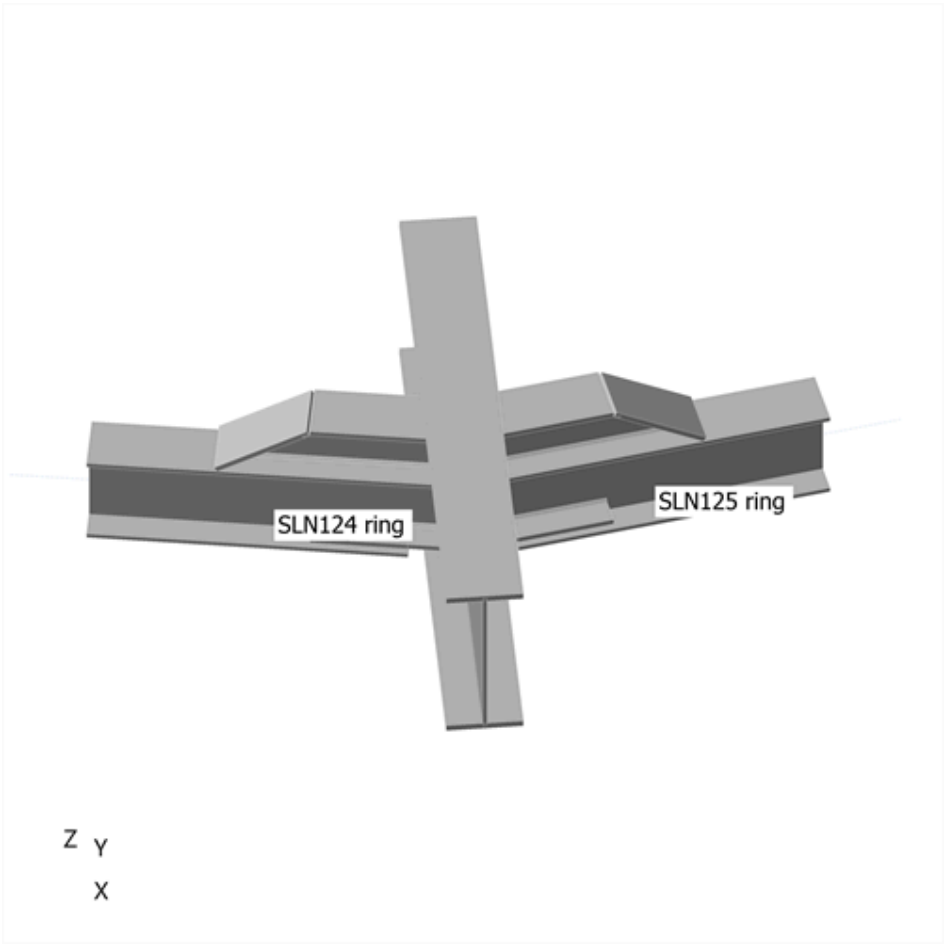
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN123b+SLN123a radial (continuous)-bfl 1	17.7	C LC_122	22.3	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-tfl 1	17.7	C LC_122	18.7	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-w 1	10.5	C LC_147	26.7	0.0	1.5	OK
SLN124 ring-bfl 1	12.7	C LC_148	43.8	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_120	50.0	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_120	49.0	0.0	0.0	OK
SLN125 ring-bfl 1	12.7	C LC_148	44.3	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_122	50.5	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_120	49.0	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_147	7.7	0.0	4.1	OK
Bottom flange plate SLN125 ring	15.0	C LC_106	14.8	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_147	8.0	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_148	7.0	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_122	33.7	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_147	5.8	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_120	15.1	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_119	7.0	0.0	3.2	OK
Bottom flange plate SLN124 ring	15.0	C LC_106	14.4	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_148	7.9	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_106	8.0	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_120	34.0	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_119	5.6	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_120	15.1	0.0	0.0	OK

Design data

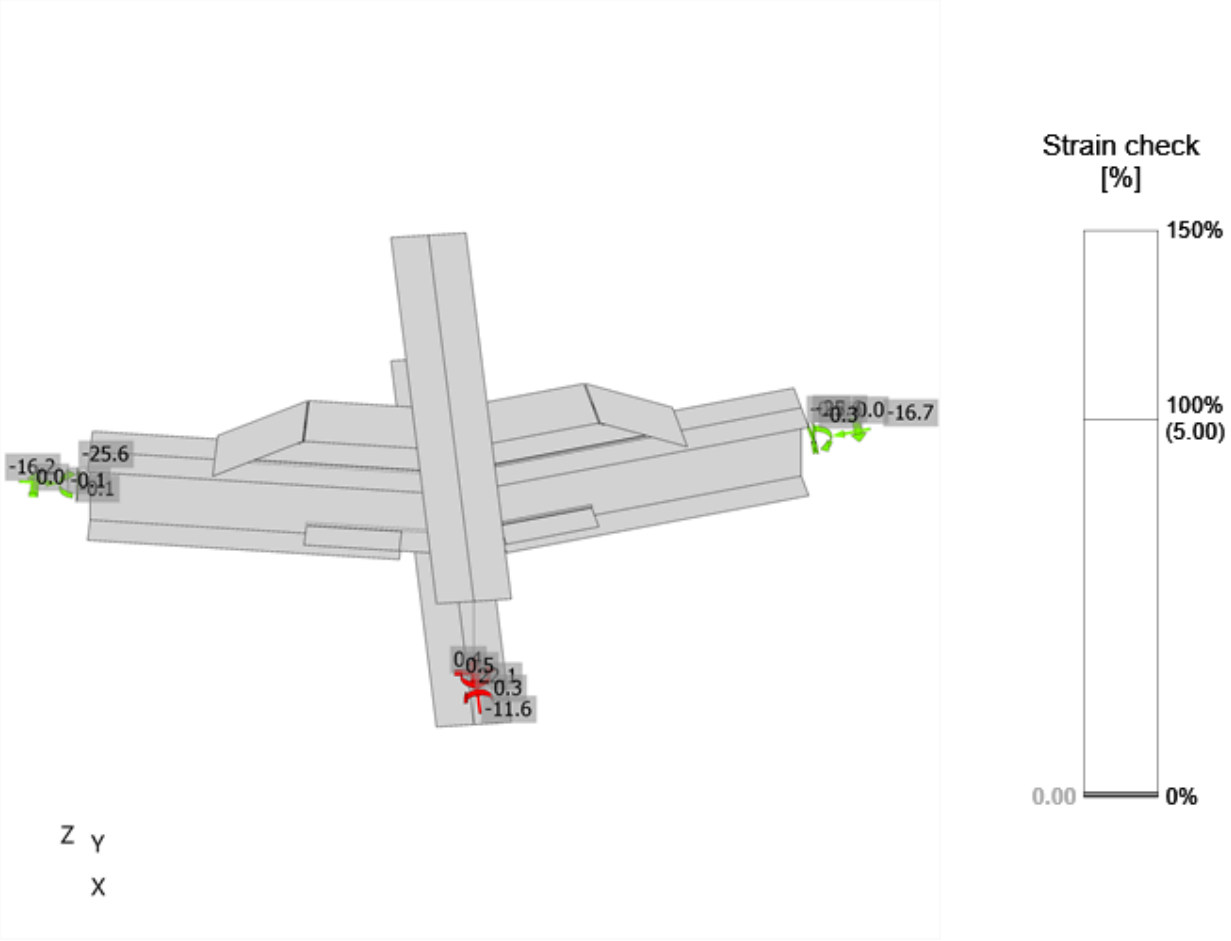
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

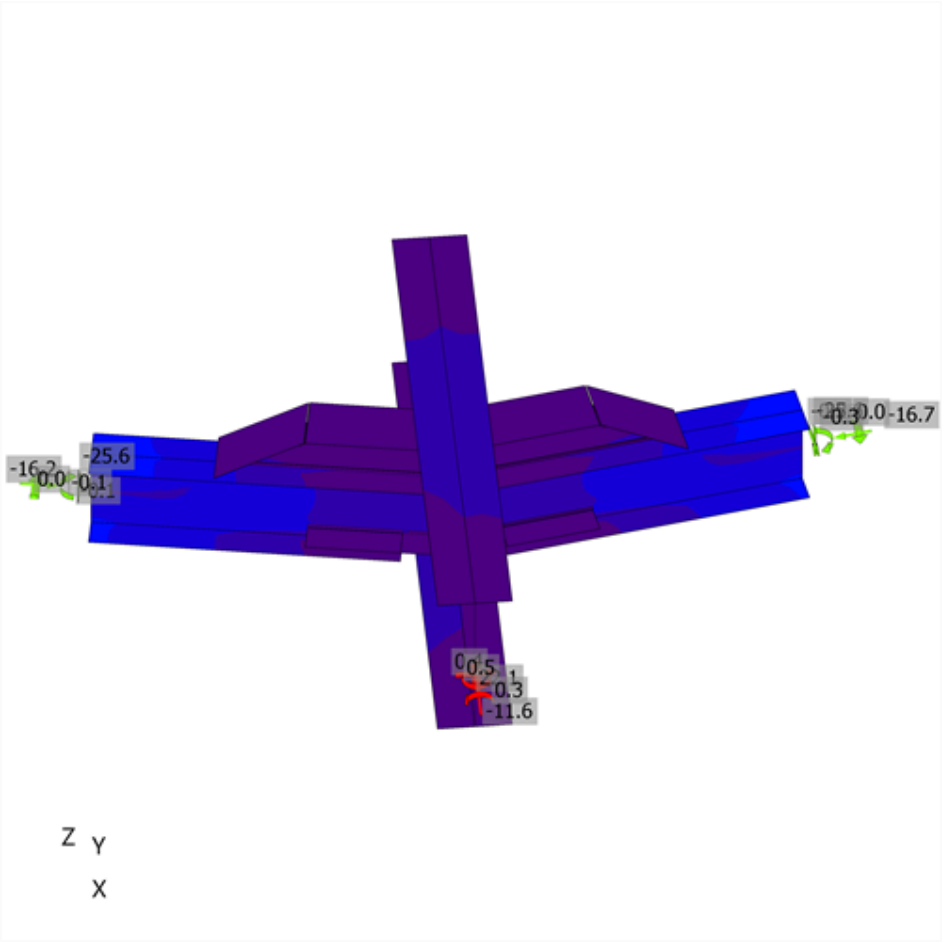
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

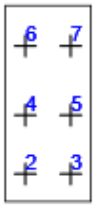


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_119	0.2	0.4	208.4	0.1	0.4	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_148	0.5	0.5	176.5	0.2	0.5	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.4	213.9	0.0	0.4	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_147	2.5	0.8	229.5	0.8	0.7	1.0	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.4	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 10.5	149	C LC_120	14.1	0.0	-7.3	6.7	1.7	3.9	3.9	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 10.5	149	C LC_147	15.7	0.0	-8.2	7.6	-1.5	3.9	3.9	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_121	10.1	0.0	-0.1	-1.6	5.6	2.5	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_119	8.1	0.0	-1.6	0.6	-4.6	2.0	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_119	18.5	0.0	1.0	0.9	10.6	4.6	4.6	OK	OK
		◢ 6.0 ◣	100	C LC_147	20.8	0.0	1.5	-2.4	-11.8	5.2	5.2	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_119	9.3	0.0	1.9	2.0	4.9	2.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_148	14.6	0.0	5.3	1.8	-7.6	3.6	3.6	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_147	12.8	0.0	2.8	2.8	-6.6	3.2	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_147	13.6	0.0	3.0	-3.0	7.0	3.4	3.4	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_147	74.2	0.0	-16.6	-17.6	37.9	18.3	10.5	OK	OK
		◢ 6.0 ◣	572	C LC_120	75.3	0.0	-19.7	19.3	-37.3	18.6	10.5	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_145	8.3	0.0	-1.4	-0.6	4.7	2.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_119	10.1	0.0	0.0	1.4	-5.7	2.5	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_119	19.2	0.0	1.3	1.2	11.0	4.7	4.7	OK	OK
		◢ 6.0 ◣	100	C LC_122	20.0	0.0	1.3	-2.2	-11.3	4.9	4.9	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_119	10.4	0.0	2.5	2.3	5.4	2.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_148	13.7	0.0	5.4	1.7	-7.1	3.4	3.4	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_122	12.0	0.0	2.2	2.7	-6.2	3.0	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_106	14.8	0.0	3.9	-3.3	7.6	3.7	3.7	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_120	76.1	0.0	-19.9	-19.5	37.7	18.8	10.6	OK	OK
		◢ 6.0 ◣	572	C LC_121	73.3	0.0	-17.1	17.5	-37.3	18.1	10.3	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_151	14.0	0.0	-4.0	-5.7	5.2	3.5	3.5	OK	OK
		◢ 6.0 ◣	149	C LC_106	18.7	0.0	-8.5	5.8	-7.7	4.6	4.6	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_120	7.0	0.0	-0.1	0.0	4.1	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	10.4	0.0	1.6	-1.3	-5.8	2.6	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_151	6.8	0.0	-3.0	-2.7	2.4	1.7	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_151	6.5	0.0	-2.4	2.7	-2.3	1.6	0.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_147	4.4	0.0	1.7	1.5	1.8	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_147	3.9	0.0	-0.3	0.3	2.2	1.0	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_119	57.7	0.0	-9.5	-8.6	31.7	14.2	8.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	61.9	0.0	-11.5	12.5	-32.8	15.3	9.3	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_106	12.9	0.0	-5.2	-6.6	1.5	3.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_106	19.9	0.0	-10.5	9.1	-3.4	4.9	4.9	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_119	10.2	0.0	1.4	1.1	5.7	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	7.5	0.0	0.1	-0.2	-4.3	1.9	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_151	4.4	0.0	-2.5	-1.6	1.4	1.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_119	4.0	0.0	2.4	-1.1	1.5	1.0	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_119	3.7	0.0	-0.2	-0.2	-2.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_105	3.9	0.0	1.5	-1.4	-1.6	1.0	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_120	62.3	0.0	-11.5	-11.8	33.3	15.4	9.4	OK	OK
		◢ 6.0 ◣	271	C LC_147	57.1	0.0	-9.4	9.3	-31.2	14.1	8.8	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_122	36.3	0.0	-3.8	-1.1	20.8	9.0	8.8	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_120	36.2	0.0	-4.3	0.7	-20.7	8.9	8.8	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.4	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_147	3.2	0.0	-0.7	1.7	0.4	0.8	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_147	7.2	0.0	-5.2	2.9	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_105	2.8	0.0	-0.6	1.5	-0.4	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-A-340
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring

Name	SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

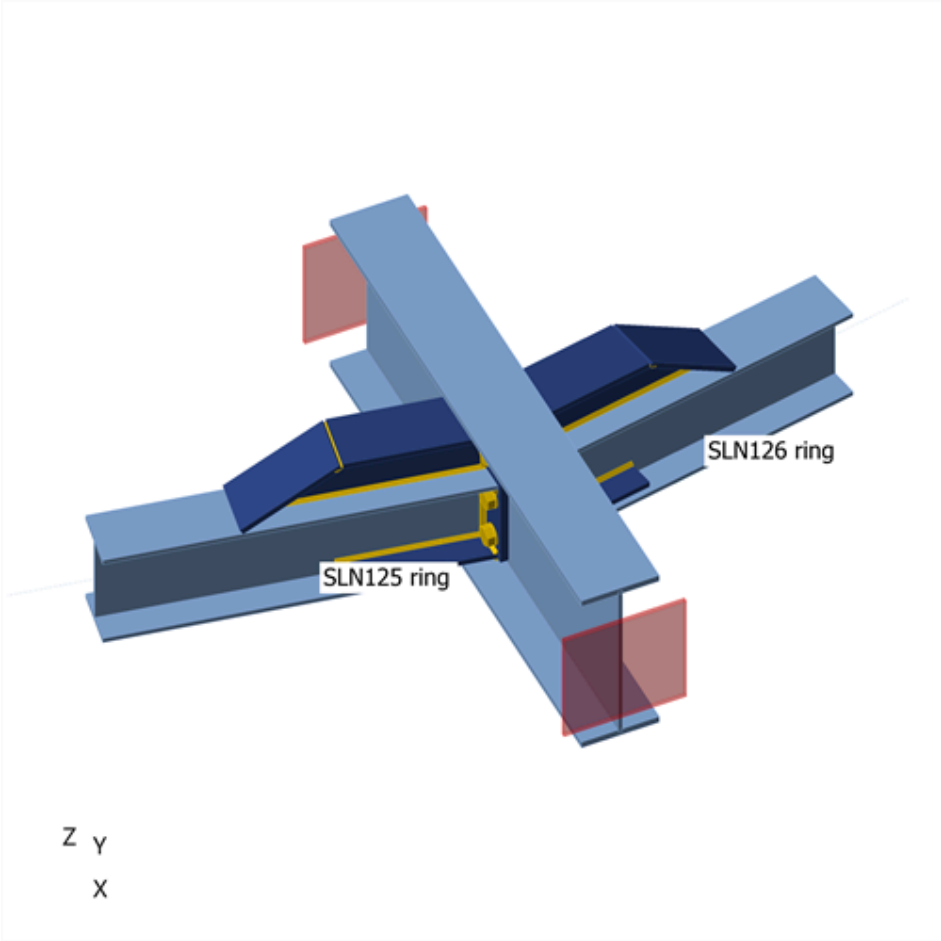
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN127a+SLN127b radial (continuous)	1 - UB457x191x89	-20.0	0.0	0.0	0	0	0
SLN125 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN127a+SLN127b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN127a+SLN127b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN125 ring / end		Position	0
SLN126 ring / end		Position	0



SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN125 ring / End	-15.7	-0.2	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-16.1	0.0	-25.6	0.0	0.0	-0.1
C LC_105	SLN125 ring / End	-18.0	0.0	-24.0	0.0	0.0	0.0
	SLN126 ring / End	-17.4	0.1	-24.0	0.0	0.0	0.2
C LC_106	SLN125 ring / End	-18.1	-0.3	-24.0	0.0	0.0	-0.5
	SLN126 ring / End	-19.3	-0.1	-24.0	0.0	0.0	-0.3
C LC_114	SLN125 ring / End	-15.5	-0.4	-16.2	0.0	0.0	-0.7
	SLN126 ring / End	-17.5	-0.3	-16.2	0.0	0.0	-0.6
C LC_135	SLN125 ring / End	-14.1	-0.2	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-14.5	0.0	-25.6	0.0	0.0	-0.1
C LC_145	SLN125 ring / End	-16.4	-0.2	-25.6	0.0	0.0	-0.3
	SLN126 ring / End	-16.9	-0.1	-25.6	0.0	0.0	-0.1
C LC_146	SLN125 ring / End	-15.0	-0.3	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-15.4	0.0	-25.6	0.0	0.0	0.0
C LC_147	SLN125 ring / End	-16.0	-0.2	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-16.2	-0.1	-25.6	0.0	0.0	-0.1
C LC_148	SLN125 ring / End	-15.4	-0.3	-25.6	0.0	0.0	-0.4
	SLN126 ring / End	-16.0	0.0	-25.6	0.0	0.0	0.0
C LC_152	SLN125 ring / End	-19.2	-0.2	-14.0	0.0	0.0	-0.3
	SLN126 ring / End	-19.5	0.0	-14.1	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	18.7 < 100%	OK
Buckling	Not calculated	

Plates

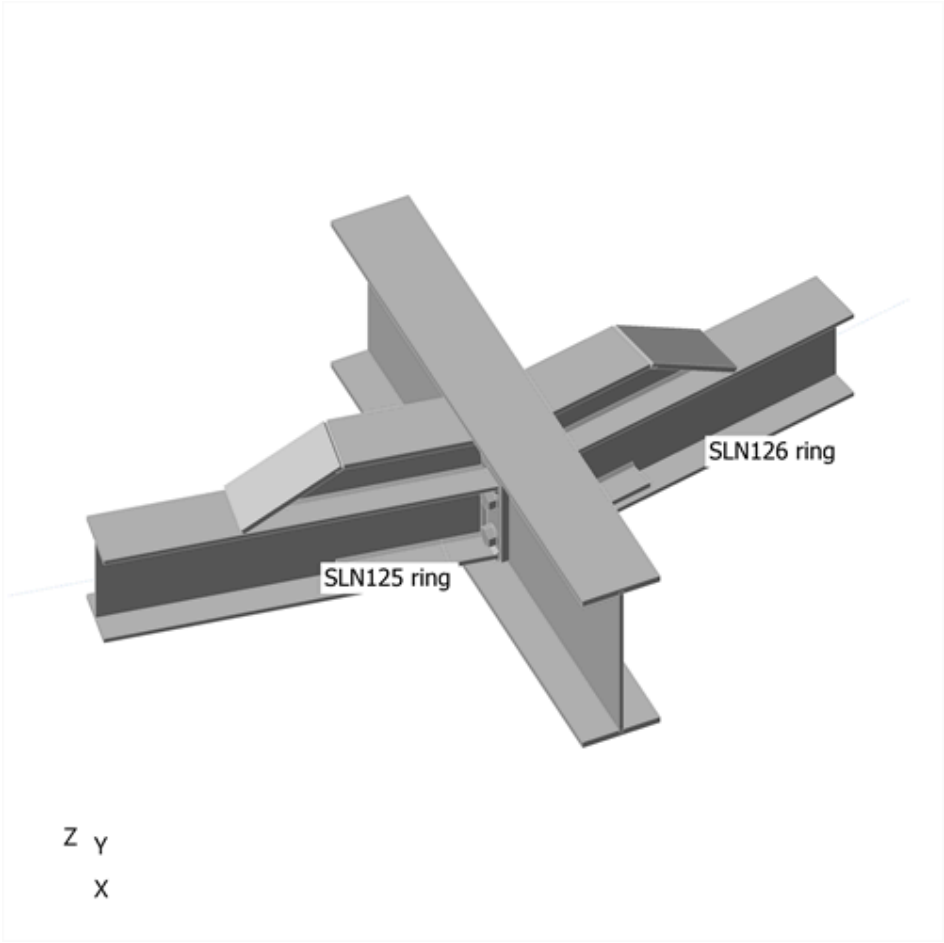
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN127a+SLN127b radial (continuous)-bfl 1	17.7	C LC_106	22.9	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-tfl 1	17.7	C LC_106	19.0	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-w 1	10.5	C LC_147	27.5	0.0	1.5	OK
SLN125 ring-bfl 1	12.7	C LC_135	44.8	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_147	50.9	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_146	43.9	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_145	49.4	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_145	48.7	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_114	12.3	0.0	13.2	OK
Bottom flange plate SLN126 ring	15.0	C LC_114	29.2	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_114	9.6	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_147	8.3	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_145	33.5	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_114	10.8	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_145	15.1	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_146	7.4	0.0	6.0	OK
Bottom flange plate SLN125 ring	15.0	C LC_114	23.0	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_135	8.8	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_135	7.5	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_147	33.8	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_114	7.4	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_145	15.0	0.0	0.0	OK

Design data

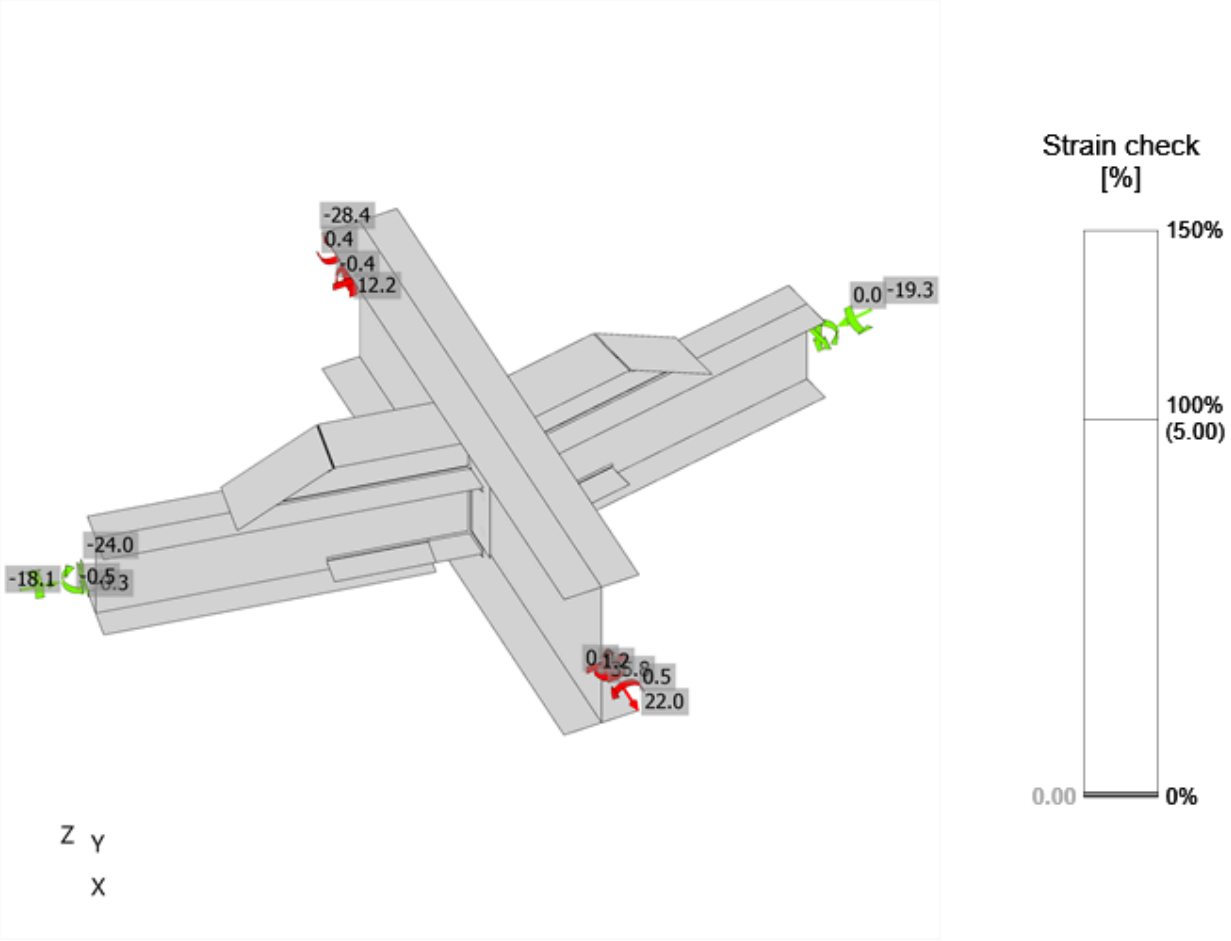
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

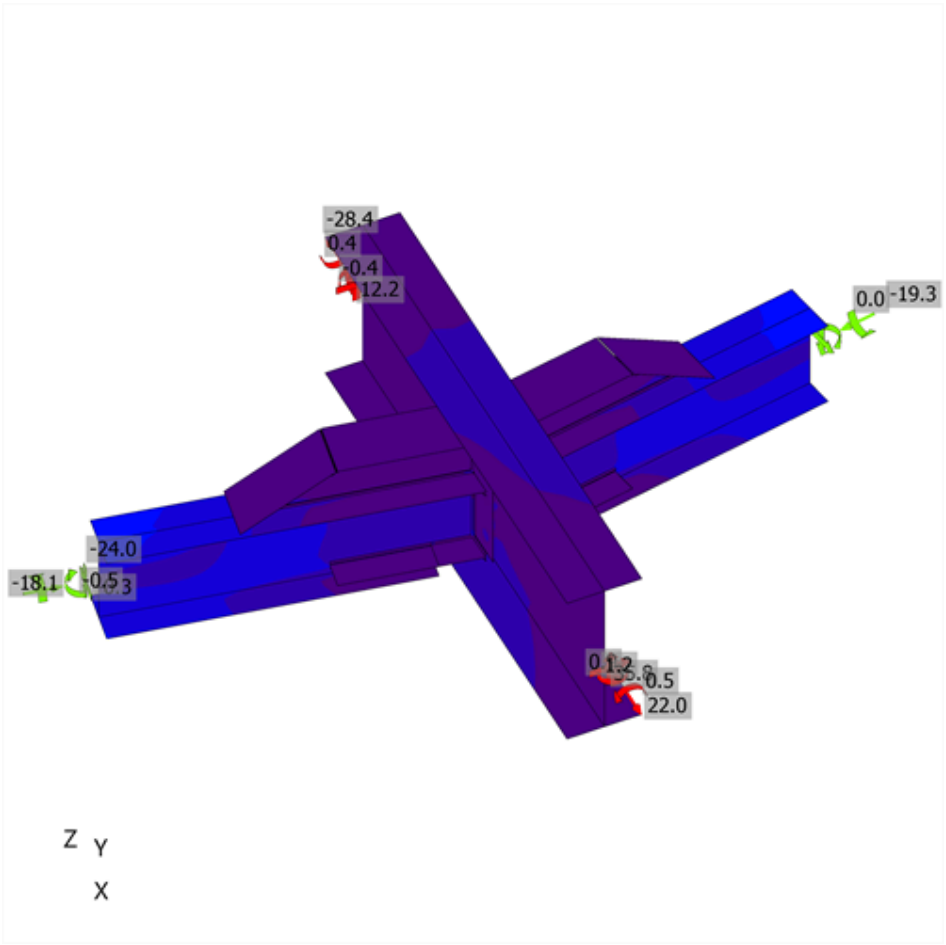
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

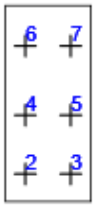


Strain check, C LC_106



Equivalent stress, C LC_106

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_145	0.5	0.5	208.4	0.2	0.4	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.5	0.5	176.5	0.2	0.5	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_145	0.2	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.1	0.4	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_147	1.8	0.9	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_146	2.4	0.8	229.5	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 10.5	149	C LC_146	16.1	0.0	-8.4	7.6	-2.2	4.0	4.0	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 10.5	149	C LC_145	13.6	0.0	-12.1	-3.5	-0.1	3.9	3.9	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_148	10.7	0.0	-0.2	-2.0	5.8	2.6	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	7.2	0.0	-1.3	-0.1	-4.1	1.8	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_148	18.4	0.0	0.9	0.0	10.6	4.5	4.5	OK	OK
		◢ 6.0 ◣	100	C LC_147	20.8	0.0	1.4	-2.6	-11.7	5.1	5.1	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_105	9.4	0.0	2.2	2.0	4.9	2.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	14.3	0.0	6.0	1.5	-7.4	3.5	3.5	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_135	12.7	0.0	2.4	2.9	-6.6	3.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_147	15.3	0.0	3.9	-3.4	7.8	3.8	3.8	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_145	75.8	0.0	-19.0	-19.1	37.8	18.7	10.6	OK	OK
		◢ 6.0 ◣	572	C LC_148	73.1	0.0	-17.6	17.8	-36.9	18.1	10.3	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_135	7.7	0.0	-1.1	0.1	4.4	1.9	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	10.2	0.0	-0.1	1.7	-5.7	2.5	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_105	18.1	0.0	1.1	1.6	10.3	4.5	4.5	OK	OK
		◢ 6.0 ◣	100	C LC_147	21.3	0.0	1.4	-2.4	-12.0	5.3	5.3	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_105	9.2	0.0	1.7	2.0	4.8	2.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	15.3	0.0	6.1	1.5	-8.0	3.8	3.8	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_135	13.7	0.0	3.1	3.0	-7.1	3.4	3.4	OK	OK
		◢ 6.0 ◣	243	C LC_146	14.5	0.0	3.2	-3.2	7.5	3.6	3.6	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_147	74.1	0.0	-16.4	-17.4	37.9	18.3	10.5	OK	OK
		◢ 6.0 ◣	572	C LC_145	74.9	0.0	-20.2	19.4	-36.8	18.5	10.4	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	36.1	0.0	-10.5	-14.9	13.3	8.9	6.5	OK	OK
		◢ 6.0 ◣	149	C LC_114	43.0	0.0	-18.6	14.2	-17.3	10.6	7.0	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_105	6.5	0.0	0.4	0.3	3.7	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	10.8	0.0	1.1	-1.4	-6.0	2.7	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	18.6	0.0	-9.2	-6.0	7.1	4.6	4.6	OK	OK
		◢ 6.0 ◣	149	C LC_114	13.3	0.0	-3.3	6.5	-3.7	3.3	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_114	4.8	0.0	1.5	1.2	2.3	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_106	4.1	0.0	0.1	0.1	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_145	61.0	0.0	-10.9	-10.6	33.0	15.1	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_145	57.7	0.0	-9.6	9.9	-31.3	14.3	8.8	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_114	25.2	0.0	-10.1	-12.9	3.5	6.2	5.6	OK	OK
		◢ 6.0 ◣	149	C LC_114	33.7	0.0	-18.2	15.4	-5.5	8.3	6.2	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_148	10.2	0.0	1.0	1.1	5.7	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	7.2	0.0	0.7	-0.7	-4.1	1.8	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_114	14.8	0.0	-7.4	-6.3	3.8	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_114	12.2	0.0	-4.3	5.4	-3.7	3.0	3.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_114	5.0	0.0	1.6	1.3	2.4	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_106	3.9	0.0	-0.3	0.3	2.2	1.0	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_145	56.8	0.0	-9.1	-7.9	31.4	14.0	8.8	OK	OK
		◢ 6.0 ◣	271	C LC_145	62.0	0.0	-11.5	12.7	-32.8	15.3	9.3	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_145	34.9	0.0	-5.0	-1.9	19.8	8.6	8.6	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_147	34.8	0.0	-4.0	1.8	-19.9	8.6	8.6	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_147	7.3	0.0	-5.0	3.0	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.9	0.0	-0.8	2.2	-0.2	1.0	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_147	7.3	0.0	-5.5	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	4.4	0.0	-0.8	2.5	-0.6	1.1	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-B-060
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN87+stub radial (ends 200 past the node) - SLN121 column below - SLN84 ring - SLN88 ring

Name	SLN87+stub radial (ends 200 past the node) - SLN121 column below - SLN84 ring - SLN88 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

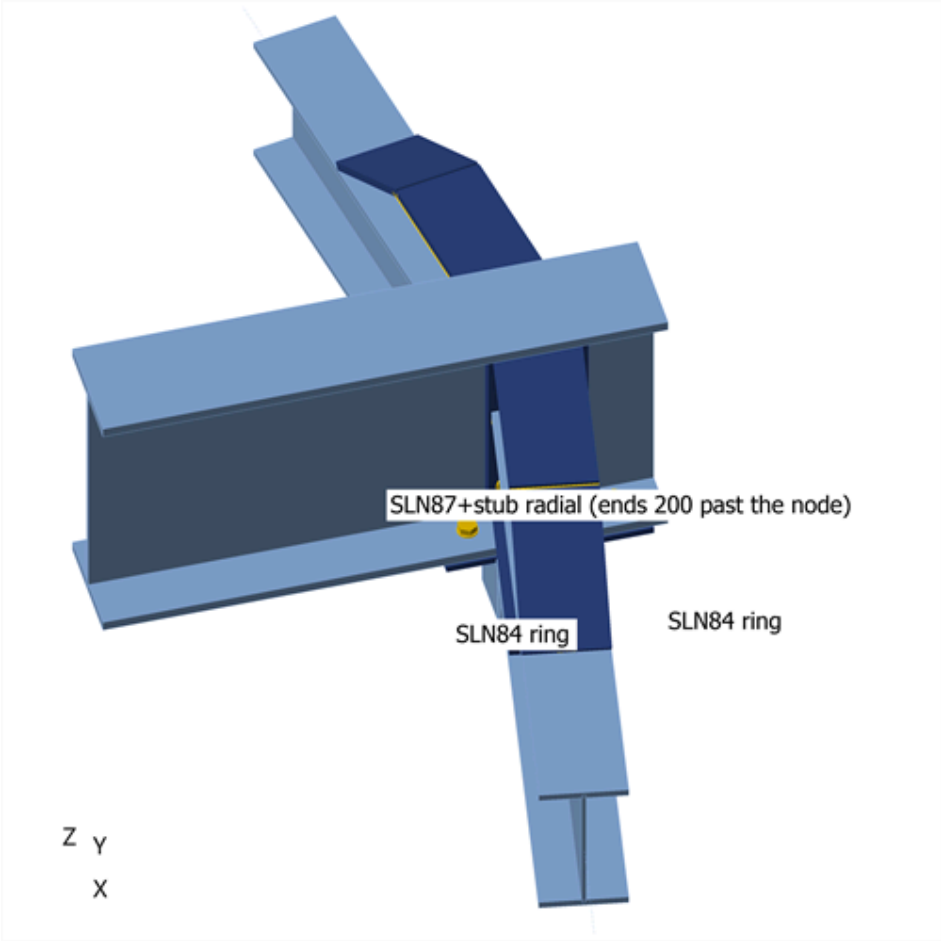
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN87+stub radial (ends 200 past the node)	1 - UB457x191x89	60.0	0.0	0.0	200	0	0
SLN121 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN84 ring	3 - UB254x146x43	-40.0	0.0	0.0	0	0	-102
SLN88 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN87+stub radial (ends 200 past the node) / begin		Position	0
SLN121 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN84 ring / end		Position	0
SLN88 ring / end		Position	0



SLN87+stub radial (ends 200 past the node) - SLN121 column below - SLN84 ring - SLN88 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN87+stub radial (ends 200 past the node) / Begin	-2.5	0.2	-48.2	0.0	-6.9	-0.6
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-18.6	-0.4	-8.9	0.0	0.0	-0.6
	SLN88 ring / End	-18.6	1.4	-8.8	0.0	0.0	1.2
C LC_106	SLN87+stub radial (ends 200 past the node) / Begin	-4.2	0.2	-45.2	0.0	-6.8	-0.6
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-23.2	-0.4	-8.4	0.0	0.0	-0.6
	SLN88 ring / End	-23.2	1.4	-8.3	0.0	0.0	1.1
C LC_120	SLN87+stub radial (ends 200 past the node) / Begin	-3.1	0.3	-48.1	0.0	-6.7	-0.8
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-20.0	-0.3	-8.9	0.0	0.0	-0.5
	SLN88 ring / End	-19.9	1.5	-8.8	0.0	0.0	1.3
C LC_122	SLN87+stub radial (ends 200 past the node) / Begin	-3.0	0.2	-48.4	0.0	-7.4	-0.6
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-20.6	-0.4	-8.9	0.0	0.0	-0.6
	SLN88 ring / End	-20.7	1.4	-8.8	0.0	0.0	1.2
C LC_145	SLN87+stub radial (ends 200 past the node) / Begin	-2.6	0.0	-48.3	0.0	-6.9	-0.5
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-19.1	-0.4	-8.9	0.0	0.0	-0.7
	SLN88 ring / End	-18.9	1.4	-8.8	0.0	0.0	1.2
C LC_146	SLN87+stub radial (ends 200 past the node) / Begin	-2.3	0.3	-48.2	0.0	-6.8	-0.7
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-18.2	-0.4	-8.9	0.0	0.0	-0.6
	SLN88 ring / End	-18.3	1.5	-8.8	0.0	0.0	1.3
C LC_151	SLN87+stub radial (ends 200 past the node) / Begin	0.9	0.1	-48.4	0.0	-10.7	-0.5
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-16.9	-0.4	-8.9	0.0	0.0	-0.5
	SLN88 ring / End	-16.9	1.2	-8.8	0.0	0.0	1.0
C LC_152	SLN87+stub radial (ends 200 past the node) / Begin	-8.3	0.2	-11.3	0.0	2.2	-0.7
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-21.7	-0.4	-2.2	0.0	0.0	-0.6
	SLN88 ring / End	-21.7	1.5	-2.2	0.0	0.0	1.3
C LC_154	SLN87+stub radial (ends 200 past the node) / Begin	1.0	0.1	-49.2	0.0	-10.9	-0.5
	SLN87+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN84 ring / End	-16.9	-0.4	-8.9	0.0	0.0	-0.5
	SLN88 ring / End	-16.8	1.2	-8.8	0.0	0.0	1.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	6.0 < 100%	OK
Welds	31.2 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN87+stub radial (ends 200 past the node)-bfl 1	17.7	C LC_145	39.7	0.0	16.1	OK
SLN87+stub radial (ends 200 past the node)-tfl 1	17.7	C LC_120	22.5	0.0	0.0	OK
SLN87+stub radial (ends 200 past the node)-w 1	10.5	C LC_120	59.5	0.0	16.3	OK
SLN121 column below	10.0	C LC_120	67.7	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_145	14.4	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_122	22.3	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_122	19.4	0.0	0.0	OK
SLN88 ring-bfl 1	12.7	C LC_154	14.2	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_122	19.8	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_122	19.1	0.0	0.0	OK
End plate column below SLN121 column below	12.0	C LC_152	59.4	0.0	10.4	OK
End plate SLN88 ring	20.0	C LC_120	5.8	0.0	13.4	OK
Bottom flange plate SLN88 ring	15.0	C LC_120	23.9	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_152	9.9	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_152	13.4	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_120	15.8	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_120	8.4	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_120	7.1	0.0	0.0	OK
End plate SLN84 ring	20.0	C LC_120	8.1	0.0	5.4	OK
Bottom flange plate SLN84 ring	15.0	C LC_152	13.7	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_152	8.7	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_152	8.0	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_120	15.1	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_120	7.3	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_122	6.4	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

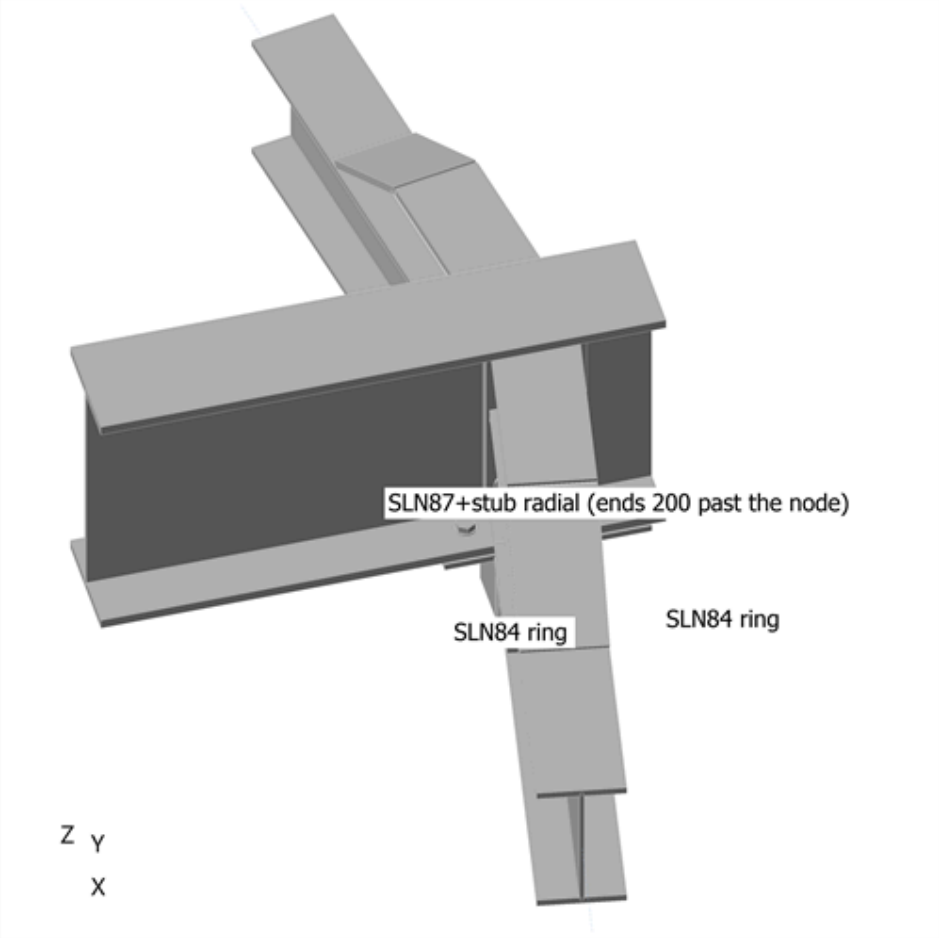
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

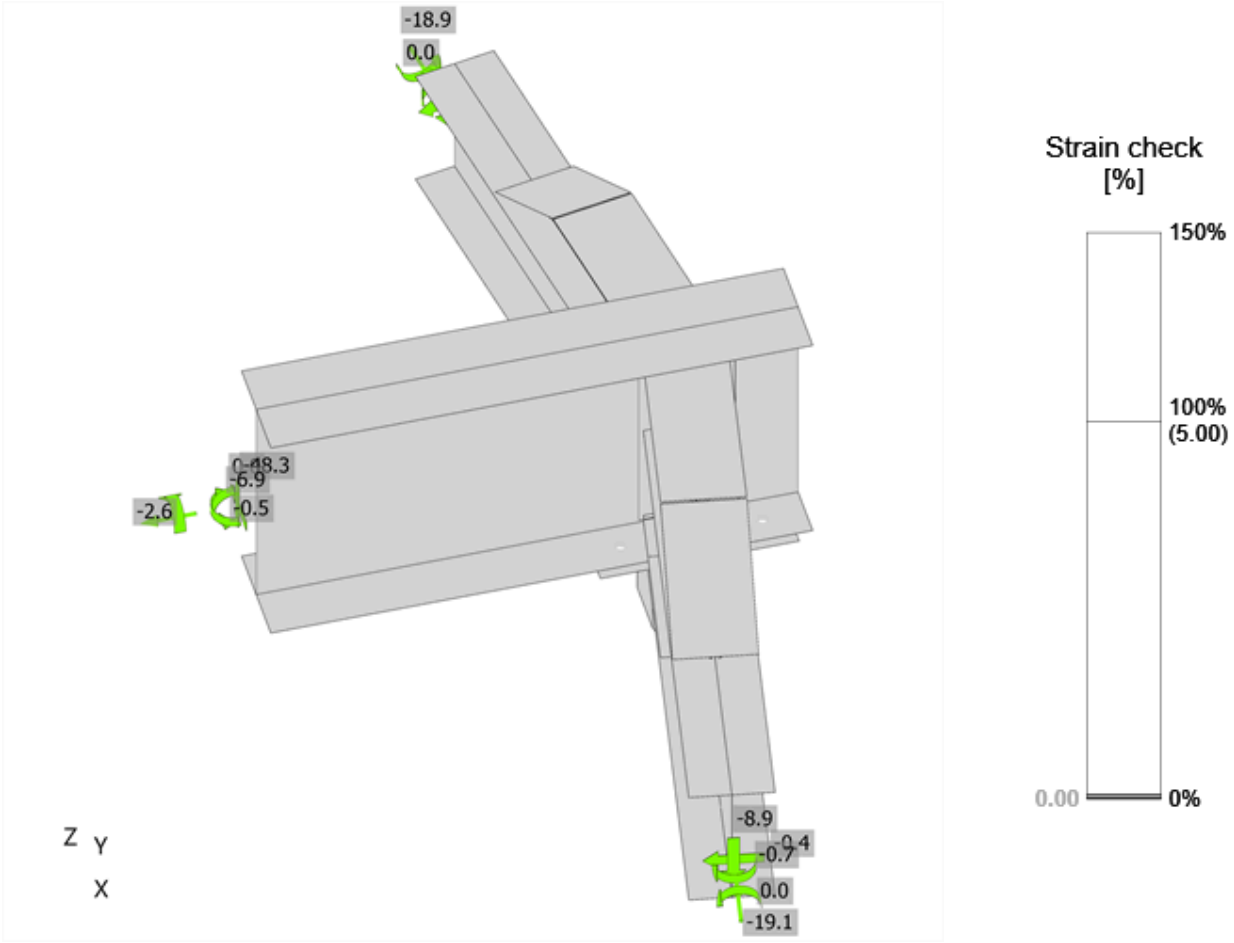
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN121 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

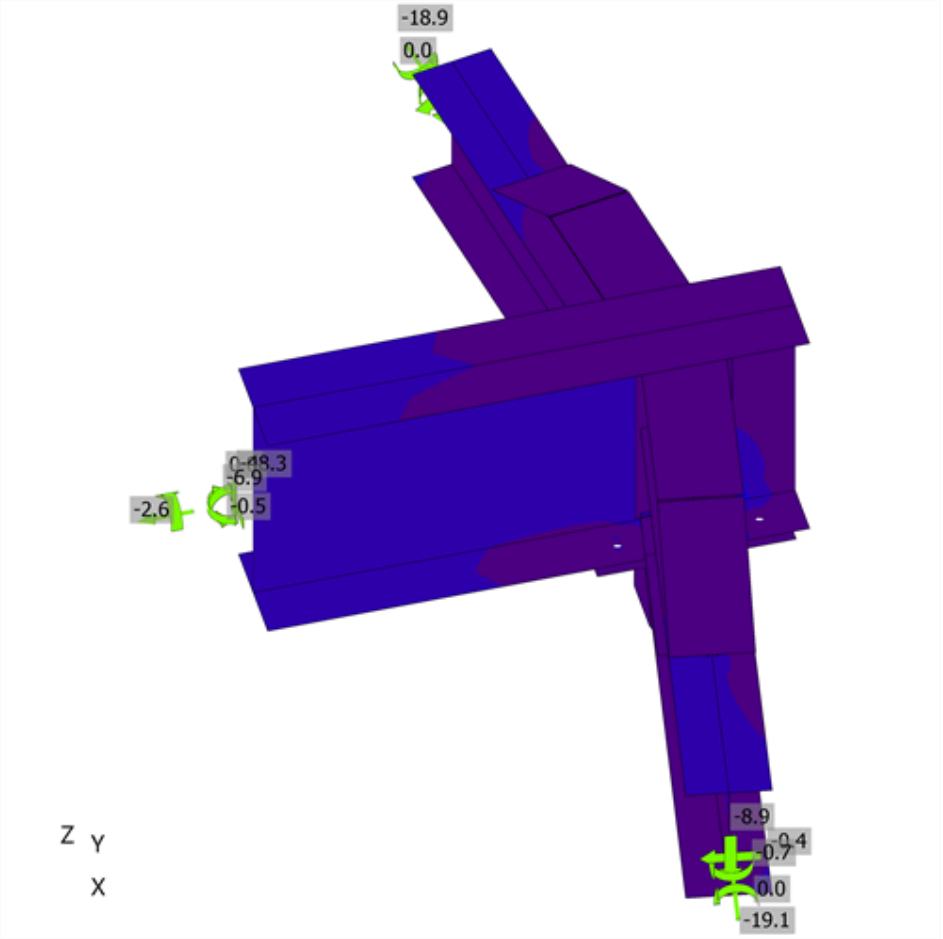
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

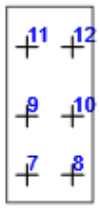


Strain check, C LC_145



Equivalent stress, C LC_145

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	7.8	0.8	156.4	5.6	0.9	4.9	OK	OK
	B3	M20 8.8 - 1	C LC_154	0.5	1.1	156.4	0.4	1.2	1.4	OK	OK
	B4	M20 8.8 - 1	C LC_154	0.5	1.1	156.4	0.3	1.2	1.4	OK	OK
	B5	M20 8.8 - 1	C LC_152	8.5	0.9	156.4	6.0	1.0	5.3	OK	OK
	B7	M30 8.8 - 2	C LC_120	0.0	0.4	213.9	0.0	0.3	0.2	OK	OK
	B8	M30 8.8 - 2	C LC_146	1.3	1.3	176.5	0.4	1.4	0.9	OK	OK
	B9	M30 8.8 - 2	C LC_120	0.2	0.6	229.5	0.1	0.5	0.3	OK	OK
	B10	M30 8.8 - 2	C LC_120	0.2	0.5	213.9	0.1	0.4	0.3	OK	OK
	B11	M30 8.8 - 2	C LC_120	0.6	0.8	229.5	0.2	0.6	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_120	1.8	0.2	142.1	0.6	0.2	0.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN87+stub radial (ends 200 past the node)-w 1	SLN84 ring-tfl 1	◢ 10.5	149	C LC_145	16.9	0.0	-8.1	8.5	1.0	4.2	4.2	OK	OK
SLN87+stub radial (ends 200 past the node)-w 1	SLN88 ring-tfl 1	◢ 10.5	149	C LC_146	20.3	0.0	-9.7	10.3	0.1	5.0	5.0	OK	OK
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	207	C LC_154	22.8	0.0	-1.1	1.0	13.1	5.6	5.6	OK	OK
		◢ 6.0 ◣	207	C LC_120	16.0	0.0	1.8	1.2	-9.1	3.9	3.9	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	100	C LC_120	12.4	0.0	0.7	3.8	6.1	3.1	3.1	OK	OK
		◢ 6.0 ◣	100	C LC_146	10.3	0.0	5.2	-4.9	-1.7	2.5	2.5	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	242	C LC_152	8.0	0.0	-1.1	-0.7	-4.5	2.0	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	11.8	0.0	3.7	-2.7	-5.9	2.9	2.9	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_152	15.7	0.0	-2.9	-4.0	7.9	3.9	3.9	OK	OK
		◢ 6.0 ◣	243	C LC_152	22.6	0.0	-6.9	5.8	-11.0	5.6	5.6	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_122	27.5	0.0	-10.5	-7.2	12.8	6.8	6.0	OK	OK
		◢ 6.0 ◣	572	C LC_120	35.3	0.0	-4.3	8.1	-18.5	8.7	6.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	207	C LC_154	14.7	0.0	1.4	-0.5	8.4	3.6	3.6	OK	OK
		◢ 6.0 ◣	208	C LC_154	24.2	0.0	-3.2	-0.5	-13.9	6.0	6.0	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_146	11.8	0.0	5.3	6.1	0.6	2.9	2.9	OK	OK
		◢ 6.0 ◣	100	C LC_154	11.2	0.0	5.3	-5.5	-1.6	2.8	2.8	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_152	10.6	0.0	-2.0	-2.8	-5.3	2.6	2.6	OK	OK
		◢ 6.0 ◣	243	C LC_152	15.0	0.0	-4.7	3.9	7.3	3.7	3.7	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	242	C LC_152	6.5	0.0	-1.3	-1.1	3.5	1.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_152	9.0	0.0	-4.3	1.7	-4.3	2.2	0.0	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_122	35.6	0.0	-7.5	-9.4	17.8	8.8	6.6	OK	OK
		◢ 6.0 ◣	572	C LC_122	27.0	0.0	-7.9	6.0	-13.6	6.7	6.0	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	37.7	0.0	-18.4	-18.8	-2.9	9.3	7.2	OK	OK
		◢ 6.0 ◣	149	C LC_120	38.8	0.0	-20.1	18.8	3.6	9.6	7.3	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_154	5.5	0.0	-0.3	-0.4	3.2	1.4	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_145	8.6	0.0	-1.4	1.4	-4.7	2.1	0.0	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_120	16.3	0.0	-7.7	-7.5	-3.6	4.0	4.0	OK	OK
		◢ 6.0 ◣	149	C LC_120	15.4	0.0	-6.3	6.5	4.8	3.8	3.8	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_120	4.6	0.0	-1.1	-2.2	-1.4	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	5.3	0.0	1.7	-1.3	-2.6	1.3	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	27.1	0.0	-4.9	-10.3	11.5	6.7	6.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	25.9	0.0	-3.2	-2.1	-14.7	6.4	6.2	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	148	C LC_152	19.2	0.0	-10.5	-9.2	1.0	4.7	4.7	OK	OK
		◢ 6.0 ◣	149	C LC_120	19.8	0.0	-7.2	5.7	9.0	4.9	4.9	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_120	10.0	0.0	-1.4	-0.8	5.6	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_152	4.8	0.0	0.9	-0.9	2.6	1.2	0.0	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_120	10.7	0.0	-5.5	-2.5	-4.7	2.6	2.6	OK	OK
		◢ 6.0 ◣	149	C LC_120	5.7	0.0	-0.1	3.2	0.9	1.4	0.0	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_145	2.5	0.0	1.1	0.9	0.9	0.6	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_152	2.1	0.0	-0.8	0.4	-1.1	0.5	0.0	OK	OK
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_120	28.0	0.0	-5.3	-2.8	15.6	6.9	6.2	OK	OK
		◢ 6.0 ◣	271	C LC_122	21.9	0.0	-3.2	6.0	-11.0	5.4	5.4	OK	OK
End plate column below SLN121 column below	SLN121 column below	◢ 4.2 ◣	521	C LC_120	120.2	0.0	-59.7	-60.0	5.3	29.7	15.8	OK	OK
		◢ 4.2 ◣	521	C LC_120	126.4	0.0	-63.1	62.8	-7.3	31.2	16.0	OK	OK
SLN87+stub radial (ends 200 past the node)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_154	65.5	0.0	4.3	-2.1	-37.7	16.2	10.1	OK	OK
SLN87+stub radial (ends 200 past the node)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_154	68.1	0.0	4.2	4.1	39.0	16.8	10.5	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_120	7.1	0.0	-6.5	-1.6	0.2	2.1	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	5.1	0.0	-0.9	2.8	0.7	1.2	0.0	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_145	3.3	0.0	-2.4	1.3	0.1	0.8	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_145	2.1	0.0	-0.5	1.0	0.6	0.5	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-B-080
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN88 ring - SLN91 ring

Name	SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN88 ring - SLN91 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

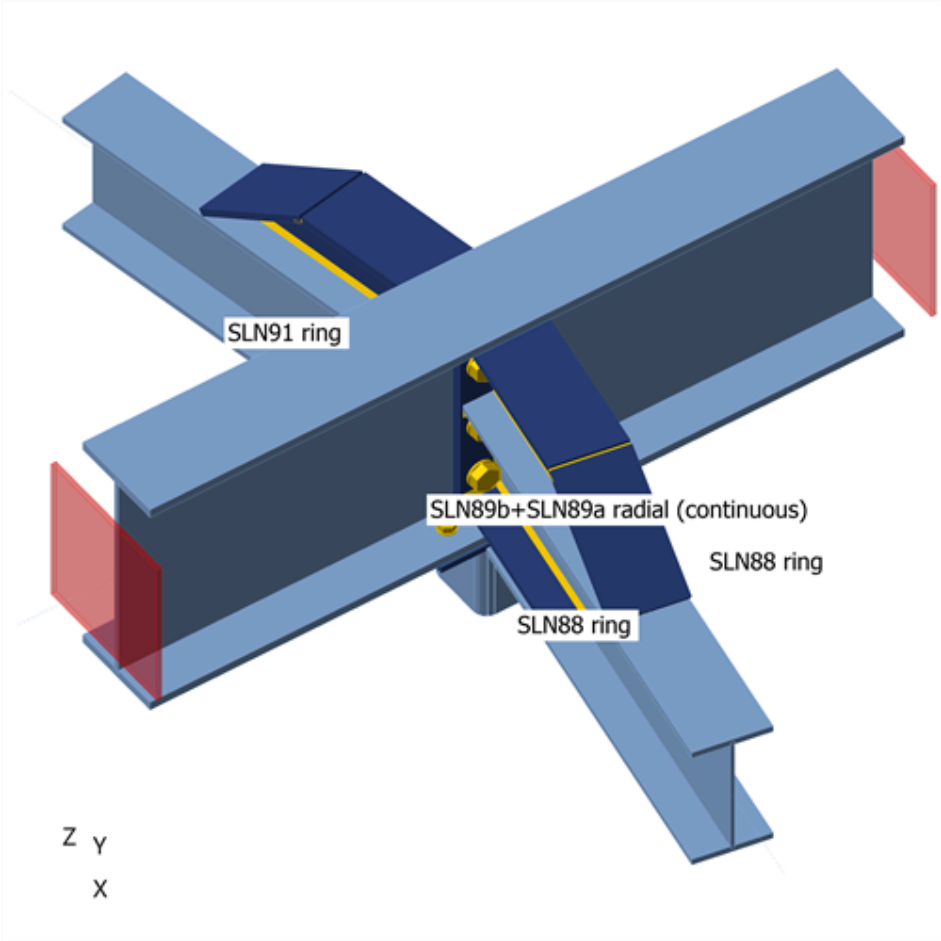
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN89b+SLN89a radial (continuous)	1 - UB457x191x89	80.0	0.0	0.0	0	0	0
SLN131 column below	2 - SHS150x150x10	0.0	90.0	-100.0	0	0	0
SLN88 ring	3 - UB254x146x43	-20.0	0.0	0.0	0	0	-102
SLN91 ring	3 - UB254x146x43	180.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN89b+SLN89a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN89b+SLN89a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN131 column below / begin		Position	0
SLN88 ring / end		Position	0
SLN91 ring / end		Position	0



SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN88 ring - SLN91 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN131 column below / Begin	192.4	0.0	2.3	-0.9	6.7	0.2
	SLN88 ring / End	-18.6	1.4	-8.8	0.0	0.0	3.0
	SLN91 ring / End	33.4	1.3	-25.6	0.0	0.0	2.8
C LC_104	SLN131 column below / Begin	191.2	0.4	2.2	-0.9	6.6	0.2
	SLN88 ring / End	-19.1	1.4	-8.3	0.0	0.0	2.9
	SLN91 ring / End	31.1	1.3	-24.0	0.0	0.0	2.8
C LC_120	SLN131 column below / Begin	197.1	0.4	2.3	-0.9	6.9	0.2
	SLN88 ring / End	-19.9	1.5	-8.8	0.0	0.0	3.1
	SLN91 ring / End	32.3	1.4	-25.6	0.0	0.0	3.0
C LC_122	SLN131 column below / Begin	196.4	0.0	2.1	-0.9	6.4	0.2
	SLN88 ring / End	-20.7	1.4	-8.8	0.0	0.0	3.0
	SLN91 ring / End	32.2	1.2	-25.6	0.0	0.0	2.7
C LC_146	SLN131 column below / Begin	192.5	0.0	2.3	-0.9	6.7	0.2
	SLN88 ring / End	-18.4	1.5	-8.8	0.0	0.0	3.1
	SLN91 ring / End	33.7	1.3	-25.6	0.0	0.0	2.9
C LC_147	SLN131 column below / Begin	192.0	0.0	2.3	-0.9	6.7	0.2
	SLN88 ring / End	-17.6	1.4	-8.8	0.0	0.0	3.0
	SLN91 ring / End	34.4	1.4	-25.6	0.0	0.0	2.9
C LC_148	SLN131 column below / Begin	192.7	0.0	2.3	-0.9	6.7	0.2
	SLN88 ring / End	-19.6	1.5	-8.8	0.0	0.0	3.1
	SLN91 ring / End	32.5	1.2	-25.6	0.0	0.0	2.8
C LC_152	SLN131 column below / Begin	150.8	0.0	5.1	-1.0	11.5	0.1
	SLN88 ring / End	-21.7	1.5	-2.2	0.0	0.0	3.2
	SLN91 ring / End	34.5	1.4	-14.1	0.0	0.0	3.1
C LC_153	SLN131 column below / Begin	192.4	0.0	2.2	-0.9	6.6	0.2
	SLN88 ring / End	-18.8	1.4	-8.8	0.0	0.0	3.0
	SLN91 ring / End	33.4	1.3	-21.0	0.0	0.0	2.9

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	8.6 < 100%	OK
Welds	86.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN89b+SLN89a radial (continuous)-bfl 1	17.7	C LC_152	259.9	0.0	31.5	OK
SLN89b+SLN89a radial (continuous)-tfl 1	17.7	C LC_122	135.6	0.0	0.0	OK
SLN89b+SLN89a radial (continuous)-w 1	10.5	C LC_152	231.0	0.0	31.5	OK
SLN131 column below	10.0	C LC_152	195.0	0.0	0.0	OK
SLN88 ring-bfl 1	12.7	C LC_152	35.0	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_120	37.3	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_147	93.4	0.0	0.0	OK
SLN91 ring-bfl 1	12.7	C LC_146	64.6	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_120	58.0	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	69.9	0.0	0.0	OK
End plate column below SLN131 column below	12.0	C LC_152	116.2	0.0	57.4	OK
End plate SLN91 ring	20.0	C LC_120	39.6	0.0	34.3	OK
Bottom flange plate SLN91 ring	15.0	C LC_152	62.4	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_120	42.1	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_120	36.0	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_120	33.5	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_152	31.7	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_120	23.5	0.0	0.0	OK
End plate SLN88 ring	20.0	C LC_152	101.2	0.0	72.3	OK
Bottom flange plate SLN88 ring	15.0	C LC_152	195.1	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_152	35.5	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_152	53.4	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_120	37.7	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_152	83.1	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_152	27.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

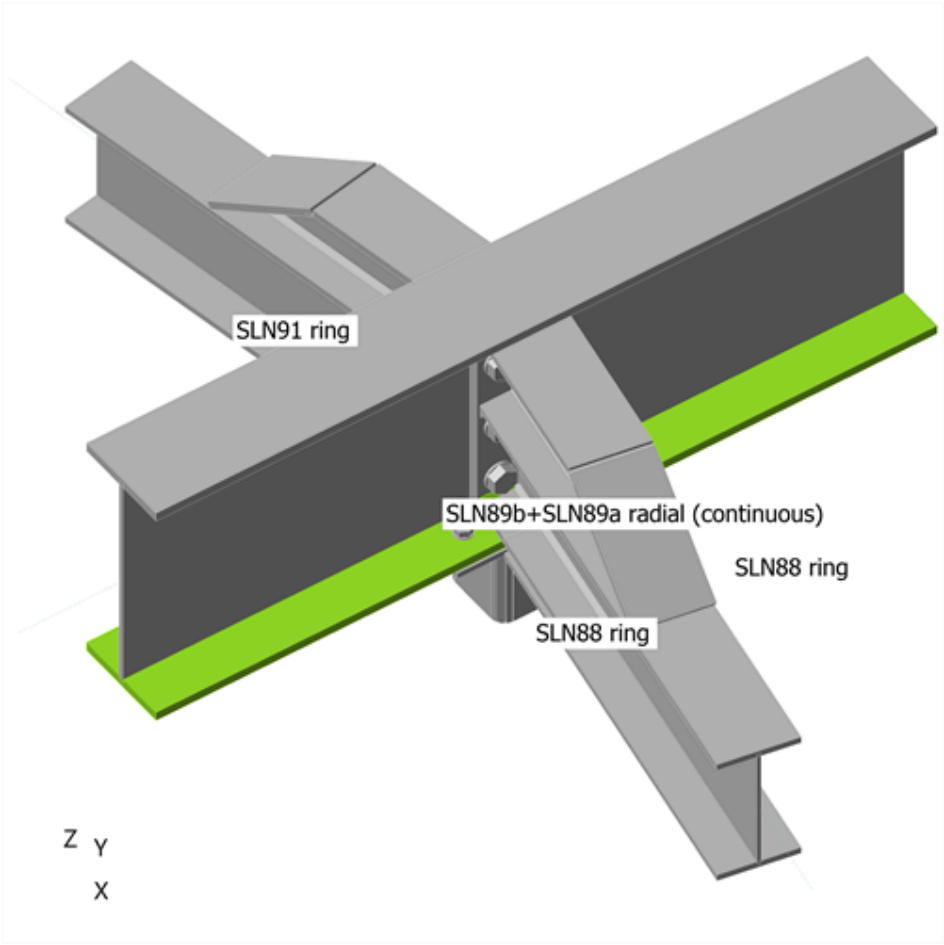
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

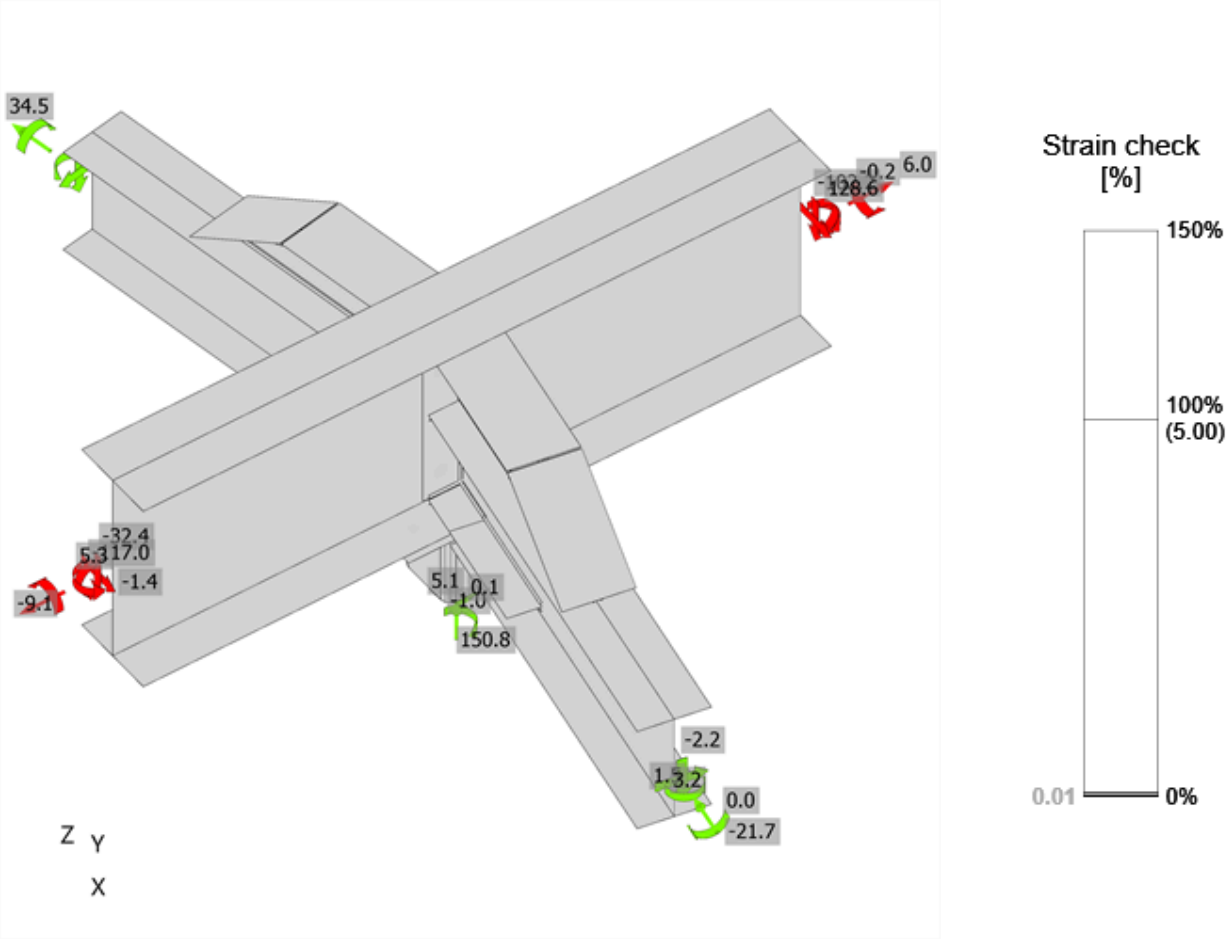
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN131 column below	150	C LC_153	0	5	0.0	OK

Symbol explanation

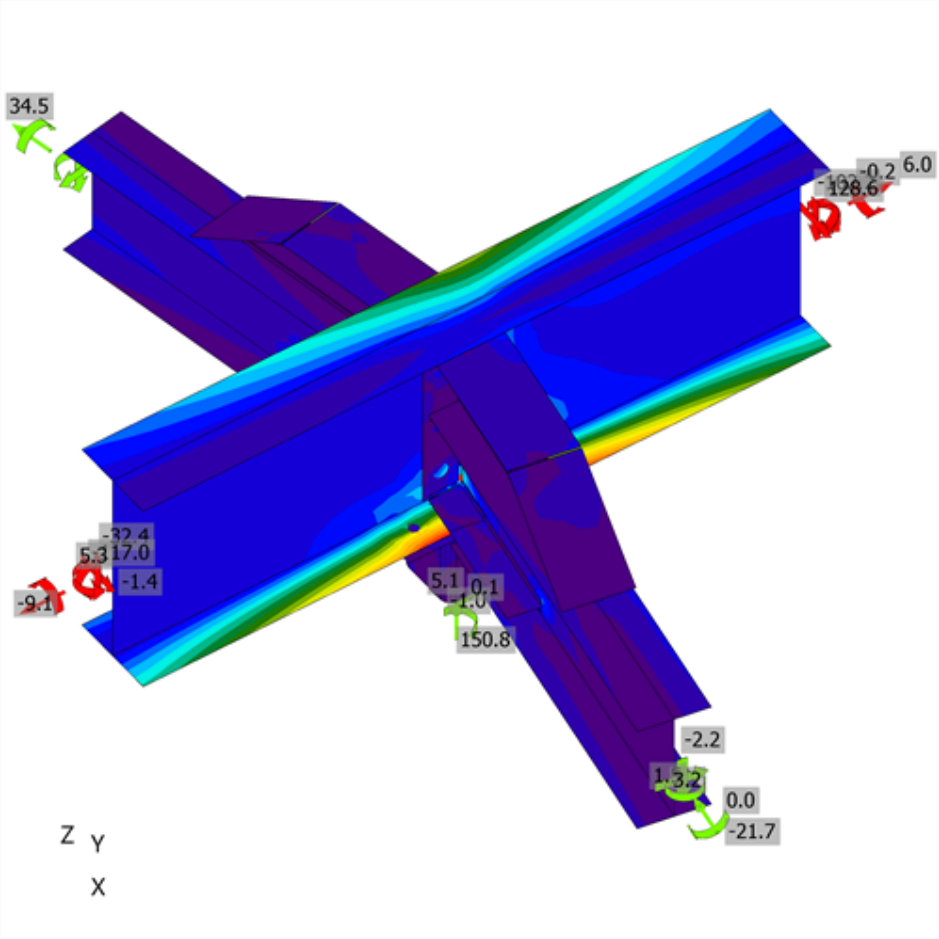
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

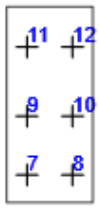


Strain check, C LC_152



Equivalent stress, C LC_152

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.5	7.4	206.4	0.3	7.9	8.1	OK	OK
	B3	M20 8.8 - 1	C LC_152	1.0	7.6	206.4	0.7	8.1	8.6	OK	OK
	B4	M20 8.8 - 1	C LC_120	1.2	3.1	172.0	0.9	3.3	3.9	OK	OK
	B5	M20 8.8 - 1	C LC_152	3.2	3.5	156.4	2.3	3.8	5.4	OK	OK
	B7	M30 8.8 - 2	C LC_152	0.2	1.5	151.1	0.1	0.7	0.7	OK	OK
	B8	M30 8.8 - 2	C LC_152	17.6	3.6	176.5	5.5	3.2	5.6	OK	OK
	B9	M30 8.8 - 2	C LC_152	1.9	1.2	229.5	0.6	0.8	1.0	OK	OK
	B10	M30 8.8 - 2	C LC_152	3.2	0.5	270.9	1.0	0.2	0.9	OK	OK
	B11	M30 8.8 - 2	C LC_120	4.0	2.1	229.5	1.2	1.6	1.9	OK	OK
	B12	M30 8.8 - 2	C LC_152	10.6	1.5	142.1	3.3	1.0	3.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-tfl 1	◢ 10.5	149	C LC_153	24.5	0.0	6.3	-5.3	12.6	6.0	6.0	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 10.5	149	C LC_152	49.6	0.0	15.1	-16.2	22.0	12.3	8.3	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	61.9	0.0	-25.9	-31.8	6.5	15.3	9.6	OK	OK
		◢ 6.0 ◣	208	C LC_147	53.8	0.0	-29.5	25.9	-2.3	13.3	8.9	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_120	53.0	0.0	7.1	18.3	24.1	13.1	11.2	OK	OK
		◢ 6.0 ◣	100	C LC_147	50.3	0.0	14.4	-22.2	-16.7	12.4	9.2	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_120	48.2	0.0	7.7	11.9	24.8	11.9	8.0	OK	OK
		◢ 6.0 ◣	243	C LC_120	73.1	0.0	22.3	-18.1	-35.9	18.1	13.2	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	242	C LC_147	27.5	0.0	3.8	2.5	-15.5	6.8	6.7	OK	OK
		◢ 6.0 ◣	243	C LC_146	44.4	0.0	26.7	-5.5	19.8	11.0	8.2	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_122	37.7	0.0	-22.1	-8.1	15.6	9.3	6.7	OK	OK
		◢ 6.0 ◣	572	C LC_120	83.9	0.0	-7.4	22.3	-42.8	20.7	11.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	208	C LC_152	63.6	0.0	-25.4	-33.5	-3.1	15.7	10.8	OK	OK
		◢ 6.0 ◣	208	C LC_147	80.4	0.0	-41.5	33.1	-21.9	19.9	12.4	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	100	C LC_147	63.3	0.0	16.4	28.2	21.2	15.6	12.6	OK	OK
		◢ 6.0 ◣	100	C LC_146	68.8	0.0	-1.9	6.4	-39.2	17.0	11.3	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	243	C LC_146	25.8	0.0	3.0	5.5	13.7	6.4	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_152	46.3	0.0	-25.4	15.5	16.2	11.4	8.2	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_152	47.5	0.0	-9.0	-12.0	24.1	11.7	7.6	OK	OK
		◢ 6.0 ◣	243	C LC_152	67.0	0.0	-20.1	17.1	-32.7	16.6	10.2	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_120	38.5	0.0	-25.7	-14.0	8.7	9.5	6.6	OK	OK
		◢ 6.0 ◣	572	C LC_120	47.0	0.0	6.6	5.1	-26.4	11.6	7.6	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	149	C LC_152	113.5	0.0	-57.9	-49.7	-26.6	28.0	18.9	OK	OK
		◢ 6.0 ◣	149	C LC_120	112.1	0.0	-49.9	56.2	14.2	27.7	17.5	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	16.1	0.0	7.0	5.6	6.2	4.0	4.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.1	0.0	-2.3	2.0	-5.3	2.5	2.5	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_152	62.4	0.0	-30.9	-27.2	-15.5	15.4	10.3	OK	OK
		◢ 6.0 ◣	149	C LC_152	51.7	0.0	-18.5	22.1	17.0	12.8	8.3	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_152	15.0	0.0	-2.1	-3.2	-7.9	3.7	3.7	OK	OK
		◢ 6.0 ◣	342	C LC_152	17.7	0.0	2.0	-3.1	-9.7	4.4	4.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_120	54.6	0.0	-6.8	-28.1	13.7	13.5	9.1	OK	OK
		◢ 6.0 ◣	271	C LC_120	67.8	0.0	-10.1	-11.2	-37.1	16.8	10.6	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	194.2	0.0	-56.1	-90.1	-58.5	48.0	30.3	OK	OK
		◢ 6.0 ◣	149	C LC_152	250.4	0.0	-119.9	85.8	93.5	61.9	36.0	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_120	22.6	0.0	4.8	9.1	8.9	5.6	5.6	OK	OK
		◢ 6.0 ◣	145	C LC_152	26.2	0.0	7.8	-4.6	13.7	6.5	6.5	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	137.3	0.0	-67.7	-45.4	-51.9	33.9	20.5	OK	OK
		◢ 6.0 ◣	149	C LC_152	97.0	0.0	-24.9	47.3	26.3	24.0	15.0	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_152	17.0	0.0	-2.1	-8.2	-5.3	4.2	4.2	OK	OK
		◢ 6.0 ◣	342	C LC_152	29.9	0.0	-2.9	1.0	-17.2	7.4	7.4	OK	OK
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	60.3	0.0	-13.2	-30.0	15.9	14.9	10.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	43.5	0.0	3.2	-20.0	-15.1	10.8	8.4	OK	OK
End plate column below SLN131 column below	SLN131 column below	◢ 4.2 ◣	520	C LC_120	319.8	0.0	-160.9	-152.3	-47.6	79.0	42.7	OK	OK
		◢ 4.2 ◣	521	C LC_152	348.9	0.0	-166.4	172.8	38.5	86.2	46.6	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_147	200.9	0.0	73.8	96.3	-48.7	49.6	30.4	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_147	240.0	0.0	103.3	-94.3	82.2	59.3	35.2	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_120	23.4	0.0	-21.6	-5.0	1.1	7.0	6.4	OK	OK
		◢ 6.0 ◣	147	C LC_152	16.7	0.0	-3.1	8.9	3.1	4.1	4.1	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_152	26.7	0.0	-24.6	-5.9	1.1	7.9	6.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_152	19.9	0.0	-3.7	10.7	3.5	4.9	4.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-B-260
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring

Name	SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

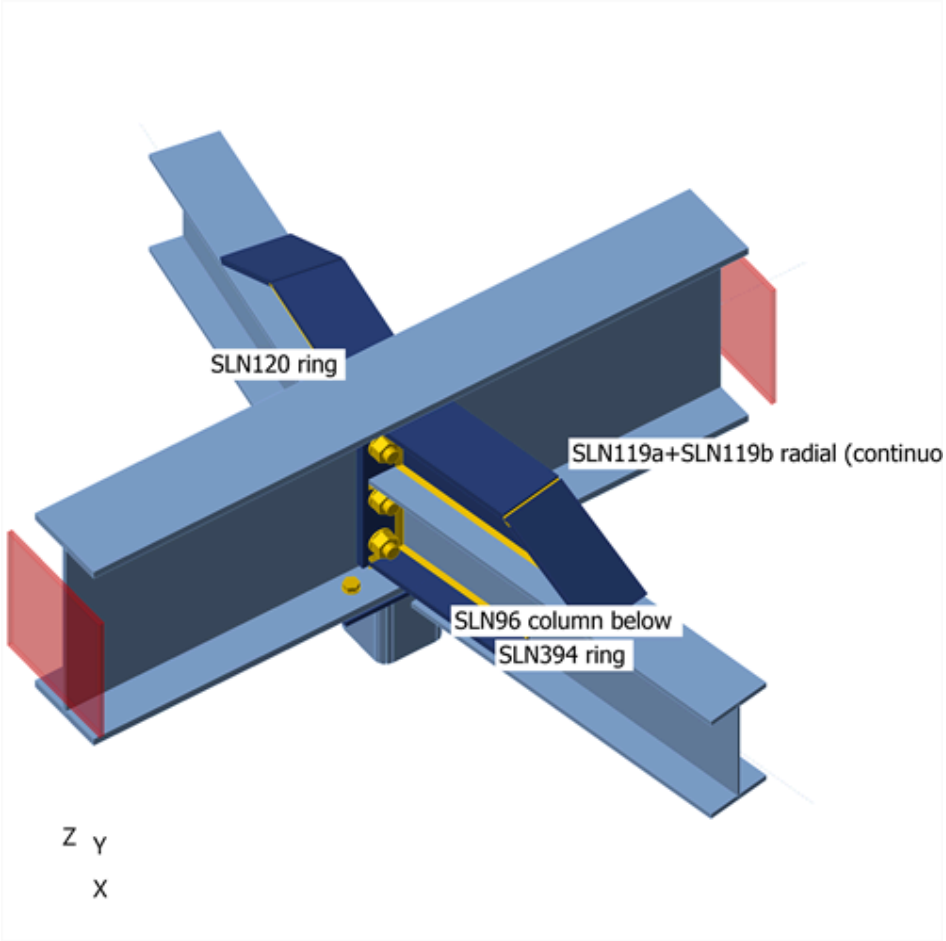
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN119a+SLN119b radial (continuous)	1 - UB457x191x89	-100.0	0.0	0.0	0	0	0
SLN96 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN120 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102
SLN394 ring	3 - UB254x146x43	0.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN119a+SLN119b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN119a+SLN119b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN96 column below / begin		Position	0
SLN120 ring / end		Position	0
SLN394 ring / end		Position	0



SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN96 column below / Begin	165.5	0.1	1.3	1.0	2.1	0.0
	SLN120 ring / End	33.2	-1.7	-25.8	0.0	0.0	-3.5
	SLN394 ring / End	-15.5	-1.5	-9.0	0.0	0.0	-3.1
C LC_104	SLN96 column below / Begin	161.5	0.1	1.4	1.0	2.4	0.0
	SLN120 ring / End	29.2	-1.6	-24.2	0.0	0.0	-3.4
	SLN394 ring / End	-19.1	-1.5	-8.5	0.0	0.0	-3.2
C LC_120	SLN96 column below / Begin	168.4	0.1	1.4	1.0	2.3	0.0
	SLN120 ring / End	31.6	-1.7	-25.8	0.0	0.0	-3.5
	SLN394 ring / End	-18.0	-1.5	-9.0	0.0	0.0	-3.3
C LC_122	SLN96 column below / Begin	163.5	0.0	1.7	0.9	2.7	0.0
	SLN120 ring / End	31.2	-1.8	-25.8	0.0	0.0	-3.5
	SLN394 ring / End	-16.2	-1.5	-9.0	0.0	0.0	-3.2
C LC_145	SLN96 column below / Begin	165.3	0.1	1.3	1.0	2.1	0.0
	SLN120 ring / End	33.3	-1.7	-25.8	0.0	0.0	-3.4
	SLN394 ring / End	-15.4	-1.4	-9.0	0.0	0.0	-3.1
C LC_146	SLN96 column below / Begin	165.7	0.1	1.3	1.0	2.1	0.0
	SLN120 ring / End	33.1	-1.7	-25.8	0.0	0.0	-3.5
	SLN394 ring / End	-15.6	-1.5	-9.0	0.0	0.0	-3.2
C LC_147	SLN96 column below / Begin	165.7	0.1	1.3	1.0	2.0	0.0
	SLN120 ring / End	32.2	-1.6	-25.8	0.0	0.0	-3.4
	SLN394 ring / End	-16.3	-1.5	-9.0	0.0	0.0	-3.1
C LC_148	SLN96 column below / Begin	165.3	0.1	1.3	1.0	2.2	0.0
	SLN120 ring / End	34.2	-1.8	-25.8	0.0	0.0	-3.5
	SLN394 ring / End	-14.8	-1.4	-9.0	0.0	0.0	-3.2
C LC_152	SLN96 column below / Begin	114.4	0.1	4.4	1.1	7.0	0.0
	SLN120 ring / End	34.1	-1.8	-13.9	0.0	0.0	-3.7
	SLN394 ring / End	-19.3	-1.6	-2.2	0.0	0.0	-3.5
C LC_153	SLN96 column below / Begin	164.7	0.1	1.2	1.0	2.1	0.0
	SLN120 ring / End	33.2	-1.7	-20.9	0.0	0.0	-3.5
	SLN394 ring / End	-15.7	-1.5	-9.0	0.0	0.0	-3.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.7 < 100%	OK
Welds	65.4 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN119a+SLN119b radial (continuous)-bfl 1	17.7	C LC_152	240.9	0.0	21.1	OK
SLN119a+SLN119b radial (continuous)-tfl 1	17.7	C LC_120	124.4	0.0	0.0	OK
SLN119a+SLN119b radial (continuous)-w 1	10.5	C LC_152	213.5	0.0	21.1	OK
SLN96 column below	10.0	C LC_152	128.1	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_146	66.9	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_120	61.2	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	73.8	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_152	38.3	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_120	38.4	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_148	95.9	0.0	0.0	OK
End plate column below SLN96 column below	12.0	C LC_122	84.1	0.0	40.7	OK
End plate SLN394 ring	20.0	C LC_152	107.8	0.0	77.0	OK
Bottom flange plate SLN394 ring	15.0	C LC_152	208.6	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_152	58.1	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_152	38.8	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_120	38.9	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_152	90.6	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_152	29.4	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_148	42.5	0.0	40.1	OK
Bottom flange plate SLN120 ring	15.0	C LC_152	73.7	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	38.5	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_148	45.1	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_122	36.1	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_152	39.9	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_152	27.9	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

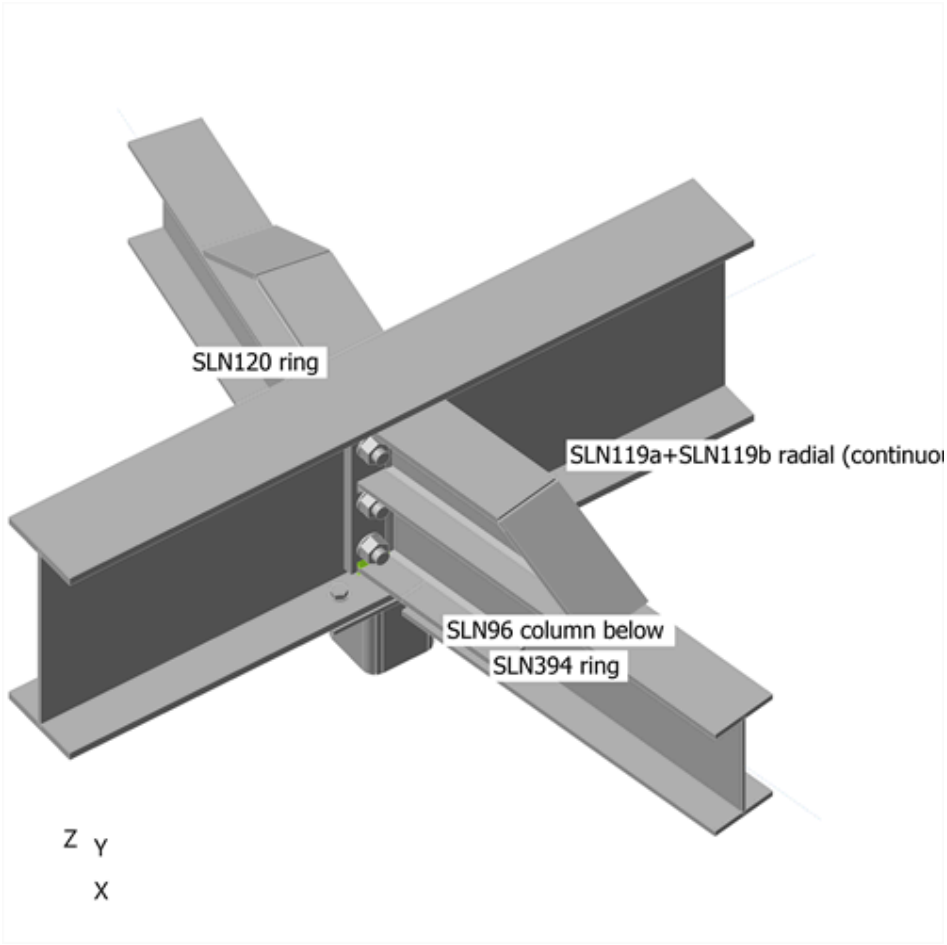
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

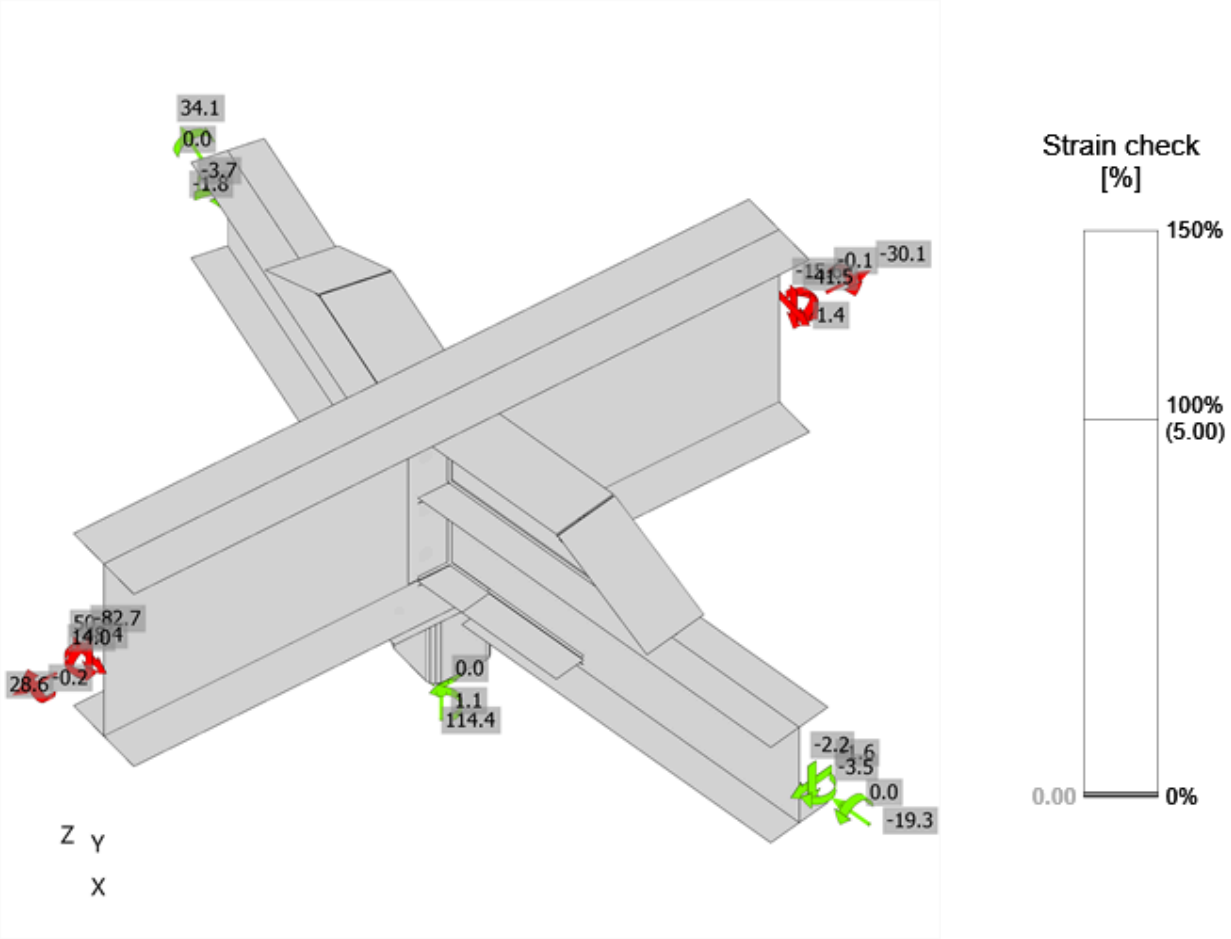
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN96 column below	150	C LC_153	0	5	0.0	OK

Symbol explanation

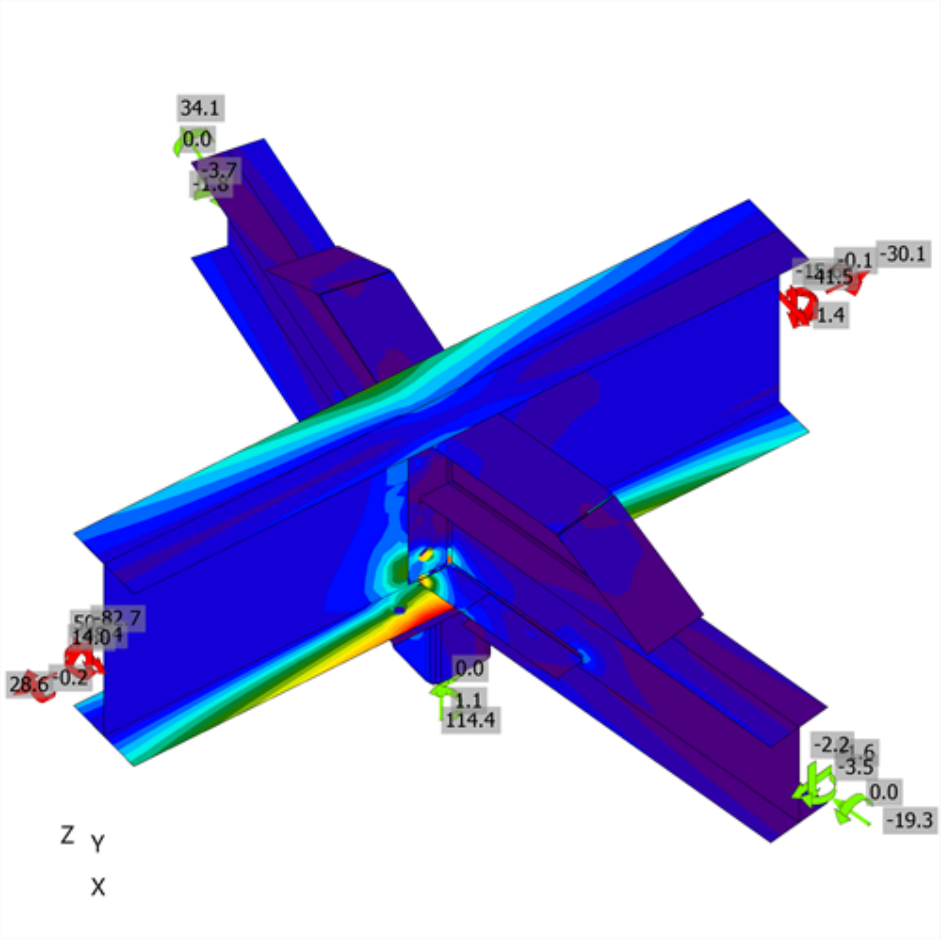
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

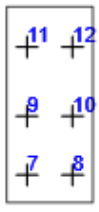


Strain check, C LC_152



Equivalent stress, C LC_152

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	2.2	3.8	156.4	1.5	4.1	5.2	OK	OK
	B3	M20 8.8 - 1	C LC_152	0.7	2.7	172.0	0.5	2.9	3.3	OK	OK
	B4	M20 8.8 - 1	C LC_152	1.2	6.4	206.4	0.8	6.8	7.4	OK	OK
	B5	M20 8.8 - 1	C LC_120	0.5	7.0	206.4	0.3	7.5	7.7	OK	OK
	B7	M30 8.8 - 2	C LC_146	1.0	1.4	208.4	0.3	0.7	0.9	OK	OK
	B8	M30 8.8 - 2	C LC_152	19.0	3.1	208.4	5.9	2.0	5.6	OK	OK
	B9	M30 8.8 - 2	C LC_152	2.3	0.9	229.5	0.7	0.5	0.9	OK	OK
	B10	M30 8.8 - 2	C LC_152	2.8	0.3	270.9	0.9	0.1	0.8	OK	OK
	B11	M30 8.8 - 2	C LC_120	4.6	1.9	146.3	1.4	1.3	1.9	OK	OK
	B12	M30 8.8 - 2	C LC_152	11.4	1.5	146.4	3.5	0.8	3.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 10.5	149	C LC_152	59.8	0.0	17.6	-18.8	-27.2	14.8	10.8	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-tfl 1	◢ 10.5	149	C LC_120	24.6	0.0	5.0	-4.5	-13.1	6.1	6.1	OK	OK
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	208	C LC_148	73.3	0.0	-39.2	-30.2	19.2	18.1	11.6	OK	OK
		◢ 6.0 ◣	208	C LC_152	66.7	0.0	-25.6	34.7	7.8	16.5	11.3	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	100	C LC_148	70.2	0.0	-2.8	-7.8	39.7	17.3	11.4	OK	OK
		◢ 6.0 ◣	100	C LC_148	62.2	0.0	14.9	-27.7	-21.1	15.4	12.7	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_152	49.7	0.0	-9.5	-12.5	-25.2	12.3	8.4	OK	OK
		◢ 6.0 ◣	243	C LC_152	71.2	0.0	-21.6	18.6	34.5	17.6	13.0	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_148	31.2	0.0	3.8	6.7	-16.6	7.7	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_146	50.8	0.0	14.6	-11.6	25.6	12.6	8.9	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_120	48.0	0.0	7.1	-5.0	26.9	11.9	7.8	OK	OK
		◢ 6.0 ◣	572	C LC_120	38.9	0.0	-26.4	14.3	-8.3	9.6	6.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	51.8	0.0	-28.4	-25.0	-0.8	12.8	8.7	OK	OK
		◢ 6.0 ◣	208	C LC_148	60.9	0.0	-26.2	31.5	-3.7	15.0	9.5	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_148	46.9	0.0	12.8	20.4	16.2	11.6	9.3	OK	OK
		◢ 6.0 ◣	100	C LC_148	57.2	0.0	7.4	-20.0	-25.9	14.1	11.8	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_148	27.4	0.0	3.6	2.2	15.5	6.8	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_148	47.3	0.0	29.4	-6.8	-20.2	11.7	7.9	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	52.3	0.0	8.0	12.9	-26.9	12.9	7.7	OK	OK
		◢ 6.0 ◣	243	C LC_148	79.5	0.0	24.1	-19.2	39.3	19.6	11.8	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_122	87.2	0.0	-5.3	-22.4	45.0	21.6	11.4	OK	OK
		◢ 6.0 ◣	572	C LC_122	38.1	0.0	-25.7	8.6	-13.8	9.4	6.6	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	203.4	0.0	-57.5	-95.4	59.9	50.3	30.4	OK	OK
		◢ 6.0 ◣	149	C LC_152	264.5	0.0	-128.0	90.1	-98.7	65.4	37.5	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_152	29.1	0.0	8.6	5.2	-15.2	7.2	7.1	OK	OK
		◢ 6.0 ◣	145	C LC_148	24.9	0.0	5.1	-9.9	-10.0	6.2	5.8	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	149.4	0.0	-74.0	-48.9	56.8	36.9	18.8	OK	OK
		◢ 6.0 ◣	149	C LC_152	104.7	0.0	-26.3	51.4	-27.9	25.9	13.5	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_152	32.2	0.0	-3.2	-0.9	18.5	8.0	7.6	OK	OK
		◢ 6.0 ◣	342	C LC_152	17.5	0.0	-2.0	8.4	5.5	4.3	4.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_152	45.9	0.0	6.0	24.9	8.3	11.3	8.6	OK	OK
		◢ 6.0 ◣	271	C LC_120	62.1	0.0	-13.5	31.1	-16.0	15.3	10.6	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	136.4	0.0	-68.5	-60.8	30.7	33.7	20.7	OK	OK
		◢ 6.0 ◣	149	C LC_152	136.9	0.0	-61.2	68.7	-16.9	33.8	19.0	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_152	11.9	0.0	2.8	5.5	-3.9	2.9	2.9	OK	OK
		◢ 6.0 ◣	145	C LC_122	17.1	0.0	7.5	-6.2	-6.3	4.2	4.2	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	78.2	0.0	-38.5	-34.1	19.5	19.3	10.7	OK	OK
		◢ 6.0 ◣	149	C LC_152	64.2	0.0	-23.2	27.7	-20.7	15.9	8.8	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_152	21.4	0.0	2.5	3.7	11.7	5.3	5.3	OK	OK
		◢ 6.0 ◣	342	C LC_152	17.6	0.0	-2.6	3.8	9.3	4.3	4.3	OK	OK
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_122	71.4	0.0	-9.3	15.0	38.0	17.6	11.1	OK	OK
		◢ 6.0 ◣	271	C LC_122	61.5	0.0	-8.1	32.3	-14.0	15.2	10.0	OK	OK
End plate column below SLN96 column below	SLN96 column below	◢ 4.2 ◣	520	C LC_120	217.3	0.0	-109.9	-108.1	-4.8	53.7	30.2	OK	OK
		◢ 4.2 ◣	520	C LC_152	227.1	0.0	-113.7	112.4	16.0	56.1	30.7	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_148	227.8	0.0	84.6	-97.4	73.6	56.3	34.1	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	166.5	0.0	89.9	70.1	-40.4	41.1	25.5	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_152	28.5	0.0	-26.3	-6.2	-1.1	8.5	7.0	OK	OK
		◢ 6.0 ◣	147	C LC_152	21.2	0.0	-3.9	11.4	-3.8	5.2	5.2	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_152	27.5	0.0	-25.5	-5.7	-1.6	8.2	6.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_152	20.4	0.0	-3.7	10.9	-4.0	5.0	5.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	L2-R8-B-280
Project number	
Author	Claude for Wing
Description	
Date	11/3/2026
Code	EN

Material

Steel	S 275

Project item SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring

Name	SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

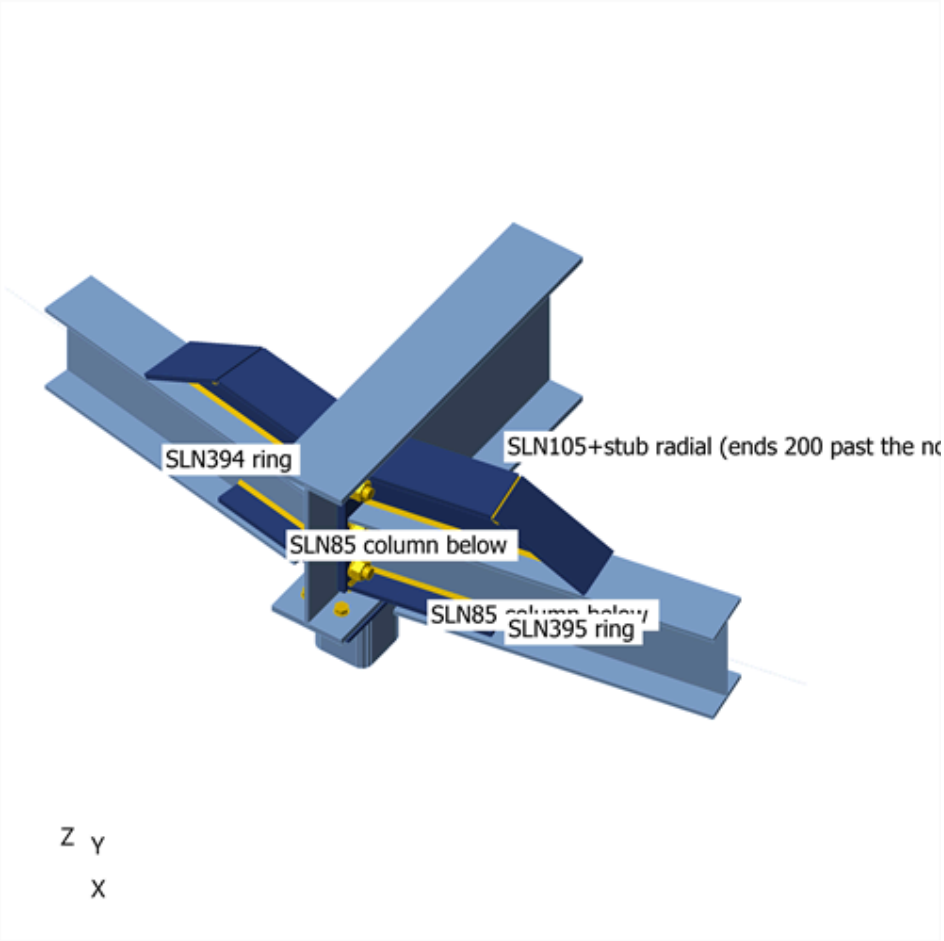
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN105+stub radial (ends 200 past the node)	1 - UB457x191x89	-80.0	0.0	0.0	200	0	0
SLN85 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN394 ring	3 - UB254x146x43	180.0	0.0	0.0	0	0	-102
SLN395 ring	3 - UB254x146x43	20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN105+stub radial (ends 200 past the node) / begin		Position	0
SLN85 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN394 ring / end		Position	0
SLN395 ring / end		Position	0



SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_102	SLN105+stub radial (ends 200 past the node) / Begin	-0.3	-0.3	-47.9	0.0	-8.2	0.9
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-15.5	-1.5	-9.1	0.0	0.0	-1.2
	SLN395 ring / End	-15.5	0.1	-8.7	0.0	0.0	0.2
C LC_103	SLN105+stub radial (ends 200 past the node) / Begin	-1.5	-0.1	-44.9	0.0	-7.4	0.7
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-18.2	-1.4	-8.5	0.0	0.0	-1.1
	SLN395 ring / End	-18.4	0.2	-8.2	0.0	0.0	0.4
C LC_104	SLN105+stub radial (ends 200 past the node) / Begin	-1.8	-0.3	-45.0	0.0	-7.2	0.8
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-19.0	-1.5	-8.5	0.0	0.0	-1.2
	SLN395 ring / End	-19.3	0.2	-8.2	0.0	0.0	0.4
C LC_106	SLN105+stub radial (ends 200 past the node) / Begin	-0.9	-0.3	-45.2	0.0	-7.5	0.9
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-16.9	-1.6	-8.5	0.0	0.0	-1.3
	SLN395 ring / End	-16.8	0.1	-8.2	0.0	0.0	0.3
C LC_112	SLN105+stub radial (ends 200 past the node) / Begin	-2.6	-0.2	-31.2	0.0	-4.5	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-16.7	-1.1	-6.0	0.0	0.0	-0.9
	SLN395 ring / End	-17.5	0.3	-5.7	0.0	0.0	0.5
C LC_120	SLN105+stub radial (ends 200 past the node) / Begin	-1.2	-0.3	-47.8	0.0	-7.8	0.9
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-17.9	-1.5	-9.1	0.0	0.0	-1.2
	SLN395 ring / End	-18.2	0.1	-8.7	0.0	0.0	0.3
C LC_121	SLN105+stub radial (ends 200 past the node) / Begin	-0.9	-0.3	-48.0	0.0	-8.5	0.9
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-17.6	-1.4	-9.1	0.0	0.0	-1.2
	SLN395 ring / End	-17.5	0.1	-8.7	0.0	0.0	0.2
C LC_146	SLN105+stub radial (ends 200 past the node) / Begin	-0.3	-0.4	-47.9	0.0	-8.2	1.0
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-15.5	-1.5	-9.1	0.0	0.0	-1.2
	SLN395 ring / End	-15.4	0.1	-8.7	0.0	0.0	0.2
C LC_151	SLN105+stub radial (ends 200 past the node) / Begin	1.9	-0.2	-47.9	0.0	-10.4	0.7
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-13.0	-1.1	-9.1	0.0	0.0	-0.9
	SLN395 ring / End	-12.9	0.0	-8.7	0.0	0.0	0.2
C LC_154	SLN105+stub radial (ends 200 past the node) / Begin	2.0	-0.2	-48.7	0.0	-10.6	0.7
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-13.0	-1.1	-9.1	0.0	0.0	-0.9
	SLN395 ring / End	-12.9	0.0	-8.7	0.0	0.0	0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	1.8 < 100%	OK
Welds	27.4 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN105+stub radial (ends 200 past the node)-bfl 1	17.7	C LC_120	42.3	0.0	14.6	OK
SLN105+stub radial (ends 200 past the node)-tfl 1	17.7	C LC_102	22.3	0.0	0.0	OK
SLN105+stub radial (ends 200 past the node)-w 1	10.5	C LC_120	55.5	0.0	14.6	OK
SLN85 column below	10.0	C LC_120	58.5	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_154	15.7	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_120	20.4	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_106	19.2	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_154	14.4	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_120	21.0	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_120	18.9	0.0	0.0	OK
End plate column below SLN85 column below	12.0	C LC_120	42.7	0.0	16.0	OK
End plate SLN395 ring	20.0	C LC_146	10.3	0.0	6.9	OK
Bottom flange plate SLN395 ring	15.0	C LC_146	17.0	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_112	4.5	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_112	6.2	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_120	15.4	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_146	9.3	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_120	6.3	0.0	0.0	OK
End plate SLN394 ring	20.0	C LC_146	6.6	0.0	14.9	OK
Bottom flange plate SLN394 ring	15.0	C LC_106	26.2	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_106	12.8	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_106	8.1	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_120	15.6	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_146	8.9	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_106	7.2	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

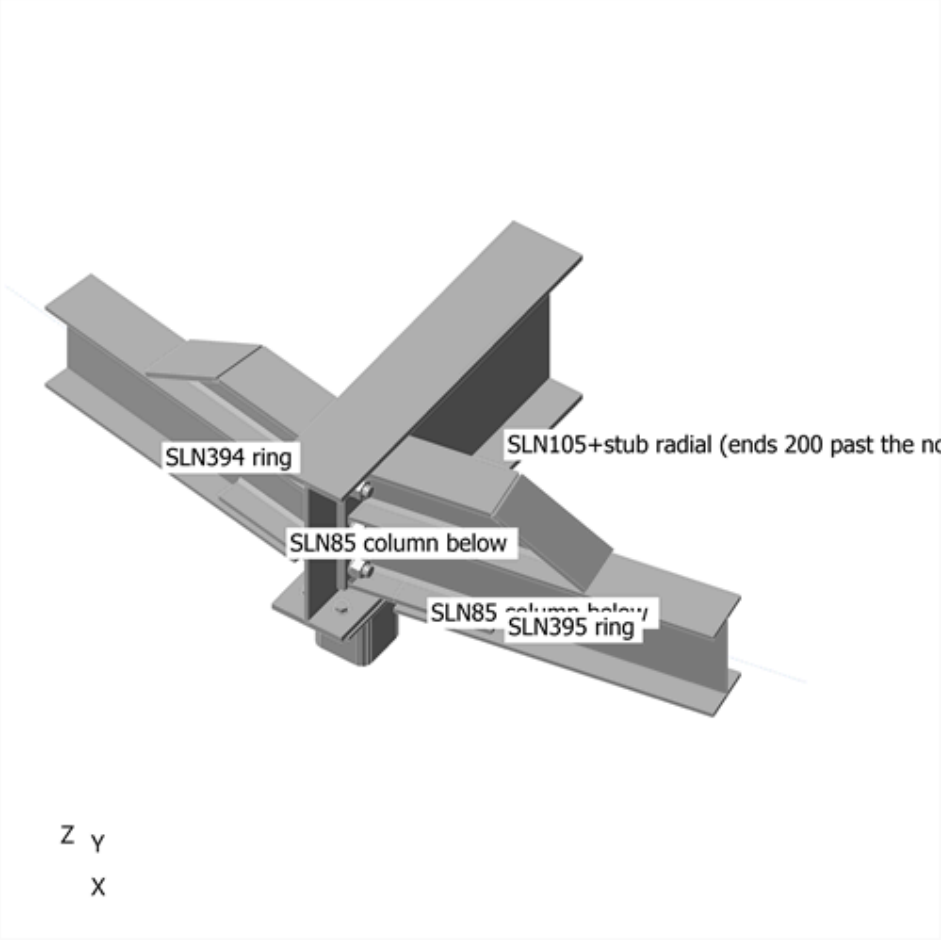
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

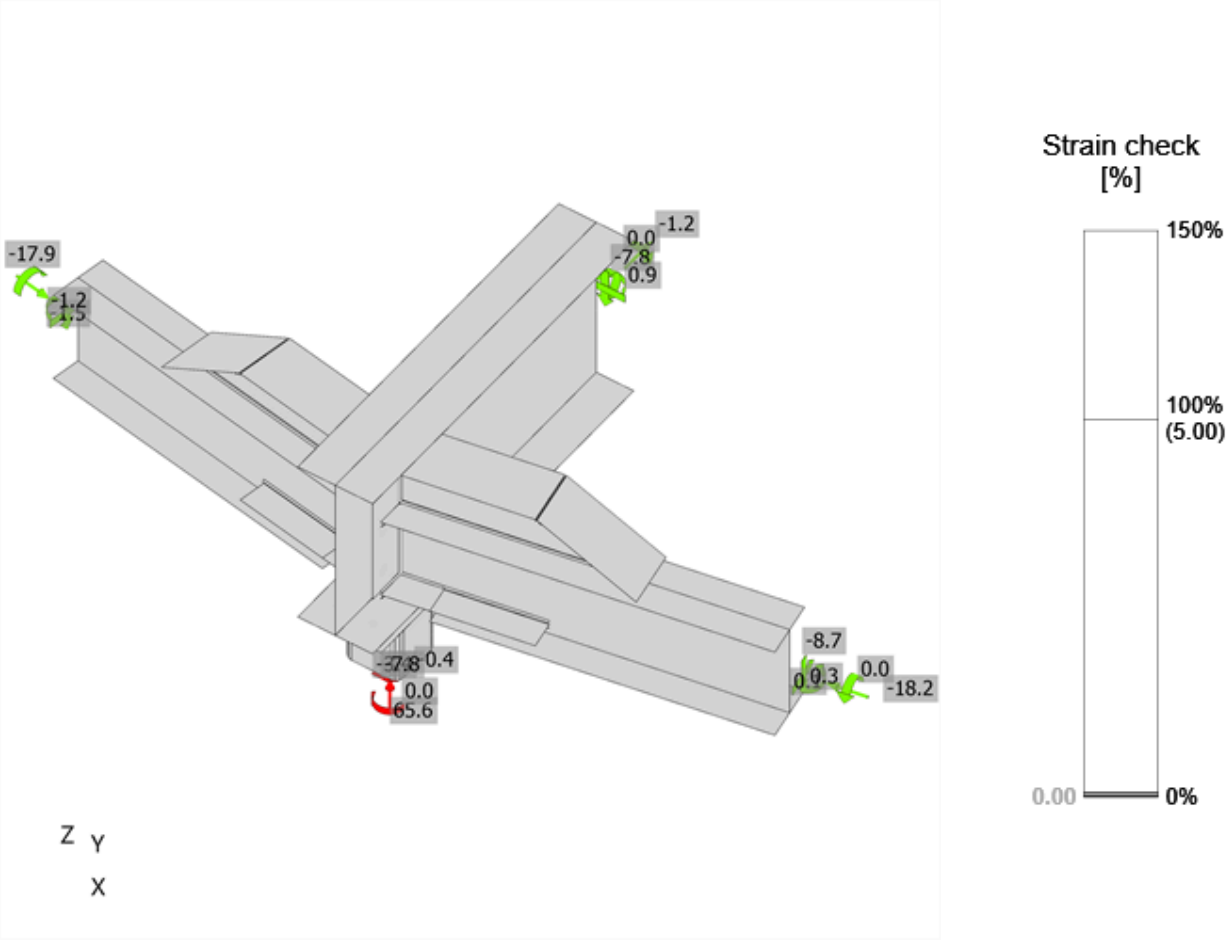
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN85 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

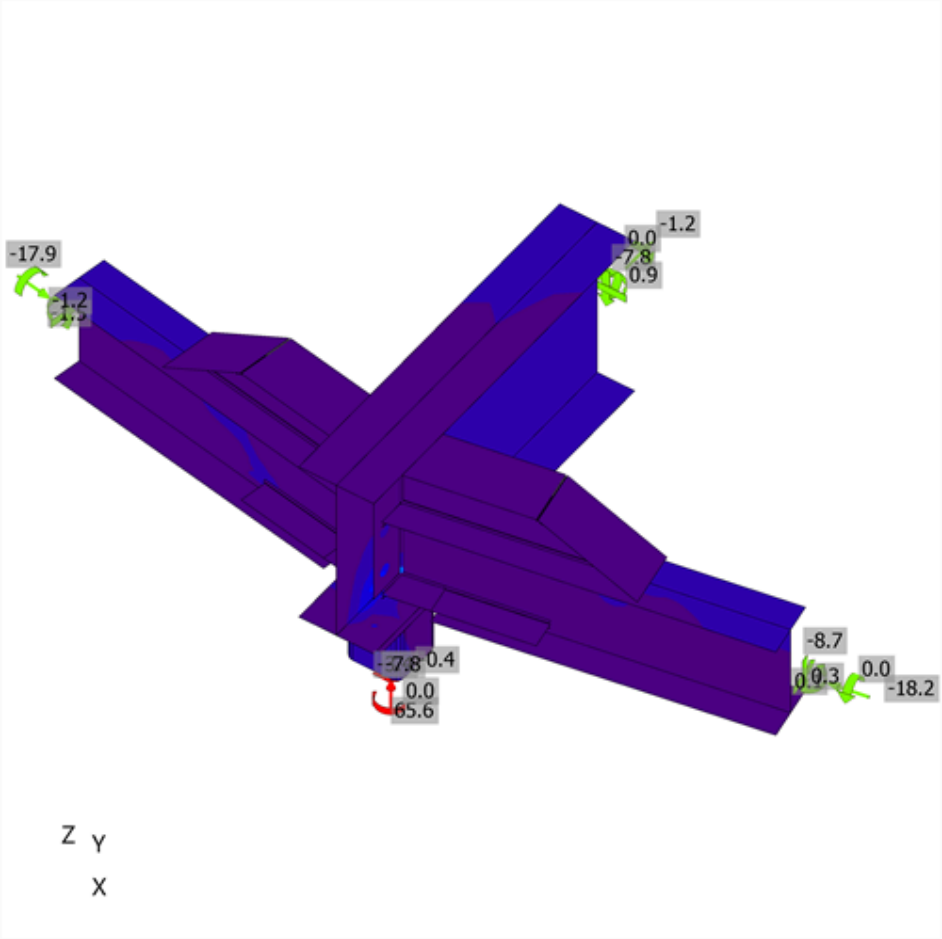
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

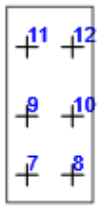


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_154	0.2	1.6	206.4	0.1	1.7	1.8	OK	OK
	B3	M20 8.8 - 1	C LC_154	0.4	1.1	156.4	0.3	1.1	1.4	OK	OK
	B4	M20 8.8 - 1	C LC_154	0.6	1.1	156.4	0.4	1.2	1.5	OK	OK
	B5	M20 8.8 - 1	C LC_154	0.1	1.7	206.4	0.1	1.8	1.8	OK	OK
	B7	M30 8.8 - 2	C LC_104	0.1	0.3	213.9	0.0	0.2	0.1	OK	OK
	B8	M30 8.8 - 2	C LC_120	1.5	1.2	176.5	0.5	1.3	0.9	OK	OK
	B9	M30 8.8 - 2	C LC_120	0.2	0.5	229.5	0.1	0.4	0.3	OK	OK
	B10	M30 8.8 - 2	C LC_120	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B11	M30 8.8 - 2	C LC_120	0.8	0.7	229.5	0.2	0.6	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_146	1.8	0.2	146.8	0.5	0.1	0.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-tfl 1	◢ 10.5	149	C LC_106	18.5	0.0	-8.8	9.3	-0.5	4.6	4.6	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-tfl 1	◢ 10.5	149	C LC_104	14.1	0.0	-6.7	7.0	-1.6	3.5	3.5	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	207	C LC_121	22.9	0.0	-2.4	2.4	12.9	5.7	5.7	OK	OK
		◢ 6.0 ◣	208	C LC_154	13.0	0.0	-0.3	2.8	-7.0	3.2	3.2	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_146	10.1	0.0	0.2	-0.3	5.8	2.5	2.5	OK	OK
		◢ 6.0 ◣	100	C LC_146	12.8	0.0	5.0	-6.7	-1.1	3.2	3.2	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_104	4.4	0.0	-0.7	-0.5	-2.5	1.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_112	4.8	0.0	-2.6	1.8	1.5	1.2	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_112	5.7	0.0	-0.9	-1.6	2.8	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_112	9.7	0.0	-3.3	2.6	-4.6	2.4	0.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_121	25.6	0.0	-6.1	-5.1	13.4	6.3	5.9	OK	OK
		◢ 6.0 ◣	572	C LC_120	34.8	0.0	-8.6	9.9	-16.8	8.6	6.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	207	C LC_154	14.7	0.0	1.1	-0.5	8.5	3.6	3.6	OK	OK
		◢ 6.0 ◣	207	C LC_154	20.8	0.0	-0.5	-1.6	-11.9	5.1	5.1	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	100	C LC_102	10.6	0.0	5.2	5.1	1.5	2.6	2.6	OK	OK
		◢ 6.0 ◣	100	C LC_146	12.9	0.0	0.8	-3.8	-6.4	3.2	3.2	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_104	10.4	0.0	-1.7	-2.7	-5.3	2.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_104	16.9	0.0	-5.4	4.4	8.1	4.2	4.2	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_146	7.1	0.0	0.5	1.5	-3.8	1.8	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_106	13.6	0.0	4.3	-3.1	6.7	3.4	3.4	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_120	35.0	0.0	-4.6	-8.1	18.3	8.6	6.6	OK	OK
		◢ 6.0 ◣	572	C LC_121	27.6	0.0	-10.6	7.3	-12.7	6.8	6.0	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	19.6	0.0	-10.2	-9.5	-2.0	4.8	4.8	OK	OK
		◢ 6.0 ◣	149	C LC_146	24.5	0.0	-10.2	7.2	-10.7	6.1	6.1	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_106	4.5	0.0	-0.2	0.2	-2.6	1.1	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_120	9.3	0.0	-1.4	0.6	-5.3	2.3	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_146	14.7	0.0	-7.5	-4.1	6.1	3.6	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_146	9.2	0.0	-1.5	4.8	-2.1	2.3	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_146	2.3	0.0	0.4	0.4	1.2	0.6	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_106	1.7	0.0	0.8	-0.4	-0.8	0.4	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_120	18.8	0.0	-2.2	-4.0	10.0	4.6	4.6	OK	OK
		◢ 6.0 ◣	271	C LC_120	28.5	0.0	-5.8	3.9	-15.6	7.0	6.2	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_106	42.1	0.0	-19.9	-20.9	4.5	10.4	7.0	OK	OK
		◢ 6.0 ◣	149	C LC_106	44.8	0.0	-22.8	21.7	-4.7	11.1	7.2	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_120	7.8	0.0	-1.4	-1.2	4.2	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_154	5.6	0.0	-0.3	0.5	-3.2	1.4	0.0	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_146	18.3	0.0	-8.9	-8.1	4.5	4.5	4.5	OK	OK
		◢ 6.0 ◣	149	C LC_106	16.1	0.0	-6.2	6.9	-5.1	4.0	4.0	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_106	5.4	0.0	1.7	1.3	2.7	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_106	4.4	0.0	-0.6	0.9	2.3	1.1	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_120	25.6	0.0	-3.3	1.6	14.6	6.3	6.2	OK	OK
		◢ 6.0 ◣	271	C LC_120	26.9	0.0	-4.9	9.8	-11.7	6.6	6.2	OK	OK
End plate column below SLN85 column below	SLN85 column below	◢ 4.2 ◣	520	C LC_120	106.6	0.0	-52.9	-53.0	-6.4	26.3	14.2	OK	OK
		◢ 4.2 ◣	521	C LC_120	110.8	0.0	-54.9	54.8	8.8	27.4	14.6	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_154	60.0	0.0	-2.6	-4.5	34.3	14.8	9.2	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_154	59.0	0.0	-1.4	3.1	-33.9	14.6	9.1	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_146	3.4	0.0	-2.1	1.5	0.2	0.8	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_106	1.6	0.0	-0.3	0.2	-0.9	0.4	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_106	7.2	0.0	-6.6	-1.6	-0.3	2.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_106	5.2	0.0	-0.9	2.8	-0.9	1.3	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5